

where λ_s = latent heat of 1 lb of pure water from solution at the temperature t and pressure p_s

λ_w = latent heat of 1 lb of pure water at the temperature t'_w but at p_s , the same pressure as t

t'_s, t'_w = boiling points of the solution and water at the same pressure, p_s , °R

$\Delta t'_s / \Delta t'_w$ = rate of change of the two boiling-point curves over the same pressure range

According to Othmer's method and based on the Clausius-Clapeyron equation,

$$\frac{\lambda_s}{\lambda_w} = \frac{d \log p_s}{d \log p_w} \quad (14.6)$$

where p_s and p_w are the respective absolute vapor pressures of the solution and pure water over an identical range of temperatures.

The BPR can be computed but only for dilute solutions which are relatively ideal. For real solutions the data on boiling-point elevation must be obtained experimentally by measuring the vapor-pressure curve for a given concentration at two different temperatures. Additional determinations can be made for other concentrations if more than a single effect is used.

Fluid Properties. In the power-plant evaporator the water-softening process is modified in different localities so that the evaporator feed composition causes a minimum of foaming and other operating difficulties. In the chemical evaporator the residue, a concentrated solution, is the desired product, and usually no adjustment can be made to the solution to prevent foaming or to eliminate the deposition of scale. This must be taken into consideration in the design of the equipment. Furthermore, concentrated solutions, as discussed in Chap. 7, produce liquors of high viscosity. Particularly since boiling is a combination of vaporization and free convection the overall coefficient of heat transfer is a function of both the concentration and the temperature at which evaporation occurs. The influence of viscosity may be so great that a negligible Grashof group, $D^3 \rho^2 g \beta \Delta t / \mu^2$, results for evaporators operating with natural circulation. Under these circumstances the liquor cannot be relied upon to circulate very rapidly about the heating element and it is necessary to use *forced circulation* instead of *natural circulation* as presumed heretofore.

CHEMICAL EVAPORATORS

Chemical evaporators fall into two classes: natural circulation and forced circulation. Natural-circulation evaporators are used singly or in multiple effect for the simpler evaporation requirements. Forced-

circulation evaporators are used for viscous, salting, and scale-forming solutions. Natural-circulation evaporators fall into four main classes:

1. Horizontal tube
2. Calandria vertical tube
3. Basket vertical tube
4. Long tube vertical

The discussion of evaporator design in this chapter deals only with those which are designed on a basis of flux and accepted overall coefficients. Those employing film coefficients are treated in the next chapter.

Horizontal-tube Evaporators. Horizontal-tube evaporators as shown in Fig. 14.22 are the oldest type of chemical evaporator. Although they once enjoyed widespread use, they have given way to other types. They consist of a round or square shell and a horizontal tube bundle which is usually square. They do not take very good advantage of the thermal currents induced by heating and therefore are not so acceptable as the types which have replaced them. The horizontal evaporator is the only distinct type of chemical evaporator employing steam in the tubes. The principal advantages of horizontal evaporators lie in the relatively small headroom they require and the ability to arrange the bundle so that air brought in with the steam does not collect and blanket useful surface. The horizontal evaporator is least satisfactory for fluids which form scale or deposit salt, the deposit being on the outside of the tube, and it is therefore used only for relatively simple problems of concentration rather than for the preparation of a liquid for ultimate crystallization. It is well suited to processes in which the final product is a liquor instead of a solid, such as industrial sugar sirups, where the large volume of liquid stored in the evaporator can permit a close adjustment of the final density by changing the holdup in the evaporator. The tube length is determined by the size of the evaporator body itself. Because evaporation occurs on the outside of the tubes, eliminating a scale problem inside the tubes, the horizontal-tube evaporator uses smaller tubes than any other type, from $\frac{3}{4}$ to $1\frac{1}{4}$ in. OD.

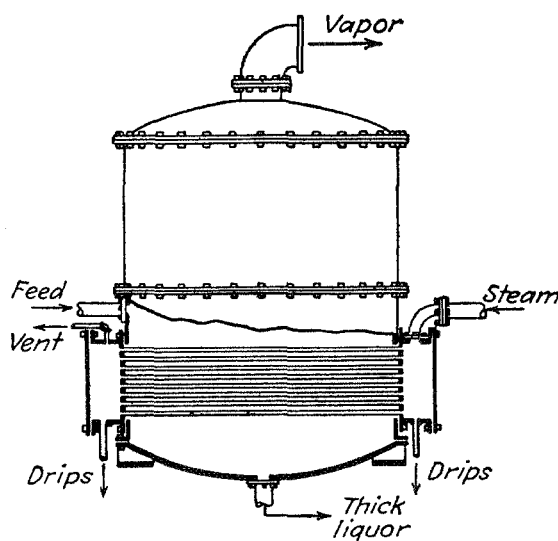


FIG. 14.22. Horizontal-tube evaporator. (Swenson Evaporator Company.)

The horizontal evaporator is least satisfactory for fluids which form scale or deposit salt, the deposit being on the outside of the tube, and it is therefore used only for relatively simple problems of concentration rather than for the preparation of a liquid for ultimate crystallization. It is well suited to processes in which the final product is a liquor instead of a solid, such as industrial sugar sirups, where the large volume of liquid stored in the evaporator can permit a close adjustment of the final density by changing the holdup in the evaporator. The tube length is determined by the size of the evaporator body itself. Because evaporation occurs on the outside of the tubes, eliminating a scale problem inside the tubes, the horizontal-tube evaporator uses smaller tubes than any other type, from $\frac{3}{4}$ to $1\frac{1}{4}$ in. OD.

Calandria-type Evaporators. The calandria evaporator is shown in Fig. 14.23. It consists of a short vertical-tube bundle, usually not more

than 6'0" high, set between two fixed tube sheets which are bolted to the shell flanges.

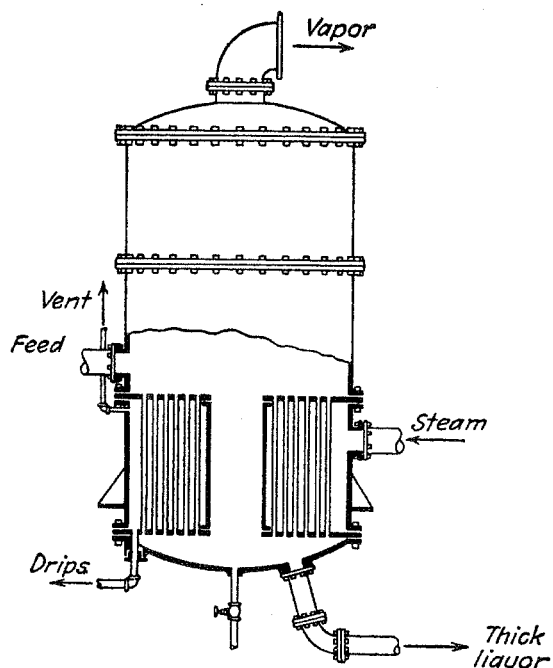


FIG. 14.23. Calandria-type evaporator. (Swenson Evaporator Company.)

The steam flows outside the tubes in the steam chest, and there is a large circular space for downtake in the center of the bundle whereby the cooler liquid circulates back to the bottom of the tubes. The flow area of the downtake is from one-half the flow area of the tubes to an area equal to it. Tubes are large, up to 3 in. OD, to reduce the pressure drop and permit rapid circulation and are installed in tube sheets with ferrule-type packing. The layout of a typical calandria is shown in Fig. 14.24. One of the problems is to baffle the steam chest so that there is a relatively uniform tube coverage.

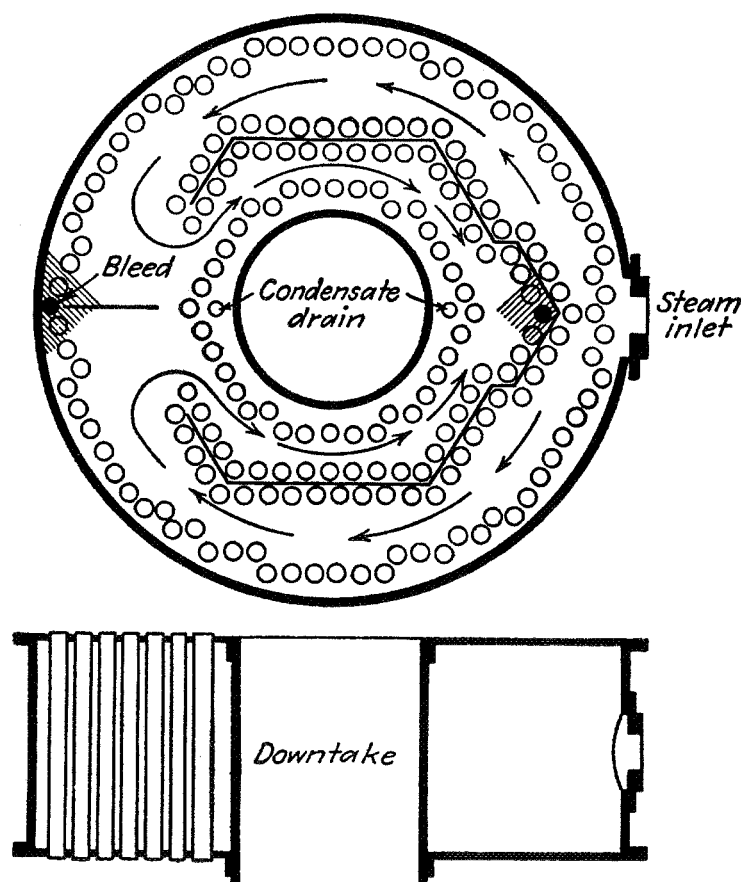


FIG. 14.24. Typical calandria baffling. Arrows indicate direction of steam flow. Shaded areas indicate location of noncondensable bleed points.

Another is to provide adequate bleed points so that no pockets of noncondensable gas develop. The condensate is removed at any convenient point. The space above the liquid level on the tube sheet serves primarily to disengage the liquid, which is carried along by the vapor. A common accessory of evaporators is a catchall which is installed on the vapor line for the purpose of removing entrained liquid and returning it to the bulk of the fluid. Two typical catchalls are shown in Fig. 14.25*a* and *b*. They operate on the principle of centrifugally removing the liquid droplets.

Calandria evaporators are so common they are often referred to as *standard* evaporators. Since scaling occurs inside the tubes, it is possible to use the standard evaporator for more rigorous services than the hori-

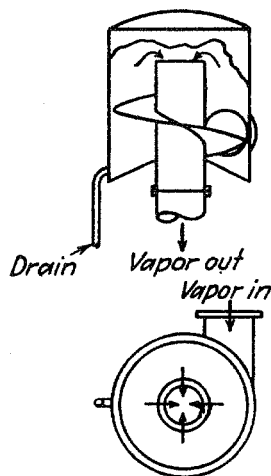


FIG. 14.25*a*. Catchall with bottom outlet.

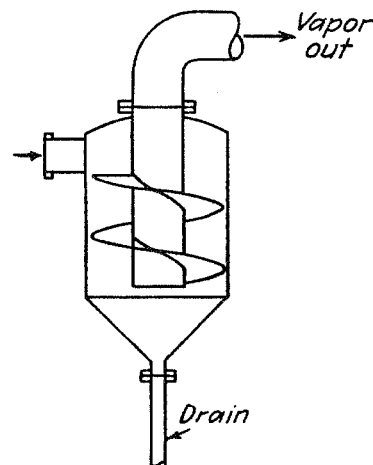


FIG. 14.25*b*. Catchall with top outlet.

zontal tube evaporator and, in addition, a propeller can be installed in the dished or conical bottom to increase the rate of circulation.

Basket-type Evaporators. A basket-type evaporator is shown in Fig. 14.26. It is similar to a calandria evaporator except that it has a removable bundle which can be cleaned quite readily. The bundle is supported on internal brackets, and the downtake occurs between the bundle and the shell instead of in a central downtake. Because the tube sheets hang freely, the problem of differential expansion between the tubes and the steam chest shell is not important. This type frequently is designed with a conical bottom and may also have a propeller installed to increase circulation. As a result of these mechanical advantages the basket evaporator can be used for liquors which have a tendency to scale, although they are not recommended for liquids with high viscosities or great rates of scaling. The selection of a basket or calandria evaporator usually follows the established policies of the different industries in which they are used after years of experience, or modifications suggested by manufacturers. Some manufacturers have preferences for the one type

in a certain application, whereas another will prefer the second type for the same service.

Long-tube Vertical Evaporators. A long-tube vertical evaporator is shown in Fig. 14.27. It consists of a long tubular heating element designed for the passage of the liquor through the tubes but once by natural circulation. The steam enters through a vapor belt as dis-

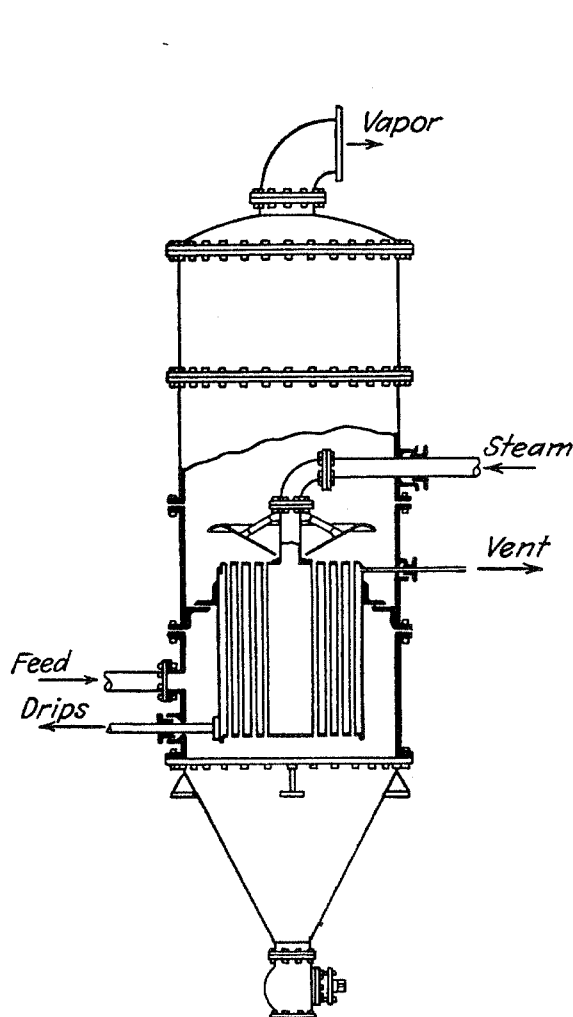


FIG. 14.26. Basket-type evaporator. (Swenson Evaporator Company.)

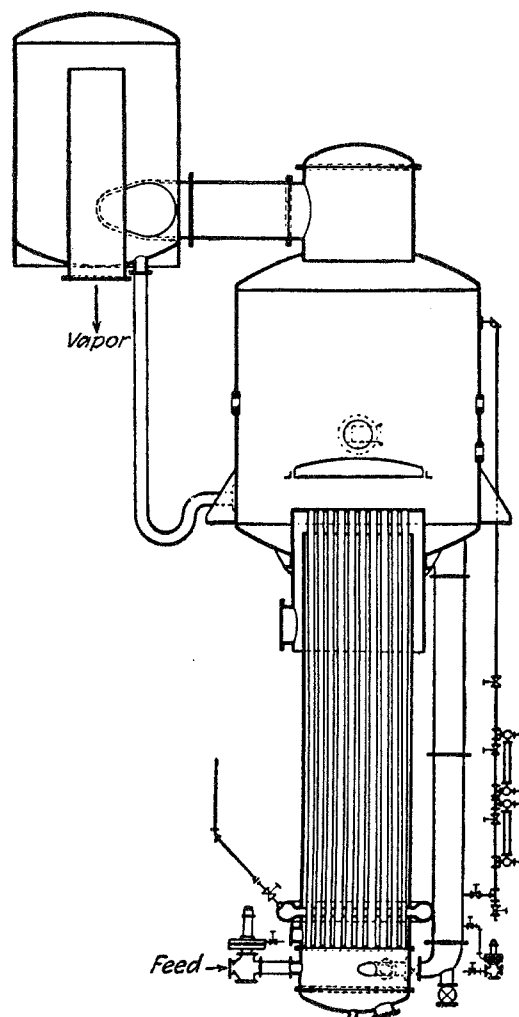


FIG. 14.27. Long-tube vertical evaporator. (General American Transportation Company.)

cussed in Chap. 12, and the bundle is baffled so that there is a free movement of steam, condensate, and noncondensable downward. The upper tube sheet is free, and just above it there is a vapor deflector to reduce the entrainment. This type of evaporator is not especially adapted to scaling or salting liquors, but it is excellent for the handling of foamy, frothy liquors. The velocity of the vapor issuing from the tubes is greater than in the short-tube vertical types. Tubes are usually $1\frac{1}{4}$ to 2 in. OD and from 12 to 24 ft in length. When arranged for recirculation the apparatus will be as shown in Fig. 14.28. In this type, disengagement

occurs outside the evaporator body. The calculation of this type of evaporator for aqueous solutions with known physical and thermal properties will be discussed in Chap. 15.

Forced-circulation Evaporators. Forced-circulation evaporators are made in a variety of arrangements as shown in Figs. 14.29 through 14.31. Forced-circulation evaporators may not be so economical to operate as natural-circulation evaporators, but they are necessary where the concen-

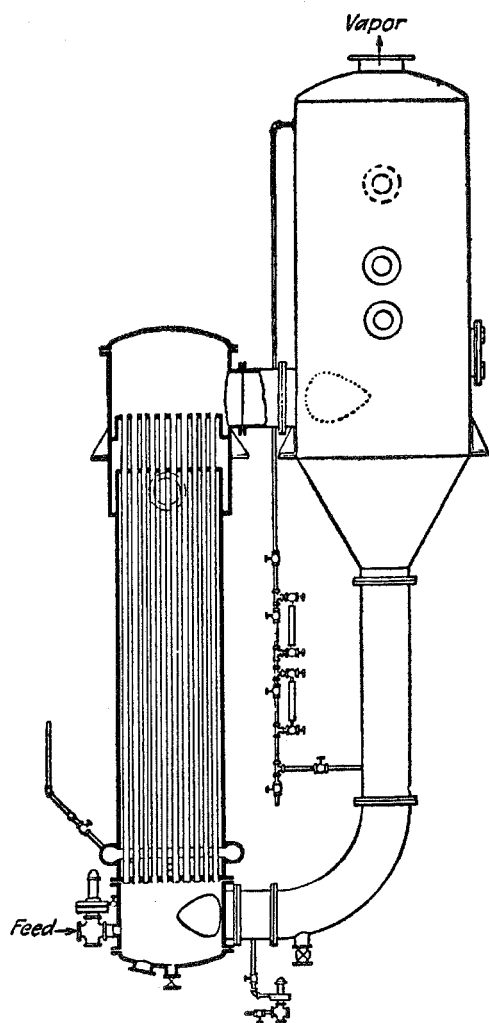


FIG. 14.28. Long-tube recirculation evaporator. (General American Transportation Company.)

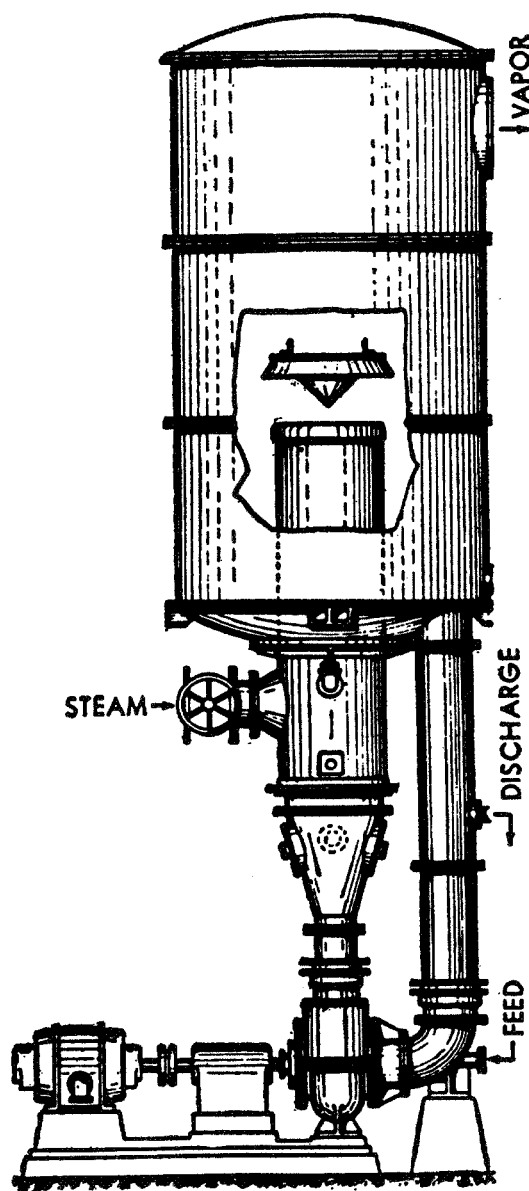


FIG. 14.29. Forced-circulation-type evaporator with inside vertical heating element. (Swenson Evaporator Company.)

tration problem involves a solution with poor flow, scale, and thermal characteristics. Since the Grashof group varies inversely with the square of the viscosity, there is a limit to the viscosities of solutions which will naturally recirculate. With very viscous materials there is no alternative but to use this type of evaporator. Also, where there is a tendency to

form scale or deposit salts, the high velocities obtainable by the use of circulating pumps are the only means of preventing the formation of excessive deposits. Forced-circulation evaporators are well adapted to a close control of flow, particularly when a long time of contact may be injurious to the chemical in solution. Tubes for forced-circulation evaporators are smaller than in natural-circulation types, usually not exceeding 2 in. OD.

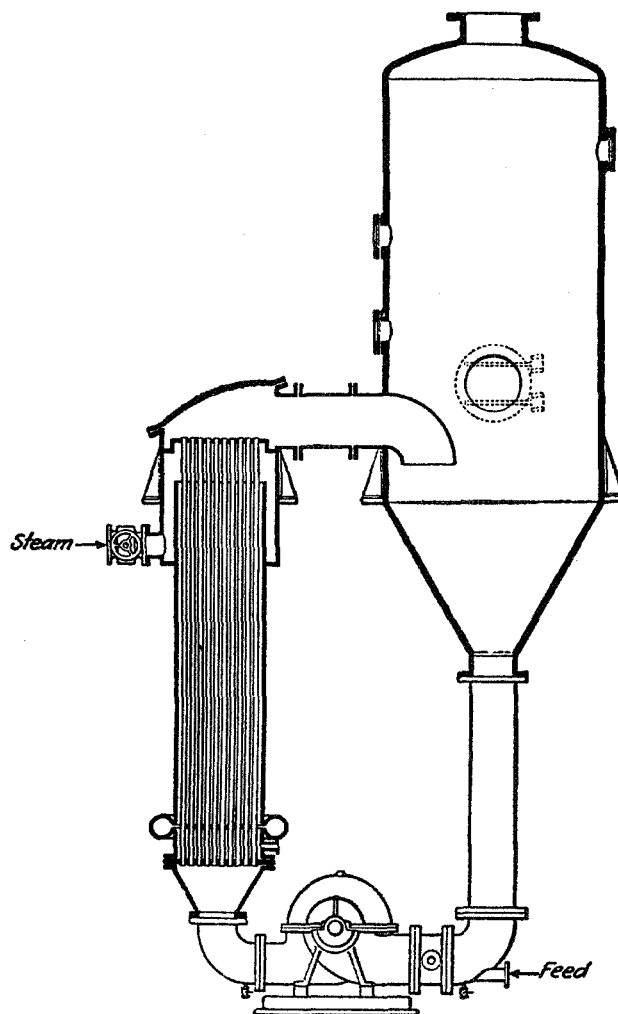


FIG. 14.30. Forced-circulation evaporator with vertical external element. (*General American Transportation Corporation.*)

In Fig. 14.27 the steam enters the bundle outside the evaporator body and contacts the tubes at the top of the bundle by means of the annular space provided. A deflector plate is installed above the upper tube sheet, and the circulating pump is installed at ground level. In Fig. 14.30 the same effect is produced by means of a vertical external bundle, which simplifies construction to a degree but which is not so compact. Figure 14.31 is a variation with a horizontal bundle which is particularly adaptable where the headroom is low.

Effect of Hydrostatic Head. Consider a pure fluid with a boiling surface somewhat above the top of a bundle of horizontal tubes. The boiling point is regarded as being set by the pressure at the liquid-vapor interface. If there is a great layer of liquid above the tube bundle, it will exert a hydrostatic pressure upon the liquid in contact with the tube surface. The added pressure upon the liquid raises the boiling temperature

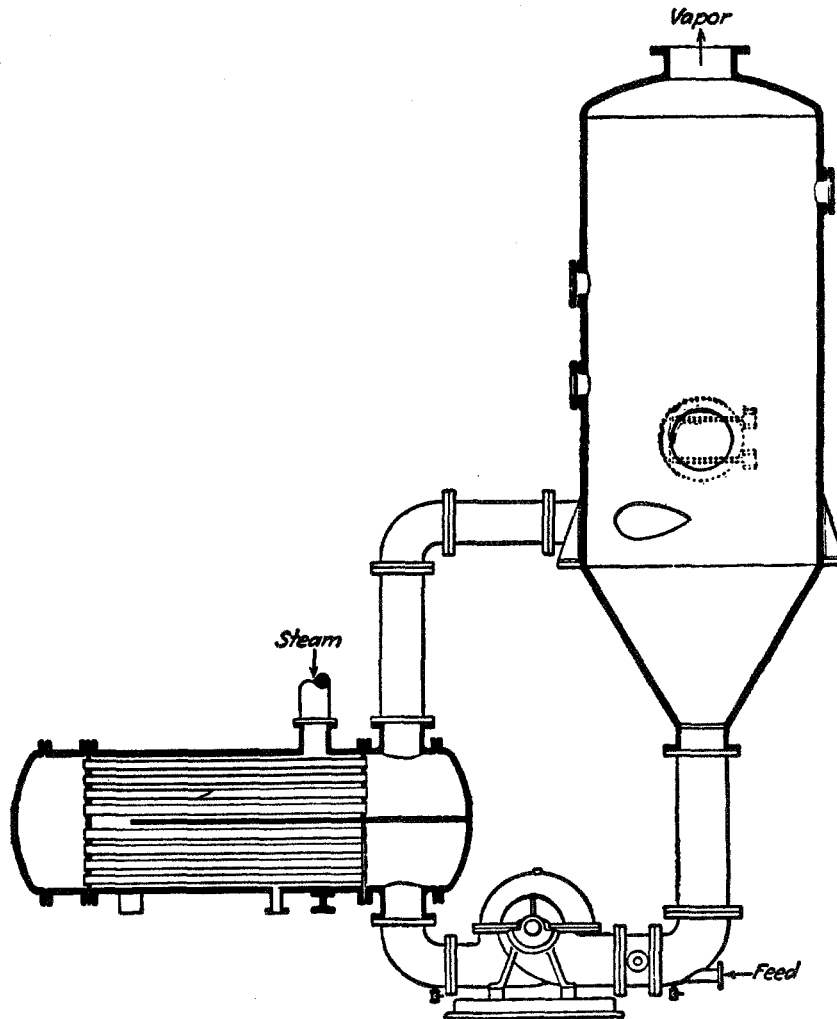


FIG. 14.31. Forced-circulation evaporator with horizontal external element. (*General American Transportation Corporation.*)

at the heat-transfer surface above that necessary to produce vapors of the saturation temperature corresponding to the pressure at the liquid-vapor interface. The effect of hydrostatic head, as in the case of BPR, reduces the useful temperature difference effective upon the heat-transfer surface. Since evaporators operate on fixed apparent temperature differences, the size of the heat-transfer surface must be increased accordingly for the presence of a hydrostatic head. The influence of temperature and pressure on the overall coefficient is essentially that shown in Fig. 14.7.

At high temperatures the liquid is less viscous and is more favorably suited to evaporation.

The effect of hydrostatic head may be estimated from

$$\Delta t_h = 0.03 \frac{T_R v}{\lambda_s} \Delta p \quad (14.7)$$

where Δt_h = hydrostatic elevation of the boiling point, °F

T_R = solution boiling temperature, °R

v = specific volume of water vapor at T_R ft³/lb

λ_s = latent heat of vaporization corresponding to the saturation pressure, Btu/lb

Δp = hydrostatic head, ft

Usually Δp may be taken as corresponding to one-half the indicated liquid level. It is apparent that the influence of hydrostatic pressure will be greater as the vacuum upon the system is increased, since v varies considerably with the pressure whereas λ_s varies but little.

For all evaporators operating with natural circulation a loss in available capacity due to hydrostatic head cannot be avoided but the loss can be reduced by maintaining the lowest liquid levels consistent with the efficient operation of the equipment. If the froth is regulated to 10 in. above the upper tube sheet of vertical heating elements, good operating control can usually be effected. The achievement of good operating control is facilitated by the choice of the tube diameter and length and in general by designing for a high fluid velocity into the disengaging space. Problems of arrangement for natural circulation will be treated in Chap. 15.

Multiple-effect Chemical Evaporation. In the study of parallel-feed multiple-effect power-plant evaporators it was shown that in a triple-effect evaporator 1 lb of steam evaporated approximately 2.25 lb of water. The use of parallel feed is by no means the most economical and is used in chemical evaporation only when the feed solution is nearly saturated to begin with and evaporation is intended for the purpose of supersaturation. In chemical evaporation it is customary to employ forward feed, backward feed, or a modification of both, known as *mixed* feed. Returning to Fig. 14.21 there are certain opposing advantages and disadvantages resulting from the use of either forward or backward feed.

In forward feed if the feed liquor is at a higher temperature than the saturation temperature of the first effect, some evaporation will occur automatically as *vapor flashing*. Since a vacuum is usually maintained on the last effect, the liquor flows by itself from effect to effect and a

liquor removal pump is required only on the last effect. Similarly, since the saturation temperature of the boiling solution in each effect is lower than the temperature of the effect preceding it, there is flashing or "free" evaporation in each succeeding effect, which reduces the overall steam requirement. In an evaporator the boiling film is the controlling resistance and the numerical value of the overall coefficient decreases with concentration because the viscosity is increasing. In forward feed the concentrated liquor is in the last effect, and obviously that effect has the lowest overall coefficient, since the liquor is most concentrated there and at the same time coldest.

When backward feed is employed, it overcomes the objection of having the most concentrated liquor in the coldest effect. Here the dilute liquor enters the last and coldest effect and leaves concentrated in the first effect, which is at the highest temperature. In this feed arrangement liquor must be heated in each effect as compared with solution flashing in each effect in forward feed. Furthermore, the feed must be pumped from effect to effect, which means that the number of places for air leakage, such as the pump and flanges, increases the maintenance and power cost. The temperature relations in backward feed usually offset these disadvantages in part, since the system is in counterflow and the economy of steam is greatest under these conditions.

If the feed liquor to a backward-feed evaporator is initially hot, its introduction into the last effect is wasteful, since the vapors which flash off in the last effect are lost to the condenser. In forward feed not only would these vapors flash but in each succeeding effect they would reevaporate additional water. The problem of the direction of feed to be employed is, as in most alternate problems of heat transfer, an economic one. Backward feed may or may not lead to smaller surface requirements, depending upon the extent of concentration and the viscosity of the desired final solution. The steam cost will be less for backward feed if the feed is cold and less for forward feed if the feed liquor is at approximately the operating temperature of the first effect or higher. The computation of problems by both methods will readily establish the most favorable operating relationship.

THE CALCULATION OF CHEMICAL EVAPORATORS

BY JOSEPH MEISLER¹

Referring to Fig. 14.21a the surface and steam requirements for multiple-effect chemical evaporation can be computed by imposing a heat balance across each effect individually and a material balance

¹ The Air Reduction Company, Inc., and the Polytechnic Institute of Brooklyn.

over the whole system. The following nomenclature will be employed for a quadruple effect:

- c_F = specific heat of feed, Btu/(lb)(°F)
 t_F = temperature of feed, °F
 w_F = feed, lb/hr
 T_S = saturation temperature of steam to first effect, °F
 W_S = steam to first effect, lb/hr
 w_{1-4} = total water removed by evaporation, lb/hr
 c_1, c_2, c_3, c_4 = specific heat of liquor in effects 1 to 4, Btu/(lb)(°F)
 t_1, t_2, t_3, t_4 = boiling points of liquor in effects 1 to 4, °F
 w_1, w_2, w_3, w_4 = water removed in effects 1 to 4, lb/hr

Assume that there are no chemical heat effects as a result of the concentration (*i.e.*, negative heats of solution) and that there is no BPR.

Forward feed

Heat balance on first effect:

$$W_S \lambda_S + w_F c_F (t_F - t_1) = w_1 \lambda_1 \quad (14.8)$$

Heat balance on second effect:

$$w_1 \lambda_1 + (w_F - w_1) c_1 (t_1 - t_2) = w_2 \lambda_2 \quad (14.9)$$

Heat balance on third effect:

$$w_2 \lambda_2 + (w_F - w_1 - w_2) c_2 (t_2 - t_3) = w_3 \lambda_3 \quad (14.10)$$

Heat balance on fourth effect:

$$w_3 \lambda_3 + (w_F - w_1 - w_2 - w_3) c_3 (t_3 - t_4) = w_4 \lambda_4 \quad (14.11)$$

Material balance:

$$w_{1-4} = w_1 + w_2 + w_3 + w_4 \quad (14.12)$$

The surface requirements will be

$$\begin{aligned}
 A_1 &= \frac{Q}{U_D \Delta t} = \frac{W_S \lambda_S}{U_1 (T_S - t_1)} \\
 A_2 &= \frac{w_1 \lambda_1}{U_2 (t_1 - t_2)} \\
 A_3 &= \frac{w_2 \lambda_2}{U_3 (t_2 - t_3)} \\
 A_4 &= \frac{w_3 \lambda_3}{U_4 (t_3 - t_4)}
 \end{aligned} \quad (14.13)$$

Let

$$A_1 = A_2 = A_3 = A_4 \quad (14.14)$$

where U_1 , U_2 , U_3 , and U_4 are the design overall coefficients in the respective effects. From the material balance and the heat balances there are five equations and five unknowns: W_s , w_1 , w_2 , w_3 , and w_4 . These may be solved for simultaneously.

Backward feed. Referring to Figure 14.21b:

Heat balance on fourth effect:

$$w_3\lambda_3 + w_F c_F (t_F - t_4) = w_4\lambda_4 \quad (14.15)$$

Heat balance on third effect:

$$w_2\lambda_2 + (w_F - w_4)c_4(t_3 - t_4) = w_3\lambda_3 \quad (14.16)$$

Heat balance on second effect:

$$w_1\lambda_1 + (w_F - w_4 - w_3)c_3(t_2 - t_3) = w_2\lambda_2 \quad (14.17)$$

Heat balance on first effect:

$$W_s\lambda_s + (w_F - w_4 - w_3 - w_2)c_2(t_1 - t_2) = w_1\lambda_1 \quad (14.18)$$

Material balance:

$$w_{1-4} = w_1 + w_2 + w_3 + w_4 \quad (14.19)$$

The surface relations will be the same as before, since it is practical to impose the restriction that the surface in each of the bodies be identical. It has also been shown by experience that under this condition the pressure differences between effects will be approximately equal. If steam enters the first effect of a quadruple-effect evaporator at atmospheric pressure and the last effect is at 26 in. Hg vacuum corresponding to 1.95 psia, the pressure difference between the steam and the first effect and from effect to effect will be $(14.7 - 1.95)/5$. This will enable selection of the saturation pressures in the individual effects. Since the heat-transfer coefficients will be different in the individual effects, it may be found that the surfaces defined by Eqs. (14.13) and (14.14) are unequal. This means that owing to the inequality in the overall coefficient in the different effects the Δt across each effect does not correspond to the assumption of an equal division of the total pressure differential. This will be particularly true when the overall coefficients in the different effects differ greatly or when there is considerable flashing in the first effect. To equalize the surface in each body the temperature differences in the individual effects can be adjusted so that a larger temperature

difference will be employed in the effect having the lowest heat transfer coefficient, the heat loads in all effects remaining nearly equal.

Multiple-effect evaporators may be designed for minimum surface or minimum initial cost. These cases have been treated by Bonilla.¹ The design of multiple-effect evaporators for optimum conditions, however, is more the exception in industry than the rule, the trend being in the direction of standardization.

Example 14.2. Calculation of a Triple-effect Forward-feed Evaporator. It is desired to concentrate 50,000 lb/hr of a chemical solution at 100°F and 10.0 per cent solids to a product which contains 50 per cent solids. Steam is available at 12 psig, and the last effect of a triple-effect evaporator with equal heat-transfer surfaces in each effect will be assumed to operate at a vacuum of 26.0 in. Hg referred to a 30-in. barometer. Water is available at 85°F for use in a barometric condenser.

Assume a negligible BPR, an average specific heat of 1.0 in all effects, the condensate from each effect leaves at its saturation temperature and that there are negligible radiation losses. Calculate: (a) Steam consumption, (b) heating surface required for each body, (c) condenser water requirement. The accepted overall coefficients of heat transfer for the different effects will be $U_1 = 600$, $U_2 = 250$, and $U_3 = 125$ Btu/(hr)(ft²)(°F),

Solution:

Total feed $w_F = 50,000$ lb/hr

Total solids in feed $= 0.10 \times 50,000 = 5000$ lb/hr

Total product $= \frac{5000}{0.50} = 10,000$ lb/hr

Total evaporation, $w_{1-3} = 50,000 - 10,000 = 40,000$ lb/hr
 $c_F = 1.0$.

The balances applying to this problem are

First effect: $W_S \lambda_S + w_F(t_F - t_1) = w_1 \lambda_1$

Second effect: $w_1 \lambda_1 + (w_F - w_1)(t_1 - t_2) = w_2 \lambda_2$

Third effect: $w_2 \lambda_2 + (w_F - w_1 - w_2)(t_2 - t_3) = w_3 \lambda_3$

Material: $w_1 + w_2 + w_3 = w_{1-3}$

$t_F = 100^\circ\text{F}$

T_S at 12 psig $= 244^\circ\text{F}$

T_3 at 26 in. Hg (1.95 psia) $= 125^\circ\text{F}$

Total temperature difference $= 119^\circ\text{F}$

When a forward-feed multiple-effect evaporator employs equal surfaces in each effect, as noted above, experience indicates that the differences in the pressures between effects will be nearly equal. This will rarely be entirely true, but it forms an excellent starting point for the calculation of the pressures in the effects. Any discrepancies can be adjusted later.

Average pressure difference $= \frac{26.70 - 1.95}{3} = 8.25$ psi/effect

¹ Bonilla, C. F., *Trans. AIChE*, **41**, 529-537 (1945).

BREAKUP OF THE TOTAL PRESSURE DIFFERENCE

	Pressure, psia	ΔP , psi	Steam or vapor, °F	λ , Btu/lb
Steam chest, 1st effect.....	26.70		$T_s = 244$	$\lambda_s = 949$
Steam chest, 2d effect.....	18.45	8.25	$t_1 = 224$	$\lambda_1 = 961$
Steam chest, 3d effect.....	10.20 (20.7 in. Hg)	8.25	$t_2 = 194$	$\lambda_2 = 981$
Vapor to condenser.....	1.95 (26 in. Hg)	8.25	$t_3 = 125$	$\lambda_3 = 1022$

$$\begin{aligned}
 949W_s + 50,000(100 - 224) &= 961w_1 \\
 961w_1 + (50,000 - w_1)(224 - 194) &= 981w_2 \\
 981w_2 + (50,000 - w_1 - w_2)(194 - 125) &= 1022w_3 \\
 w_1 + w_2 + w_3 &= 40,000
 \end{aligned}$$

Solving simultaneously,

$$\begin{aligned}
 w_1 &= 12,400 \\
 w_2 &= 13,300 \\
 w_3 &= 14,300 \\
 w_{1-3} = w_1 + w_2 + w_3 &= 40,000 \\
 W_s &= 19,100
 \end{aligned}$$

$$\begin{aligned}
 A_1 &= \frac{W_s \lambda_s}{U_1(T_s - t_1)} = \frac{19,100 \times 949}{600 \times 20} = 1510 \text{ ft}^2 \\
 A_2 &= \frac{w_1 \lambda_1}{U_2(t_1 - t_2)} = \frac{12,400 \times 961}{250 \times 30} = 1590 \\
 A_3 &= \frac{w_2 \lambda_2}{U_3(t_2 - t_3)} = \frac{13,300 \times 981}{125 \times 69} = 1510 \quad (\text{Use } 1600 \text{ ft}^2/\text{effect})
 \end{aligned}$$

$$\text{Heat to condenser} = w_3 \lambda_3 = 14,300 \times 1022 = 14,710,000 \text{ Btu/hr}$$

$$\begin{aligned}
 \text{Water requirement} &= 14,710,000 / (120 - 85) = 420,000 \text{ lb/hr or } 420,000 / 500 \\
 &= 840 \text{ gpm}
 \end{aligned}$$

$$\text{Economy, lb evaporation/lb steam} = 40,000 / 19,100 = 2.09 \text{ lb/lb}$$

Remarks. During operation the equal pressure-drop distribution may not maintain itself. This will occur if there is undue scaling in one of the effects, if a body is gas bound, or if liquor levels are not properly maintained. Another factor may be the withdrawal of a large quantity of steam from one of the effects as a source of low-pressure heating steam. Any deviation from an equal pressure-drop distribution does not mean that the entire multiple-effect assembly will fail to operate but instead that the unit will assume a new pressure distribution and operate with a reduced capacity and steam economy.

Nonalgebraic Solution of Evaporators. It will presently be shown that the use of a simultaneous algebraic solution such as that above can scarcely be applied advantageously to more complicated arrangements or to a great number of effects. In industrial problems it has been found preferable and timesaving to perform the heat and material balances on the evaporator by directly assuming the value of W_s and solving each effect by a direct balance on itself instead of through the use of simul-

taneous equations. If the total evaporation based on the assumed value of W_s does not equal the required quantity, a new value of W_s is assumed and the calculation repeated. This method is demonstrated in Example 14.2a, in which the hint that $W_s = 19,100$ from Example 14.2 is used.

Example 14.2a. Solution of Example 14.2 by Assuming W_s

	Heat balance	
1. Assume steam to 1st effect $W_s = 19,100$ lb/hr, $w_F = 50,000$ lb/hr		
19,100 lb at 12 psig = $19,100 \times 949$	= 18,100,000 Btu/hr	
Deduct for heating feed = $50,000(224 - 100)$	= 6,200,000	
Available for evaporation	= 11,900,000 Btu/hr	
λ_1 at $224^\circ\text{F} = 961$ Btu/lb, $w_1 = 11,900,000/961$	= 12,400 lb/hr	
Transfer to 2d effect = $50,000 - 12,400$		37,600 lb/hr
2. Vapors from 1st effect	= 11,900,000 Btu/hr	
Add flash, $37,600(224 - 194)$	= 1,130,000	
Available for evaporation	= 13,030,000 Btu/hr	
λ_2 at $194^\circ\text{F} = 981$ Btu/lb, $w_2 = 13,030,000/981$	= 13,300 lb/hr	
Transfer to 3d effect = $37,600 - 13,300$		24,300 lb/hr
3. Vapors from 2d effect	= 13,030,000 Btu/hr	
Add flash, $24,300(194 - 125)$	= 1,680,000	
Available for evaporation	= 14,710,000 Btu/hr	
λ_3 at $125^\circ\text{F} = 1022$ Btu/lb $w_3 = 14,710,000/1022$	= 14,300	
Product = $24,300 - 14,300$		10,000 lb/hr
4. Heat to condenser	= 14,710,000 Btu/hr	

If the quantities failed to check a new value for W_s could be assumed.

As a first trial the steam could have been estimated in the absence of the hint obtained from Exercise 14.2 by the relationship

$$W_s = \frac{w_e}{0.75 \times \text{number of effects}} \quad (14.20)$$

where w_e is the total pounds of evaporation. Equation (14.20) is based on feed entering at its boiling point. If the feed enters below its boiling point, the factor 0.75 must be reduced somewhat. With the feed at only 100°F and the boiling point at 125°F the value must be reduced to 0.70.

Example 14.3. Backward-feed Multiple-effect Evaporator. Conditions are the same as Example 14.2 except using backward feed with overall coefficients of $U_1 = 400$, $U_2 = 250$, and $U_3 = 175$ Btu/(hr)(ft²)(°F).

Solution. As before,

$$w_1 = 50,000 \text{ lb/hr} \quad w_{1-3} = 40,000 \text{ lb/hr}$$

and

$$c_F = 1.0$$

The balances applying to this problem are

$$\text{Third effect: } w_3\lambda_3 + w_F(t_F - t_3) = w_2\lambda_2$$

$$\text{Second effect: } w_1\lambda_1 + (w_F - w_3)(t_2 - t_3) = w_2\lambda_2$$

$$\text{First effect: } W_s\lambda_s + (w_F - w_3 - w_2)(t_1 - t_2) = w_1\lambda_1$$

Material:

$$\begin{aligned}
 w_1 + w_2 + w_3 &= w_{1-3} \\
 981w_2 + 50,000(100 - 125) &= 1022w_3 \\
 961w_1 + (50,000 - w_3)(125 - 194) &= 981w_2 \\
 949W_s + (50,000 - w_3 - w_2)(194 - 224) &= 961w_1 \\
 w_1 + w_2 + w_3 &= 40,000 \\
 w_1 &= 15,950 \\
 w_2 &= 12,900 \\
 w_3 &= 11,150 \\
 w_{1-3} = w_1 + w_2 + w_3 &= 40,000 \\
 W_s &= 16,950
 \end{aligned}$$

$$A_1 = \frac{16,950 \times 949}{400 \times 20} = 2010 \text{ ft}^2$$

$$A_2 = \frac{15,950 \times 961}{250 \times 30} = 2040 \text{ ft}^2$$

$$A_3 = \frac{12,900 \times 981}{175 \times 69} = 1050 \text{ ft}^2$$

Since the heating surfaces for the three effects are far from equal, the temperature differences employed in the effects must therefore be modified to meet the conditions of the problem. For the first trial the average surface was

$$\frac{(2010 + 2040 + 1050)}{3} = 1700 \text{ ft}^2$$

With a better distribution of temperatures and pressures, however, less than

$$3 \times 1700 = 5100 \text{ ft}^2$$

of surface may be expected, since the Δt in both of the first two effects will be improved at the expense of only the last effect.

Recalculation:

Assume an average surface of 1500 ft²/effect, and find the temperature differences which will provide these surfaces. Assume

First effect: $T_s - t_1 = 28^\circ\text{F}$, $A_1 = 2\frac{1}{2} \times 2010 = 1450 \text{ ft}^2$

Second effect: $t_1 - t_2 = 41^\circ\text{F}$, $A_2 = 3\frac{1}{4} \times 2040 = 1490 \text{ ft}^2$

Third effect: $t_2 - t_3 = 50^\circ\text{F}$, $A_3 = 6\frac{2}{5} \times 1050 = 1440 \text{ ft}^2$

$T_s - t_3 = 119^\circ\text{F}$.

The new temperature-pressure distribution is

	Pressure, psia	Steam or vapor, °F	λ , Btu/lb
Steam chest, 1st effect.....	26.7	$T_s = 244$	949
Steam chest, 2d effect.....	16.0	$t_1 = 216$	968
Steam chest, 3d effect.....	16.4 in. Hg	$t_2 = 175$	992
Vapor to condenser.....	26.0 in. Hg	$t_3 = 125$	1022

Solving again for w_1 , w_2 , w_3 , and W_s ,

$$\begin{aligned} w_1 &= 15,450 & A_1 &= 1450 \\ w_2 &= 13,200 & A_2 &= 1470 \\ w_3 &= 11,350 & A_3 &= 1490 & (\text{Use } 1500 \text{ ft}^2/\text{effect}) \\ W_s &= 16,850 \end{aligned}$$

Heat to condenser $= 11,350 \times 1022 = 11,600,000 \text{ Btu/hr}$

Water requirement $= 11,600,000 / (120 - 85) = 332,000 \text{ lb/hr} = 332,000 / 500$
 $= 664 \text{ gpm}$

Economy, lb evaporation/lb steam $= 40,000 / 16,850 = 2.37 \text{ lb/lb}$

COMPARISON OF FORWARD AND BACKWARD FEED

	Forward	Backward
Total steam, lb/hr.....	19,100	16,850
Cooling water, gpm.....	840	664
Total surface, ft ²	4,800	4,500

The operating conditions for both forward and backward feed are shown in Fig. 14.32a and b. Substantiating the simple reasoning, backward feed is more effective thermally than forward feed. Omitted, however, is the maintenance and investment on backward-feed pumps between each effect and the problems of air leakage and flow control. All of these are considerably greater in backward feed.

Optimum Number of Effects. The greater the number of effects the larger the amount of evaporation per pound of steam admitted to the first effect. The operating costs will be less the larger the number of effects. This is offset, however, by the increased first cost of the apparatus and increased maintenance charges for cleaning and replacement, both of which enter as fixed charges. Supervisory labor will be the same for the operation of any number of effects. The cost of condenser water must also be included, and it, too, will decrease the greater the number of effects employed. The optimum number of effects may be obtained by computing the process requirements with two, three, four, or up to six or eight effects and determining the fixed charges and operating cost resulting from each arrangement. When the total cost is plotted against the number of effects, a minimum will occur corresponding to the optimum number of effects. Actually, however, the number of effects in the various heavy chemical industries are fairly standardized. For example, table salt is concentrated in four effects in which parallel feed of liquor is used, caustic soda with two or three effects and backward feed, and sugar with five or six effects and forward feed. Except when introducing an entirely new chemical process it will rarely be necessary to carry out a complete economic analysis.

Bleed Steam. In certain industries and particularly in the production of sugar, there is a great need for low-pressure steam, say 10 to 15 psia, for the large amount of liquor preheating required throughout the plant. For the preheating it is found advantageous to use some of the steam from the first or subsequent effects for various preheating services such as in the decolorization of sugar sirup. Since any vapor formed in the first and later effects will have already been used one or more times for

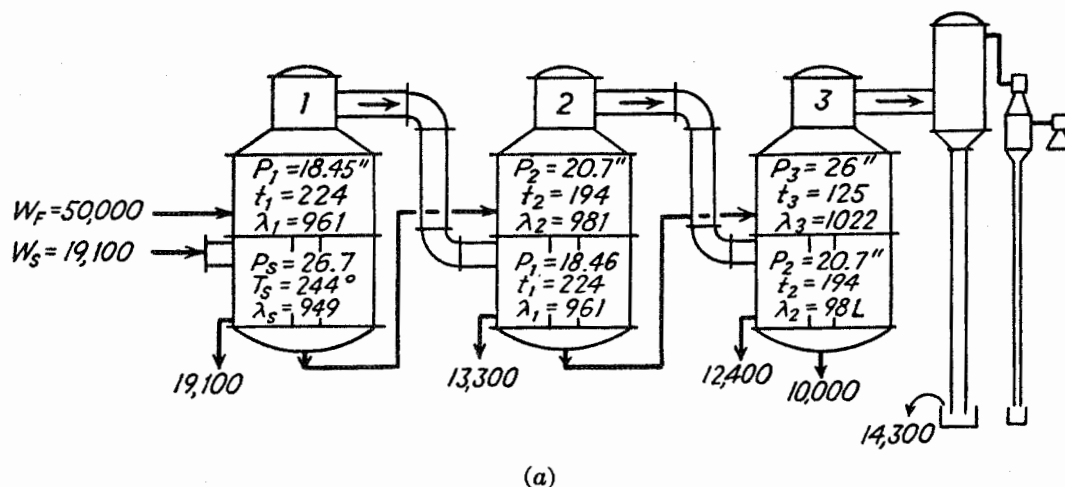


FIG. 14.32a. Example 14. Forward feed.

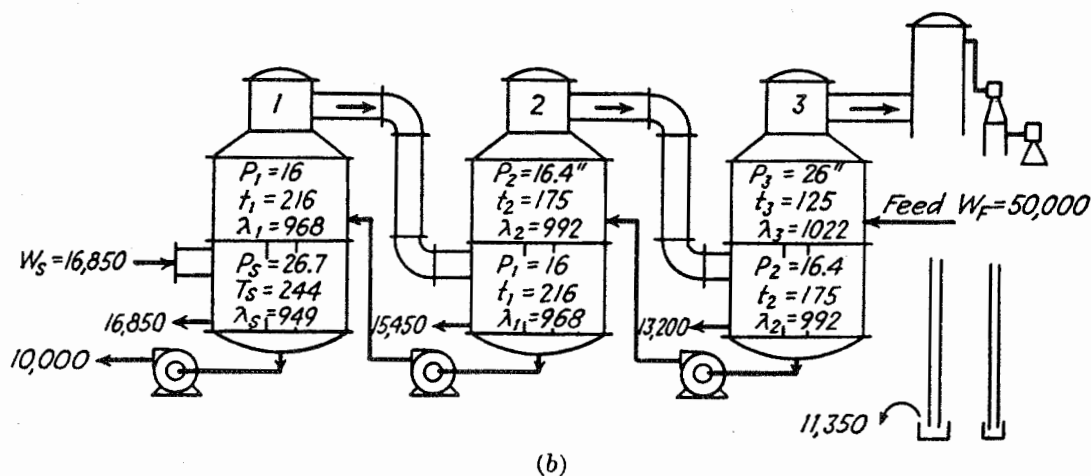


FIG. 14.32b. Example 14. Backward feed.

evaporation, the cost per Btu of the vapor is cheaper than that of fresh steam. Steam can be bled economically from one or more effects for these additional services while reducing the overall heating cost of the plant. To account for this in the heat-balance equations [Eqs. (14.8) through 14.18)] one need only include the term W_B for the pounds per hour of bleed steam in the effect in which it occurs. The quantity of bleed steam must also be introduced into the material balance of Eqs. (14.11) and (14.18). To prevent the introduction of one or more

unknowns than there are simultaneous equations, W_B should not be entered as another unknown but as a definite number of pounds of bleed steam or as a percentage of the total evaporation if the calculation is for an existing installation.

The Solution of Industrial Problems.¹ The preceding examples have served to introduce the elementary methods of calculation and the principles of multiple-effect evaporation. Actually, industrial problems are rarely so simple. Instead, the evaporator system must be integrated with the operation of the entire manufacturing process, and this complicates the calculation greatly. The element of experience is essential to the completion of a calculation within a reasonable period of time. In the remainder of this chapter several of the most widely met commercial problems will be analyzed. These are sugar concentration, waste-liquor evaporation in the paper pulp industry, and the production of caustic soda. The analyses involved in their solution should be readily adaptable to the majority of other problems. A fourth process, distillery waste concentration, has been omitted in favor of evaporation by thermocompression. The methods indicated may be employed either for the design of a new evaporator or for the calculation of performance in an existing evaporator.

THE CONCENTRATION OF CANE-SUGAR LIQUORS—FORWARD FEED

Description of the Process. It is the practice in cane-sugar production to strain the juice containing the sugar after it has been pressed from the sugar cane and clarified chemically. In the initial stage of clarification, called defecation, large amounts of colloids and inorganic and organic salts are removed. This is accomplished by the addition of lime to the juice at 200°F, forming a heavy slime which is removed by settling, decantation, and filtration. The clear solution is then fed to the multiple-effect evaporator for concentration.

Example 14.4. An evaporator installation is to have a capacity for concentrating 229,000 lb/hr of 13°Brix (degrees Brix is the per cent by weight of sugar in the solution) to 60°Brix, at which concentration it will be decolorized. This quantity of sugar solution results from milling 2300 tons of sugar cane per 20-hr day. The raw juices will be heated from 82 to 212°F by the use of exhaust vapors bled from the first and second effects. Steam will be available to the first effect at 30 psig.

Solution. The arrangement of equipment is shown in Fig. 14.33 and is justified by the fact that bled vapor is a cheaper preheating medium than fresh steam. In this analysis it is arbitrarily assumed that 37,500 lb/hr of 15 psig vapor is bled from the first effect for use in the vacuum pans in the crystallizing section of the plant. The

¹ Readers who are not directly interested in chemical evaporation problems may omit the remainder of this chapter without affecting the context of later chapters.

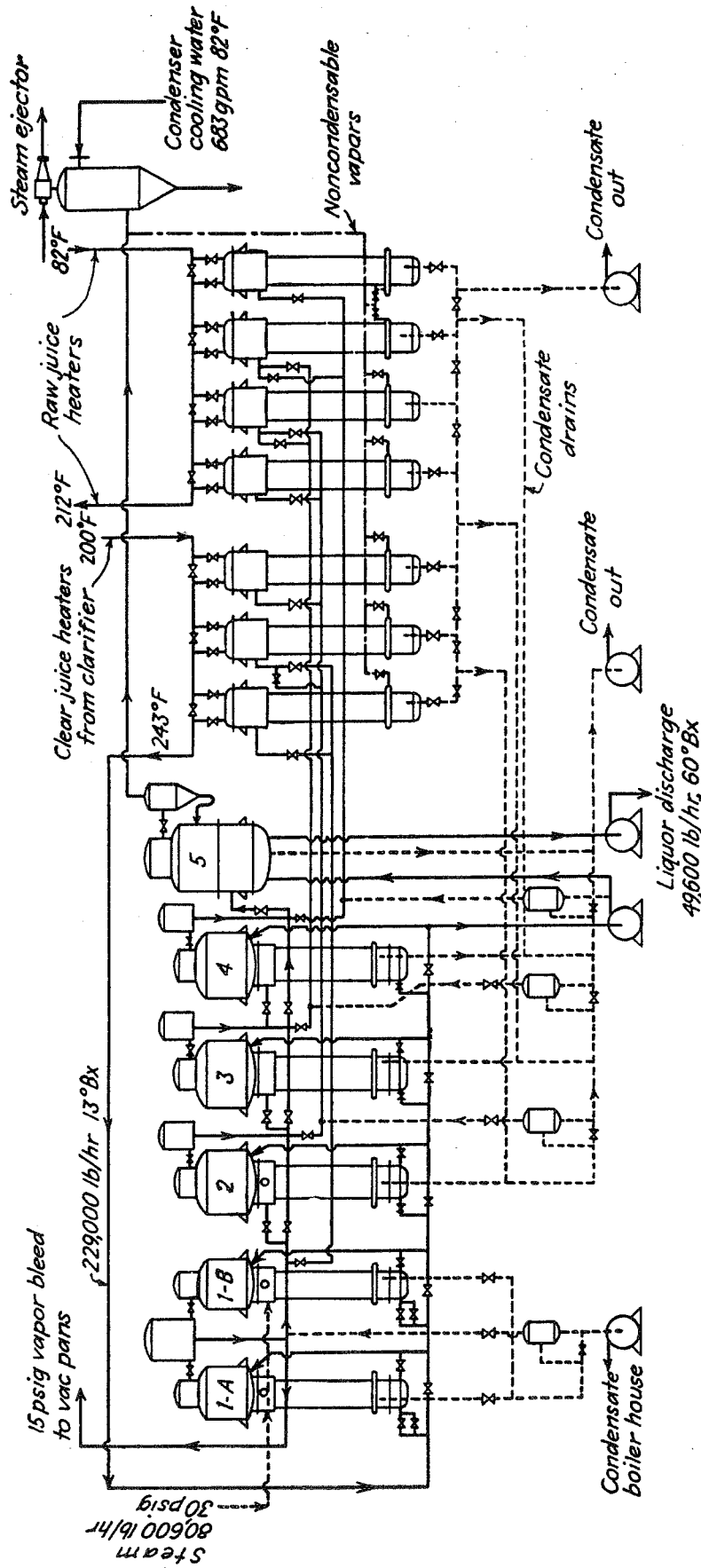


FIG. 14.33. Six-body sugar evaporator with preheaters.

BPR and specific heats of sugar solutions are given in Fig. 14.34. A flow diagram such as that in Fig. 14.33 is usually arrived at from experience or from preliminary studies of total surface requirements as in Table 14.1.

TABLE 14.1. AVERAGE EVAPORATION PER SQUARE FOOT HEATING SURFACE FOR SUGAR EVAPORATORS

Effects	Water evaporated, lb/(hr)(ft ²)
1	14-16
2	6-8
3	5-6
4	4-5
5	3-4

In the matter of choosing the number of effects from experience, it is the practice to use a straight quadruple effect if little or no vapor bleeding is required and to add a "preevaporator" as a first effect if excessive bleeding of low pressure vapor are required for preheating feed, etc., or if relatively low-pressure live steam is bled for other plant processes, *i.e.*, vacuum pans.¹

The forward-feed arrangement is common in the sugar industry, since stronger juices or sirups are sensitive to high temperatures. This procedure places a severe handicap on the last effect of a multiple-effect evaporator where the most viscous juice

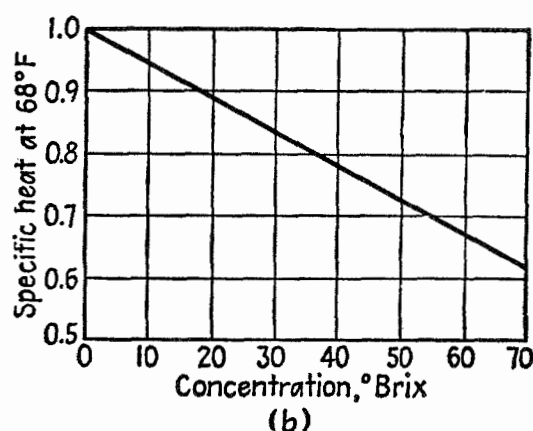
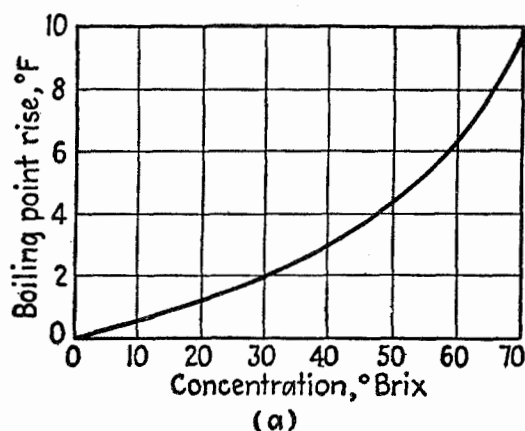


FIG. 14.34a. BPR of sugar solutions.

FIG. 14.34b. Specific heats of sugar solutions.

boils at the lowest temperature. However, because of the "caramelizing" tendency of high-concentration solutions this procedure is of paramount importance in the preparation of a product of high purity.

Table 14.2 is a convenient reference in estimating the temperature-pressure distribution and for presenting a summary of all the important process and design conditions. Items 2 to 9 inclusive are estimated first, and from these the heat balance can be made.

It is shown in item 2 of Table 14.2 that 30 psig steam is fed in parallel to effects 1A and 1B, the temperature 274°F in 3 being the saturation temperature corresponding to the steam pressure. The last-effect vapor body is designed to operate under a

¹ Further discussion of the flow diagram will be given at the conclusion of the heat balance.

vacuum of 23 in. Hg (147°F). This is shown in 7 and 8. Item 13 shows that the sirup leaving the last effect is 60°Brix. From Fig. 14.34a this concentration corresponds to a boiling temperature of 147 or 154°F as shown in 5. For 6 of Table 14.2 it can be assumed tentatively that the °Brix concentration is divided evenly among all the effects; thus, the BPR for each effect can be read from Fig. 14.34a. The ultimate material distribution determined from the heat balance, however, will give a precise

TABLE 14.2. EVAPORATOR SUMMARY

Item	Effects					
	1A	1B	2	3	4	5
1. Steam flow, lb/hr	42,600	38,000				
2. Steam pressure, psi/in. Hg	30	30	15	5	4"	14½"
3. Steam temp, °F	274	274	250	227	205	181
4. Δt , °F	23	23	21	20	20	27
5. Liquor temp, °F	251	251	229	207	185	154
6. BPR, °F	1	1	2	2	4	7
7. Vapor temp, °F	250	250	227	205	181	147
8. Vapor pressure, psi/in. Hg	15	15	5	4	14½	23
9. λ , Btu/lb	946	946	960	975	990	1,010
10. Liquor in, lb/hr	229,000	190,200	154,000	117,100	87,800	64,000
11. Liquor out, lb/hr	190,200	154,000	117,100	87,800	64,000	49,600
12. Evaporation, lb/hr	38,800	36,200	36,900	29,300	23,800	14,400
13. °Brix (out)	15.7	19.4	25.5	34.3	46.5	60.0
14. A , ft ²	3,500	3,500	5,000	5,000	5,000	3,500
15. U_D , Btu/(hr)(ft ²)(°F)	478	425	310	264	219	138
16. $U_D \Delta t$, Btu/(hr)(ft ²)	11,000	9,780	6,520	5,270	4,390	3,740

estimate of the Brix distribution in the different effects. Item 13 is a summary of the latter. In order to complete 2 to 9 inclusive, the distribution of temperature differences in each effect must be determined as in 4.

The total temperature difference in the evaporator system is $274^\circ - 147 = 127^\circ\text{F}$. From the total must be subtracted the sum of all the BPR (*i.e.*, from effect 1B to the fifth effect inclusive), a sum of 16°F . This will give $127 - 16 = 111^\circ\text{F}$ as the total "effective" temperature difference. The latter is distributed along the following lines: (a) The total is distributed in proportion to the values found in practice for a unit with the same number of effects. (b) The total is distributed on the basis of accepted average values of heat flux $U_D \Delta t = Q/A$ as found in industry and with values of U_D estimated from similar operations. Accepted values of the flux are given in (16) of Table 14.2. They may be redistributed for a different number of effects. (c) An equal distribution of the total pressure drop is assumed through the system. Either method will give a sufficiently close estimate of the temperature difference distribution for obtaining items 2 to 9 required for working the heat balance. It might be noted here that, if the estimated values of the temperature

differences are found to be very much out of line with those found in practice, they must be revised. Having estimated the distribution of the temperature difference the saturated vapor pressure above the liquor or the saturated steam pressure in the following effect may be determined from the steam tables (Table 7). The latent heats corresponding to saturated vapor pressures will be obtained similarly.

Since it is planned to bleed vapors from the different effects for preheating both the raw and clarified sugar solutions, it is necessary to determine the quantities which will be bled from each effect. The raw sugar solutions will be preheated from 82 to 212°, a range of 130°F. Vapors from the fourth effect will be used to preheat the raw sugar solution from 82 to 144°F, vapors from the third effect from 144 to 184°, and vapors from the second effect from 184 to 212°. Heat quantities and surfaces for the juice heaters are calculated as in Table 14.3. Similarly juice from the clarifier is preheated by second-effect vapors from 200 to 220° and by first-effect evaporators from 220 to 243°F. In all instances the preheaters are designed to permit a duplication of surface, although in some instances overdesign obviously results. A spare heater is usually included in the system to allow continuity of operation during cleaning. In arriving at the temperature range for the individual preheaters a minimum terminal difference of 7°F has been used. Larger terminal differences may be employed if the spread of temperatures makes them permissible.

TABLE 14.3. SUGAR-JUICE HEATERS

<i>Raw-juice heaters</i>	<i>Clear-juice heaters</i>
1. 229,000(212 - 184)(0.91) = 5,840,000 Btu/hr Vapor temp. = 227°F $\Delta t = 26.6^\circ\text{F}$ $U_D = 231$ Surface, $A = \frac{5,840,000}{26.6 \times 231} = 950 \text{ ft}^2$	1. 229,000(243 - 220)(0.91) = 4,800,000 Btu/hr Vapor temp. = 250°F $\Delta t = 15.8^\circ\text{F}$ $U_D = 234$ Surface, $A = \frac{4,800,000}{15.8 \times 234} = 1300 \text{ ft}^2$
2. 229,000(184 - 144)(0.90) = 8,250,000 Btu/hr Vapor temp. = 205°F $\Delta t = 37.6^\circ\text{F}$ Surface, $A = \frac{8,250,000}{37.6 \times 230} = 950 \text{ ft}^2$	2. 229,000(220 - 200)(0.90) = 4,120,000 Btu/hr Vapor temp. = 227°F $\Delta t = 14.8^\circ\text{F}$ Surface, $A = \frac{4,120,000}{14.8 \times 214} = 1300 \text{ ft}^2$
3. 229,000(144 - 82)(.90) = 12,800,000 Vapor temp. = 181°F $\Delta t = 62.2^\circ\text{F}$ Surface, $A = \frac{12,800,000}{62.2 \times 217} = 950 \text{ ft}^2$ Use 3 heaters at 1000 ft ² each plus 1 heater as spare	Use 2 heaters at 1300 ft ² each plus 1 heater at 1300 ft ² as spare

Table 14.4 shows the heat and material balance for the evaporator. To effect a balance it is necessary to estimate correctly the steam flow to the first effect. In the simple case where no vapors are bled for process heating an assumption is made by using a quantity of steam which is about 20 per cent above theoretical, *i.e.*, the hourly evaporation divided by the number of effects. In the present case where excessive vapors are bled, the "extrapolated evaporation" is divided by the number of effects. The extrapolated evaporation is equivalent to the actual plus that which the vapors could effect if not bled for process work. Since only an estimate of the steam is

TABLE 14.4. HEAT BALANCE

Effect	Btu/hr	Evaporation, lb/hr
1A. Heat in steam = $42,600 \times 929 \times .97$	38,400,000*	38,800
Heating liquor = $229,000(251 - 243).91$	1,670,000	
— 38,800	36,730,000/946	
Liquor to 1B = 190,200		36,200
1B. Heat in steam = $38,000 \times 929 \times .97$	34,200,000*	
Heating liquor = $190,200(251 - 251)$		
— 36,200	34,200,000/946	
Liquor to 2d effect = 154,000		36,900
2. Heat in 1A vapors.....	36,730,000	
Heat in 1B vapors.....	34,200,000	
Total heat available.....	70,930,000	36,900
First effect cond. flash = $80,600(274 - 250)$	1,940,000	
	72,870,000	
Heat to vacuum pans = $37,500 \times 946$	— 35,500,000	36,900
	37,370,000	
Heat to clear-juice heaters.....	— 4,800,000	
Heat to 2d effect.....	32,570,000*	36,900
Liquor flash = $154,000(251 - 229).85$	+ 2,880,000	
36,900	35,450,000/963	
Liquor to 3d = 117,100		29,300
3. Heat in 2d vapors.....	35,450,000	
Cond. flash = $37,500(250 - 227)$	865,000	
	36,315,000	29,300
Heat to clear-juice heaters.....	— 4,120,000	
	32,195,000	
Heat to raw-juice heaters.....	— 5,840,000	29,300
	26,355,000*	
Liquor flash = $117,100(229 - 207).83$	+ 2,150,000	
29,300	28,505,000/975	23,800
Liquor to 4th = 87,800		
4. Heat in 3d vapors.....	28,505,000	
Cond. flash = $74,200(227 - 205)$	1,630,000	23,800
	30,135,000	
Heat to raw-juice heaters.....	— 8,250,000	
	21,885,000*	23,800
Liquor flash = $87,800(207 - 185).80$	+ 1,540,000	
23,800	23,425,000/989	
Liquor to 5th = 64,000		14,400
5. Heat in 4th vapors.....	23,425,000	
Cond. flash = $101,400(205 - 181)$	2,480,000	
	25,905,000	14,400
Heat to raw-juice heaters.....	— 12,800,000	
	13,105,000*	
Liquor flash = $64,000(184 - 154).74$	+ 1,470,000	14,400
14,400	14,575,000/1010	
Prod. discharge = 49,600	Total hourly evaporation	
		179,400

* Heat supplied by heating element.

required, the following rule can be used to obtain the extrapolated evaporation: Add to the actual evaporation the equivalent evaporation of the bled vapors thus:

	<i>Lb/hr</i>
(a) Actual evaporation.....	179,400
(b) Equivalent evaporation from vapors of 1st effect used for vacuum pans (4 × 35,500,000)/977.....	145,500
(c) Equivalent evaporation from 1st effect vapors used for clarified-juice heaters, (4 × 4,800,000)/977.....	19,700
(d) Equivalent evaporation from 2d effect vapors used for clarified- and raw- juice heaters, (3 × 9,960,000)/977.....	30,600
(e) Equivalent evaporation from 3d effect vapors used for raw-juice heaters, (2 × 8,250,000)/977.....	17,900
(f) Equivalent evaporation from 4th effect vapors used for raw-juice heaters, (1 × 12,800,000)/977.....	13,100
Extrapolated evaporation.....	406,200

$$\text{Estimated steam quantity } \frac{406,200}{5} = 81,240 \text{ lb/hr}$$

$$\text{Actual steam required from final heat balance} = 80,600$$

$$\text{Error} = 640 \text{ lb/hr}$$

This method of estimating the steam quantity to the first effect gives quite consistent results and can be used as a "first" trial in Table 14.4. One or two trials will determine the exact steam flow for obtaining the required evaporation.

It will be noted from Table 14.4 that liquor and steam flow in series through all effects (in forward-feed arrangement) except for the first effect where the total steam is divided for parallel flow between effects 1A and 1B. The latter procedure is common where excessive evaporation in a single effect would require a very large vapor body. The division of the total steam in the heat balance between effects 1A and 1B is arbitrary, especially if equal heating surface will be supplied. However, it is the usual practice to divide the total steam in the ratio of 1.1/1.0 for effects 1A/1B to simplify calculations and to take into account the fact that the 1A effect will have a higher heat-transfer coefficient and thus use more steam for greater evaporation.

In effects 1A and 1B liquor must be preheated to its boiling point before evaporation commences. In all other effects the liquor flashes or vaporizes a portion of its water on entering the following body. To increase the steam economy still further, it is customary to flash the condensate to the vapor space of the following effect. Thus the total steam condensate is flashed to the vapor space in the first effect, and the vapors obtained from this flash are added to the total vapors leaving the 1A and 1B effects. Likewise the net condensate in the second effect (38,800 + 36,200 - 37,500, which was removed for vacuum pans) is flashed to the corresponding second-effect saturated vapor, etc. It will be noted that the total flash in each effect is cumulative, since the condensate flows in series to a common discharge header through a series of flash tanks. A well-designed condensate flash system may save as much as 10 per cent of the live-steam requirement.

The heat balance is completed when one obtains a product from the last effect equal in quantity and °Brix concentration to that for which the evaporator is being designed. Thus a material balance in the present case shows:

$$\text{Feed} = 229,000 \text{ lb/hr} \times 0.13 = 29,800 \text{ lb/hr sugar}$$

$$\text{Discharge} = \frac{29,800}{0.60} = 49,600 \text{ lb/hr of } 60^\circ\text{Brix sugar}$$

$$\text{Evaporation} = 229,000 - 49,600 = 179,400 \text{ lb/hr water evaporated}$$

If an incorrect quantity of steam is assumed or a very poor temperature distribution is made, the required evaporation or product discharged will not be obtained. In the latter case, an apportionment of the difference between the required evaporation and that obtained in the first trial, based upon the assumed steam quantity and a recalculation of the heat balance, will give results close enough for practical purposes. In general, if the heat balance is off by less than 1 per cent in product or evaporation, it may be considered to check.

The vapors from the last effect can be used for process heating or carried to a condenser. Evaporator vapors are used to heat the juice before and after clarification by means of two sets of heaters. In Fig. 14.33 the heaters are grouped in sets of four and three. The first set is for preheating the raw juice with vapors of the second, third, and fourth effects. The second set is for preheating the clarified juice before it enters the first effect of the evaporator and uses vapors from the first and second effects. Each set of heaters operates with one less than that indicated, the omitted unit acting as a spare.

The first four effects of the evaporator system are designed to have bodies of the long-tube vertical-film type (Fig. 14.27), while the fifth effect is provided with a calandria-type body (Fig. 14.23). The first four effects have catchalls with tangential vapor inlets and bottom outlets, whereas the fifth effect is designed with a top vapor outlet. The entire evaporator unit is arranged to allow by-passing any unit from the system for cleaning or repair.

Vacuum is maintained by a barometric condenser and ejector. The gallons per minute of water required in the barometric condenser is calculated from Eq. (14.4). Where low vacuum and cold water are combined as in the present example, a value of 15°F may be used for the approach temperature difference t_a .

$$\text{Water} = \frac{14,575,000}{500(147 - 82 - 15)} = 583 \text{ gpm} \quad (14.4)$$

Discussion. In Table 14.2 items 10 to 13 inclusive summarize the data obtained from the material and heat balance in Table 14.4. The surface design, *i.e.*, the heating element, vapor head, etc., is shown in 14 to 16 inclusive. Item 16 lists values of $U_D \Delta t$ employed commercially; dividing these values of $U_D \Delta t$ by the estimated Δt yields the values of U_D shown in 15. If the values of U_D are much out of line with correlated values for similar operation, then a redistribution of either Δt or $U_D \Delta t$ may be necessary. Both are affected simultaneously. Surfaces are found by dividing the starred heat loads in Table 14.4 by respective values of $U_D \Delta t$.

In designing heaters for both raw and clear juice in Table 14.3 the importance of equal size and duplication of units is emphasized. A careful division of the total heat load was made to allow large temperature differences when using the bled vapors of the different effects. Slightly larger transfer coefficients could have been used, but it is desirable to equalize the surfaces at the expense of arbitrarily varying the transfer coefficients.

A check of Eq. (14.7) will show that it is not imperative to account for the hydrostatic loss when setting up Table 14.2. Since the evaporator

bodies are designed for high velocity and low liquor levels, it has been assumed that any capacity loss due to hydrostatic head is compensated for by a slightly conservative design of surface.

THE EVAPORATION OF PAPER PULP WASTE LIQUORS—BACKWARD FEED

Description of Paper-pulp Production Processes. There are three standard processes for producing paper pulp, namely, the soda, sulfate, and sulfite processes. All three produce wastes which are recovered either because of valuable residues or to avoid public nuisance.

1. In the soda process wood chips are cooked in digesters with caustic soda solution of about 12°Bé. After digestion, the pulp is washed clean with a hose in tanks having screen bottoms or else passed through a filter which is used for washing it. The liquor separated from the washed pulp, called *black liquor*, contains the resin and lignin of the wood and all the alkali used in the digester. The black liquor is evaporated, and solids are recovered for the value of the alkali.

The soda process is applied to both nonresinous and resinous woods. In the latter case the black liquors tend to foam and evaporators of the long-tube, vertical-film type are used almost exclusively. The solutions are not scale forming, undergo little elevation in boiling point (BPR), and the concentrated product is not excessively viscous. Liquors are usually concentrated to about a 40 per cent solid content by backward feed in a sextuple (or higher) effect evaporator, and the concentrate is subsequently fed to rotary incinerators. The latter produces *black ash*, which is leached, and the resulting solution is recausticized for use in the digesters.

2. The sulfate pulp process uses a mixture of sodium hydroxide, sodium carbonate, and sodium sulfide for digestion of wood chips. The sulfide is renewed by adding sodium sulfate and reducing it with carbon (thus, "sulfate" process). The process is used on resinous woods, and the content of inorganic material in the waste liquor is higher. It is usually concentrated to about 50 and often 55 per cent. The liquor is more viscous than the soda black liquor of same density, foams more, and has a higher elevation in boiling point owing to the larger content of inorganic solids. The usual practice is to use long-tube vertical-film-type evaporators with six effects or more and with backward feed. Because of the high viscosity in the first effect it is customary to divide the latter into two bodies with parallel steam flow. This procedure increases the liquor rate, which offsets the detrimental effect of the high viscosity on the heat-transfer rate.

3. The sulfite pulp process uses calcium or magnesium bisulfite as the active chemical in wood chip digestion. The active material is not

recovered.¹ When a mill is located in an isolated area the waste liquor is discarded; when a mill is located in nonisolated districts, evaporation of the liquors and burning of the concentrate are required to prevent surface or subsurface water pollution. In recent years, however, much development work has been carried out on the fermentation of sulfite liquors for alcohol production.

Example 14.5. A unit concentrating soda pulp black liquor will be designed to evaporate 90,000 lb/hr of water from a feed of 144,000 lb/hr which enters at 170°F and contains 15.2 per cent solids. This is equivalent to a product discharge of 54,000 lb/hr containing 40.5 per cent solids. These quantities are based on the production of 150 tons of pulp per day, and the resulting black liquor contains 3500 lb/ton of total solids. Utilities available are dry saturated steam at 35 psig and condenser water at 60°F.

Solution. Economy studies in the industry indicate that a six-body sextuple-effect evaporator will be required. In this type of unit the liquor flow is arranged for divided feed to the fifth and sixth effects to eliminate the large vapor bodies which would otherwise be required were the total feed passed through in series. From the last two effects the liquor is pumped through the remaining bodies in series, backward flow. The discharge from the first effect, after passing through a liquid-level control valve to a liquor flash tank, is finally sent to a product storage tank. Figure 14.35 illustrates schematically the arrangement of the equipment for this process.

In this type of process all vapor, liquor, and condensate lines are designed with by-pass arrangements, permitting any one body to be cut out for cleaning or maintenance and still allowing the operation of the unit as a quintuple effect.

The vapors from the last effect are condensed in a multijet barometric condenser or are used for heating mill water to 110°F. If the latter scheme is followed, a surface condenser can be substituted for the multijet condenser.

The factors entering in making the heat balance are similar to those discussed under sugar evaporation. Referring to Table 14.5 it is important first to estimate correctly the temperature-pressure distribution, items 2 to 8.

The overall temperature difference is $280 - 125 = 155^\circ\text{F}$ (corresponding to 35 psig and 26 in. Hg). For the estimated per cent solids distribution in each body, the BPR can be obtained from Fig. 14.36a. The estimated total BPR is 41°F , and the effective temperature difference (disregarding hydrostatic head factors in accordance with the discussion under sugar evaporators) is $155 - 41$, or 114°F . Specific-heat data are given in Fig. 14.36b. The distribution of temperature differences Δt in item 4 is in accordance with general practice, using higher temperature differentials in the first and last effects owing to the larger evaporation loads and the relatively even distribution in the other effects. Slight deviations in the estimated values of the temperature-difference distribution from what may be found in actual operation will not affect the heat balance or design of the evaporator too greatly. Poor temperature-difference estimates, however, will be reflected by poorly correlatable transfer coefficients. The accepted overall coefficients for multiple-effect soda-black-liquor processes are given

¹ Recent development by the Weyerhaeuser Timber Co. of a cyclic recovery system using the magnesium bisulfite base has increased interest in this method of pulp production. From unpublished paper on sulfite waste evaporation by D. Q. Kern and J. Meisler.

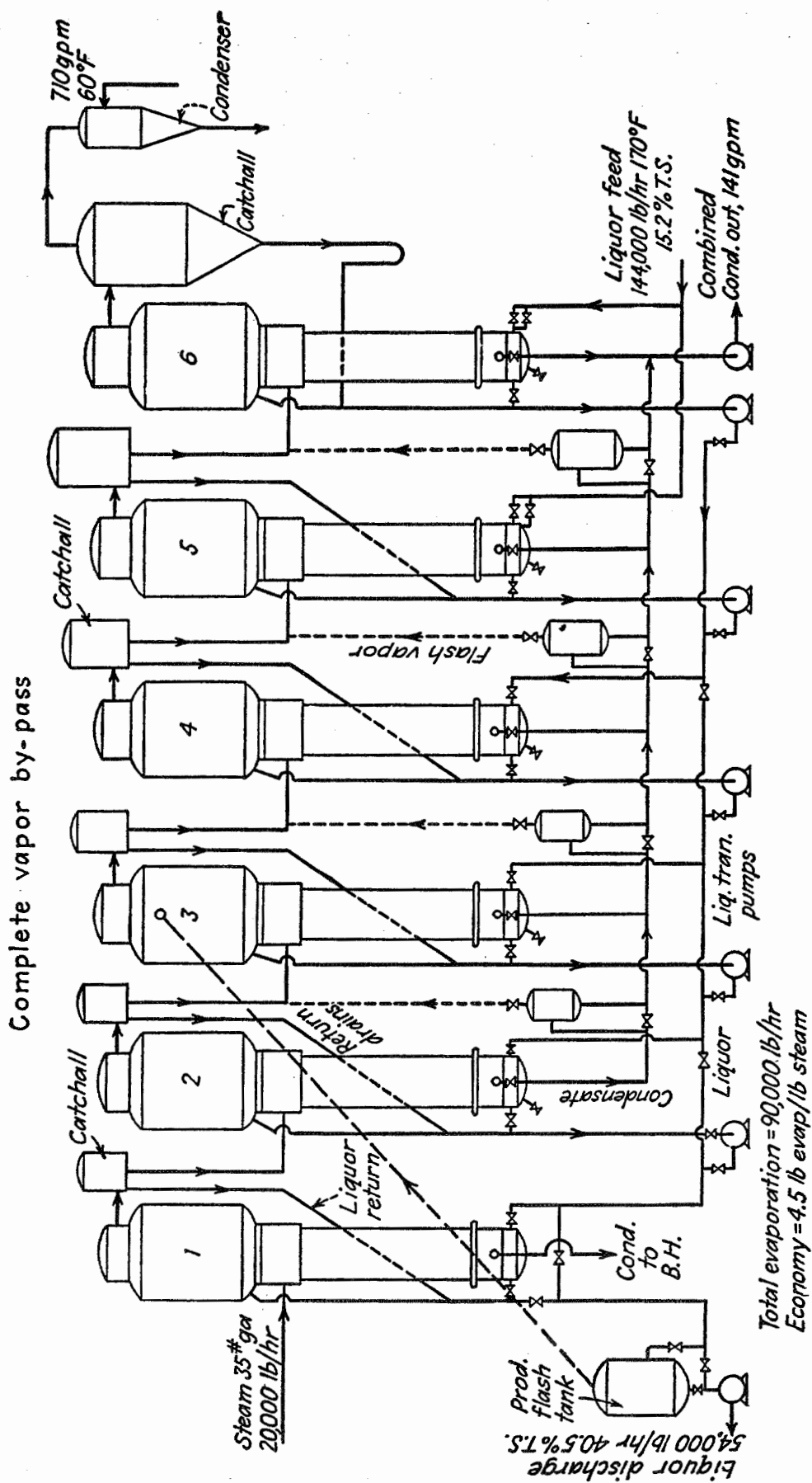


Fig. 14.35. Soda black-liquor evaporator.

TABLE 14.5. EVAPORATOR SUMMARY

All bodies will consist of 300 2 in. OD, 10 BWG tubes 24'0" long

Item	Effects					
	1	2	3	4	5	6
1. Steam flow, lb/hr.....	20,000					
2. Steam pressure, psi/in. Hg.....	35	14.5	4	7"	16.5"	22"
3. Steam temp, °F.....	280	249	224	199	174	151
4. Δt , °F.....	21	17	18	19	18	21
5. Liquor temp, °F.....	259	232	206	180	156	130
6. BPR, °F.....	10	8	7	6	5	5
7. Vapor temp, °F.....	249	224	199	174	151	125
8. Vapor pressure, psi/in. Hg.....	14.5	4	7"	16.5"	22"	26"
9. λ , Btu/lb.....	946	962	978	994	1,008	1,022
10. Liquor in, lb/hr.....	73,400	88,300	101,100	113,000	72,000	72,000
11. Liquor out, lb/hr.....	56,200	73,400	88,300	101,100	58,300	54,700
12. Evaporation, lb/hr.....	17,200	14,900	12,800	11,900	13,700	17,300
13. Total solids, %*.....	38.9	29.8	24.7	21.6	18.7	20.0
14. A , ft ²	3,250	3,250	3,250	3,250	3,250	3,250
15. U_D , Btu/(hr)(ft ²)(°F).....	262	295	252	251	221	221
16. $U_D \Delta t$, Btu/(hr)(ft ²).....	5,510	5,000	4,530	4,770	3,980	4,650

* 40.5 per cent from flash tank; 22,000 lb/hr vaporized in flash tank.

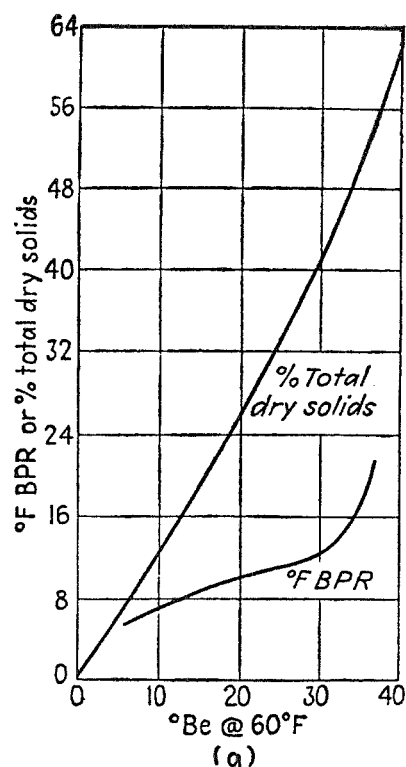


FIG. 14.36a. °Bé vs. °F BPR and total dry solids for soda black liquor.

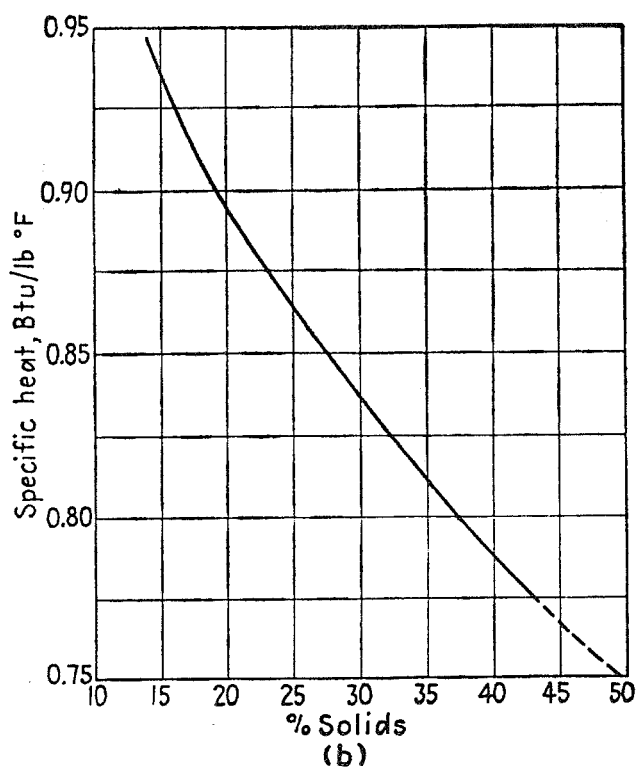


FIG. 14.36b. Specific heat vs. per cent solids for soda black liquor.

in Table 14.5. If the calculated coefficients differ appreciably from the accepted values, the Δt distribution will have to be altered accordingly.

Table 14.6 shows the procedure in determining the heat balance. It will be observed that this balance is one of trial and error. With some experience in backward-flow-evaporator calculations, however, not more than two trials are necessary to obtain the correct balance. A first trial is necessary for determining the approximate quantity of live steam to the first effect. Since the overall material balance (Table 14.5) shows that a total of 90,000 lb/hr of water is evaporated, the theoretical amount of steam for a six-effect evaporator would be 90,000/6, or 15,000 lb/hr. It is found, however, that the economy of multiple-effect black-liquor evaporators falls in the order of 75 per cent of theoretical or 6×0.75 which is 4.5. Thus, as an estimate

$$\frac{90,000 \text{ lb}}{4.5} = 20,000 \text{ lb/hr}$$

of steam will be used for a trial balance. This value is converted to its total evaporation potential by multiplying by its latent heat of vaporization and a factor of about 0.97 to take into account heat losses by radiation.

Since the feed is backward, the quantity of liquor being pumped from the second to the first effect is unknown. In fact, the only known liquor quantity is 54,000 lb/hr of product discharge leaving the flash tank. Ordinarily the black liquor in the first effect is not concentrated to the final desired solid content, but several of the last per cent of water are allowed to vaporize in a flash tank which is located ahead of the first effect and which is connected to one of the low-pressure evaporator bodies as shown in Fig. 14.35. In this instance the flash tank receiving the discharge from the first effect is piped to the vapor body of the third effect, which is at a saturation temperature of 199°F. Since the solution discharged from the first effect has a BPR of 10°F, its actual boiling temperature in the third effect is $199 + 10 = 209^\circ\text{F}$ and a flash corresponding to 259 to 209°F is effected.

If the quantity of liquor discharged from the first effect were known, the amount of additional water evaporated by flash could easily be calculated. An estimate is therefore made of this quantity as follows:

Divide the total evaporation by the number of effects, *i.e.*, $90,000/6 = 15,000$, and add to it about 15 per cent, giving a total of 17,200 lb/hr. The latter is an estimate of the amount of evaporation in the first effect. Since the flash is usually a few per cent of the product (here $2200/54,000 = 4.1$ per cent), an estimated value of the flash is added to the evaporation in the first effect and the total added to the product discharge from the flash tank: thus $54,000 + (17,200 + 2200) = 73,400$ lb/hr. This quantity is the estimated discharge from the second effect and must be preheated from 232 to 259°F, the boiling point in the first effect. The latter is indicated in item 1b of the heat-balance calculations, Table 14.6. By subtracting the preheat from 1a and dividing the difference $(16,270,000/946)$ by the latent heat of vaporization corresponding to the saturated vapor pressure in the first effect, the amount of actual evaporation in the first effect 1c is obtained. The latter value should correspond to that estimated. If it does not check, the value of the discharge liquor from the second effect 1b is revised to bring the estimated and the calculated evaporation rates in line. Moreover, since the preheat load 1b is usually in the order of 8 to 10 per cent of the steam load 1a only a small change is ordinarily required in the former to balance the estimated and calculated evaporation rates. The discharge to the flash tank is the difference between 1b and 1c. The amount of flashed vapor is equal to the heat load 1d divided by the latent heat of vaporization corresponding to

TABLE 14.6. HEAT BALANCE

$$\text{Cooling water at } 60^{\circ}\text{F} = \frac{17,750,000}{500(125 - 15 - 60)} = 710 \text{ gpm}$$

Effect	Btu/hr	Evaporation, lb/hr
1. a. Heat in steam = $20,000 \times 924 \times .97^*$	17,900,000†	
b. Heating liquor = $73,400(259 - 232).82$	1,630,000	
c. Evaporation = <u>17,200</u>	16,270,000/946	17,200
d. To flash tank = $56,200(259 - 209).78$	2,200,000/978	2,200
e. Flashed vapor = <u>2,200</u>		
f. Product = <u>54,000</u>		
2. a. Heat in 1st vapors.....	16,270,000†	
b. Heating liquor = $88,300(232 - 206).85$	- 1,940,000	
c. Evaporation = <u>14,900</u>	14,330,000/962	14,900
d. Liquor to 1b = <u>73,400</u>		
3. a. Heat in 2d vapors.....	14,330,000	
b. Condensate flash = $17,200(249 - 224)$	+ 430,000	
c. Total heat to 3d effect steam chest.....	14,760,000†	
d. Heating liquor = $101,100(206 - 180).86$	2,250,000	
e. Evaporation = <u>12,800</u>	12,510,000/978	12,800
f. Liquor to 2b = <u>88,300</u>		
4. a. Heat in 3d vapors.....	12,510,000	
b. Condensate flash = $32,100(224 - 199)$	+ 800,000	
c. Liquor flash from flash tank.....	+ 2,200,000	
d. Total heat to 4th effect.....	15,510,000†	
e. Heating liquor = $113,000(180 - 143).88$	- 3,680,000	
f. Evaporation = <u>11,900</u>	11,830,000/995	11,900
g. Liquor to 3d = <u>101,100</u>		
5. a. Heat in 4th vapors.....	11,830,000	
b. Condensate flash = $44,900(199 - 174)$	+ 1,120,000	
c. Total heat to 5th effect.....	12,950,000†	
d. Liquor flash = $72,000(170 - 156).90$	+ 900,000	
e. Evaporation = <u>13,700</u>	13,850,000/1008	13,700
f. Liquor to 4e = <u>58,300</u>		
6. a. Heat in 5th vapors.....	13,850,000	
b. Condensate flash = $56,800(174 - 151)$	+ 1,310,000	
c. Total heat to 6th effect.....	15,160,000†	
d. Liquor flash = $72,000(170 - 130).90$	+ 2,590,000	
e. Evaporation = <u>17,300</u>	17,750,000/1022	17,300
f. Liquor to 4e = <u>54,700</u>		
g. Liquor to 4e from 5f = <u>58,300</u>		
h. 4e = <u>113,000</u>		

Total hourly evaporation = 90,000 lb

$$\text{Economy} = \frac{90,000}{20,000} = 4.50 \text{ lb/lb}$$

* Radiation factor.

† For surface calculation.

saturated pressure in the third effect. Subtracting the flash $1e$ from $1d$ should give the known product in $1f$. If a deviation from the product is obtained, slight changes are required in $1d$, $1c$, and $1b$ to complete the balance in the first effect.

In the second effect the value of $2a$ corresponds to the heat load of the vapors leaving the first effect and entering the second effect steam chest. The value of $2d$ is equal to $1b$, which has already been determined. An estimate of $2b$ determines $2c$. A check on the estimated value of $2b$ is obtained if $2b - 2c = 2d$. If they are unequal, a slight change in $2b$ will establish the equality.

The value of item $3a$ corresponds to the heat load of vapors leaving the second effect. Moreover, to increase steam economy the condensate from the second effect is discharged through a flash tank into a condensate header. The flash tank is piped to the steam chest of the third effect. Item $3b$ shows the equivalent heat load added to the steam chest of the third effect by flashing the condensate of the second effect into it.

Since item $3f$ equals $2b$, an estimate of $3d$ determines the value of $3e$. A check on the estimate $3d$ is determined by the equality of the values under $3d - 3e = 3f$.

Item $4a$ is the net heat load in the third effect vapors and $4b$ is the combined condensate flash from effects 2 and 3; *i.e.*, $17,200 + 14,900 = 32,100$ lb/hr. Item $4c$ corresponds to $1d$, since it was stated above that the discharge from the first effect passed through a flash tank piped to the steam chest of the fourth effect (or to the vapor line of the third effect). The flashed vapors are therefore used as an additional source of heat for evaporation and are included in the total heat load in the vapors entering the fourth-effect steam chest $4d$. Item $4e$ is estimated similarly to $3d$, etc. It will be noted, however, that in using a temperature differential for preheating the liquor to the boiling temperature, 180°F , the liquor was preheated from 143 to 180°F . From Table 14.5 it will be observed that the boiling points in effects 5 and 6 are, respectively, 156 and 130°F . Since the feed is passed through the last two effects in parallel, the common temperature of the combined discharged liquor from these effects is taken as their arithmetic average; *i.e.*, $(156 + 130)/2 = 143^\circ\text{F}$.

Item $5b$ in Table 14.6 is the combined condensate flash of effects 2, 3, and 4; *i.e.*, $17,200 + 14,900 + 12,800 = 44,900$ lb/hr. Since feed enters above its boiling-point temperature in both the fifth and sixth effects, a certain amount will flash on entering the two bodies. Items $5d$ and $6d$ show the flash calculations. The discharge from the fifth effect, $5f$, combined with the discharge from the sixth effect, $6f$, is pumped to the fourth effect.

Item $6b$ is the combined condensate flash of effects 2 to 5 inclusive. When added to $6a$ the sum gives the total heat load to the steam chest of the sixth effect.

If the heat balance is closed so that the total calculated hourly evaporation is found to be equal to that which is required in the material balance, then the sum of $5f$ and $6f$ will be equal to $4e$ and the estimated steam requirement will be satisfactory. If the heat balance does not close on the first trial, a revision of the steam estimate is made and 1 through 6 inclusive are repeated. To revise the steam requirements it is seen that the algebraic difference between the obtained and required evaporation divided by the estimated economy need only be added to the initial estimate of steam. Thus, if the calculated evaporation were $94,500$ lb/hr, $(90,000 - 94,500)/4.5 = -1000$ or approximately $20,000 - 1000 = 19,000$ lb/hr of steam would be used for reworking the balance. Ordinarily not more than two trials need be made to obtain a closure in the balance.

Each body in Fig. 14.35 is of the long-tube vertical-film type Fig. 14.27.

The water rating of a multijet barometric condenser is quite similar to that for a

countercurrent barometric type employed on sugar evaporators. The major difference is in the allowable approach to the vapor temperature. Whereas in the countercurrent barometric condenser it is permissible to use about 5°F approach for a good design, the multijet is designed conservatively for a 15°F approach. Thus, the gallons per minute required are

$$\frac{Q}{500(T_s - 15 - t_w)} = \frac{17,750,000}{500(125 - 15 - 60)} = 710 \text{ gpm}$$

In using a surface condenser it will not be desirable to approach the saturated vapor temperature closer than 15°F; thus, 710 gpm can be heated from 60 to 110°F for use as mill water.

CAUSTIC SODA CONCENTRATION—FORCED-CIRCULATION EVAPORATORS

Description of the Process. The concentration of caustic soda solutions obtained either from the reaction of soda ash and lime or from the electrolytic decomposition of brine present certain distinct problems: (1) Caustic solutions have a high elevation in boiling point which causes a large loss of available temperature difference in multiple-effect evaporators. (2) The concentrated solutions are highly viscous, reducing sharply the heat-transfer rate in natural-circulation evaporators. (3) Caustic solutions may have detrimental effects on steel, causing caustic embrittlement. (4) Caustic solutions may require the removal of large quantities of salt as the solution becomes concentrated.

In order to avoid caustic embrittlement of steel, solid nickel tubes are used along with nickel-clad plate-steel bodies. All auxiliary equipment which comes in contact with the caustic solution is also nickel clad. Due to the costliness of nickel as a material of construction, however, it is important to design this class of equipment for as small a heating surface as possible. Since the liquor-film heat-transfer coefficient is a function of the velocity of the caustic solution through the tubes, a high velocity must be established in order to obtain a large transfer coefficient in the tubes.

The process flow sheet is shown in Fig. 14.37. Cell liquor feed, along with filtrate and wash brine from the salt filter or centrifuge, is introduced into the filtrate feed tank *a*. From *a* the liquor is pumped into the second effect *b* of the evaporator where partial concentration and partial salt precipitation occur. The resulting semiconcentrated effluent from the second effect is pumped into the weak-liquor settler *c* along with the slurry underflow from the 50 per cent caustic settler *d*. The underflow from *a* contains all the precipitated salt from the system in the form of a slurry. This slurry is pumped to the filter or centrifuge to remove the precipitated salt and to wash the salt with brine. The filtrate entering the filtrate tank *a* is recirculated through the system with cell liquor feed

as described above. The overflow from *c* is pumped into the first effect of the evaporator *f* for further concentration and further salt precipitation. The effluent from the first effect is at a temperature of 246°F and contains concentrated caustic solution and a suspension of salt. This

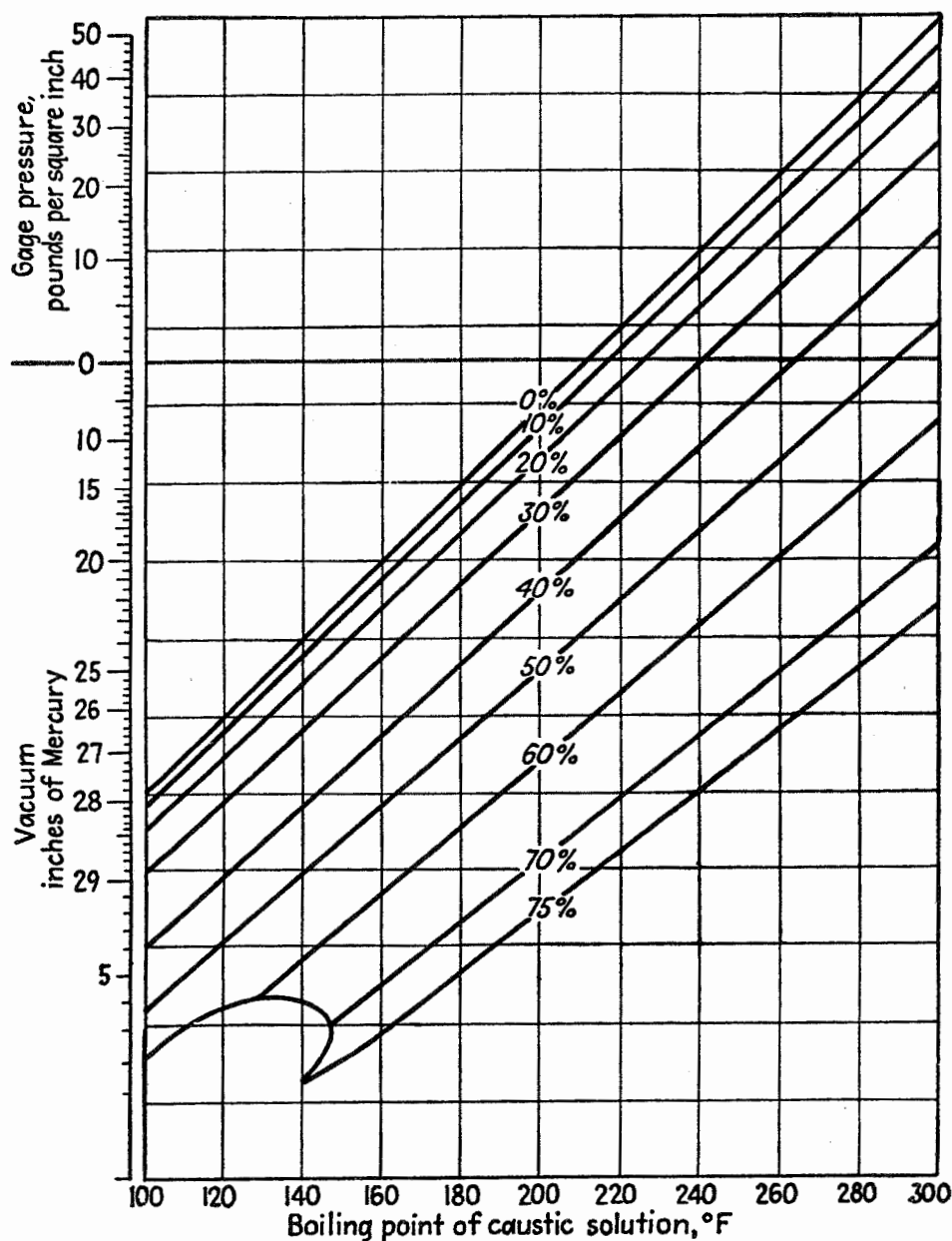


FIG. 14.38. Boiling-point-pressure relations of caustic soda solutions. (Columbia Alkali Corporation.)

slurry is pumped to the forced-circulation flash tank *g* where it is cooled to approximately 192°F and where further flash evaporation and further salt precipitation occurs. The flash-tank effluent is pumped to *d*. The underflow from this settler consists of a salt slurry which flows into *c* along with the second-effect effluent, and the salt is eventually removed

by *a* as described above. The overflow from *d* is the product liquor containing 50% NaOH, 2.7% NaCl, and 47.3% water.

Steam, usually 20 psig, is introduced into the heating element of the first effect. Here the steam condenses and gives up its latent heat to the liquor circulated through the tubes, causing this liquor to boil when it

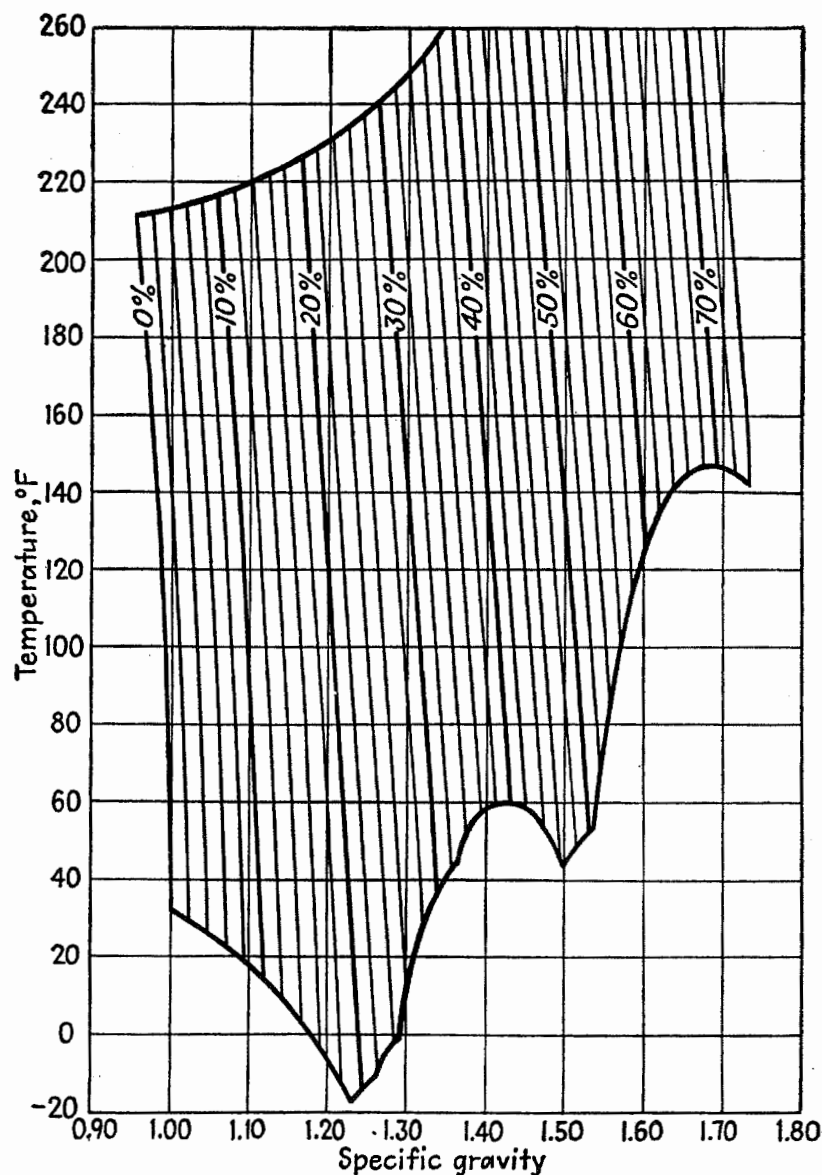


FIG. 14.39. Specific gravity of caustic soda solutions. (Columbia Alkali Corporation.)

reaches the vapor body of the first effect. The condensate from the first effect passes through a trap into a flash chamber *h*, where flashing occurs with the evolution of more vapors. The vapors from the first effect leave through a catchall *k*, returning the entrained liquor back to the bottom of the first-effect body. The stripped vapors leaving the catchall combine with the condensate flash vapors and are introduced into the second-effect heating element, where they are condensed and removed by

means of a condensate pump. The condensation of these vapors produce further boiling in the second-effect vapor body. The vapors leaving the second body pass through another catchall *l* into a barometric condenser *m*, where they are totally condensed by contact with cold water. The

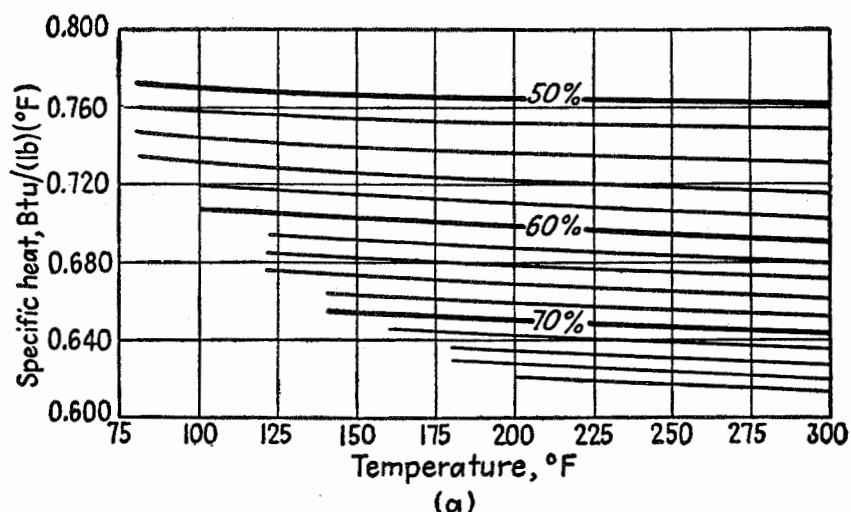


FIG. 14.40a. Specific heats of high-concentration caustic soda solutions.

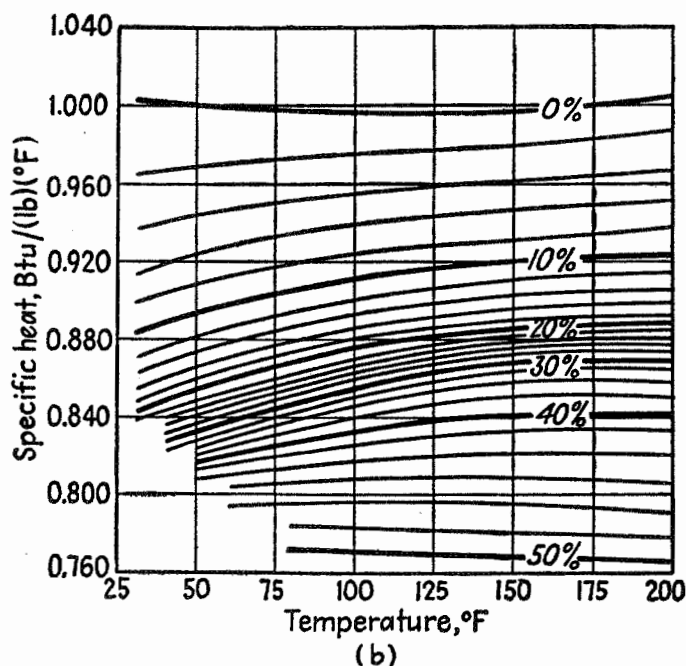


FIG. 14.40b. Specific heats of low-concentration caustic soda solutions. (Columbia Alkali Corporation.)

noncondensables from the system eventually reach the condenser through a system of vent piping, and there they are removed by a steam jet ejector *n*.

Example 14.6. Electrolytic cell liquor containing 8.75% NaOH, 16.60% NaCl, and 74.65% water at a temperature of 120°F is to be concentrated to a product containing 50% NaOH, 2.75% NaCl, and 47.25% water at a temperature of 192°F. The

capacity of the unit is 25 tons per 20-hour day based on 100% caustic soda. Physical properties are given in Figs. 14.38 to 14.41.

Solution. Table 14.7 is a material balance. It will be observed that the calculations are conveniently based on 1 ton NaOH production per hour, and therefore the numerals in Fig. 14.37 are obtained by using a factor of 1.25 greater than those shown on Table 14.7.

From the material balance and from the temperature-pressure distribution in Table 14.8 the heat balance (Table 14.9) can be worked out rather easily for this evaporator.

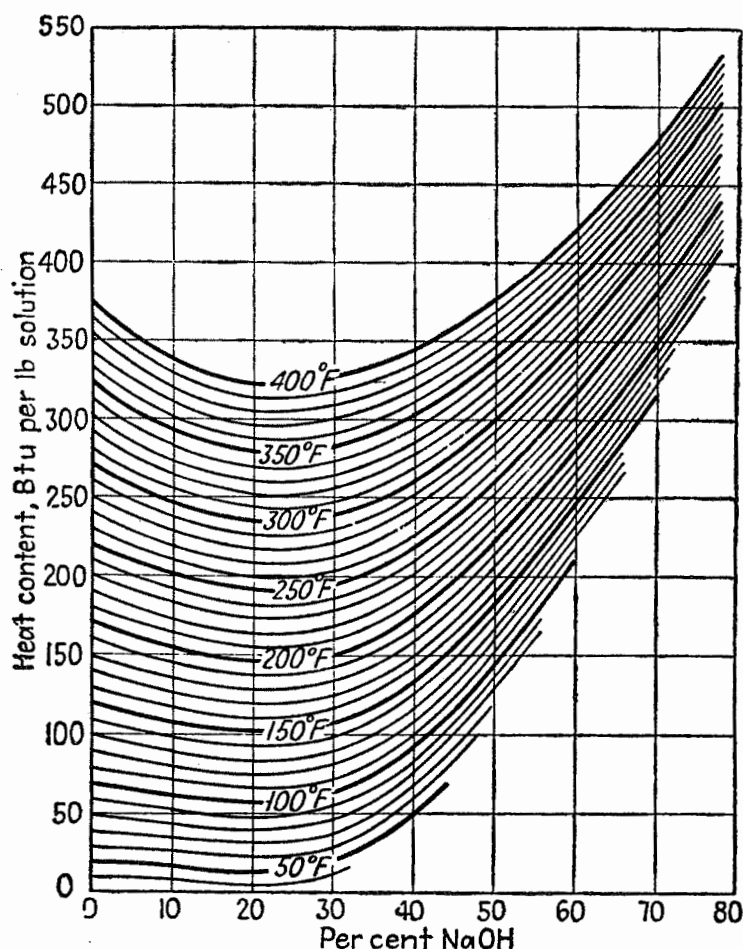


FIG. 14.41. Relative heat content of caustic soda solutions. (Columbia Alkali Corporation.)

In Table 14.8 the total temperature difference between the saturated live steam and 27 in. Hg vacuum is $274 - 115 = 159^{\circ}\text{F}$. From Fig. 14.38 the estimated BPR of the solutions in the first and second effects is 77 and 26°F , respectively, corresponding to concentrations appearing in Table 14.7. Thus, the effective temperature difference is $159 - (77 + 26) = 56^{\circ}\text{F}$. The latter is arbitrarily divided evenly between the two effects. In addition to the two effects of the evaporator, an additional amount of evaporation is obtained in the forced-circulation flash tank. The latter operates at 27 in. Hg and receives the concentrated slurry from the first effect and cools it from 246 to 192°F . The last column of Table 14.8 summarizes the thermal and material quantities for the flash tank.

TABLE 14.7. CAUSTIC EVAPORATOR MATERIAL BALANCE

Basis: 1 ton/hr NaOH

Cell liquor at 120°F		Wash at 80°F	
8.75 % NaOH =	2000	25 % NaCl =	340
16.6 % NaCl =	3800	75 % H ₂ O =	1020
74.65 % H ₂ O =	17050	Total wash =	1360
Total cell liquor =	22850		

	NaOH		NaCl		H ₂ O, Lb	Total, Lb
	%	Lb	%	Lb		
Overall operation:						
Cell liquor.....	8.75	2000	16.60	3800	17,050	22,850
Wash.....			25.00	340	1,020	1,360
Total in.....		2000		4140	18,070	24,210
Product.....	50.00	2000	2.75	110	1,890	4,000
Removed.....				4030	16,180	20,210
Weak-liquor settler operation: ^a						
From 2d effect.....		6000		5505	20,775	32,280
From 50 % settler.....	50.00	2500		1390	2,360	6,500
Total in.....		8500		6895	23,135	38,780
Solids.....	24.62	8500	8.22	2835	23,135	34,470
Salt pptd.....				4060		
Centrifuge operation: ^b						
From weak liquor settler.....		4000		5395	10,905	20,300
Wash.....			25.00	340	1,020	1,360
Total in.....		4000		5735	11,925	21,660
Solids.....	23.05	4000	9.84	1705	11,925	17,630
Salt pptd.....				4030		
2d effect operation: ^c						
From centrifuge.....	23.05	4000	9.84	1705	11,925	17,630
Cell liquor.....	8.75	2000	16.60	3800	17,050	22,850
Evaporator feed.....	14.82	6000	13.60	5505	28,975	40,480
Evaporation.....					8,200	8,200
Product.....		6000		5505	20,775	32,280
Solids.....	19.95	6000	11.22	3370	20,775	30,045
Salt pptd.....				2135		
1st effect operation: ^d						
From weak liquor settler.....	24.62	4500	8.22	1500	12,230	18,230
Evaporation.....					7,537	7,537
Product.....		4500		1500	4,693	10,693
Solids.....	47.05	4500	3.84	367	4,693	9,560
Salt pptd.....				1133		1,133
Flash-tank operation: ^e						
From 1st effect.....		4500		1500	4,693	10,693
Evaporation.....					443	443
Product.....		4500		1500	4,250	10,250
Solids.....	50.00	4500	2.75	248	4,250	8,998
Salt pptd.....				1252		1,252
From flash tank.....		4500		1500	4,250	10,250
50 % out.....	50.00	2000	2.75	110	1,890	4,000
Underflow.....		2500		1390	2,360	6,250
Salt pptd.....				1252		1,252
Solids.....	50.00	2500	2.75	138	2,360	4,998

^a Solution temperature (approximate) = 150°F.^b Solution temperature (approximate) = 145°F.^c Solution temperature (approximate) = 141°F.^d Solution temperature (approximate) = 246°F.^e Solution temperature (approximate) = 192°F.

It was noted that in a forced-circulation evaporator the surface is not a direct function of the temperature difference between the boiling solution and the saturated vapor temperature in the steam chest. The surface in a forced-circulation evaporator is determined by an economic balance between the size of the evaporator and the rate of circulation or, what amounts to the same thing, the power requirements of the pumps.

TABLE 14.8. CAUSTIC EVAPORATOR SUMMARY*

Item	Effect		Flash tank
	I	II	
1. Steam pressure, psi.....	30		
2. Steam temperature, °F.....	274	169	
3. Δt , °F.....	28	28	
4. Liquor temperature, °F.....	246	141	192
5. BPR, °F.....	77	26	77
6. Vapor temperature, °F.....	169	115	115
7. λ , Btu/lb.....	997	1,027	1,027
8. Feed, lb/hr*.....	22,788	50,602	13,367
9. Product, lb/hr*.....	13,367	40,352	12,813
10. Evaporation, lb/hr*.....	9,421	10,250	554
11. Heat flow, Btu/hr*.....	11,890,000	11,020,000	
12. U_D , Btu/(hr)(ft ²)(°F).....	700		
13. A , ft ²	683	683	
14. Tubes, OD, in. and BWG.....	1, 16	1, 16	
15. Tube length, ft.....	7	7	
16. No. tubes.....	432	432	
17. Circulating pump, gpm.....	3200 at 20 ft	3200 at 20 ft	167 at 45 ft
18. Apparent efficiency, %.....	54	64	
18. BHP.....	38	35	8.2
20. Motor, hp.....	40	40	10.0

* Correspond to actual flows in Fig. 14.37.

A method generally used to solve for surface requirements is indicated below. In designing for optimum surface it is customary to use about 8 fps velocity through the tubes of the heater. Furthermore, in this illustration tubes will be 1 in. OD, 16 BWG, solid nickel, 7'0" long arranged on a two-pass layout. The heaters must be designed for sufficient surface to transfer enough heat from the vapor or steam side to satisfy the heat quantities under 11 of Table 14.8.

For effect I. $0.8w(t_x - 246) = 11,840,000 = U_D A \Delta t$. That is, by circulating w lb/hr of liquid through the tubes and raising the temperature of the latter to t_x , a small temperature differential above the saturated-solution boiling temperature in the flash chamber, a gentle flash will occur with not too rapid a crystal growth. By assuming a given value t_x , both w and Δt are determined. Furthermore, by designing for a given velocity of 8 fps, the value of U_C , the clean overall coefficient, is determined from the thermal characteristics of the solution. This is illustrated below:

$$G = V(s \times 62.5 \times 3600) = 8 \times 1.5 \times 62.5 \times 3600 = 2,700,000, \text{ lb/(hr)(ft}^2\text{)}$$

For this mass velocity a tube-film coefficient approximately equal to 1375 can be obtained from the methods employed in Chap. 7. Combining the latter with a steam film coefficient of approximately 1500 will give a value for U_C or U_D equal to 700. Setting up the following table:

t_x , °F	w , lb/hr	Δt	U_c	A , ft ²	a , flow area per pass, ft ²	G_{calc}	U_{calc}
251	2,970,000	25.4	700	670	0.87	3,420,000	700
252	2,480,000	25.0	700	680	0.88	2,820,000	
252.5	2,290,000	24.7	700	686	0.89	2,570,000	
253	2,120,000	24.5	700	695	0.90	2,520,000	

Thus the gain per minute for circulation is 3200, and from pressure-drop calculations in tubes, the static head, etc., a total dynamic head of 20 ft is found to be required.

TABLE 14.9. CAUSTIC EVAPORATOR HEAT BALANCE
Basis = 1 ton/hr NaOH

Effect	Btu/hr	Evaporation, lb/hr
1. a. Heat in steam = $10,500 \times 930 \times 0.974$	9,500,000	7,750
b. Heating liquor = $18,230 \times (246 - 150) \times 0.83$	1,470,000	
c. Resultant heat.....	8,030,000	
d. Heat of concentrate.....	300,000	
e. Heat of vapors.....	7,730,000/997	
2. a. Heat of vapors.....	7,730,000	8,200
b. Condensate flash = $10,500(274 - 169)$	1,090,000	
c. Heat flow.....	8,820,000	
d. Heating liquor = $40,480(141 - 130)0.92$	410,000	
e. Heat of vapors.....	8,410,000/1027	
Flash tank:		
a. Liquor flash = $10,693(246 - 192)0.79$..	460,000	443
b. Heat of concentrate.....	10,000	
c. Heat of vapors.....	450,000/1027	
Total evaporation.....		16,393
Condenser:		
Heat of flash tank vapors (above 110°F).	452,000	8,902,000/20 = 445,000 lb = 890 gpm/(ton)(hr) of NaOH
Heat of vapors 2 (above 110°F).....	8,450,000	
Total heat in (above 110°F).....	8,902,000/20 = 445,000 lb	
Water $\Delta t = (110 - 90) = 20^\circ$		

For effect II. $0.92w(t_x - 141) = 11,020,000 = U_D A \Delta t$. For a velocity of 8 fps $G = 8 \times 1.35 \times 62.5 \times 3600 = 2,430,000$ lb/(hr)(ft²). Using thermal characteristics for this solution gives an approximate $U_D = 700$ Btu/(hr)(ft²)(°F).

As for effect I:

t_x , °F	w , lb/hr	Δt	U_c	A , ft ²	a , flow area per pass, ft ²	G_{calc}	U_{calc}
146	2,400,000	25.4	700	620	0.80	2,790,000	700
146.5	2,160,000	25.2	700	683	0.89	2,430,000	

From the design coefficient above a smaller surface could have been used. However, since it is desirable to duplicate both the surface and pump of the first effect, the larger surface is chosen.

THERMOCOMPRESSION

The principle of thermocompression is continuously finding wider application in industry. Consider a single-effect evaporator. It is fed with steam and generates vapors which have nearly as much heat content

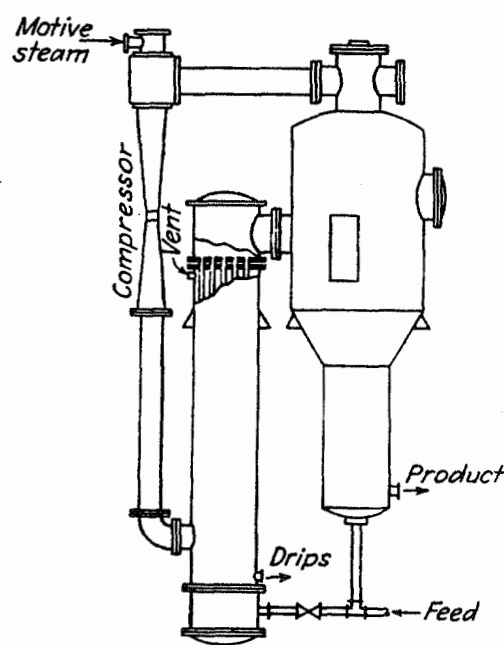


FIG. 14.42. Thermocompression evaporator. (Buffalo Foundry & Machine Company.)

as was originally present in the steam. These vapors are condensed with water as a convenient means of removal, but this is a severe waste of both Btu and water. Were it not for the fact that the temperature of the generated vapors is lower than the steam, it would be possible to circulate the vapor back into the steam chest and evaporate continuously without supplying additional steam. But apart from the heat balance, a temperature difference must exist between the steam and the generated vapors or no heat would have been transferred originally. If the vapors from the evaporator were compressed to the saturation pressure of the steam, however, the temperature of the vapors could be raised to that

of the original steam. The cost of supplying the necessary amount of compression is usually small compared with the value of the latent heat in the vapors.

The practice of recompressing a vapor to increase its temperature and permit its reuse is *thermocompression*. Where fuel is costly, the compression may be accomplished with a centrifugal compressor as in the Kleinschmidt still for the production of distilled water. Where steam is available at a pressure higher than that required in the evaporator, the

recompression can be effected in a steam jet booster. The latter operates on the principle of an ejector and is used to compress vapors instead of a noncondensable gas. A thermocompression evaporator of the type using a booster ejector is shown in Fig. 14.42.

The pressure at which steam is discharged from a thermocompressor depends upon the pressure and proportion at which the live steam and the vapors are supplied. It will be apparent from the analysis below that the higher the desired discharge pressure the greater the percentage of live steam required. Table 14.10 compares the proportions of live steam to entrained vapors for steam-chest pressures of 18 to 26 psig when live steam is available to the jet at 150 psig and the entrained vapors are taken off at 15 psig.

TABLE 14.10. THERMOCOMPRESSOR ECONOMY

Steam-chest pressure, psig.....	18	20	22	24	26
Entrainment ratio, lb entrained vapor/lb live steam.....	3.33	2	1.43	1.08	.87
Amount of live steam saved, %.....	66	67	59	52	46

Table 14.10 shows substantial savings of live steam for fixed quantities of low-pressure steam discharged from the compressor. Figure 14.43 represents the variation of the reciprocal of the entrainment ratio with steam-chest pressure for entrained vapors at different pressures.

Derivation of charts similar to Fig. 14.43 can be made by analyzing the thermodynamic conditions imposed on the jet thermocompressor. Referring to Fig. 14.44, high-pressure steam is caused to expand in the nozzle *a* from which it issues with a high velocity into the mixing space *b* where it transfers some of its momentum to the vapor sucked in. In the diffuser section *d*, which is the reverse of a nozzle, the mixed vapors are compressed to the steam-chest pressure p_3 . The work of compression results from the conversion of kinetic energy of the high-velocity mixture into pressure head. Thus, vapor at low pressure p_2 is entrained and compressed to a higher pressure p_3 at the expense of energy in the motive steam. Using the treatment of Kalustian,¹

let H_1 = enthalpy of steam at p_1 , Btu/lb

H_2 = enthalpy of steam after isentropic expansion in nozzle to pressure p_2 , Btu/lb

H_2' = enthalpy of steam after actual expansion in nozzle to pressure p_2 , Btu/lb

e_1 = efficiency of the nozzle

¹ Kalustian, P., *Refrig. Eng.*, **28**, 188-193 (1934).

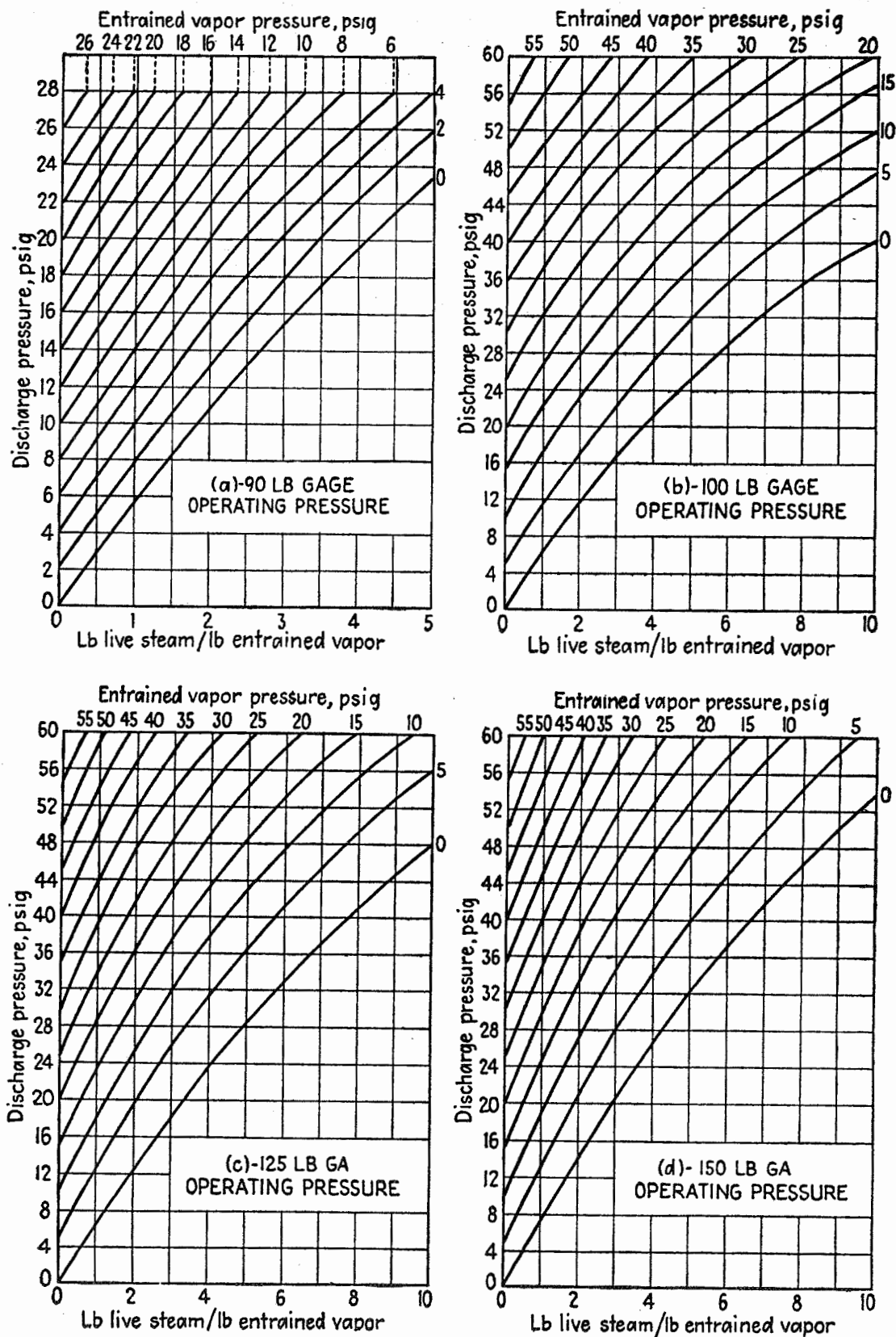


FIG. 14.43. Thermocompressor steam entrainment. (Schutte & Koerting Company.)

Then

$$e_1 = \frac{H_1 - H_2'}{H_1 - H_2} = \frac{\text{actual work of expansion}}{\text{theoretical work of expansion}} \quad (14.21)$$

Let H_3 = enthalpy of the mixture at start of compression in the diffuser section at p_2 , Btu/lb

H_4 = enthalpy of the mixture after isentropic compression from p_2 to the discharge pressure p_3 , Btu/lb

e_2 = efficiency of compression in the diffuser

Then

$$e_2 = \frac{H_4 - H_3}{\text{actual work of compression}}$$

Let M_2 = entrained vapor at pressure p_2 lb

M_1 = motive high-pressure steam at pressure p_1 , lb

The actual work required for compressing the mixture of vapors in the diffuser is given by $(M_1 + M_2)(H_4 - H_3)/e_2$. The actual work obtain-

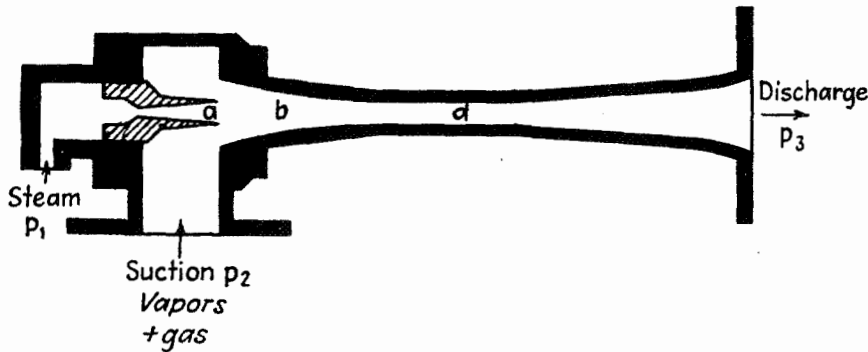


FIG. 14.44. Ejector.

able from the expansion of the high-pressure steam is less than the theoretical value $(H_1 - H_2)$ because of friction in the nozzle, which is taken into account by the nozzle efficiency e_1 . There is a further loss in available energy in the transfer of momentum from the high-velocity jet to the relatively slow-moving entrained vapor. If e_3 is the efficiency of momentum transfer, the net available work from the jet is

$$M_1 e_1 e_3 (H_1 - H_2).$$

By equating the net available work obtained from the jet to the actual work required for compressing the mixture in the diffuser, one obtains

$$\text{Entrainment ratio, } \frac{M_2}{M_1} = \left[\frac{(H_1 - H_2)}{(H_4 - H_3)} e_1 e_2 e_3 - 1 \right],$$

lb vapor entrained/lb steam (14.22)

Equation (14.22) must be solved by trial and error, since H_3 and H_4 are functions of x_3 , the quality of the steam at p_3 which is in itself a func-

tion of M_2 and M_1 . If one assumes no heat and no enthalpy change due to the mixing a balance on the mixing process gives

$$x_3(M_1 + M_2) = x_{2''}M_1 + x_4M_2 \quad (14.23)$$

$$x_3 = \frac{x_{2''}M_1 + x_4M_2}{M_1 + M_2} \quad (14.24)$$

where $x_{2''}$ is the quality of motive steam after expansion to p_2 and after it has lost its kinetic energy in the entraining process, x_4 is the quality of the entrained steam, and $x_{2'}$ is related to x_2 , the quality after expansion but before the entrainment step, by the equation

$$(1 - e_3)(H_1 - H_2) = (x_{2''} - x_{2'})\lambda \quad (14.25)$$

where λ is the latent heat of vaporization of the fluid at p_2 . $x_{2'}$ is related to x_2 , the quality after isentropic expansion, by

$$(1 - e_1)(H_1 - H_2) = (x_{2'} - x_2)\lambda$$

As an illustration, take the case where $p_1 = 150$ psig, $p_2 = 9$ psig, $H_1 = 1196$ Btu/lb (from Table 7), $H_2 = 1050$ Btu/lb (after isentropic expansion from p_1 to p_2), and $x_2 = 0.885$. It will be assumed that the values of e_1 , e_2 , e_3 are, respectively, 0.98, 0.95, and 0.85.

$$H_1 - H_{2'} = 0.98(1196 - 1050) = 143$$

and therefore $H_{2'} = 1053$. From Table 7 $\lambda = 954$ and from Eq. (14.22)

$$954(x_{2'} - x_2) = (1 - 0.98)(1196 - 1050)$$

and therefore $x_{2'} = 0.89$. From Eq. (14.25)

$$(1 - 0.85)(143) = 954(x_{2''} - 0.89)$$

and therefore $x_{2''} = 0.91$. Assuming $x_3 = 0.95$,

$$H_3 = 1159(0.95) = 1100.$$

Since the entrained steam at p_2 is assumed saturated, $x_4 = 1$ and therefore $H_4 = 1164$; hence from Eq. (14.22)

$$\frac{M_2}{M_1} = \frac{146}{64} (0.79) - 1 = 1.80 - 1 = 0.8$$

As a check on the assumed value of x_3 , from Eq. (14.24) above

$$x_3 = \frac{x_{2''} + x_4(M_2/M_1)}{1 + M_2/M_1} = \frac{0.91 + 0.80}{1 + 0.80} = \frac{1.71}{1.8} = 0.95$$

(which checks the assumed value).

Since the entrainment ratio M_2/M_1 is a function of the thermal (expansion and compression) and mechanical (mixing) efficiencies of the compressor, specialized and good design of the different elements of the

unit will increase the overall efficiency $e_1 e_2 e_3$. The closer this product approaches unity the higher the entrainment ratio or the greater the saving in live steam. In good design, overall efficiencies approach 0.75 to 0.8, with individual efficiencies in the order of $e_1 = 0.95 - 0.98$, $e_2 = 0.90 - 0.95$, and $e_3 = 0.80$ to 0.85.

To study the saving in steam, the reduction in water requirements and condenser size, a comparison will be made of a straight triple-effect evaporator and one involving the use of a thermocompressor in conjunction with a triple-effect evaporator.

THERMOCOMPRESSION SUGAR EVAPORATOR

Example 14.7. It is required to concentrate 100,000 lb/hr of a 10°Brix sugar solution to 30°Brix. Exhaust steam at 20 psig is available in limited quantity. Water is available at 85°F and also in limited quantity owing to cooling-tower capacity. The feed will enter at 230°F. The thermocompressor evaporator will recompress vapors from the second effect as shown in Fig. 14.45 using saturated steam at 150 psig.

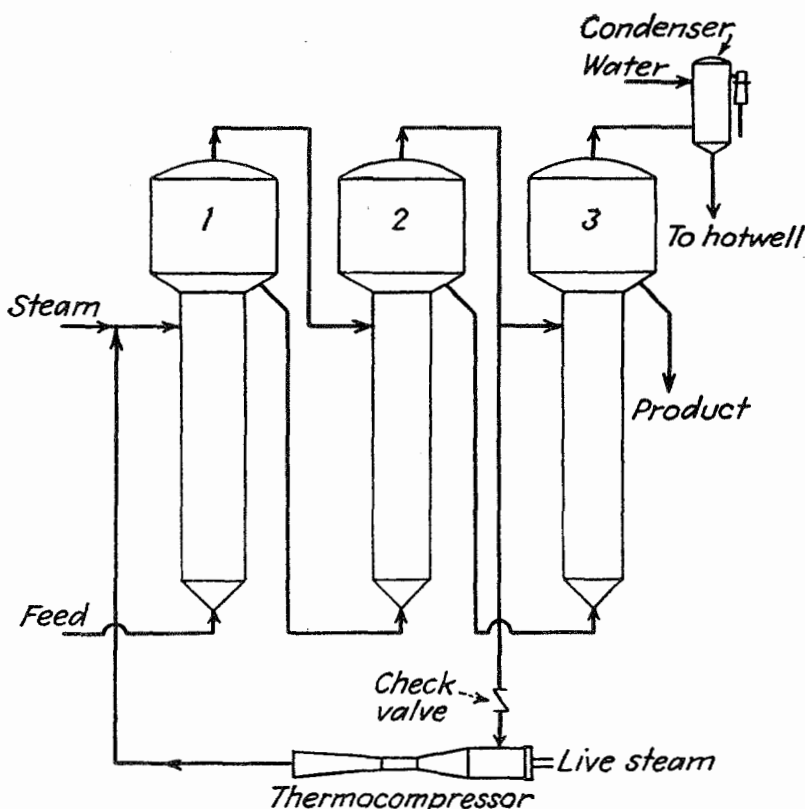


Fig. 14.45. Thermocompression sugar evaporator.

Solution. The analysis is shown in Tables 14.11 to 14.13. The live steam required for the thermocompressor is $M_1 = 32,200 - 14,300^* = 17,900$.

Saving in live steam over straight triple effect = 20 per cent

Saving in water over straight triple effect = 20 per cent

* $M_1/M_2 = 1.25$ from Fig. 14.43. $M_1 + M_2 = 32,200$ from heat balance above, and therefore $M_2 = 32,200/(1.25 + 1) = 14,300$.

TABLE 14.11. EVAPORATOR SUMMARY

Effect	Straight triple effect			Thermocompression		
	1	2	3	1	2	3
Steam flow, lb/hr.....	22,400			17,900*		
Steam pressure, psi.	20	9	2	20	9	2
Steam temp, °F.....	258	237	217	258	237	217
Δt , °F.....	20	18	22	20	18	22
Liquor temp, °F.....	238	219	195	238	219	195
BPR, °F.....	1	2	3	1	2	3
Vapor temp, °F.....	237	217	192	237	215	192
Vapor pressure, psi/in. Hg.....	9	2	10''	9	2	10''
λ , Btu/lb.....	954	965	983	954	965	983
Liquor in, lb/hr.....	100,000	79,400	56,900	109,000	70,000	52,400
Liquor out, lb/hr.....	79,400	56,900	33,300	70,000	52,400	33,300
Evaporation, lb/hr.....	20,600	22,500	23,500	30,000	17,600	19,100
°Brix (out).....						30
Condenser water, gpm.....		455			365	

* Live steam to thermocompressor.

TABLE 14.12. HEAT BALANCE—STRAIGHT TRIPLE EFFECT

$$\text{Condenser water} = \frac{23,232,000}{500(192 - 5 - 85)} = 455 \text{ gpm}$$

Effect	Btu/hr	Evaporation, lb/hr
1. a. Heat in steam = $(22,400)(940)(.97)$	20,400,000	20,600
b. Heating liquor = $100,000(238 - 230).92$	735,000	
c. Evaporation = <u>20,600</u>	19,665,000/954	
d. Liquor to 2d = <u>79,400</u>		
2. a. Heat in 1st vapors.....	19,665,000	22,500
b. Steam cond. flash = $22,400(258 - 237)$	470,000	
c. Liquor flash = $79,400(238 - 217).89$	1,485,000	
d. Heat for evaporation.....	21,620,000/965	
e. Evaporation = <u>22,500 lb/hr</u>		23,600
3. a. Heat in 2d vapors.....	21,620,000	
b. Condensate flash = $20,600(237 - 217)$	412,000	
c. Liquor flash = $56,900(217 - 192).85$	1,200,000	
d. Heat for evaporation.....	23,232,000/983	66,700
e. Product = $56,900 - 23,600 = 33,300$ lb/hr		
f. Total evaporation.....		

TABLE 14.13. HEAT BALANCE—THERMOCOMPRESSION EVAPORATOR

$$\text{Condenser water} = \frac{18,662,000}{500(192 - 5 - 85)} = 365 \text{ gpm}$$

Effect	Btu/hr	Evaporation lb/hr
1. a. Heat in steam = (32,200)(940)(.97).....	29,300,000	30,000
b. Heating liquor = 100,000(238 - 230).92.....	735,000	
c. Heat for evaporation.....	28,565,000/954	
d. Liquor to 2d = 70,000 lb/hr		
2. a. Heat in 1st vapors.....	28,565,000	17,600
b. Heat removed entrained vapors by thermo- compression = $\left(\frac{32,200}{1.25 + 1}\right) 954$	13,600,000	
c. Heat in 1st vapors.....	14,965,000	
d. Condensate flash = 32,200(258 - 237).....	677,000	
e. Liquor flash = 70,000(238 - 217).89.....	1,310,000	
f. Heat for evaporation.....	16,952,000/965	
g. Liquor to 3d = 70,000 - 17,600 = 52,400 lb/hr		
3. a. Heat in 2d vapors.....	16,952,000	19,100
b. Condensate flash = 30,000(237 - 217).....	600,000	
c. Liquor flash = 52,400(217 - 192).85.....	1,110,000	
d. Heat for evaporation.....	18,662,000/983	
e. Product = 52,400 - 19,100 = 33,300 lb/hr		
f. Total evaporation.....		66,700

The saving in utilities illustrated in the above problem is of importance only if there is a shortage of them. In all circumstances, however, installation of a thermocompressor in conjunction with a multiple-effect evaporator involves balancing the initial outlay of the additional equipment against the reduction in utilities and in condenser size, change in evaporator body sizes, etc. If auxiliary motive power such as a turbine-driven pump is required in the plant, the available power from 150 psig live steam exhausting to 20 psig should be included in the economy.

Although the above treatment of thermocompression is of interest because of the large savings in utilities which it effects when used in conjunction with a multiple-effect evaporator, it is recommended also in applications where multiple-effect evaporation cannot be used. Thus the use of thermocompression evaporators is particularly advantageous in concentrating liquids which are sensitive to high temperatures. Low operating temperatures are assured because a high vacuum can be maintained. The steam used in a steam chest is usually below atmospheric pressure.

PROBLEMS

14.1. It is desired to concentrate a caustic soda solution from 12.5 to 40 per cent in a double-effect evaporator. 50,000 lb/hr of feed solution enters the evaporator at 120°F, and steam at 15 psig is available for effecting the concentration. Vacuum in the second effect is held at 24 in. Hg, and water from a cooling tower is available at 85°F. Estimate (a) the quantity of steam and water required for forward feed operation; (b) the quantity of steam and condenser water required for backward feed operation; (c) the surface required for operation in (a), assuming overall coefficients of heat transfer of 400 and 250 Btu/(hr)(ft²)(°F) for the first and second effects; (d) the surface in (b) for values of U_D equal to 450 and 350 in the two effects respectively; (e) if equal bodies were required for (c) and (d), what steam economy could be expected.

14.2. In Prob. 14.1 it is desired to concentrate the caustic soda solution in a forced-circulation double-effect evaporator unit. Only a limited amount of steam at 15 psig is available, and it is therefore decided to use turbine-driven circulation pumps taking 150 psig steam and exhausting at 15 psig. Assuming feed rate, vacuum, and water temperature as in Prob. 14.1 and forward-feed operation and equal surface for both effects, estimate (a) the water rate required for the turbine if its efficiency is taken at 70 per cent, (b) the pounds of 15 psig steam required for the desired concentration, (c) the make-up steam at 15 psig required if the turbine exhaust is used for the concentration and a solution circulation rate of 6 fps is assumed in the heaters of the two effects. (*Hint.* Assume $U_D = 400$; estimate surface, cross-sectional flow area, total circulation, and total head and power required.) (d) Estimate surface required for each effect.

14.3. A 20,000 lb/hr sugar solution at 180°F is to be concentrated from 12 to 30°Brix in a double-effect, forward-feed, calandria-type evaporator. Assuming that exhaust steam from a steam engine is available at 5 psig and the vacuum in the second effect is held at 23 in. Hg, estimate (a) steam economy; (b) surface of each body, assuming U_D of 500 and 200 for first and second effects, respectively, and equal surface for each effect; (c) water required for a barometric condenser, assuming that 90°F water is available and a 10°F approach to condensing temperature is permitted.

14.4. A triple-effect sugar evaporator is to be designed to concentrate 100,000 lb/hr of solution at 150°F from 14 to 50°Brix. Steam is available at 150 psig, and water at 75°F. Steam to the first effect will be 25 psig, and the vacuum in the last effect will be held at 24 in. Hg. Vapors will leave the first effect at 15 psig and will partly be compressed by a thermocompressor using 150 psig steam and partly enter the second effect as steam. Assuming equal surfaces for the three bodies, forward-feed arrangement, and a 5°F approach in the condenser, estimate (a) the total steam required to the first effect, (b) the total 150 psig steam used, (c) economy, (d) water required in the condenser, (e) surface required, assuming $U_D = 500, 300$, and 150 for the three effects, respectively.

14.5. A paper mill producing 300 tons of pulp per 24 hr by the magnesium sulfite process concentrates a 12 per cent waste liquor to 55 per cent in a quintuple-effect long-tube evaporator. The solution entering the last two effects in parallel at 135° is evaporated by 45 psig saturated steam. Assume the following: (a) Feed contains 2800 lb total solids per ton pulp produced; (b) first effect is divided into two parallel bodies of 4500 and 7000 ft², respectively; (c) water is available at 75°F; (d) vacuum in the last-effect vapor body is 26 in. Hg; (e) 7°F water approach to condenser saturation temperature; (f) BPR is 18, 16, 13, 10, 5, 7°F, respectively, for all bodies; (g) product from first effect is flashed to fourth-effect steam chest; (h) vapor condensate flashes between effects.

Find (a) material and heat balance for the evaporator, (b) total water required (gallons per minute), (c) the total surface required, (d) steam economy of the evaporator.

14.6. Residual liquor from the electrolysis of brine in chlorine cells flows to a caustic concentration system at the rate of 1176 tons per 21-hr day. Concentration to 50% NaOH is effected in a triple-effect, forced-circulation evaporator. The latter boils off the water content in three successive stages, thereby crystallizing the salt content and increasing the percentage of caustic soda contained in the circulating solution.

The liquor flow is as follows:

Order of flow	Analysis of outgoing liquor flow			
	% crystal- lized salt	% dis- solved salt	% dis- solved NaOH	% water
1. Cell liquor at 160°F.....	0	14.95	11.00	74.05
2. 2d effect evaporator.....	3.56	15.47	13.97	67.00
3. 3d effect evaporator.....	7.55	8.08	22.43	61.94
4. Weak-liquor settler.....	0*	8.00*	25.70*	66.30*
5. 1st effect evaporator.....	8.23	4.84	42.10	44.83
6. Flash tank.....	11.35	2.40	44.30	41.95
7. 50% caustic settler.....	0*	2.70*	50.00*	47.30*

* Per cent changes owing to removal of crystallized salt at these stages to salt-recovery system.

Utilities:

Steam at 75 psig

Water to barometric condenser at 80°F

Condenser held at 27¼ in. Hg vacuum (third effect at 27 in. Hg vacuum)

Determine (a) the material balance in the system, (b) the heat balance in the system; estimate (c) the surface required (Ans. three 1800 ft² external heaters), (d) total water required, (e) steam economy.

14.7. A 15°Brix sugar solution is to be concentrated to 60°Brix in a quadruple-effect calandria-type evaporator. Part of the vapors from the first, second, and third effects will heat the cold solution from 100 to 220°F in heat exchangers prior to injection into the first effect of the evaporator system. Dilute juice is concentrated at the rate of 500 gpm, using live steam at 25 psig. Estimate the following: (a) material balance through the evaporator system, (b) heat balance on the evaporator system assuming condensate flashing between effects and 26½ in. Hg vacuum in fourth effect, (c) steam economy, (d) water rate to a barometric condenser assuming 75°F water available, (e) surface required for the heaters, (f) surface required for the calandria bodies, assuming equal surface for each body.

NOMENCLATURE FOR CHAPTER 14

A	Heat-transfer surface, ft ²
a_t	Tube-side flow area, ft ²
BPR	Boiling point rise, °F
C	Specific heat of hot fluid in derived equations, Btu/(lb)(°F)
c	Specific heat, Btu/(lb)(°F)

D	Inside diameter of tubes, ft
e_1, e_2, e_3	Efficiency of nozzle, diffuser, and momentum transfer, respectively, dimensionless
G	Mass velocity, lb/(hr)(ft ²)
g	Acceleration of gravity, ft/hr ²
H	Enthalpy; H_1 , motive steam, H_2 after isentropic expansion in nozzle, H'_2 after actual expansion in nozzle; H_3 at start of compression in diffuser; H_4 after compression in diffuser, Btu/lb
M_1, M_2	High-pressure motive steam and entrained vapor, respectively, lb
p_1, p_2, p_3	Pressure of motive steam, suction, and ejector discharge, respectively, psi
p_s, p_w	Vapor pressure of solution and or pure water, respectively, psi
Δp	Hydrostatic pressure, ft
Q	Heat flow, Btu/hr
T	Temperature of the hot fluid, °F
T_R	Boiling temperature, °R
t	Temperature of fluid in general, °F
t_a, t_s, t_w	Approach temperature, temperature of solution and of pure water, °F
t'_s, t'_w	Boiling point of solution and of pure water, °R
$\Delta t, (\Delta t)_a$	True temperature difference, apparent temperature difference, °F
$(\Delta t)_w$	Temperature difference between the wall and liquid, °F
Δt_h	Temperature rise due to hydrostatic heat, °F
$\Delta t_s, \Delta t_w$	Temperature change of boiling points of solution and pure water, °F
U	Overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
U_C, U_D	Clean and design overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
w_e	Evaporation, lb/hr
v	Specific volume, ft ³ /lb
x	Quality, subscripts correspond to p_1, p_2 , and p_3 , per cent
z	Height, ft
β	Coefficient of thermal expansion, per cent/°F
λ	Latent heat of vaporization, Btu/lb
λ_s, λ_w	Latent heat of vaporization of solution and pure water, Btu/lb
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$

Subscripts (except as noted above)

F	Feed
S	Steam
v	Vapor or vaporization
1, 2, 3, 4	Effect

CHAPTER 15

VAPORIZERS, EVAPORATORS, AND REBOILERS

In the preceding chapter calculations were included for only those types of evaporators which are usually designed on the basis of "accepted" overall heat-transfer coefficients. It is unfortunate that so important a class of apparatus is designed in this way, but the properties of concentrated aqueous solutions present problems which leave little alternative. When applied to solutions of liquids in water rather than solids in water, the evaporator may be calculated by methods similar to those of earlier chapters. Heat-transfer coefficients in the basket, calandria, and horizontal-tube evaporators are excluded from further discussion.

For the sake of clarity it is well to repeat the definitions adopted in Chap. 14 for dealing with *vaporizing exchangers*. Any unfired exchanger in which one fluid undergoes vaporization and which is not a part of an evaporation or distillation process is a *vaporizer*. If the vapor formed is steam, the exchanger is referred to as an *evaporator*. If a vaporizing exchanger is used to supply the heat requirements of a distillation process as vapors at the bottom of the distilling column, it is a *reboiler* whether or not the vapor produced is steam. The process requirements of evaporators have already been treated, and those of vaporizers and reboilers will be discussed presently. The principles and limitations which apply to vaporizers and vaporizer processes specifically apply also to evaporators and reboilers and should be regarded accordingly.

Vaporizing Processes. Vaporizers are called upon to fulfill the multitude of latent-heat services which are not a part of evaporative or distillation processes. The heat requirements are usually very simple to compute. Perhaps the commonest type of vaporizer is the ordinary horizontal 1-2 exchanger or one of its modifications, and vaporization may occur in the shell or in the tubes. If steam is the heating medium, the corrosive action of air in the hot condensate usually makes it advantageous to carry out the vaporization in the shell.

There are some fundamental differences between the operation and calculation of vaporization as treated in Chap. 14 and in the 1-2 horizontal or vertical vaporizer. In the power-plant evaporator, for example, the upper 50 or 60 per cent of the shell is used for the purpose of disengaging the liquid entrained by the bursting bubbles at the surface of

the pool. The disengagement is further implemented by the use of a steam separator in the shell. The mechanical design and thickness of the evaporator shell, flanges, and tube sheets are based upon the product of the shell-side pressure and the diameter of the shell. In the majority of instances the pressure or vacuum is not great and the shell, flange, and tube-sheet thicknesses are not unreasonable. In the case of a vaporizer, however, operation is often at high pressure, and it is usually too expensive to provide disengagement space in the shell, since the inclusion of disengagement space at high pressures correspondingly increases the shell thickness. For this reason vaporizers are not usually designed for internal disengagement. Instead some external means, such as an inexpensive welded drum, is connected to the vaporizer wherein the entrained liquid is separated from the vapor.

When a vapor evaporates from the surface of a pool, as in the power-plant evaporator, it is possible to evaporate 100 per cent of the liquid fed to it without reducing the level of the pool provided the evaporator was originally filled to operating level with liquid. The reason less than 100 per cent of the feed is normally vaporized is because residue accumulates and it is necessary to provide a blowdown connection for its removal.

When a 1-2 exchanger is used as a vaporizer, it is filled with tubes and cannot be adapted for blowdown, since all the feed to a vaporizer is usually of value and a rejection as blowdown is prohibitive. If the feed were completely vaporized in the vaporizer, it would emerge as a vapor and any dirt which was originally present would be left behind on the tube surface over which total vaporization occurred, fouling it rapidly. If the 1-2 exchanger (vaporizer) were over-designed, that is, if it contained too much surface, disengagement would have to occur on the tubes and due to the excess surface the vapor would superheat above its saturation temperature. The latter is undesirable in most processes, since superheated vapors subsequently require surface elsewhere for desuperheating. In the case of a reboiler it will be shown later that superheat actually reduces the performance of the distilling column. These factors establish a rule which should always be employed for the calculation of vaporization processes: *The feed to a vaporizer should not be vaporized completely.* The value of this rule is apparent. If less than 100 per cent of the feed is vaporized in a 1-2 exchanger, the residual liquid can be counted on to prevent the accumulation of dirt directly on the surface of the heating element. A maximum of about 80 per cent vaporization appears to provide favorable operation in 1-2 exchangers, although higher percentages may be obtained in vessels having internal disengagement space.

Forced- and Natural-circulation Vaporizer. When liquid is fed to a vaporizer by means of a pump or gravity flow from storage, the vaporizer

is fed by *forced circulation*. A typical example is shown in Fig. 15.1. The circuit consists of a 1-2 exchanger serving as the vaporizer and a disengaging drum from which the unvaporized liquid is withdrawn and recombined with fresh feed. The generated vapor is removed from the top of the drum.

Since it is desirable to vaporize only 80 per cent of the liquid entering the vaporizer, the total liquid entering will be 125 per cent of the quantity of vapor required. Even better cleanliness will result if less than 80 per cent of the liquid is vaporized, and this can be accomplished by *recirculation*. Suppose 8000 lb/hr of vapor is required. The liquid entering the vaporizer will be $1.25 \times 8000 = 10,000$ lb/hr. In this manner 80 per cent will be vaporized when 10,000 lb/hr of liquid passes into the vapor-

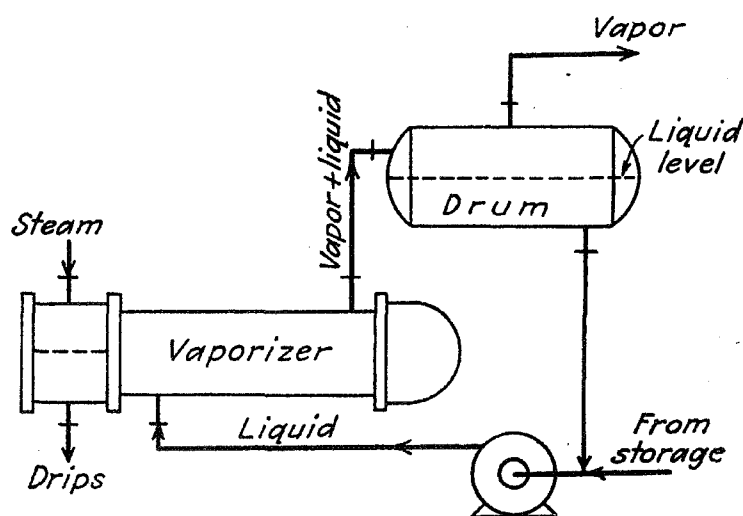


FIG. 15.1. Vaporizing process with forced circulation.

izer. The 2000 lb/hr of liquid which was not vaporized will be recombined with 8000 lb/hr of fresh liquid coming from storage or, if the storage is hot, returned directly to storage.

Now suppose that, for greater cleanliness and the attainment of a higher coefficient, the liquid is speeded through the vaporizer at a faster flow rate by means of a larger pump. It may be desired that liquid enter the vaporizer at a rate of 40,000 instead of 10,000 lb/hr suggested by the process. Since only 8000 lb/hr of vapor will be produced, only 8000 lb/hr of fresh liquid will continue to come from storage. The difference, 32,000 lb/hr, will have to be supplied by recirculation again and again of unvaporized liquid through the vaporizer. For 40,000 lb/hr entering the vaporizer, only 8000 lb/hr, or 20 per cent will actually be vaporized, and the higher velocity over the heat-transfer surface and the low percentage vaporized will permit longer operation without excessive fouling.

The advantage of recirculation can be computed economically. The high recirculating rate will increase the power cost but decrease the size of the equipment and the maintenance cost.

The vaporizer may also be connected with a disengaging drum without the use of a recirculating pump. This scheme is natural circulation and is shown in Fig. 15.2. It requires that the disengaging drum be elevated above the vaporizer. Recirculation is effected by the hydrostatic head difference between the column of liquid of height z_1 and the column of mixed vapor and liquid of height z_3 . The head loss in the vaporizer itself due to frictional pressure drop corresponds to z_2 . The hydrostatic head difference between z_3 and z_1 is available to cause liquid to circulate at

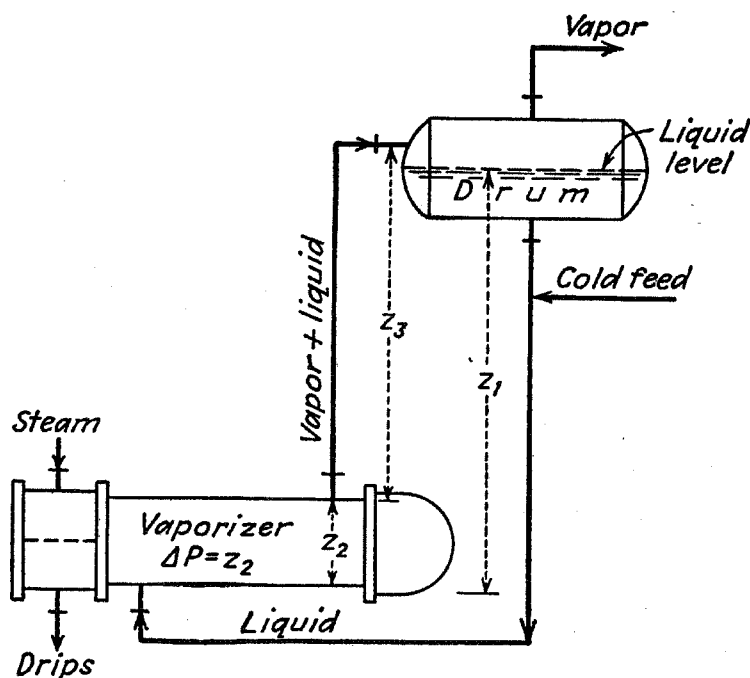


FIG. 15.2. Vaporizing process with natural circulation.

such a velocity that it produces a pressure drop z_2 in the vaporizer equal to the hydrostatic difference between z_3 and z_1 . The cold feed is usually fed high on the return pipe so that the leg z_1 will have as great a density and hydrostatic pressure $z_1\rho$ as possible. If the pressure of the feed is greater than the operating pressure of the system, the feed will have to join the return pipe at a lower point unless a jet is used, so that there will be a sufficient liquid head above the junction to prevent feed from backing up directly into the drum. Piping the feed directly to the disengaging drum usually makes control of the operating variables more difficult. If there is a sufficiently great pressure difference between the feed and operating pressure, the feed can be used as the motive fluid in an ejector to increase the recirculation of liquid through the vaporizer. This is

inefficient if the fresh liquid is compressed purposely to serve as the motive fluid, but if the fresh liquid is available under pressure for other reasons, the ejector permits recovery of some of the pressure head.

The advantages of forced circulation or natural circulation are in part economic and in part dictated by space. The forced-circulation arrangement requires the use of a pump with its continuous operating cost and fixed charges. As with forced-circulation evaporators, the rate of feed recirculation can be controlled very closely. If the installation is small, the use of a pump is preferable. If a natural-circulation arrangement is used, pump and stuffing-box problems are eliminated but considerably more headroom must be provided and recirculation rates cannot be controlled so readily. Natural-circulation steam generators are frequently laid out in accordance with Fig. 15.2 but employing a vertical

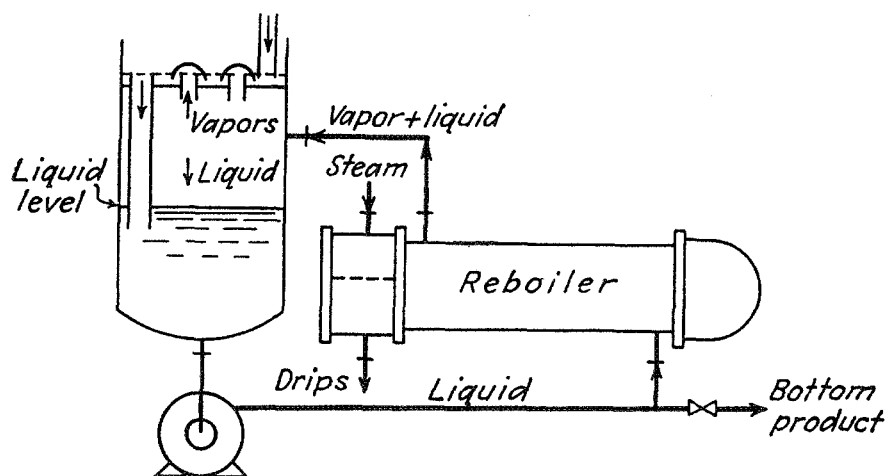


FIG. 15.3. Forced-circulation reboiler arrangement.

1-1 exchanger with vaporization in the tubes. Water is especially adaptable to natural-circulation arrangements, since the density differences between the liquid and vapor at a given temperature are very large.

Reboiler Arrangements. When reboilers are used, the space at the bottom of the column between the liquid level and the bottom plate is employed for disengagement and the bubble caps serve as separators. A typical arrangement for a forced-circulation reboiler is shown in Fig. 15.3. This type is called a *pump-through* reboiler. All the liquid on the bottom plate, frequently called *trapout* to distinguish it from bottom product, is carried by the downcomer below the liquid level of the column. The liquid can be recirculated through the reboiler as many times as is economically feasible, so that the percentage vaporized per circulation is kept low while the bottom product is withdrawn from a separate connection. In general, forced-circulation or pump-through reboilers are used only on small installations or those in which the bottoms liquid is so viscous and

the pressure drop through the piping and reboiler so high that natural circulation is impeded.

By far the greater number of large reboiler installations employ natural circulation. This can be achieved in either of two simple ways as shown in Fig. 15.4a and b. In Fig. 15.4a all the liquid on the bottom plate is circulated directly to the reboiler, whence it is partially vaporized. The unvaporized portion, on being disengaged under the bottom plate, is withdrawn as bottom product. In Fig. 15.4b the liquid passes through the downcomer below the liquid level of the column as in forced circulation. The bottom liquid is free to recirculate through the reboiler as many times as the hydrostatic pressure difference between z_1 and z_3 will permit. Because there is no opportunity for recirculation in the arrange-

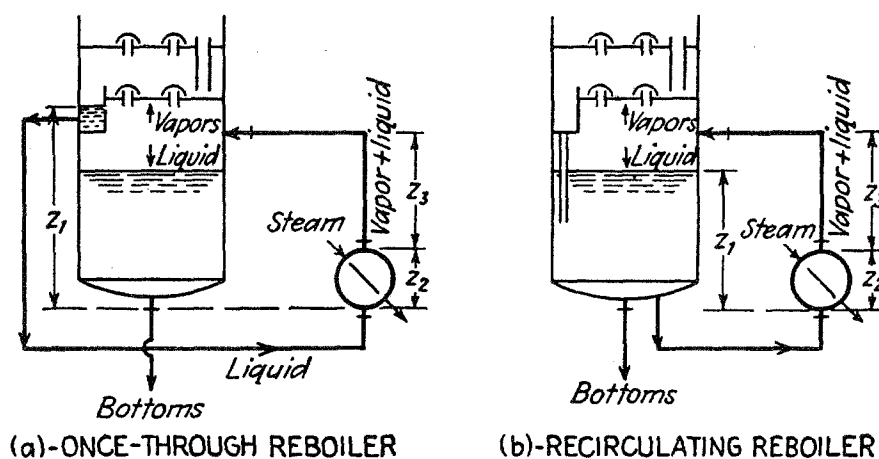


FIG. 15.4. Natural-circulation reboiler arrangements.

ment in Fig. 15.4a, it is called a *once-through* reboiler arrangement. Figure 15.4b is referred to as a *recirculating* reboiler.

Classification of Vaporizing Exchangers. There is a greater hazard in the design of vaporizing exchangers than any other type of heat exchanger. For this reason it is convenient to set up a classification based on the method of calculation as employed for each distinct type of service. Each of the common classes below is distinguishable by some difference in calculation.

A. Forced-circulation vaporizing exchangers

1. Vaporization in the shell

- a. Vaporizer or pump-through reboiler with isothermal boiling
- b. Vaporizer or pump-through reboiler with boiling range
- c. Forced-circulation evaporator or aqueous-solution reboiler

2. Vaporization in the tubes

- a. Vaporizer or pump-through reboiler with or without boiling range
- b. Forced-circulation evaporator or aqueous solution reboiler

B. Natural-circulation vaporizing exchangers

1. Vaporization in the shell
 - a. Kettle reboiler
 - b. Chiller
 - c. Bundle-in-column reboiler
 - d. Horizontal thermosyphon reboiler
2. Vaporization in the tubes
 - a. Vertical thermosyphon reboiler
 - b. Long-tube vertical evaporator

Heat-flux and Temperature-difference Limitations. It may be assumed that process conditions will always be established which provide for the vaporization of only part of the liquid fed to a vaporizer. When vaporizing liquids from pools, extremely high maximum fluxes have been obtained. For water a maximum flux is reported at 400,000 Btu/(hr)(ft²) and for organics from 70,000 to 125,000 Btu/(hr)(ft²), although these have been obtained only in laboratory-scale apparatuses with perfectly clean surfaces. It is again pointed out that the maximum flux occurs at the critical temperature difference and is a limitation of the maximum coefficient which may be attained. Beyond the critical temperature difference both the coefficient and the flux decrease, the decrease being due to the formation of a layer of gas on the tubes. It is the phenomenon of vapor blanketing which poses the principal difficulty in the design and operation of vaporizing exchangers.

Fluxes of magnitudes as high as those above are of little practical value in design. It will be recognized that vaporization in a 1-2 exchanger, occurring as it does without continuous disengagement, is very different from vaporization out of liquid pools. By restricting disengagement in the 1-2 exchanger, the possibility of vapor blanketing is increased greatly, so that it is necessary also to restrict the flux to an allowable value safely out of the range in which blanketing might occur. The flux is defined by Q/A or $U_D \Delta t$ but not by $h_v(\Delta t)_w$, where h_v is the vaporization coefficient and $(\Delta t)_w$ is the temperature difference between the tube wall and the boiling temperature. $h_v(\Delta t)_w$ is the flux based on the clean surface A_c , while Q/A is the flux based on the actual surface A , and A is greater than A_c in a vaporizer designed with a dirt factor. However, it is customary to restrict both Q/A and h_v to safe maxima, the two also serving to prevent the presence of too great a temperature difference $(\Delta t)_w$. The following restrictions will be observed throughout this chapter:

I. Flux

- a. The maximum allowable flux for forced circulation vaporizers and reboilers vaporizing organics is 20,000 Btu/(hr)(ft²) and for natural circulation 12,000 Btu/(hr)(ft²).

- b. The maximum allowable flux for the vaporization of water or aqueous solutions of low concentration using forced or natural circulation is 30,000 Btu/(hr)(ft²).

II. Film coefficient

- a. The maximum allowable vaporizing film coefficient for the forced- or natural-circulation vaporization of organics is 300 Btu/(hr)(ft²)(°F).
- b. The maximum vaporizing film coefficient for the forced- or natural-circulation vaporization of water and aqueous solutions of low concentration is 1000 Btu/(hr)(ft²)(°F).

Relationship between Maximum Flux and Maximum Film Coefficient.

The objects of the limitations above are the elimination of all vapor blanketing possibilities. Suppose it is desired partially to vaporize an organic compound boiling at 200°F in a forced-circulation vaporizer using steam at a temperature of 400°F so that $\Delta t = 200^\circ\text{F}$ and the flow is such that a vaporizing coefficient of 300 Btu/(hr)(ft²)(°F) can be obtained. If the condensing steam coefficient is 1500, $U_c = 250$, and if $R_d = 0.003$, $U_D = 142$. The flux will be $142 \times 200 = 28,400$ Btu/(hr)(ft²), which exceeds limitation Ia. Since Q/A or $U_D \Delta t$ may not exceed 20,000, any change to permit compliance with Ia amounts to an increase in the total surface for vaporization. If the original steam and vapor temperatures are retained, the new overall coefficient U_D will be $20,000/200 = 100$ Btu/(hr)(ft²)(°F). The temperature difference $(\Delta t)_w$ may be greater than the critical temperature difference, since it does not occur at the maximum attainable flux and under these conditions the critical temperature difference can be exceeded within limits without the danger of vapor blanketing. There is no advantage to the use of very high temperature differences, however, since at the maximum allowable flux any increase in Δt must be offset by a decrease in the allowable value of U_D . Only when U_D is naturally small may the use of a high Δt be partially justified.

The test of whether or not a vaporizer exceeds the allowable flux is always determined by dividing the total vaporizing heat load by the total available surface for vaporization. By the same token the maximum value of U_D which may be anticipated is given by $U_D = (Q/A)(1/\Delta t)$ regardless of the dirt factor which results. When setting the temperature of the heating medium, it is seen that the use of a large Δt and corresponding $(\Delta t)_w$ also requires decreasing U_D which in turn gives a large value of $R_d = 1/U_D - 1/U_c$. The large dirt factor is not essential to the continued operation of the vaporizer from the standpoint of dirt but only as a preventive against vapor blanketing. Accordingly, when the temperature of the heating medium may be selected independently such as by setting the pressure in the case of steam, it need not be set at a value of Δt greater than that which gives a U_D corresponding to the desired dirt factor.

FORCED-CIRCULATION VAPORIZING EXCHANGERS

1. VAPORIZATION IN THE SHELL

a. Vaporizer or Pump-through Reboiler with Isothermal Boiling. The calculations employed in the solution of this type of vaporizer are common to the many simple vaporizing problems found in a plant whether or not connected with a distilling column. If a liquid is substantially pure or a constant-boiling mixture, it will boil isothermally. This usually applies to the bottom liquid of a distilling column separating a binary mixture into relatively pure compounds. For utility boiling operations, such as the vaporization of a cold liquid coming from storage, the liquid may not be at its boiling point and may require preheating to the boiling point. Since the shell of a forced-circulation vaporizer is essentially the same as any other 1-2 exchanger, the preheating can be done in the same shell as the vaporization. If the period of performance of a vaporizer is to be measured by a single overall dirt factor, it is necessary to divide the shell surface into two successive zones, one for preheating and one for vaporization, in much the same manner employed in condenser-subcoolers. The true temperature difference is the weighted temperature difference for the two zones, and the clean coefficient is the weighted clean coefficient as given by Eqs. (12.50) and (12.51).

If steam is the heating medium, only two tube passes are required and these need not be equally divided, since the return pass carries considerably less vapor than the first pass. If a hot stream such as gas oil is the heating medium, there is a problem in determining the true temperature difference in each zone. If the approach between the heating-medium outlet temperature and the vapor outlet temperature is not too small, the true temperature difference can be approximated by considering the temperature fall in each zone proportional to the heat removed from the heating medium. The method of using zones has also been discussed in Chap. 12 along with condenser-subcoolers.

Film Coefficients. Where there is a preheating zone, it can be computed by the use of Fig. 28 like any other heater with the cold fluid in the shell. The isothermal boiling film coefficient is also obtained by the use of Fig. 28 based on the premise that the heat must first be absorbed by the liquid by forced convection before passing into the vapor bubbles and that the liquid heating coefficient is the controlling coefficient in the sequence.

Pressure Drop. The shell-side pressure drop for the vaporizing zone is computed by introducing the average specific gravity into the denominator of Eq. (7.44). If the vaporizing liquid boils isothermally at t_b and receives heat from a medium having a range $T_1 - T_2$, two possibilities may be considered:

Case I: The vapor and heating medium may be in counterflow.

Case II: The vapor and heating medium may be in parallel flow.

CASE I: VAPORS AND HEATING MEDIUM IN COUNTERFLOW. Referring to Fig. 15.5, if W is the weight flow of the heating medium, C its specific heat, and T is the temperature of the heating medium at any tube

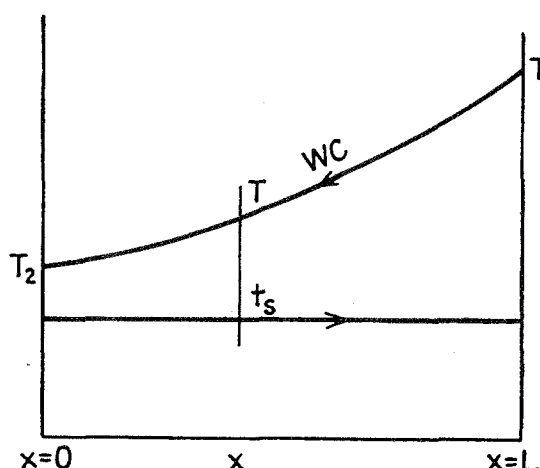


FIG. 15.5. Temperatures during vaporization of an isothermal fluid by a non-isothermal heating medium.

length x ,

$$WC dT = Ua'' dx(T - t_s) \quad (15.1)$$

where $a'' dx$ is the surface. Integrating T with respect to x ,

$$\ln(T - t_s) = \frac{Ua''x}{WC} + C_1 \quad (15.2)$$

At $x = 0$,

$$C_1 = \ln(T_2 - t_s) \quad (15.3)$$

$$\ln \frac{T - t_s}{T_2 - t_s} = \frac{Ua''x}{WC} \quad (15.4)$$

or

$$T = t_s + (T_2 - t_s)e^{Ua''x/WC} \quad (15.5)$$

When $x = L$, $T = T_1$,

$$Q = WC(T_1 - T_2) = WC(T_2 - t_s)(e^{Ua''L/WC} - 1) \quad (15.6)$$

The pounds evaporated w_e at any point x are given by

$$w_e = \frac{WC}{\lambda_s} (T_2 - t_s)(e^{Ua''x/WC} - 1) \quad (15.7)$$

If w is the weight flow of all the isothermal fluid and v_{av} its average specific volume, then for the entire mixture in passing from inlet to outlet

$$wv_{av} = \frac{WC}{\lambda_s} (T_2 - t_s)(e^{Ua''x/WC} - 1)v_v + \left[w - \frac{WC}{\lambda_s} (T_2 - t_s)(e^{Ua''x/WC} - 1) \right] v_l \quad (15.8)$$

where v_v is the specific volume of the vapor and v_l is the specific volume of the liquid.

For the pressure drop by assuming $1/\rho_{av} = v_{av}$

$$\int d\Delta P = \int \frac{fG^2}{2g\rho_{av}} dx = \int \frac{fG^2}{2g} v_{av} dx \quad (15.9)$$

$$\frac{2gw}{fG^2} \int d\Delta P = \int \left[\frac{WC}{\lambda_s} (T_2 - t_s)(e^{Ua''x/WC} - 1)v_v dx + wv_l dx - \frac{WC}{\lambda_s} (e^{Ua''x/WC} - 1)v_l dx \right]$$

Integrating and simplifying,

$$\frac{2g \Delta P}{fG^2L} = v_{av} = \frac{WC}{\lambda_s w} [(v_v - v_l)(\text{LMTD}) - (T_2 - T_s)(v_v - v_l)] + v_l \quad (15.10)$$

Generally, however, it is simpler to apply the law of mixtures as given below, which closely resembles Eq. (15.10).

Case I: Vapors in counterflow to heating medium:

$$v_{av} = \frac{WC}{UA} (v_2 - v_l) - \frac{WC(T_2 - t_s)}{\lambda_s w} (v_v - v_l) + v_l \quad (15.11)$$

$$s_{av} = \frac{1}{v_{av}(62.5)} \quad (15.12)$$

Case II: Vapors in parallel flow to heating medium:

$$v_{av} = \frac{WC}{UA} (v_l - v_2) + \frac{WC}{\lambda_s w} (T_1 - t_s)(v_v - v_l) + v_l \quad (15.13)$$

$$s_{av} = \frac{1}{v_{av}(62.5)}$$

With the average gravity based on Eqs. (15.11) and (15.13), Eq. (7.44) is only part of the pressure drop, since no term has been included to account for the acceleration of the vapor through the vaporizer or the contraction loss at the outlet. Instead of using Eqs. (15.11) and (15.13), the pressure drop can be computed more rapidly by an arbitrary assumption of the value of the average specific gravity. At the start of the vaporizing zone the gravity is that of the liquid, whereas at the outlet it is considerably less, even if the percentage converted to vapor is not very high. Likewise with constant-flow area the velocity through the shell increases toward the outlet owing to the large volume of the exit fluid, although the mass velocity is presumably constant. On the other hand, the viscosity effective upon the pressure drop probably does not vary much over the vaporizing zone, being that of the liquid throughout. The pressure drop can be computed using the mass velocity as heretofore, the Reynolds number based on liquid properties at inlet conditions and the *mean* gravity between inlet and outlet. Similar reasoning can be applied to vaporization on the tube side. The comparison between the true average gravity defined by Eqs. (15.11) and (15.13) and the mean of the inlet and outlet gravities is shown below.

Example 15.1. Calculation of the Average Specific Volume. A vaporizer is to generate 10,000 lb/hr of 15 psig steam by removing heat from a bottom product. The bottom product is 150,000 lb/hr of approximately 42°API kerosene leaving a column at 400°F. The design coefficient in the vaporizer will be about 100.

Compare the true average specific volume with that obtained by the approximate method.

$$t_s = 250^\circ\text{F} \quad \lambda_s = 945.3 \text{ Btu/lb} \quad (\text{Table 7})$$

$$\text{Steam, } Q = 10,000 \times 945.3 = 9,450,000 \text{ Btu/hr}$$

$$\text{Kerosene, } Q = 150,000 \times 0.63(400 - 300) = 9,450,000 \text{ Btu/hr}$$

$$\Delta t = \text{LMTD} = 91^\circ\text{F} \quad (5.14)$$

$$A = \frac{Q}{U_D \Delta t} = \frac{9,450,000}{100 \times 91} = 1040 \text{ ft}^2$$

$$WC = 150,000 \times 0.63 = 94,500 \text{ Btu}/^\circ\text{F}$$

From Table 7

$$v_l = 0.017 \quad v_v = 13.75 \text{ ft}^3/\text{lb}$$

By the law of mixtures

Assume 80 per cent of the outlet fluid is vapor.

$$v_2 = 0.80 \times 13.75 + 0.20 \times 0.017 = 11.0 \text{ ft}^3/\text{lb}$$

$$v_{av} = \frac{94,500(11.0 - 0.017)}{100 \times 1040} - \frac{94,500(300 - 250)}{945.3 \times 10,000} (13.75 - 0.017) + 0.017$$

$$= 3.04 \text{ ft}^3/\text{lb}$$

By the approximate method

$$v_l = 0.017 \quad v_2 = 11.0 \text{ ft}^3/\text{lb}$$

$$v_{av} = \frac{0.017 + 11.0}{2} = 5.5 \text{ ft}^3/\text{lb}$$

$$\text{Actual density} = \frac{1}{3.04} = 0.329 \text{ lb/ft}^3 \quad s = \frac{0.329}{62.5} = 0.0053$$

$$\text{Approximate density} = \frac{1}{5.5} = 0.182 \text{ lb/ft}^3 \quad s = 0.0029$$

Since the pressure drop is inversely proportional to the gravity, the approximate method is safe. The acceleration loss is usually taken at about two velocities heads and can be omitted when using the approximate method.

While the condenser is usually the cleanest part of a distillation or vaporization-condensation assembly, the vaporizer is usually quite the opposite. Vaporizers tend to accumulate dirt, and for this reason high recirculation rates and large dirt factors will often be desirable. Preference should be given to the use of square pitch and a removable tube bundle. Although it may reduce the possibility of using a 1-2 vaporizing exchanger for other services, the baffle spacing can be increased or staggered from inlet to outlet to reduce the pressure drop of the fluid vaporizing in the shell.

Example 15.2. Vaporizer or Pump-through Reboiler with Isothermal Boiling. It is desired to provide 19,750 lb/hr of butane vapor at 285 psi using cold feed from storage at 75°F. The butane will boil isothermally at 235°F, and steam will be available at 100 psi.

Available for the service is a 15¼ in. ID 1-2 exchanger with 76 1 in. OD, 16 BWG tubes 16'0" long laid out on 1¼-in. square pitch for two passes. The baffles are spaced 5 in. apart.

What will the dirt factor and pressure drops be?

Solution. To prevent total vaporization in the shell, the total liquid entering the vaporizer will be $19,750/0.80 = 24,700$ lb/hr. The vapor should then be disengaged from the liquid in a drum, and the excess liquid returned to the pump suction for recombination with 19,750 lb/hr of new feed. Thus for every 19,750 lb/hr coming from storage at 75°F, 4950 lb/hr at 235°F will be mixed with it so that the inlet temperature will be 108°F.

Solution:

Exchanger:

Shell side	Tube side
ID = 15¼ in.	Number and length = 76, 16'0"
Baffle Space = 5 in.	OD, BWG, pitch = 1 in., 16 BWG, 1¼-in. square
Passes = 1	Passes = 2

(1) Heat balance:

Preheat:

Enthalpy of liquid at 108°F and 300 psia = 162 Btu/lb (Fig. 9)

Enthalpy of liquid at 235°F and 300 psia = 248 Btu/lb

$q_p = 24,700(248 - 162) = 2,120,000$ Btu/hr

Vaporization:

Enthalpy of vapor at 235°F = 358 Btu/lb

$q_v = 19,750(358 - 248) = 2,170,000$ Btu/lb

Butane, $Q = 2,120,000 + 2,170,000 = 4,290,000$ Btu/hr

Steam, $Q = 4880 \times 880.6 = 4,290,000$ Btu/hr (Table 7)

(2) Δt weighted: (Subscripts p and v indicate preheating and vaporizing.)

$(\Delta t)_p = \text{LMTD} = 158.5^\circ\text{F}$ (5.14)

$(\Delta t)_v = \text{LMTD} = 103.0^\circ\text{F}$ (5.14)

$$\frac{q_p}{(\Delta t)_p} = 2,120,000/158.5 = 13,400$$

$$\frac{q_v}{(\Delta t)_v} = 2,170,000/103.0 = 21,100$$

$$\sum \frac{q}{\Delta t} = 34,500$$

$$\text{Weighted } \Delta t = \frac{Q}{\sum q/\Delta t} = \frac{4,290,000}{34,500} = 124.5^\circ\text{F} \quad (12.51)$$

(3) T_c and t_c : Average values of temperatures will be satisfactory for preheat zone.

Hot fluid: tube side, steam	Cold fluid: shell side, butane
Preheating:	
(4) $a'_t = 0.594$ in. ² [Table 10]	(4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)]
$a_t = N_t a'_t/144n$ [Eq. (7.48)]	$= 15.25 \times 0.25 \times 5/144 \times 1.25$
$= 76 \times 0.594/2 \times 144 = 0.157$ ft ²	$= 0.106$ ft ²
(5) $G_t = W/a_t$	(5') $G_s = w/a_s$ [Eq. (7.2)]
$= 4880/0.157$	$= 24,700/0.106$
$= 31,100$ lb/(hr)(ft ²)	$= 233,000$ lb/(hr)(ft ²)

Hot fluid: tube side, steam	Cold fluid: shell side, butane
(6) At $T_s = 338^\circ\text{F}$, $\mu = 0.015 \times 2.42 = 0.0363 \text{ lb/(ft)(hr)}$ [Fig. 15]	(6') At $T_a = 172^\circ\text{F}$ (average of 108 and 235°F) $\mu = 0.115 \times 2.42 = 0.278 \text{ lb/(ft)(hr)}$ [Fig. 14]
$D = 0.87/12 = 0.0725 \text{ ft}$ (Re_t is for pressure drop only)	$D_s = 0.99/12 = 0.0825 \text{ ft}$ [Fig. 28]
$Re_t = DG_t/\mu$ $= 0.0725 \times 31,100/0.0363 = 62,000$	$Re_s = D_s G_s/\mu$ $= 0.0825 \times 233,000/0.278 = 69,200$ (7') $j_H = 159$ [Fig. 28]
	(8') At 172°F (114°API) $k(c\mu/k)^{1/2} = 0.12 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$ [Fig. 16]
	$\phi_s = 1.0$
(9) h_{io} for condensing steam $= 1500 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$	(9') $h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k}\right)^{1/2}$ [Eq. (6.15b)] $= 159 \times 0.12/0.0825$ $= 231 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$

Clean overall coefficient for preheating U_p :

$$U_p = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1500 \times 231}{1500 + 231} = 200 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

Clean surface required for preheating A_p :

$$A_p = \frac{q_p}{U_p(\Delta t)_p} = \frac{13,400}{200} = 67.0 \text{ ft}^2$$

(9) h_{io} for condensing steam = 1500

Vaporization:
(6') At 235°F , $\mu = 0.10 \times 2.42 = 0.242 \text{ lb/(ft)(hr)}$ [Fig. 14]
$Re_s = 0.0825 \times 233,000/0.242 = 79,500$ (7') $j_H = 170$ [Fig. 28]
(8') At 235°F $k(c\mu/k)^{1/2} = 0.115 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$ $\phi_s = 1.0$ (Fig. 16)
(9') $h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k}\right)^{1/2}$ [Eq. (6.15)] $= 170 \times 0.115/0.0825 = 237$

Clean overall coefficient for vaporization U_v :

$$U_v = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1500 \times 237}{1500 + 237} = 205 \quad (6.38)$$

Clean surface required for vaporization A_v :

$$A_v = \frac{q_v}{U_v(\Delta t)_v} = \frac{21,100}{205} = 103 \text{ ft}^2$$

Total clean surface A_C :

$$A_C = A_p + A_v = 67.0 + 103 = 170 \text{ ft}^2$$

(13) Weighted clean overall coefficient U_c :

$$U_c = \frac{\Sigma UA}{A_c} = \frac{13,400 + 21,100}{170} = 203 \quad (12.50)$$

(14) Design overall coefficient:

$$\begin{aligned} \text{Surface/lin ft of tube} &= 0.2618 && \text{(Table 10)} \\ \text{Total surface} &= 76 \times 16'0'' \times 0.2618 = 318 \text{ ft}^2 \\ U_D &= \frac{Q}{A \Delta t} = \frac{4,290,000}{318 \times 124.5} = 108.5 \end{aligned}$$

Check of maximum flux:

A total of 170 ft² are required of which 103 are to be used for vaporization. For the total surface required 318 ft² will be provided. It can be assumed, then, that the surface provided for vaporization is

$$103/170 \times 318 = 193 \text{ ft}^2$$

The flux is $Q/A = 2,170,000/193 = 10,700 \text{ Btu}/(\text{hr})(\text{ft}^2)$. (Satisfactory)

(15) Dirt factor:

$$R_d = \frac{U_c - U_D}{U_c U_D} = \frac{203 - 108.5}{203 \times 108.5} = 0.0043 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Pressure Drop

(1) For $Re_t = 62,000$,

$$f = 0.000165 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 26}]$$

From Table 7, specific vol of steam at 115 psia = 3.88 ft³/lb

$$s = \frac{1}{3.88 \times 62.5} = 0.00413$$

$$(2) \Delta P_t = \frac{1}{2} \frac{f G^2 L n}{5.22 \times 10^{10} D s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$\begin{aligned} &= \frac{1}{2} \times \\ &\frac{0.000165 \times 31,100^2 \times 16 \times 2}{5.22 \times 10^{10} \times 0.0725 \times 0.00413 \times 1} \\ &= 0.16 \text{ psi} \end{aligned}$$

Preheat:

$$(1') Re_s = 69,200, f = 0.00145 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 29}]$$

(2') Length of preheat zone

$$\begin{aligned} L_p &= LA_p/A_c \\ &= 16 \times 67.0/170 = 6.3 \text{ ft} \end{aligned}$$

$$(3') \text{ No. of crosses, } N + 1 = 12L_p/B \quad [\text{Eq. (7.43)}]$$

$$= 12 \times 6.3/5 = 15$$

$$s = 0.50 \quad [\text{Fig. 6}]$$

$$D_s = 15.25/12 = 1.27 \text{ ft}$$

$$(4') \Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad [\text{Eq. (7.44)}]$$

$$\begin{aligned} &= \frac{0.00145 \times 233,000^2 \times 1.27 \times 15}{5.22 \times 10^{10} \times 0.0825 \times 0.50 \times 1.0} \\ &= 0.70 \text{ psi} \end{aligned}$$

Vaporization:

$$(1') Re_s = 79,500, f = 0.00142 \text{ ft}^2/\text{in.}^2$$

(2') Length of vaporization zone

$$L_v = 16 - 6.3 = 9.7 \text{ ft}$$

$$(3') \text{ No. of crosses, } N + 1 = 12L/B \quad [\text{Eq. (7.43)}]$$

$$= 9.7 \times 1\frac{2}{5} = 23$$

$$\text{Mol. wt.} = 58.1$$

Pressure Drop

$$\begin{aligned}
 \text{Density, } \rho &= \frac{58.1}{359 \times \frac{695}{492} \times 14.7/300} \\
 &= 2.34 \text{ lb/ft}^3 \\
 s_{\text{outlet liquid}} &= 0.43 \quad [\text{Fig. 6}] \\
 \rho_{\text{outlet liquid}} &= 0.43 \times 62.5 = 26.9 \text{ lb/ft}^3 \\
 s_{\text{outlet mix}} &= \frac{24,700/62.5}{19,750/2.34 + 4950/26.9} = 0.046 \\
 s_{\text{inlet}} &= 0.50 \\
 s_{\text{mean}} &= (0.50 + 0.046)/2 = 0.28 \\
 \Delta P_s &= \frac{0.00142 \times 233,000^2 \times 1.27 \times 23}{5.22 \times 10^{10} \times 0.0825 \times 0.28 \times 1.0} \\
 &= 1.9 \text{ psi} \\
 \Delta P_s (\text{total}) &= 0.7 + 1.9 = 2.6 \text{ psi}
 \end{aligned}$$

Summary

1500	h outside	$23\frac{1}{2}/237$
U_c	203	
U_D	108.5	
R_d Calculated 0.0043		
R_d Required		
0.16	Calculated ΔP	2.6
Neg	Allowable ΔP	5.0

b. Vaporizer or Pump-through Reboiler with Boiling Range. If a liquid undergoing vaporization is a mixture of a number of miscible compounds, it does not boil isothermally. Instead it has an initial boiling temperature (bubble point) and a final boiling temperature (dew point) at which the last bit of liquid is vaporized. When the mixture starts to boil at its bubble point, the more volatile compounds are driven out of the solution at a greater rate and as the volatile compounds enter the vapor phase the boiling temperature of the residual liquid rises. This means that throughout the vaporizer there is a temperature range over which boiling occurs, and the greater the percentage of the total liquid vaporized the more nearly the range extends from the bubble point to the dew point of the inlet liquid.

Because of the boiling range, sensible as well as latent heat must be absorbed simultaneously by the liquid as it proceeds through the vapor-

izer so that it will possess a range of boiling temperatures. Furthermore the sensible heat is absorbed over the same surface as the heat for vaporization in contrast to the preheater-isothermal vaporizer in which the two occur in separate zones. The calculation of the boiling coefficient in this case, however, is the same as for a preheater-vaporizer as calculated in Example 15.2. Here the heat from the tube wall is first absorbed by the liquid as sensible heat before it is transformed into vaporization. Since the rate of heat transfer from a hot liquid into an incipient vapor bubble is very great, it may be assumed that the sensible-heat-transfer coefficient as calculated from Fig. 28 for either direct vaporization or simultaneous sensible-heat transfer is the limiting resistance. The coefficient for the combined sensible-heat transfer and vaporization is calculated as if the entire vaporizing heat load were transferred as sensible heat to the liquid over its boiling range in the vaporizer.

The true temperature difference may be taken as the LMTD if the heating medium is isothermal. This assumes that the heat transferred is proportional to the change in temperature, namely, that one-half of the total load is delivered while the temperature rises one-half of the total temperature range for vaporization. If the bulk of a mixture consists of closely related compounds with some more or less volatile compounds, the assumption that the heat and temperature proportions are equal may cause considerable error. The true temperature difference can be obtained by graphical integration as treated in Example 13.3.

Film Coefficients. The sensible-heat-transfer coefficient should be regarded as the boiling coefficient when applying the restrictions of allowable flux and allowable coefficient even though it is computed from Fig. 28. When a liquid has a boiling range, the *average* flux, Q/A may be less than 20,000 but because of the variation in temperature difference, $U_D \Delta t_1$ at the greater terminal temperature difference may exceed 20,000. Discrepancies of this sort may cause bumping or erratic vaporization if the initial compounds to vaporize are very volatile compared with the bulk and tend to separate from the liquid too readily. A check of the inlet flux can prevent this difficulty. If the inlet flux does not exceed, say, 25,000 Btu/(hr)(ft²), there is no need to penalize the entire design by providing excess surface simply because of an excessive flux over the first few per cent of the vaporizing surface.

Pressure Drop. The pressure drop is computed in the same manner as for isothermal boiling, using the Reynolds number based on inlet conditions and a gravity which is the mean of inlet and outlet gravities. It is also quite possible that a fluid with a boiling range may enter a vaporizer below its bubble point. In such cases the surface is again divided into two consecutive zones, one for pure preheating and the other for vapori-

zation of a mixture with a boiling range. The weighted coefficients and temperature difference may be obtained as before through the use of Eqs. (12.50) and (12.51).

c. Forced-circulation Evaporator or Aqueous-solution Reboiler. As seen in Chap. 14, the shells of 1-2 exchangers are not used in forced-circulation-evaporator processes, since the properties of water are excellent for natural-circulation equipment. However, a 1-2 exchanger can easily serve as a forced-circulation evaporator. In distillation processes such as the distillation of an acetone-water or alcohol-water solution the bottom product is nearly pure water. It may be advantageous in smaller operations of this nature to use a pump-through reboiler in preference to natural circulation, since the losses in the connecting piping may be unduly high and the use of larger connecting pipes does not assure smooth operation. The aqueous-solution reboiler can be calculated in the same manner as the pump-through reboiler, with or without a boiling range, except that the allowable flux and film coefficient is greater.

This type of equipment is usually designed with the dirt factor as the controlling resistance. The applicability of a method of computing water-vaporization rates is therefore of value only at low mass velocities. Since water vapor has a very low vapor density, low mass velocities must be employed whenever the allowable pressure drop is small. Film coefficients for boiling water and aqueous solutions can be obtained by the use of Fig. 28, although these will be about 25 per cent lower than those which have been obtained experimentally. Where data on the physical properties of aqueous solutions are lacking, they can be approximated by the methods of Chap. 7. If the mass velocity is very low, the value of the coefficient so obtained may be multiplied by 1.25 and the overall value of U_c should rarely exceed 600 Btu/(hr)(ft²)(°F).

2. VAPORIZATION IN THE TUBES

a. Pump-through Vaporizer or Reboiler with or without Boiling Range. The coefficients for vaporization with or without a boiling range can be obtained from Fig. 24 for organic liquids. The number of tube passes may be as great in horizontal exchangers as the pressure drop will allow. If the number of tubes in the final passes are greater than the number in the initial passes, it is possible to obtain a reduced pressure drop. The shell side, when employing steam, may be laid out on triangular pitch, since cleaning will be infrequent and the shell side can be cleaned by boiling out. The pressure drop can be computed using Eq. (7.45) with a Reynolds number based on the inlet properties and a specific gravity which is the mean between inlet and outlet. The tube fluid should flow upward.

b. **Forced-circulation Evaporator or Aqueous-solution Reboiler.** The tube-side data for the evaporation of water and aqueous solutions can also be obtained from Fig. 24. The boiling coefficients will be about 25 per cent greater than the computed values, and at low mass velocities the coefficient can be multiplied by a correction factor of 1.25.

This class also includes the forced-circulation evaporators similar to 1-1 exchangers. The calculation of the vertical long-tube evaporator will be treated as a natural-circulation vaporizer.

NATURAL-CIRCULATION VAPORIZING EXCHANGERS

1. VAPORIZATION IN THE SHELL

a. **Kettle Reboiler.** The kettle reboiler is shown in Fig. 15.6. It is a modification of the power-plant evaporator. The relation of the bundle to the shell is seen better from an end elevation. Another form of the kettle reboiler employing a tube sheet which covers the entire shell is

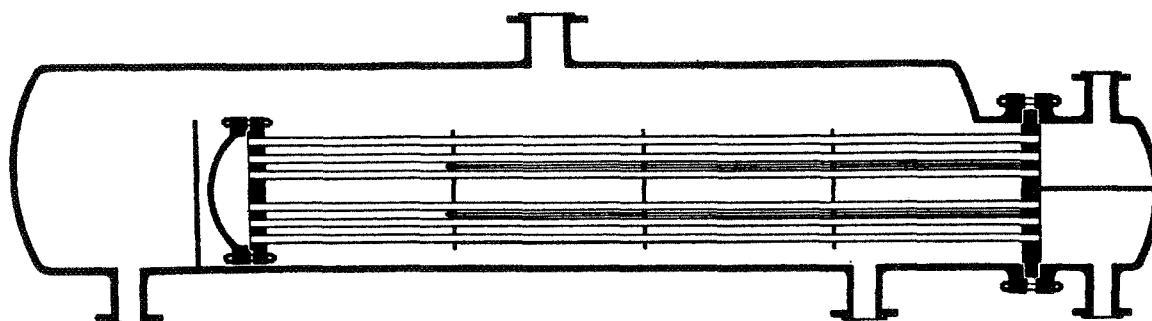


FIG. 15.6. Kettle-type reboiler. (Patterson Foundry and Machine Co.)

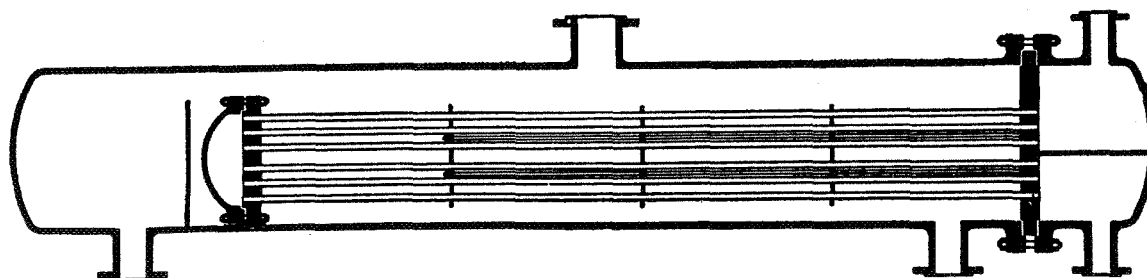


FIG. 15.7. Kettle-type reboiler with integral tube sheet. (Patterson Foundry and Machine Co.)

shown in Fig. 15.7. In this type the bundle is not circular but conforms to the shell as seen from an end elevation. The method of connecting this type of reboiler to the distilling column is shown in Fig. 15.8. Kettle reboilers are fitted with a weir to ensure that the liquid level in the reboiler is maintained and that the tube surface is not exposed. Since only about 80 per cent of the liquid bottoms entering at the inlet are vaporized, provision must be made for the removal of the bottoms product which is on the discharge side of the weir. There are a number of

arbitrary rules on the volume required above the liquid level for disengagement and the maximum number of pounds per hour which may be vaporized from the liquid surface. If the top row of tubes is not higher than 60 per cent of the shell diameter, adequate disengaging space will

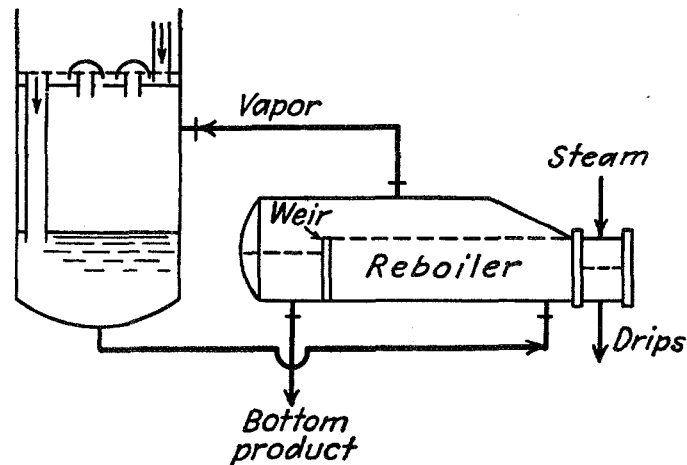


FIG. 15.8. Kettle reboiler arrangement.

be provided when a liquid level covering the top row of tubes is ensured by the weir.

b. Chiller. The chiller is shown in Fig. 15.9. It is a typical kettle reboiler, except for the weir, and the bundle has tubes to a height of about 60 per cent of the diameter. The vapor space above is for disengagement

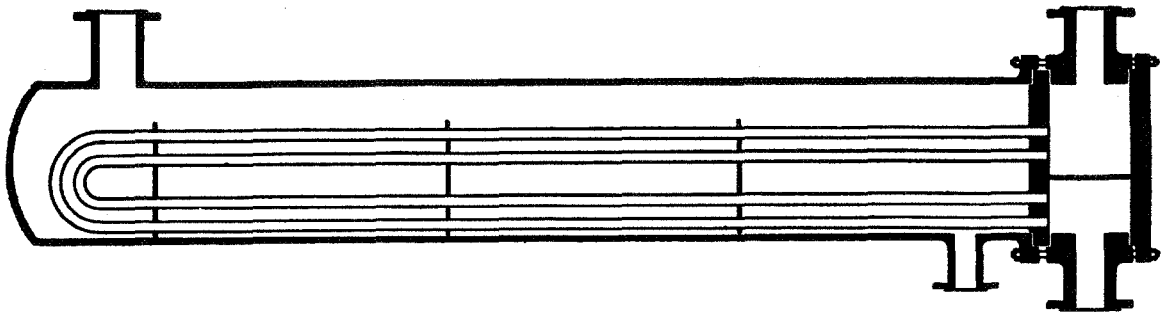


FIG. 15.9. Chiller. (Patterson Foundry and Machine Co.)

of the vapor from the liquid. Chillers are used in refrigeration processes of the vapor-compression type as shown in Fig. 15.10. The refrigeration cycle begins at the point *a*, where liquid refrigerant at a temperature higher than that of the condenser water and at high pressure passes to a constant-enthalpy throttle valve where its pressure is reduced. The pressure and temperature of the liquid on the downstream side of the valve are naturally less than on the high-pressure side. The expansion is adiabatic, and some of the liquid flashes into vapor cooling the remainder of the refrigerant on the low-pressure side at *b*. If the cold refrigerant is to be circulated directly to a refrigerator, the saturation tempera-

ture at *b* is often about 5 or 10°F lower than the temperature ultimately desired in the chamber being refrigerated. The partially vaporized refrigerant may enter the shell of a chiller, where the remainder is vaporized isothermally at low temperature by the fluid being chilled as it flows through the tubes. The vapor then passes to the compressor between *c* and *d*, where it is recompressed to the higher pressure (and temperature) such that it can be recondensed with the available cooling water.

Cold brine is often circulated to the chamber being refrigerated in preference to the refrigerant itself. Brines are usually sodium chloride

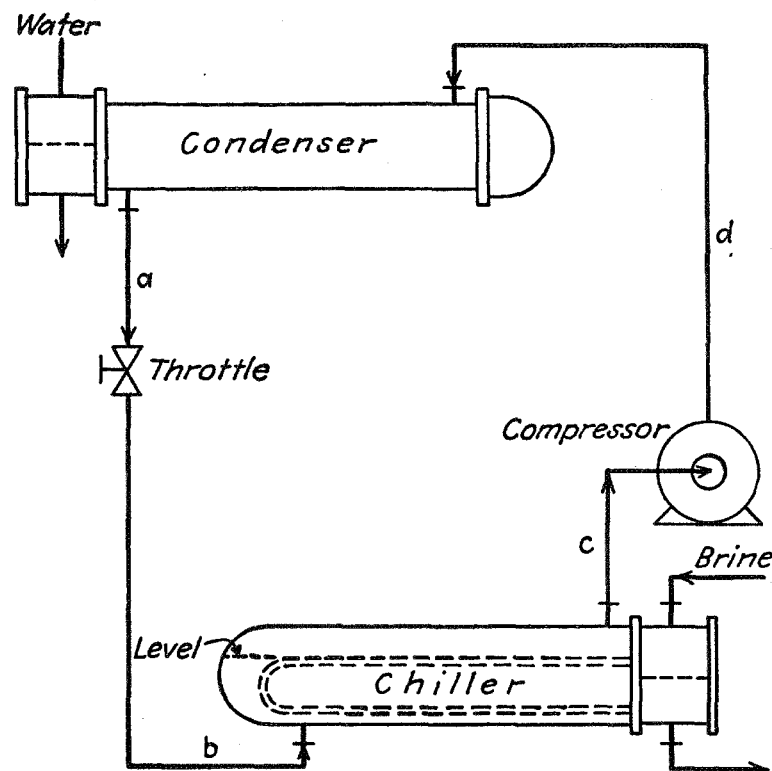


FIG. 15.10. Compression refrigeration system.

or calcium chloride solutions in concentrations up to 25 per cent by weight, depending upon the coldness of temperature to be maintained. They are cheap and have little susceptibility toward leakage. On accomplishing refrigeration by sensible heat absorption, brines remain in the liquid phase without developing high pressures and eliminate the need for costly vapor piping through the refrigerating system. In this way contaminants which might find their way into the refrigerant, particularly under vacuum, are kept out of it and from the compressor, condenser, and throttle. Brines, on the other hand, require that an additional temperature difference be maintained. When the refrigerant is circulated directly to the cold chamber, there is but one temperature difference between the chamber and the refrigerant. Using brines, however, there is one temperature difference between the refrigerant and

brine and one between the brine and cold chamber, and this arrangement increases the cost of the refrigeration. For other aspects of refrigerating cycles, such as the selection of optimum conditions, reference may be made to standard thermodynamics texts. The chiller is frequently called the evaporator in the refrigeration process, although this use of the term is at variance with the nomenclature used herein. When used in large

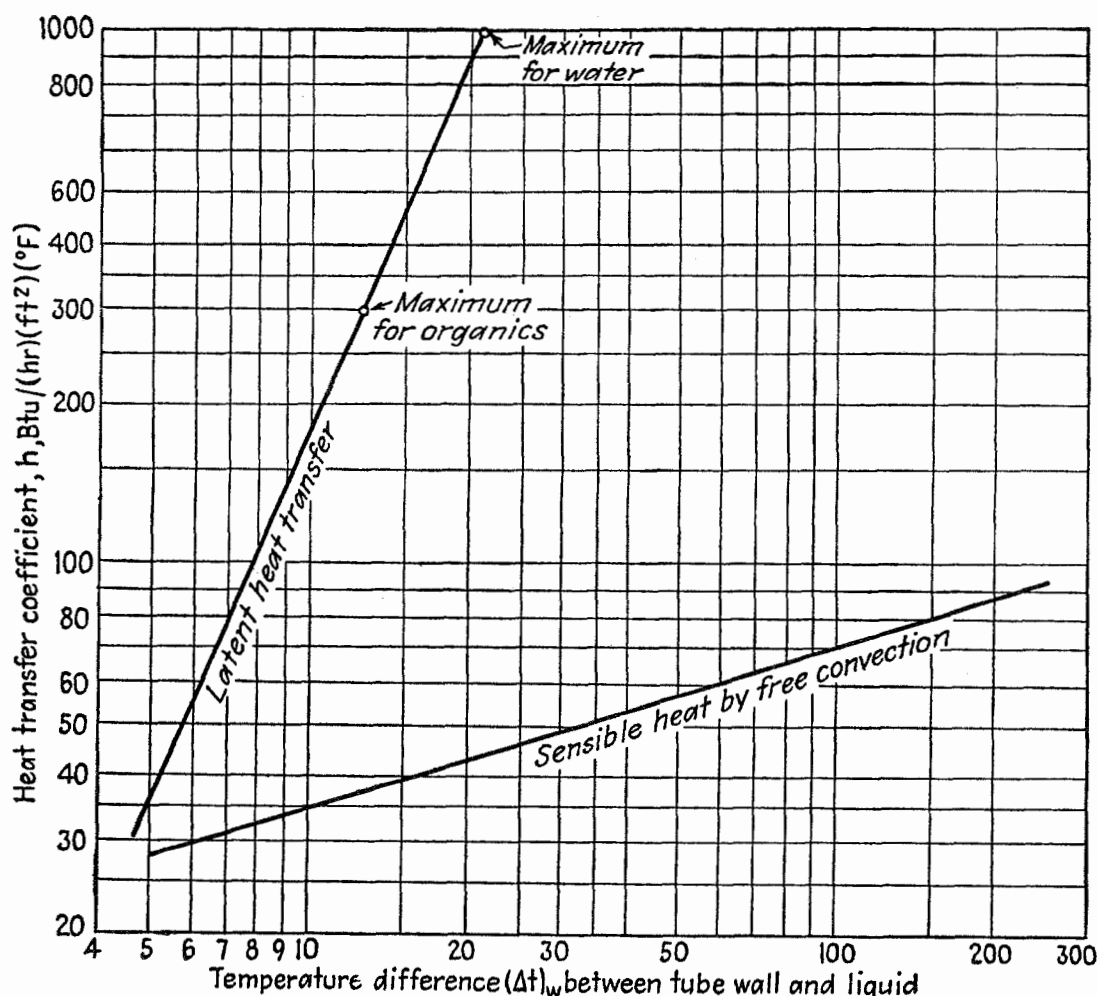


FIG. 15.11. Natural-circulation boiling and sensible film coefficients.

installations, the refrigerant may disengage into a separate drum instead of in the chiller shell.

Film Coefficients in Chillers and Kettle Reboilers. When a liquid is vaporized from a pool, the velocity of the liquid flowing over the heat-transfer surface is very low. At these low velocities the film coefficient for boiling is independent of the velocity and dependent instead upon the temperature difference between the tube wall and the saturation temperature of the boiling fluid. A curve showing this relationship is represented by the upper line in Fig. 15.11. The coefficients are again limited to 300 for organics and 1000 for water, except that the maximum

allowable flux for organics is 12,000 Btu/(hr)(ft²). The latter does not permit the use of large temperature differences for natural-circulation vaporizers and reboilers, which require more surface as a class than do forced-circulation types. The difference in cost of surface is usually more than offset by the saving in circulating pump power.

Chillers operate isothermally unless the vapor space of the chiller also serves as a flash chamber for the expansion, in which case the inlet temperature will be that at the throttle valve and the outlet temperature the saturation temperature of the refrigerant. The vaporization out of the liquid surface, however, is isothermal. Kettle reboilers also operate under nearly isothermal conditions, particularly when employed at the bottom of an aqueous-solution distilling column. When used with organic bottoms, however, there is usually a boiling range, and it is necessary to provide for the introduction of sensible heat. The sensible heat is regarded as entering the liquid by a modification of *free convection*,¹ and the curve for free convection to organics in natural circulation is given by the lower line of Fig. 15.11. It corresponds fairly closely to the results obtained from Fig. 10.4. In the case of kettle reboilers the total heat load is divided into sensible- and latent-heat fractions, and the surface required for each fraction is computed separately at its respective boiling or sensible coefficient. While there are several ways in which this procedure can be justified, it is nonetheless an empirical means of calculating coefficients comparable to those obtained in practice. This method is demonstrated in Example 15.4.

Pressure Drop in Kettle Reboilers and Chillers. The height of bottoms maintained in a distilling column naturally seeks its own level in the reboiler. If the reboiler is not mounted much below the liquid level in the column, there is a negligible hydrostatic head for fluid flow from the column to the reboiler, and hence the circulation rate is relatively small. This results in a low fluid velocity across the reboiler surface, and the pressure drop, as well as that of the connecting piping, may be considered negligible. The kettle reboiler, in fact, is the most suitable of all natural-circulation reboilers where the reboiler cannot be mounted sufficiently below the liquid level in the column to provide a large recirculation rate.

Example 15.3. Calculation of a Kettle Reboiler. 45,500 lb/hr of bottoms of 65°API gravity and small boiling range at 400°F enter a kettle reboiler from which 28,100 lb/hr of vapor is formed at an operating pressure of 200 psig. Heat is supplied by 28°API gas oil in the range from 575 to 475°F and 120 psig operating pressure. A pressure drop of 10 psi is permissible.

Available for the service is a 25 in. ID kettle reboiler containing a six-pass 15¼-in.

¹ The terms natural convection and natural circulation sometimes cause confusion. The former refers to heat transfer, and the latter to fluid flow.

circular bundle. The bundle contains 68 1 in. OD, 14 BWG tubes 12'0" long on 1½-in. square pitch. The bundle is baffled only by quarter-circle support plates.

Will the reboiler be satisfactory?

What are the dirt factor and pressure drops?

Solution:

Kettle:

<i>Shell side</i>	<i>Tube side</i>
15¼-in. circular bundle in 25 in.	Number and length = 68, 12'0"
ID	OD, BWG, pitch = 1 in., 14 BWG, 1½-in. square
¼-circle support plates	Passes = 6

(1) Heat balance:

Enthalpy of liquid at 400°F and 215 psia = 290 Btu/lb (Fig. 11)

Enthalpy of vapor at 400°F and 215 psia = 385 Btu/lb

Gasoline, $Q = 28,100 \times (385 - 290) = 2,670,000$ Btu/hr

Gas oil, $Q = 34,700 \times 0.77(575 - 475) = 2,670,000$

(2) Δt : Isothermal boiling

$\Delta t = \text{LMTD} = 118^\circ\text{F}$ (5.14)

(3) $T_c: \frac{\Delta t_c}{\Delta t_h} = \frac{475 - 400}{575 - 400} = 0.428$

$K_c = 0.37$ (Fig. 17)

$F_c = 0.42$

$T_c = 475 + 0.42(575 - 475) = 517^\circ\text{F}$ (5.28)

Hot fluid: tube side, gas oil

(4) Flow area, $a'_t = 0.546$ in.² [Table 10]

$a_t = N a'_t / 144n$ [Eq. (7.48)]

$= 68 \times 0.546 / 144 \times 6 = 0.043$ ft²

(5) $G_t = W/a_t$

$= 34,700 / 0.043$

$= 807,000$ lb/(hr)(ft²)

(6) At $T_c = 517^\circ\text{F}$,

$\mu = 0.27 \times 2.42 = 0.65$ lb/(ft)(hr) (extrapolated) [Fig. 14]

$D = 0.834 / 12 = 0.0694$ ft

$Re_t = DG_t / \mu$

$= 0.0694 \times 807,000 / 0.65 = 85,700$

(7) $j_H = 220$ [Fig. 24]

(8) At 517°F (28°API) [Fig. 16]

$k(c\mu/k)^{1/3} = 0.118$ Btu/(hr)(ft²)(°F/ft)

(9) $h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/3} \phi_t$ [Eq. (6.15d)]

$h_i / \phi_t = 220 \times 0.118 / 0.0694 = 374$

(10) $\frac{h_{io}}{\phi_t} = \frac{h_i}{\phi_t} \times \frac{ID}{OD}$ [Eq. (6.9)]

$= 374 \times 0.834 / 1.0 = 311$

The correction for $\left(\frac{\mu}{\mu_w} \right)^{0.14}$ is negligible.

Cold fluid: shell side, gasoline

(9') Assume $h_o = 300$ for trial.

(10') $t_w = t_c + \frac{h_{io}}{h_{io} + h_o} (T_c - t_c)$ [Eq. (5.31)]

$= 400 + \frac{311}{311 + 300} (517 - 400)$
 $= 460^\circ\text{F}$

$(\Delta t)_w = 460 - 400 = 60^\circ\text{F}$

From Fig. 15.11, $h_v > 300$;

hence use 300.

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{311 \times 300}{311 + 300} = 152 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Design overall coefficient U_D :

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft} \quad (\text{Table 10})$$

$$\text{Total surface} = 68 \times 12 \times 0.2618 = 214 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{2,670,000}{214 \times 118} = 105.5 \quad (5.3)$$

Check of maximum flux:

$$\frac{Q}{A} = \frac{2,670,000}{214} = 12,500 \text{ Btu/(hr)(ft}^2\text{)} \text{ (satisfactory)}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{152 - 105.5}{152 \times 105.5} = 0.0029 \quad (6.12)$$

Pressure Drop

(1) For $Re_t = 85,700$, $f = 0.00015 \text{ ft}^2/\text{in.}^2$ Negligible

[Fig. 26]

 $s = 0.71$

[Fig. 6]

$$\begin{aligned} (2) \Delta P_t &= \frac{f G_t^2 L n}{5.22 \times 10^{10} D s \phi_t} \\ &= \frac{0.00015 \times 807,000^2 \times 12 \times 6}{5.22 \times 10^{10} \times 0.0694 \times 0.71} \\ &\quad \times 1.0 \end{aligned}$$

$$= 2.8 \text{ psi}$$

(3) $G_t = 807,000$, $V^2/2g = 0.090$

[Fig. 27]

$$\begin{aligned} \Delta P_r &= \frac{4n}{s} \frac{V^2}{2g} \quad [\text{Eq. (7.46)}] \\ &= \frac{4 \times 6}{0.71} \times 0.09 = 3.1 \text{ psi} \end{aligned}$$

$$\begin{aligned} (4) \Delta P_T &= \Delta P_t + \Delta P_r \quad [\text{Eq. (7.47)}] \\ &= 2.8 + 3.1 = 5.9 \text{ psi} \end{aligned}$$

Summary

311	h outside	300
U_C	152	
U_D	105.5	
R_d Calculated 0.0029		
R_d Required		
5.9	Calculated ΔP	Neg
10.0	Allowable ΔP	Neg

c. **Bundle-in-column Reboiler.** It will probably occur while examining the kettle reboiler (Fig. 15.6) that, if the bundle is to be submerged in the trapout, it can be inserted directly into the bottom of the column, as in Fig. 15.12. There is no objection from a heat-transfer standpoint. As seen in Example 15.3 only 214 ft² of surface was required for the trans-

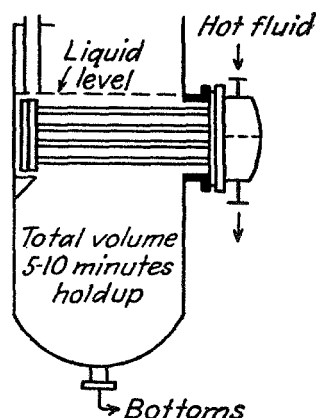


FIG. 15.12. Bundle-in-column reboiler.

fer of 2,670,000 Btu/hr and this in a 15¼-in. circular bundle 16'0" long. The column required for 28,100 lb/hr of the vapor at 200 psig has a diameter of less than 3 ft. If the bundle is inserted in the bottom of such a column, many short tubes will be required and the height of the bottom of the column must be increased so as to maintain the same holdup space. Another obvious disadvantage lies in the size of the flanged connection which must be welded to the side of the column to accommodate the larger bundle. Internal supports are also required to keep the weight of the bundle from forming a cantilever with the column connection flange. These difficul-

ties are usually surmountable when the column diameter is greater than 6 ft, but overall experience favors the use of external reboilers in preference to the small savings realized by eliminating the shell.

The calculations for the bundle-in-column reboiler are identical with those for the kettle reboiler, using coefficients from Fig. 15.11.

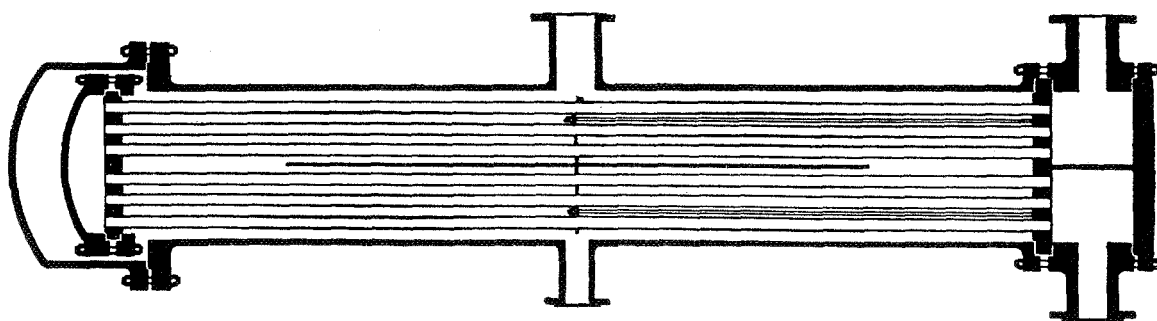


FIG. 15.13. Horizontal thermosyphon reboiler. (Patterson Foundry and Machine Co.)

d. **Horizontal Thermosyphon Reboiler.** This is perhaps the commonest type of reboiler. Figure 15.13 shows a horizontal thermosyphon reboiler. It consists of centrally located inlet and outlet nozzles, a vertical circular support plate between the nozzles, and a horizontal longitudinal baffle. Horizontal thermosyphons operate on the principle of divided flow as first outlined under condensers in Fig. 12.17, with halves of the entering fluid flowing away from each other below the longitudinal baffle and toward each other above it. Disengagement takes place in the column, and the reboiler may be connected by the

arrangement of Fig. 15.4a or b. In Fig. 15.4a, as stated previously, all the liquid from the bottom plate is led directly to the reboiler. The rate of feed to the reboiler is the hourly trapout rate, which passes through the reboiler but once. In Fig. 15.4b the reboiler is connected to the bottoms, which are free to recirculate at a rate such that the frictional pressure drop in the reboiler and other resistances of the circuit just balance the hydrostatic head difference between the liquid and liquid-vapor legs. The hydrostatic head available in the latter arrangement, however, is less than in the once-through arrangement although a greater head is required for recirculation. The head is provided by raising the bottom liquid level in the column or by raising the column itself. Occasionally the reboiler may be set in a well, but this practice finds little favor in modern plants.

Film Coefficients in Horizontal Reboilers. The coefficients used for thermosyphons are substantially the same as those employed for kettle reboilers and are given by Fig. 15.11. When there is a boiling range, it is imperative that the overall clean coefficient be weighted for sensible- and latent-heat loads individually, although the procedure differs from the weighting of successive zones, since both sensible heating and boiling occur in the same temperature range. This problem was not encountered in forced-circulation reboilers and vaporizers because the rates of boiling and sensible-heat transfer are ordinarily very nearly identical. However, in a shell without forced convection the rate of sensible-heat transfer by free convection is usually less than one-sixth the natural-circulation boiling rate. In natural circulation, however, where both sensible-heat transfer and boiling occur over the same surface, the free convection coefficient is undoubtedly modified by the bubble movements which far exceed the agitation derived from ordinary free convection currents. To account for this modification, the sensible portion of the heat load is assumed to be transferred by ordinary free convection and the boiling portion is assumed to be transferred as natural-circulation vaporization.

Although the flow is not counterflow, it usually does not deviate greatly from it, because one or both of the fluids is substantially isothermal. If steam is the heating medium, the counterflow temperature difference applies directly. If a liquid is the heating medium rather than a vapor, the counterflow temperature difference applies only if the range on the material being vaporized is small and the approach between the heating medium and vaporizing medium inlet temperatures is appreciable. If F_T for a 1-2 exchanger exceeds 0.90 an insignificant error may be anticipated from the use of the 1-2 parallel flow-counterflow temperature difference.

Since the temperature differences for sensible heating and vaporization are the same, there is no weighted temperature difference. But the sensible heat q_s is transferred with a free convection coefficient of h_s , and the latent heat q_v is transferred with the considerably higher coefficient h_v . To permit obtaining a single dirt factor, as the index of performance or maintenance of the reboiler, the weighted coefficient may be obtained as follows: From $q = hA \Delta t$,

$$A_s(\Delta t)_s = \frac{q_s}{h_s}$$

$$A_v(\Delta t)_v = \frac{q_v}{h_v}$$

The weighted coefficient is then

$$h = \frac{Q}{q_s/h_s + q_v/h_v}$$

Since neither h_v nor h_s is influenced by the velocity through the reboiler, it will be of no consequence in the calculation whether the reboiler has been connected for once-through or recirculation operation.

Pressure Drop. There is an obvious need in the recirculation arrangement to keep the pressure drop through the thermosyphon as small as possible. When studying condensers it was observed that the greater the pressure drop through the condenser the higher the condenser must be elevated above the column to permit the gravity return of condensate. The effect of pressure drop on the elevation of the bottom tower liquid level above the reboiler is even more critical. The greater the pressure drop through the reboiler the higher the entire column and auxiliaries must be elevated above the ground level to produce sufficient hydrostatic head to overcome the pressure drop. Pressure drops of about 0.25 psi are generally allowed for the reboiler and attendant losses. If the column is small in diameter or height, a pressure drop as high as 0.50 psi may be allowed, but concessions of this nature to the design of the reboiler are rare. For a reboiler vaporizing a small fraction of the liquid entering it, the required elevation is greater, since the return leg to the column contains more liquid than vapor and the difference in density of the streams to and from the reboiler is small.

While half baffles may occasionally be used to increase the turbulence in the shell, the tubes are usually prevented from sagging by the vertical support plate between the inlet and outlet nozzles and additional quarter-circle support plates. The liquid entering the horizontal thermosyphon flows *one-half of the tube length* on the underside of the longitudinal baffle and one-half the tube length on the upper side, so that all the liquid travels the entire tube length but in each instance with a mass-velocity based on one-half the total flow. The length of path of each parallel stream equals

the tube length, and it is sufficiently accurate to treat the pressure drop the same as for a shell without baffles and with axial flow as in Example 7.8. The diameter of a horizontal thermosyphon reboiler is greater than that which corresponds to the tube count of a conventional 1-2 exchanger because of the free space which must be provided in the upper half to allow the light mixture of vapor and liquid readily to reach the outlet nozzle. If the surface of a 25 in. ID layout sufficed for heat transfer, the tubes would be relocated in a 27 in. ID shell while retaining the tube pitch so as to leave a free vapor flow channel at the top of the shell and a lesser entrance channel at the bottom.

The equivalent diameter is computed directly by Eq. (7.3) from the wetted perimeter of the tubes, half the shell, and the width of the longi-

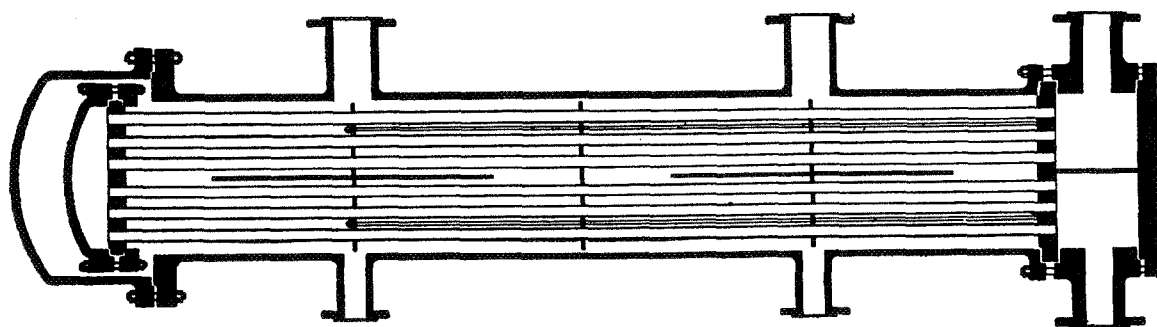


FIG. 15.14. Horizontal thermosyphon with double nozzles. (Patterson Foundry and Machine Co.)

tudinal baffle. The flow area is the difference between half a circle and the number of tubes in the upper or lower shell pass. In the absence of the actual tube layout these may be assumed to be equal. The Reynolds number is computed from the inlet liquid viscosity and the equivalent diameter. The pressure drop is based on the mean specific gravity between inlet and outlet, using a friction factor obtained from Fig. 26 for the *tube side*.

When there is a single inlet nozzle on the shell, it is customary not to use a tube length more than five times the shell diameter. Long, thin reboilers do not thermosyphon well. When a long, thin reboiler is indicated, it is usually equipped with two inlet nozzles as shown in Fig. 15.14 with a mass velocity based on one-fourth the flow rate in one-half the flow area. The following table will serve as a guide to well-proportioned horizontal thermosyphons:

Shell ID, in.	Tube length
12-17 $\frac{1}{4}$	8'0"
19 $\frac{1}{4}$ -29	12'0"
31 and over	16'0"

When using a recirculating arrangement with a horizontal thermosyphon, recirculation can be computed approximately as the rate at

which the pressure drop through the reboiler equals the hydrostatic difference of z_1 and z_3 in Fig. 15.4b although the rate does not affect the film coefficient. Instead of the recirculation rate it is preferable in reboilers to specify the *recirculation ratio*, which is defined as the pounds per hour of *liquid* leaving the reboiler compared with the pounds per hour of vapor alone. This should not be confused with the conventional definition of recirculation rates, which is defined as the ratio of the total hourly throughput to the hourly requirement.

In a recirculating reboiler the temperature range is not identical with that of a once-through arrangement. If the liquid recirculates, only a small amount of vapor is formed in each circulation and the vaporization occurs over a smaller temperature range, although the outlet temperatures in both arrangements are identical. In the recirculating arrangement the temperature difference is somewhat smaller. Usually the reduction in temperature difference is not significant unless the range of the heating medium is very close to that of the vaporization. A recirculation ratio four times or greater than the hourly once-through vapor rate is considered favorable from a cleanliness standpoint. The method of calculating recirculation ratios will be discussed in connection with the vertical thermosyphon reboiler, where a higher order of recirculation is usually obtained.

Example 15.4. Calculation of a Once-through Horizontal Thermosyphon Reboiler. 38,500 lb/hr of 60°API naphtha in a once-through arrangement is to enter a horizontal thermosyphon reboiler and produce 29,000 lb/hr of vapor in the temperature range from 315 to 335°F at an operating pressure of 5.0 psig. Heat will be supplied by 28°API gas oil with a range from 525 to 400°F.

Available for the service is a 21¼ in. ID reboiler containing 116 1 in. OD, 14 BWG tubes 12'0" long laid out on 1¼-in. square pitch. The bundle has a support plate above the single inlet nozzle and is arranged for eight passes.

What are the dirt factor and the pressure drops?

Solution:

Reboiler:

<i>Shell side</i>	<i>Tube side</i>
ID = 21¼ in.	Number and length = 116, 12'0"
Support plates = ¼ circles	OD, BWG, pitch = 1 in., 14 BWG, 1¼-in. square
Passes = divided	Passes = 8

(1) Heat balance:

Enthalpy of liquid at 315°F and 19.7 psia = 238 Btu/lb

(Fig. 11)

Enthalpy of liquid at 335°F and 19.7 psia = 252 Btu/lb

Enthalpy of vapor at 335°F and 19.7 psia = 378 Btu/lb

Naphtha $q_v = 29,000(378 - 252) = 3,650,000$

$q_s = 38,500(252 - 238) = 540,000$

$Q = 4,190,000$ Btu/hr

Gas oil, $Q = 51,000 \times 0.66(525 - 400) = 4,190,000$

(2) Δt :

Hot Fluid		Cold Fluid	
525	Higher Temp	335	190
400	Lower Temp	315	85
125	Differences	20	105

$$\text{LMTD} = 131^\circ\text{F}$$

$$R = \frac{125}{20} = 6.25 \quad S = \frac{20}{525 - 315} = 0.095$$

$$F_T = 0.97$$

$$\Delta t = F_T \times \text{LMTD} = 0.97 \times 131 = 127^\circ\text{F}$$

(3) T_c :

$$\frac{\Delta t_c}{\Delta t_h} = \frac{85}{190} = 0.447$$

(Fig. 17)

$$K_c = 0.42$$

$$F_c = 0.41$$

$$T_c = 400 + 0.41(525 - 400) = 451^\circ\text{F} \quad (5.28)$$

$$t_c = 315 + 0.41(335 - 315) = 323^\circ\text{F} \quad (5.29)$$

Hot fluid: tube side, gas oil
(4) $a'_t = 0.546 \text{ in.}^2$ [Table 10]

$$a_t = N_t a'_t / 144n$$

$$= 116 \times 0.546 / 144 \times 8 = 0.055 \text{ ft}^2$$

(5) $G_t = W/a_t$
 $= 51,000 / 0.055$
 $= 928,000 \text{ lb}/(\text{hr})(\text{ft}^2)$

(6) At $T_c = 451^\circ\text{F}$,
 $\mu = 0.45 \times 2.42 = 1.09 \text{ lb}/(\text{ft})(\text{hr})$
 [Fig. 14]

$D = 0.834/12 = 0.0695 \text{ ft}$ [Table 10]
 $Re_t = DG_t/\mu$
 $= 0.0695 \times 928,000 / 1.09 = 59,200$

(7) $j_H = 168$ [Fig. 24]

(8) At $T_c = 451^\circ\text{F}$ (28°API) [Fig. 16]

$$k(c\mu/k)^{1/3} = 0.142 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

(9) $h_i = (j_H k / D)(c\mu/k)^{1/3} \phi_i$ [Eq. (6.15a)]

$$\frac{h_i}{\phi_i} = 168 \times \frac{0.142}{0.0695} = 343$$

$$\phi_i = 1.0$$

(10) $h_{io} = h_i \times \text{ID}/\text{OD}$ [Eq. (6.6)]
 $= 343 \times 0.834/1.0$
 $= 286 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Cold fluid: shell side, naphtha

Assume weighted $h_o = 200$ h_{io} from (10) = 286

$$t_w = t_c + \frac{h_{io}}{h_{io} + h_o} (T_c - t_c) \quad [\text{Eq. (5.31)}]$$

$$= 323 + \frac{286}{286 + 200} (451 - 323)$$

$$= 382^\circ\text{F}$$

$$(\Delta t)_w = 382 - 323 = 59^\circ\text{F}$$

From Fig. 15.11, $h_v = > 300$, use 300 $h_s = 60$

$$q_v/h_v = 3,650,000/300 = 12,150$$

$$q_s/h_s = 540,000/60 = 9,000$$

$$21,150$$

(10') $h_o = 4,190,000/21,150$
 $= 198 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Checks 200 assumed for h_o .(13) Overall clean coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{286 \times 198}{286 + 198} = 116 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Design overall coefficient U_D :

$$\begin{aligned}\text{Surface per linear foot} &= 0.2618 && \text{(Table 10)} \\ \text{Total surface} &= 116 \times 12'0'' \times 0.2618 = 364 \text{ ft}^2 \\ U_D &= \frac{Q}{A \Delta t} = \frac{4,190,000}{364 \times 127} = 90.7\end{aligned}$$

Check of maximum flux (based on total transfer through the surface):

$$\frac{Q}{A} = \frac{4,190,000}{364} = 11,500 \text{ vs. } 12,000 \text{ allowable}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{116 - 90.7}{116 \times 90.7} = 0.0024 \quad (6.13)$$

Pressure Drop

(1) $Re_t = 59,200$, $f = 0.000168 \text{ ft}^2/\text{in.}^2$ [Fig. 26] $s = 0.73$ [Fig. 6] (2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D s \phi_t}$ $= \frac{0.000168 \times 928,000^2 \times 12 \times 8}{522 \times 10^{10} \times 0.0695 \times 0.73 \times 1}$ $= 5.3 \text{ psi}$ (3) $G_t = 928,000 \text{ V}^2/2g' = 0.11$ [Fig. 27] $\Delta P_r = 4 \frac{n}{s} \frac{V^2}{2g'} = \frac{4 \times 8}{0.73} \times 0.11 = 4.8 \text{ psi}$ [Eq. (7.46)] (4) $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)] $= 5.3 + 4.8 = 10.1 \text{ psi}$	(1') $D'_e = 4 \times \text{flow area/frictional wetted perimeter}$ Assume half of tubes above and half of tubes below longitudinal baffle. Flow area = $\frac{1}{2}$ shell cross section - $\frac{1}{2}$ tube cross section $= \frac{\pi}{8} (21.25^2 - 1.0 \times 116) = 132 \text{ in.}^2$ $a_s = 132/144 = 0.917 \text{ ft}^2$ Wetted perimeter = $\frac{\pi \times 21.25}{2} + \frac{\pi}{2} \times 1$ $\times 116 + 21.25 = 236.7 \text{ in.}$ $d'_e = 4 \times 132/236.7 = 2.23 \text{ in.}$ [Eq. (7.3)] $D'_e = 2.23/12 = 0.186 \text{ ft}$ $G_s = (w/2)/a_s$ $= \frac{1}{2} 38,500 \times 0.917$ $= 21,000 \text{ lb}/(\text{hr})(\text{ft}^2)$ For 60°API at 315 use data in Fig. 14 for 56°API gasoline as an approximation. $\mu = 0.18 \times 2.42 = 0.435 \text{ lb}/(\text{ft})(\text{hr})$ $Re_s = D'_e G_s / \mu$ $= 0.186 \times 21,000 / 0.435 = 8950$ $f = 0.00028 \text{ ft}^2/\text{in.}^2$ [Fig. 26] From Fig. 13.14, mol. wt. = 142 Density, $\rho = \frac{142}{359 \times 795/492 \times 14.7/19.7}$ $= 0.337 \text{ lb}/\text{ft}^3$ Soutlet liquid at 335°F = 0.61 $\rho_{\text{outlet liquid}} = 0.61 \times 62.5 = 38.1 \text{ lb}/\text{ft}^3$ $\text{Soutlet mix} = \frac{38,500/62.5}{29,000/0.337 + 9500/38.1}$ $= 0.071$ $s_{\text{inlet at } 315^\circ\text{F}} = 0.625$ $s_{\text{av}} = \frac{1}{2}(0.625 + 0.071) = 0.35$
---	--

Pressure Drop

$$\begin{aligned} \Delta P_s &= \frac{fG_s^2(L_{\text{total}})}{5.22 \times 10^{10} D_s' s \phi_s} \quad [\text{Eq. (7.45)}] \\ &= \frac{0.00028 \times 21,000^2 \times 12}{5.22 \times 10^{10} \times 0.186 \times 0.35 \times 1.0} \\ &= 0.0004 \text{ psi} \end{aligned}$$

Summary

286	<i>h</i> outside	30%
U_C	116	
U_D	90.7	
R_d Calculated 0.0024		
R_d Required		
10.1	Calculated ΔP	Neg
10.0	Allowable ΔP	0.25

The dirt factor is somewhat low for continued service. The high pressure drop on the gas oil line is insignificant.

When a reboiler is oversized, it may operate by *breathing*. As liquid enters the reboiler, it may be completely vaporized very quickly because of the overdesign. New liquid replaces it and cools the surface down. The new liquid remains in the reboiler momentarily and is heated and completely vaporized also, so that intermittent bursts of vapor issue from the reboiler outlet instead of a smooth continuous flow of vapor and liquid mixture. This can be overcome by reducing the pressure on the steam if steam is the heating medium or by placing an orifice on the shell outlet flange so as to cause an increased pressure drop on the vapor.

Horizontal Thermosyphons with Baffles. Horizontal thermosyphons are occasionally designed with vertically cut baffles such as were discussed in connection with the 2-4 exchanger. The baffles do not affect the boiling film, but they do affect the sensible-heating coefficient, increasing it beyond the free-convection value. If the baffles are the usual 25 per cent, vertically cut, segmental baffles arranged for side-to-side flow, the sensible film coefficient can be computed from shell-side data as given in Fig. 28 and as discussed under the 2-4 exchanger. If the baffles are cut 50 per cent corresponding to quarter-circle support plates, the coefficient is treated on the basis of axial flow as above, using the equivalent diameter as calculated in Example 7.8 and the tube-side data of Fig. 24.

2. VAPORIZATION IN THE TUBES

The members of this class are vertical units operating with a relatively large hydrostatic head and a low pressure drop. For this reason the vaporization usually occurs in the tubes of a one-tube-pass exchanger, which permits a greater recirculation rate than is common to horizontal units with vaporization in the shell. The three main classes of equipment employing this arrangement are the vertical long-tube evaporator, the vertical thermosyphon reboiler, and the unfired steam generator. The

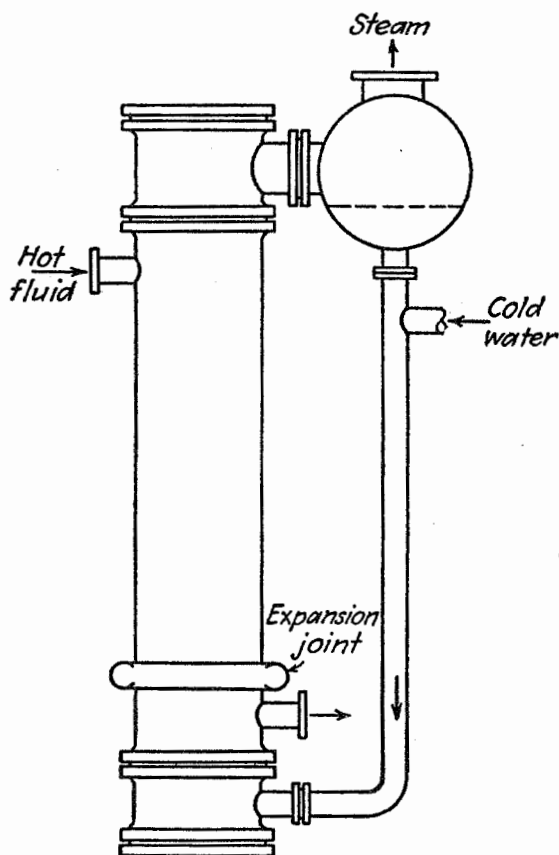


FIG. 15.15. Natural-circulation steam generator. This is similar to Fig. 14.27 but for foaming and sealing provisions.

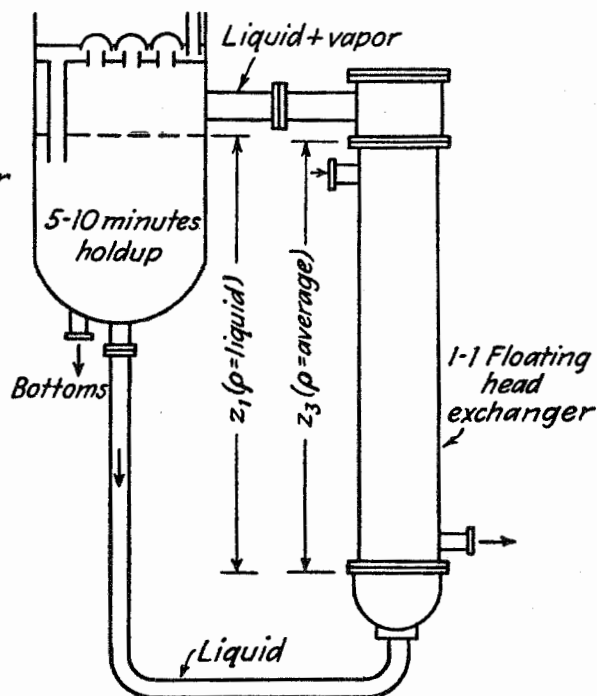


FIG. 15.16. Vertical thermosyphon reboiler connected to tower.

steam generator is shown in Fig. 15.15, and it is similar to the vertical long-tube evaporator in Fig. 14.27. The vertical thermosyphon reboiler is shown in Fig. 15.16, and it will presently be treated in detail to demonstrate the calculation common to natural-circulation units.

Recirculation Ratios. The recirculation ratio is attained when the sum of the resistances in the vaporization circuit is equal to the hydrostatic driving force on the vaporizing fluid. Referring to the vertical thermosyphon in Fig. 15.16, there are five principal resistances:

1. Frictional pressure drop through the inlet piping
2. Frictional pressure drop through the reboiler (z_2)
3. Expansion or acceleration loss due to vaporization in the reboiler

4. Static pressure of a column of mixed liquid and vapor (z_3) in the reboiler

5. Frictional pressure drop through the outlet piping

Expansion Loss Due to Vaporization. This is taken as two velocity heads based on the mean of the inlet and outlet specific gravities.

$$\Delta P_1 = \frac{G^2}{144g\rho_{av}} \quad \text{psi} \quad (15.14)$$

Particularly where the recirculation ratio and the operating pressure are great, the difference in the densities between the inlet and outlet are not very large and the expansion loss is negligible.

Weight of a Column of Mixed Liquid and Vapor. This is difficult to evaluate if precision is required, since the expansion of the vapor is a function of the recirculation ratio, average specific volume of the vapor, coefficient of expansion of the liquid, etc. For nearly all practical cases it may be assumed that the variation of the specific gravity is linear between the inlet and the outlet. If v is the specific volume at any height x in the vertical tube of Fig. 15.17 whose total length is L and whose inlet and outlet specific volumes are v_i and v_o ,

$$v = v_i + \frac{(v_o - v_i)x}{L} \quad (15.15)$$

If the weight of the column of mixture is m , the change in weight with height is dm , and if a is the cross-section flow area,

$$dm = \frac{a}{v} dx \quad (15.16)$$

If the static pressure of the column of liquid and vapor is designated by $z_3\rho_{av}$ and the cross-section area a is unity,

$$z_3 \rho_{av} = \int^L \frac{dx}{v} = \int \frac{dx}{v_i + (v_o - v_i)x/L} \quad \text{psi} \quad (15.17)$$

Integrating and dividing by 144 to obtain the head per square inch

$$\frac{z_3 \rho_{av}}{144} = \frac{2.3L}{144(v_o - v_i)} \log \frac{v_o}{v_i} \quad \text{psi} \quad (15.18)$$

Rational solutions for the recirculation ratio can be established by taking all the heads in the circuit into account as functions of the mass velocity G , and upon solution for G the recirculation rate can be obtained directly. Because the gravity of the reboiler outlet mixture also varies

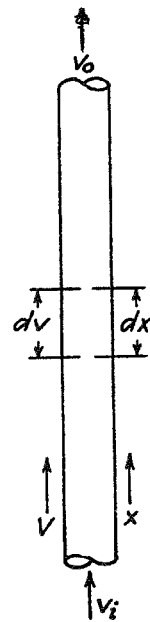


FIG. 15.17.
V o l u m e
c h a n g e i n a
s i n g l e t u b e

with the recirculation ratio, the expression becomes complex and it is simpler to solve by trial and error. If the height of an existing reboiler is given, the recirculation ratio can be computed. If the recirculation ratio is given, the required head $z_1\rho_l$ may be computed.

a. Vertical Thermosyphon Reboiler. The vertical thermosyphon is usually a conventional 1-1 exchanger with the channel end up. The upper tube sheet is placed close to the liquid level of the bottoms in the distilling column. Since the reboiler can be set close to the column, the frictional loss in the inlet and outlet piping is usually negligible. The recirculation ratio is determined from the difference between the hydrostatic head in the distilling column corresponding to the tube length of the reboiler and the weight of the vapor-liquid mixture. Recirculation ratios exceeding 4:1 are usually employed.

Example 15.5. Calculation of a Vertical Thermosyphon Reboiler. A vertical thermosyphon reboiler is to provide 40,800 lb/hr of vapor which is almost pure butane. In an arrangement identical with Fig. 15.16 the column operates at a pressure of 275 psig corresponding to a nearly isothermal boiling point of 228°F. Heat will be supplied by steam at 125 psig.

A recirculation ratio of 4:1 or greater should be employed.

What is the optimum exchanger to fulfill this requirement? $\frac{3}{4}$ in. OD, 16 BWG tubes on 1-in. triangular pitch will be used.

Solution:

(1) Heat balance:

Enthalpy of liquid at 228°F and 290 psia = 241 Btu/lb (Fig. 9)

Enthalpy of vapor at 228°F and 290 psia = 338 Btu/lb

Butane, $Q = 40,800(338 - 241) = 3,960,000$ Btu/hr

Steam, $Q = 4570 \times 868 = 3,960,000$ Btu/hr (Table 7)

(2) Δt : Isothermal boiling

$\Delta t = \text{LMTD} = 125^\circ\text{F}$ (5.14)

(3) T_c and t_c : Both streams are isothermal.

Trial 1 (see Chap. 11 for method of approach):

(a) When establishing reboiler surface the first trial should always be taken for the maximum allowable flux

$$A = \frac{Q}{Q/A} = \frac{3,960,000}{12,000} = 330 \text{ ft}^2$$

Assume 16'0" long tubes. These will reduce the shell diameter and provide the cheapest reboiler. However, it will also require the greatest elevation of the column.

Number of tubes = $330/16'0" \times 0.1963 = 105$ (Table 10)

(b) Since this will be a 1-1 exchanger, only one tube pass

From the tube counts: 105 tubes, 1 pass, $\frac{3}{4}$ in. OD, 1-in. triangular pitch

Nearest count: 109 tubes in a $13\frac{1}{4}$ in. ID shell

(c) Corrected coefficient U_D :

$A = 109 \times 16'0" \times 0.1963 = 342 \text{ ft}^2$

$U_D = \frac{3,960,000}{342 \times 125} = 92.5$

Recirculation ratio: Assume 4:1 recirculation ratio.

$$\text{Static pressure of reboiler leg, } \frac{z_3 \rho_{av}}{144} = \frac{2.3L}{144(v_o - v_i)} \log \frac{v_o}{v_i} \quad \text{psi} \quad (15.18)$$

$$\text{Vapor density, } \rho_v = \frac{58}{359 \times 688/492 \times 14.7/290} = 2.27 \text{ lb/ft}^3$$

$$v_{\text{vapor}} = \frac{1}{2.27} = 0.44 \text{ ft}^3/\text{lb}$$

$$v_{\text{liquid}} = v_i = \frac{1}{62.5 \times 0.43} = 0.0372 \text{ ft}^3/\text{lb} \quad (\text{Fig. 6})$$

Weight flow of recirculated liquid = $4 \times 40,800 = 163,200 \text{ lb/hr}$

Total volume out of reboiler:

$$\text{Liquid, } 163,200 \times 0.0372 = 6,100 \text{ ft}^3$$

$$\text{Vapor, } 40,800 \times 0.44 = 17,950 \text{ ft}^3$$

$$\text{Total} = 24,050 \text{ ft}^3$$

$$v_o = \frac{24,050}{(163,200 + 40,800)} = 0.1175 \text{ ft}^3/\text{lb}$$

$$\text{Pressure of leg, } \frac{z_3 \rho_{av}}{144} = \frac{2.3 \times 16}{144(0.1175 - 0.0372)} \times \log \frac{0.1175}{0.0372} = 1.60 \text{ psi}$$

Frictional resistance:

Flow area:

$$a_t = N_t \frac{a'_t}{144n} = 109 \times \frac{0.302}{144} = 0.229 \text{ ft}^2 \quad (\text{Table 10})$$

$$G_t = \frac{w}{a_t} = \frac{163,200 + 40,800}{0.229} = 891,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$\text{At } 228^\circ\text{F, } \mu = 0.10 \times 2.42 = 0.242 \text{ lb/(ft)(hr)} \quad (\text{Fig. 14})$$

$$D = \frac{0.62}{12} = 0.0517 \text{ ft}$$

$$Re_t = \frac{DG}{\mu} = \frac{0.0517 \times 891,000}{0.242} = 190,000$$

$$f = 0.000127 \text{ ft}^2/\text{in.}^2 \quad (\text{Fig. 26})$$

$$s_{\text{mean}} = \frac{(0.43 + 1/0.1175 \times 62.5)}{2} = 0.285$$

$$\Delta P_t = \frac{fG^2Ln}{5.22 \times 10^{10}Ds\phi_t} = \frac{0.000127 \times 891,000^2 \times 16}{5.22 \times 10^{10} \times 0.0517 \times 0.285 \times 1.0} = 2.09 \text{ psi} \quad (7.45)$$

Total resistance = $1.60 + 2.09 = 3.69 \text{ psi}$

Driving force, $\frac{z_1 \rho_l}{144} = 16 \times 0.43 \times 62.5/144 = 2.98 \text{ psi}$ no check

The resistances are greater than the hydrostatic head can provide; hence the recirculation ratio will be less than 4:1. Of the resistances the frictional pressure drop may be reduced by the square of the mass velocity if the tubes are made shorter. The other alternative is to raise the liquid level in the column above the upper tube sheet.

Trial 2: Assume 12'0" tubes and 4:1 recirculation ratio:

(a) Number of tubes $\approx 330 \text{ ft}^2 = 330/12'0'' \times 0.1963 = 140$

(b) from the tube counts: 140 tubes, one pass, $\frac{3}{4}$ in. OD, 1-in. triangular pitch
Nearest count: 151 tubes in a $15\frac{1}{4}$ in. ID shell

(c) Corrected coefficient U_D :

$$A = 151 \times 12'0'' \times 0.1963 = 356 \text{ ft}^2$$

$$U_D = 3,960,000/356 \times 125 = 89.0$$

Recirculation ratio: Assume 4:1 recirculation ratio

$$v_i = 0.0372 \text{ as before}$$

$$v_o = 0.1175$$

$$\text{Static pressure of leg, } \frac{z_3 \rho_{av}}{144} = \frac{2.3 \times 12}{144(0.1175 - 0.0372)} \log \frac{0.1175}{0.0372} = 1.20 \text{ psi}$$

Frictional resistance:

$$a_t = 151 \times \frac{0.302}{144} = 0.316 \text{ ft}^2$$

$$G_t = \frac{204,000}{0.316} = 645,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$Re_t = 0.0517 \times \frac{645,000}{0.242} = 138,000$$

$$f = 0.000135 \text{ ft}^2/\text{in.}^2$$

$$\Delta P_t = \frac{0.000135 \times 645,000^2 \times 12}{5.22 \times 10^{10} \times 0.0517 \times 0.285 \times 1.0} = 0.88 \text{ psi}$$

$$\text{Total resistance} = 1.20 + 0.88 = 2.08 \text{ psi}$$

$$\text{Driving force, } \frac{z_1 \rho_l}{144} = 12 \times 0.43 \times \frac{62.5}{144} = 2.24 \text{ psi}$$

Since the driving force is slightly greater than the resistances, a recirculation ratio better than 4:1 is assured. With a mass velocity of 645,000 lb/(hr)(ft²) equivalent to an inlet velocity ($V = G_t/3600\rho$) of $645,000/3600 \times 62.5 \times 0.43 = 6.7$ fps the butane boiling coefficient may be computed as for forced circulation.

Hot fluid: shell side, steam

(9') Condensing steam

$$h_o = 1500 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Cold fluid: tube side, butane

$$(4), (5), (6) \quad Re_t = 138,000$$

$$(7) \quad j_H = 330 \quad [\text{Fig. 24}]$$

$$(8) \quad k(c\mu/k)^{1/2} = 0.115 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$$

$$(9) \quad h_i = (j_H k/D)(c\mu/k)^{1/2} \\ = 330 \times 0.115/0.0517 = 735$$

This exceeds the maximum. Use 300.

$$(10) \quad h_{io} = h_o \times \text{ID/OD} = 300 \\ \times 0.62/0.75 = 248 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{1500 \times 248}{1500 + 248} = 213 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Dirt factor R_d :

U_D has been obtained above.

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{213 - 89}{213 \times 89} = 0.0065 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Pressure Drop: The pressure drop through the reboiler has been computed, 0.88 psi. The head elevation z_1 will be 12 ft. The pressure drop on the shell using half-circle support plates is negligible.

Summary

1500	h outside	248
U_C	213	
U_D	89	
R_d Calculated 0.0065		
R_d Required 0.004–0.006		
Neg	Calculated ΔP	0.88
Neg	Allowable ΔP	0.88

The large dirt factor must be retained because of the flux requirements. This is clearly an instance in which the high temperature of the steam yields no advantage. If the steam temperature were lower, a higher value of U_D could be used and the surface would remain the same.

The final reboiler will be

Shell side	Tube side
ID = 15¼ in.	Number and length = 151, 12'0"
Baffle space = ½ circles	OD, BWG, pitch = ¾ in., 16 BWG, 1-in. tri.
Passes = 1	Passes = 1

b. Long-tube Vertical Evaporator. The calculation in Example 15.5 can be applied directly to the long-tube vertical evaporator and the steam generator. The method of computing the recirculation ratio can also be applied directly to horizontal thermosyphons, although it is seen in practice that the recirculation ratio is usually quite low.

CALCULATIONS FOR DISTILLATION PROCESSES

The Reboiler Heat Balance. The heat requirements for a reboiler can be determined readily from heat balances on any continuous distillation column. A typical column is shown in Fig. 15.18 along with a condenser and reboiler. The function of reflux has already been discussed in Chap. 12. If R is the reflux ratio, *i.e.*, the number of mols of condensate poured back into the column *per mol* of product withdrawn, the heat balance on the condenser is

$$(R + 1)W_D H_{D(v)} - (R + 1)W_D H_{D(l)} = Q_C \quad (15.19)$$

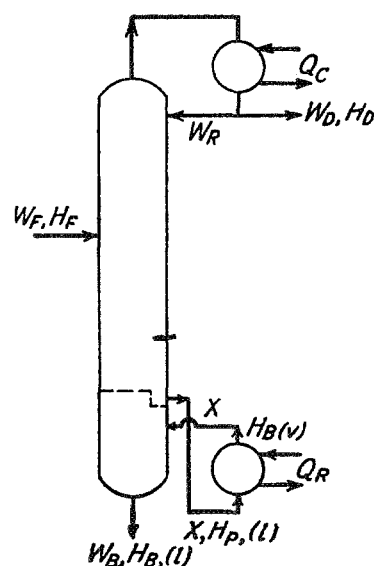


FIG. 15.18. Distilling column with condenser and reboiler.

where W_D = distillate (overhead product), lb/hr

H_D = enthalpy of the distillate (overhead product), Btu/lb

Q_C = heat removed in the condenser, Btu/hr

Subscripts (l) and (v) refer to the liquid and vapor, respectively.

The heat balance on the entire column where the feed may be liquid or vapor is

$$\begin{array}{ccc} \text{In} & & \text{Out} \\ W_F H_{F(l \text{ or } v)} + Q_R & = & Q_C + W_B H_{B(l)} + W_D H_{D(l)} \end{array} \quad (15.20)$$

where Q_R is the reboiler duty in Btu per hour and subscripts F and B refer to the feed and bottoms, respectively.

Rearranging,

$$Q_R = (R + 1)W_D H_{D(v)} - R W_D H_{D(l)} + W_B H_{B(l)} - W_F H_F \quad (15.21)$$

Assuming that enthalpy data are available, the heat load for the reboiler can be determined if the distillate, feed and bottom quantities, and temperatures are known and if the reflux ratio is also given.

Example 15.6. Calculation of the Reboiler Duty. 20,000 lb/hr of a 50-50 mixture by weight of benzene and toluene is to be distilled at 5 psig total pressure to produce a distillate or overhead product containing 99.0 per cent by weight of benzene, the more volatile component, and a bottom product containing not more than 5 per cent of benzene. A reflux ratio of 2.54 mols of reflux per mol of distillate will be used. What heat load must the reboiler deliver?

First, how much distillate and product will be formed? Two balances may be applied to obtain this information: the overall material balance and a balance to determine how the total benzene in the feed is distributed between the distillate and the bottoms, thus:

Overall balance, $20,000 = W_D + W_B$

Benzene balance, $20,000 \times 0.50 = 0.99W_D + 0.05W_B$

Solving simultaneously, $W_D = 9570$ lb/hr

$W_B = 10,430$ lb/hr

The enthalpies are obtained from Figs. 3 and 12 weighted for chemical composition at the respective temperatures

$$\begin{array}{ll} H_{B(l)} = 108.0 \text{ Btu/lb} & \text{Latent heat} = 153.0 \text{ Btu/lb} \\ H_{D(l)} = 85.8 & H_{D(v)} = 253.8 \\ H_{F(l)} = 92.0 & \end{array}$$

Substituting in Eq. (15.21),

$$Q_R = (2.54 + 1)9570 \times 253.8 - 2.54 \times 9570 \times 85.8 + 10,430 \times 108.0 - 20,000 \times 92.0 = 5,800,000 \text{ Btu/hr}$$

Vapor which must be generated in the reboiler = $5,800,000/153 = 37,900$ lb/hr

The liquid entering the reboiler in a once-through arrangement is the total quantity leaving the bottom plate or trapout. The trapout, in

turn, is equal to the sum of the material vaporized in the reboiler and the bottoms product. The total quantity entering the reboiler is

$$37,900 + 10,430 = 48,330 \text{ lb/hr}$$

The percentage vaporized is $37,900 \times 100/48,330 = 78.5$ per cent and the vapor and liquid are disengaged under the bottom plate. The temperature during vaporization does not remain constant in the reboiler, although the boiling range for a binary mixture with a fairly pure bottom product is very small. The vapor-liquid equilibrium relationships have already been treated in Chap. 13. The general method of determining the boiling range will be demonstrated presently with a multicomponent mixture.

Temperature Potentials in Distillation. The operating pressure of a distilling column is usually determined by the approach of the bubble point of the overhead product to the average temperature of the cooling water available to the condenser. If condenser water is available from 80 to 100°F, it is impossible to remove the latent heat of condensation from a vapor at such a pressure that it condenses at 60°F. In such a case it is necessary to raise the operating pressure and obtain a higher saturation temperature at which the heat can flow into the cooling water. The higher the pressure the greater the temperature difference across the condenser and the smaller the condenser. This is offset by the increased cost of the reboiler, column, and condenser, all of which must be designed for a greater pressure. If steam is available in a plant at a definite maximum pressure, the greater the temperature difference in the condenser the smaller the temperature difference attainable in the reboiler.

The Quantitative Relationships in Fractional Distillation.¹ Only a brief discussion of the elements of fractional distillation are within the scope of this chapter, but they are of considerable value in analyzing the influence of the process variables on the size of the heat-transfer equipment and particularly the reboiler. Excellent treatments of distillation theory will be found in a number of standard references.² Con-

¹ Pages 493 to 502 are preliminary to the calculation of the reboiler duty in Examples 15.7 and 15.8.

² Badger, W. L., and W. L. McCabe, "Elements of Chemical Engineering," McGraw-Hill Book Company, Inc., New York, 1936. Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950. Robinson and Gilliland, "Elements of Fractional Distillation," McGraw-Hill Book Company, Inc., New York, 1939. Walker, W. H., W. K. Lewis, W. H. McAdams, and E. R. Gilliland, "Principles of Chemical Engineering," McGraw-Hill Book Company, Inc., New York, 1937.

sider the distillation arrangement in Fig. 15.19 enclosed by the dash line 0-0'-3'-3. The upper horizontal dash line 1-1' cuts the distilling column between any two plates above the feed arbitrarily designated as the n th plate and the $n + 1$ th plate. There is a quantity of vapor V mols/hr entering 1-1'-0'-0 with a vapor composition corresponding to that above the liquid of the composition of the n th plate and designated by V_n . The liquid L mols/hr leaving 1-1'-0'-0 are those coming from the $n + 1$ th plate of the column and are designated by L_{n+1} . In addition some distillate W_D mols/hr is withdrawn. The material balance on 1-1'-0'-0 is the equation of the quantities in and out

$$V_n = L_{n+1} + W_D \quad (15.22)$$

The material balance applied to the mols of one individual component also holds.

$$V_n y_n = L_{n+1} x_{n+1} + W_D x_D \quad (15.23)$$

where y_n is the vapor mol fraction of one component in V_n , x_{n+1} is the liquid mol fraction of one component in L_{n+1} , and x_D is the mol fraction of the same component in the distillate. Solving for y_n , the vapor composition on any plate in the section above the feed plate, and eliminating V_n ,

$$y_n = \left(\frac{L_{n+1}}{L_{n+1} + W_D} \right) x_{n+1} + \left(\frac{W_D}{L_{n+1} + W_D} \right) x_D \quad (15.24)$$

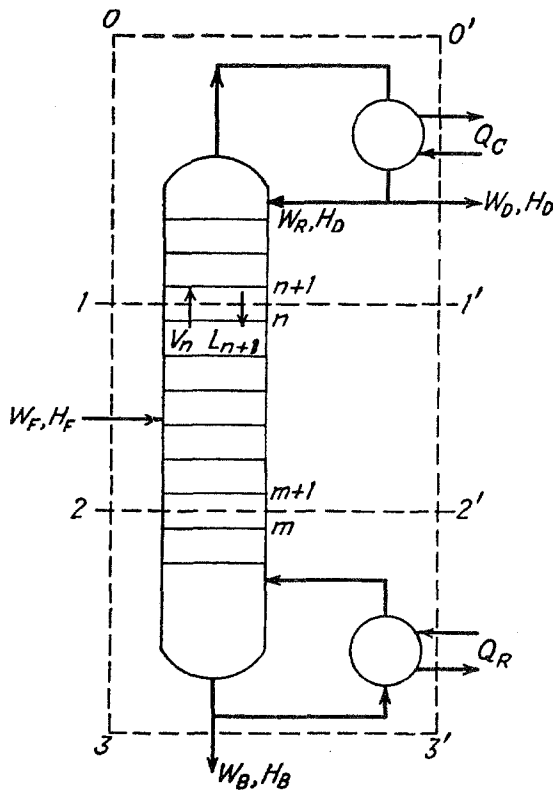


FIG. 15.19. Analysis of balances about a distilling column.

This equation defines the composition of the vapor throughout the upper section of the column. When a separation of components is to be accomplished and a distilling column is to be designed for it, certain data must be given, namely, the quantity and composition of the feed (W_F , x_F), the desired overhead distillate composition (x_D), and the bottoms-product composition (x_B). The quantities of distillate and bottoms (W_D and W_B) can be readily obtained by an overall material and component balance on the column as in Eqs. (15.22) and (15.23). In Eq. (15.24) there then remain three unknown variables, y_n , L_{n+1} , and x_{n+1} . If one of these variables can be determined at some point in the column, only two will remain unknown. Since the liquid composition

of the feed and of the distillate (top plate) must be given at the start, only y_n and L_{n+1} are ordinarily unknown. Equation (15.24) is soluble when another relationship is introduced containing y_n and L_{n+1} . Referring to Fig. 15.19 a heat balance can be made about the upper portion of the column alone *without* including the condenser.

$$V_n H_{n(v)} + W_R H_{D(l)} = (W_R + W_D) H_{D(v)} + L_{n+1} H_{n+1(l)} \quad (15.25)$$

Similarly for the section below the feed, 2-2'-3'-3, the analogues of Eqs. (15.22) to (15.24), where any plate below the feed is taken as the m th plate and the plate above it as the $m + 1$ th plate, the balances are

$$V_m + W_B = L_{m+1} \quad (15.26)$$

$$L_{m+1} x_{m+1} = V_m y_m + W_B x_B \quad (15.27)$$

$$y_m = \left(\frac{L_{m+1}}{L_{m+1} - W_B} \right) x_{m+1} - \left(\frac{W_B}{L_{m+1} - W_B} \right) x_B \quad (15.28)$$

From a heat balance at the bottom

$$Q_R + L_{m+1} H_{m+1(l)} = V_m H_{m(v)} + W_B H_{B(l)} \quad (15.29)$$

It has been observed that the ratio of the molar latent heats of vaporization to the absolute boiling points of related compounds is a constant. This is Trouton's rule. If it is further assumed that the latent heats of all compounds are equal, the latent heats of any ideal solutions they may form are also equal per mol of mixture. By assuming that the sensible-heat change from plate to plate is negligible compared with the latent-heat changes, the calculation of distillation processes is greatly simplified. If the temperature difference between the distillate and bottoms is several hundred degrees, however, this assumption cannot be made. Following this assumption, if the molar latent heats of the vaporization and condensation throughout the tower are constant, the mols of material on each plate must also be constant. Then

$$H_{n(v)} = H_{B(v)} \quad (15.30)$$

and

$$V_n = W_R + W_D$$

And from Eq. (15.25) as a result of the equality in vapor heat content,

$$W_R H_{R(l)} = L_{n+1} H_{n+1(l)} \quad (15.31)$$

By assumption,

$$H_{R(l)} = H_{n+1(l)} \quad (15.32)$$

and

$$W_R = L_{n+1}$$

The mols of liquid and vapor flow have thus been proved constant above the feed. It is now necessary to determine whether it is also constant below the feed and, if constant, whether or not the value of V_m is also equal to V_n . Combining the heat balance for the section below the feed [Eq. (15.29)] with the overall heat balance [Eq. (15.21)],

$$V_m H_{m(v)} - L_{m+1} H_{m+1(l)} = (W_R + W_D) H_{D(v)} - W_F H_{F(l)} - W_D H_{D(l)} \quad (15.33)$$

and using the assumptions established previously,

$$V_m H_{m(v)} = (W_R + W_D) H_{D(v)} \quad (15.34)$$

$$\begin{aligned} H_{m(v)} &= H_{D(v)} = H_{n(v)} \\ V_m &= W_R + W_D = V_n \end{aligned} \quad (15.35)$$

Equation (15.34) establishes the equality of vapor flow in both portions of the column.

Similarly

$$L_m H_{m(l)} = W_F H_{F(l)} + W_R H_{R(l)} \quad (15.36)$$

$$\begin{aligned} H_{m(l)} &= H_{F(l)} = H_{R(l)} \\ L_m &= W_F + W_R \end{aligned} \quad (15.37)$$

Equation (15.24) may be generalized for the composition at any plate above the feed by

$$y = \left(\frac{L}{L + W_D} \right) x + \left(\frac{W_D}{L + W_D} \right) x_D = \frac{L}{V} x + \frac{W_D}{V} x_D \quad (15.38)$$

Equation (15.28) may be generalized for any plate below the feed by

$$y = \left(\frac{L}{L - W_B} \right) x - \left(\frac{W_B}{L - W_B} \right) x_B = \frac{L}{V} x - \frac{W_B}{V} x_B \quad (15.39)$$

The McCabe-Thiele¹ Diagram. If the composition of a binary mixture can be traced by the analysis of only one component throughout the column, the solutions of Eqs. (15.38) and (15.39) can be obtained rather simply, since these apply only to the more volatile component. The solution can be accomplished by plotting a vapor-composition curve at constant total pressure which relates the mol fraction of the more volatile component in the vapor arising from and in equilibrium with a given composition of liquid as calculated by Eq. (13.7). Such an equilibrium vapor-composition curve of ordinate y vs. the values of x as the abscissa for a benzene-toluene mixture, is plotted in Fig. 15.20. These are the equilibrium changes which would occur if an individual mixture of, say, 20 mol-per cent of benzene was maintained at its boiling point in a closed vessel and shows the difference in the vapor and liquid compositions

¹ McCabe, W. L., and E. W. Thiele, *Ind. Eng. Chem.*, **17**, 605 (1925).

required at equilibrium. The same is true at the other compositions, 40, 60 per cent, etc. These are also the changes which would occur to the liquid on the plates of a distilling column if no liquid entered and no vapor left the plates. Thus, a 40 mol-per cent benzene solution on a plate or $x_n = 0.40$ is in equilibrium with a vapor of 61.4 mol-per cent or

$$y_n = 0.614$$

If the vapor were completely condensed on the plate above ($n + 1$) by a cooling coil and without the presence of liquid reflux, the concentration

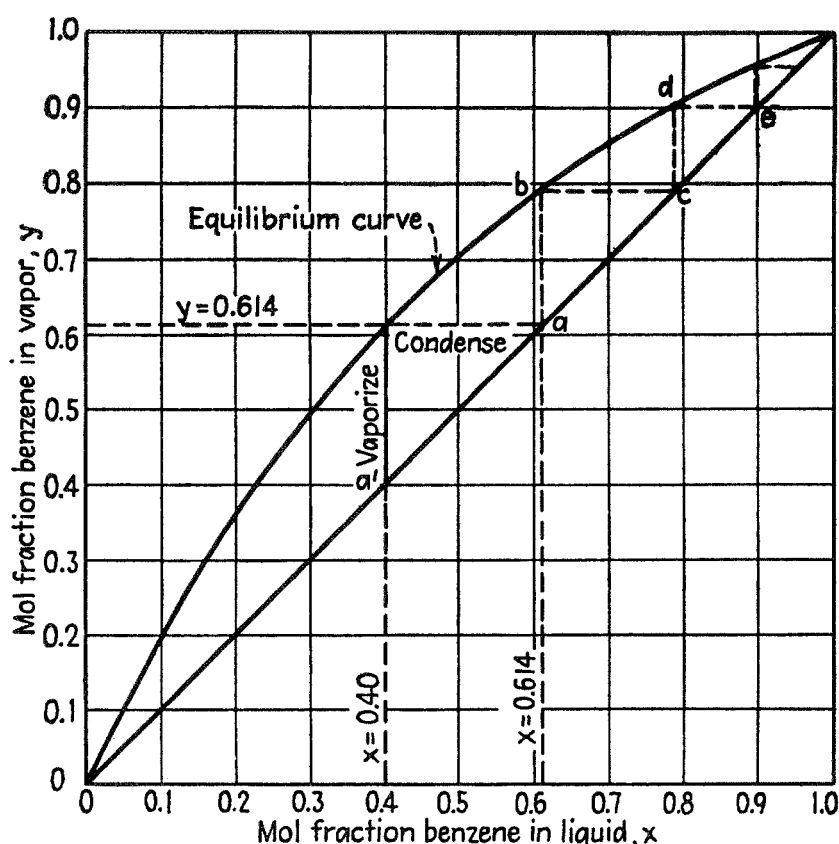


FIG. 15.20. Vapor-composition curve for mixtures of benzene and toluene at 5 psig.

of benzene in the condensed liquid would be $x_{n+1} = 0.614$. But the liquid reflux constantly enters and leaves the plates, and vapor constantly enters through the bubble cap and arises from it with a different composition. Although an equilibrium change occurs on each plate, there is never so broad an equilibrium change from plate to plate as indicated in Fig. 15.20 due to the continuous addition and removal of compounds from the plates. These have been accounted for in the balances from which Eqs. (15.38) and (15.39) were determined.

It is interesting to note that, if a diagonal is drawn across Fig. 15.20, it is the locus of the points at which a horizontal line joins a vapor and a

liquid of the same composition. Thus if the 61.4 mol-per cent vapor were condensed by heat removal using a cooling coil rather than cold reflux, the new condensate at a would also be 61.4 mol-per cent. If the 61.4 mol-per cent condensate were vaporized again, it would describe the path ab and its condensation would be bc . The function of a plate in a distilling column is to provide a space for equilibrium to take place, and the number of abc paths required one after the other, as abc , cde , etc., to permit the separation of a volatile component of concentration x_F in the feed to a concentration x_D in the distillate is the number of theoretical

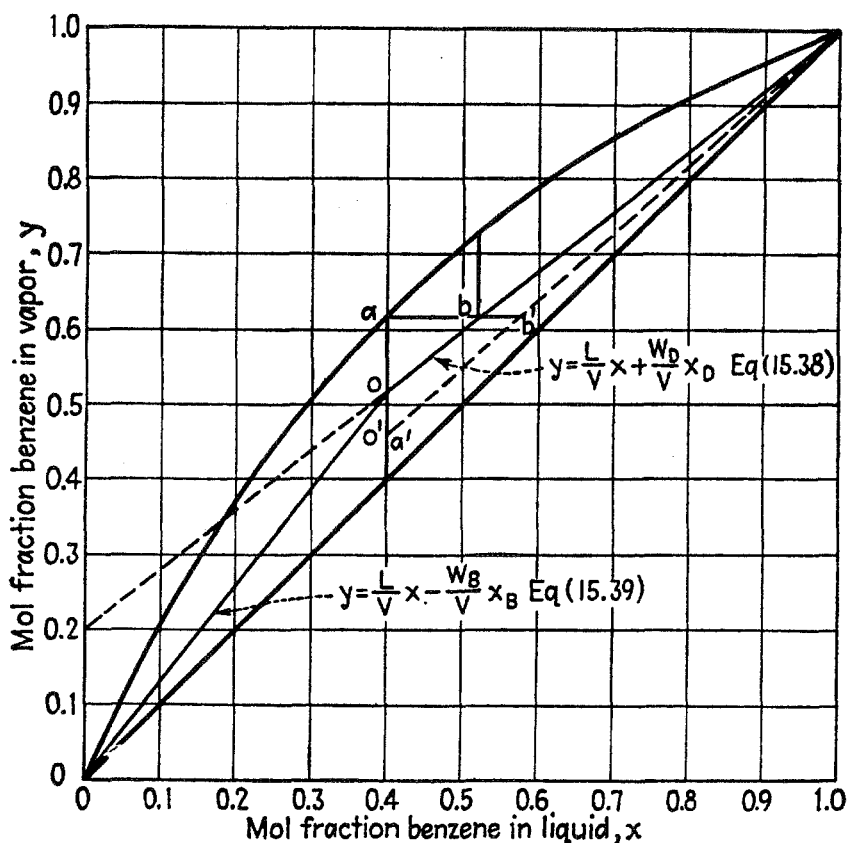


FIG. 15.21. Effect on the operating line of varying reflux ratio.

plates required for the separation from x_F to x_D . If the number of plates required to separate x_B from x_F is added to it, one obtains the total number of required theoretical plates to produce a bottoms and distillate of desired compositions from a feed of a given composition. Instead of following the diagonal line in Fig. 15.20 the composition between the liquid and the vapor passing throughout the column may be represented by the two lines in Fig. 15.21 with their identifying equations [Eqs. (15.38) and (15.39)] written directly thereon. Since these lines describe the compositions with which the column operates with reflux being poured back, they are the *operating lines*. The upper line represents the enrichment of volatile component above the feed plate and is called the *rectifying* sec-

tion. The lower line represents the depletion of volatile component and is the *stripping* section.

The influence of the reflux can be seen more clearly with the aid of Fig. 15.21. If the amount of reflux is increased while x_D is fixed, the slope of the operating line for the rectification above the feed plate also increases, moving the point o downward to o' . When the vapor at a is condensed at b' instead of b , a greater composition change occurs per plate and a smaller number of plates are required for the separation. In terms of height a shorter column can be used. But again it must be pointed out that the greater the quantity of reflux and the shorter the column the greater the diameter of the column and the greater the heat load on the reboiler and condenser. For this reason the determination of the optimum reflux ratio is an economic consideration obtained by adding the operating costs and fixed charges of the equipment for different reflux ratios and determining the minimum operating cost as a function of the reflux ratio.

It is now apparent that the 45° diagonal line in Fig. 15.21 represents the smallest possible number of plates with which the separation of feed into distillate and bottoms can be effected. But the slope of the diagonal is 1.0, and in Eq. (15.38) if the slope is unity, $L/(L + W_D) = 1.0$ and $W_D = 0$, corresponding to a primed column in equilibrium from top to bottom but without feed and producing no distillate. On the other hand, suppose o were moved up to a . The number of equilibrium changes necessary to go from x_F to x_D with an operating line between x_D on the diagonal and a would be infinite, although the reflux would be a minimum. This last would require an infinitely high column, and if o were moved to a point even higher than a , it would be impossible to achieve the separation.

Minimum Reflux. The determination of the minimum reflux is a useful limitation for establishing a practical reflux ratio. The optimum reflux ratio, as mentioned before, is simply one of economic costs. If the operating line connects a and x_D , corresponding to minimum reflux, the value of the total mols of reflux at minimum reflux may be designated by W'_R . Then Eq. (15.38) may be written

$$y_F = \frac{W'_R}{V} x_F + \frac{W_D}{V} x_D \quad (15.40)$$

but

$$W_D = V - W'_R \quad (15.41)$$

or

$$\frac{W'_R}{V} = \frac{x_D - y_F}{x_D - x_F} \quad (15.42)$$

where W'_R/V is the ratio of the mols of reflux at minimum reflux to the total vapor.

Thermal Condition of the Feed—Effect upon the Reboiler. In all the heat balances employed so far in this chapter it has been assumed that the feed was a liquid entering at its boiling point. This is not a requisite of a distillation process. The feed may enter the column directly from storage as a liquid colder than the boiling point corresponding to its composition, or it may come from the partial condenser of a preceding distilling column as a saturated vapor. It may even come from a preceding column as a liquid or vapor at a higher pressure than the operating pressure of the column to which it is fed. If a high-pressure liquid, it will flash on being released to the low-pressure column, giving a feed which is a mixture of liquid and vapor. If it is a high-pressure vapor and passes through a reducing valve before entering the low-pressure column, it may be superheated. How do these possibilities affect the column and the reboiler?

For the case of a liquid feed at its boiling point a liquid balance around the feed plate is

$$W_F = L_{m+1} - L_{n+1} \quad (15.43)$$

Define a new equation to correct for deviations from the balance in Eq. (15.43).

$$q = \frac{L_{m+1} - L_{n+1}}{W_F} \quad (15.44)$$

If $q = 1$, then as in Eq. (15.43) all the feed enters as liquid at its boiling point.

If $q < 1$, some of the feed is vapor, accounting for the discrepancy in the liquid balance.

If $q < 0$, qW_F must be negative. This means that there is less liquid downflow in the lower portion than in the upper section. This could occur only if some of the feed was superheated so that less fluid was required for the heat transfer.

If $q = 0$, $qW_F = 0$ and the feed is vapor at its boiling point.

If $q > 1$, the feed is cold liquid.

If these are the possibilities, how do they affect the intersection of the upper and lower operating lines of the column, since any point on the vertical line from x_F to a in Fig. 15.21 represents a liquid only? Employing Eqs. (15.23) and (15.27) and calling the coordinates of the intersection x', y' ,

$$\begin{aligned} V_n y' &= L_{n+1} x' + W_D x_D \\ V_m y' &= L_{m+1} x' + W_B x_B \end{aligned}$$

and the component balance for the volatile component

$$W_F x'_F = W_D x'_D + W_B x'_B \quad (15.45)$$

And if $q = (L_{m+1} - L_{n+1})/W_F$, then $q - 1 = (V_m - V_n)/W_F$, from which

$$y' = \frac{q}{q-1} x' - \frac{x_F}{q-1} \quad (15.46)$$

If q is determined from Eq. (15.44), then Eq. (15.46) describes the point of intersection in terms of x and y . The various positions of the intersection in terms of q are shown in Fig. 15.22. Inspection of Fig. 15.22

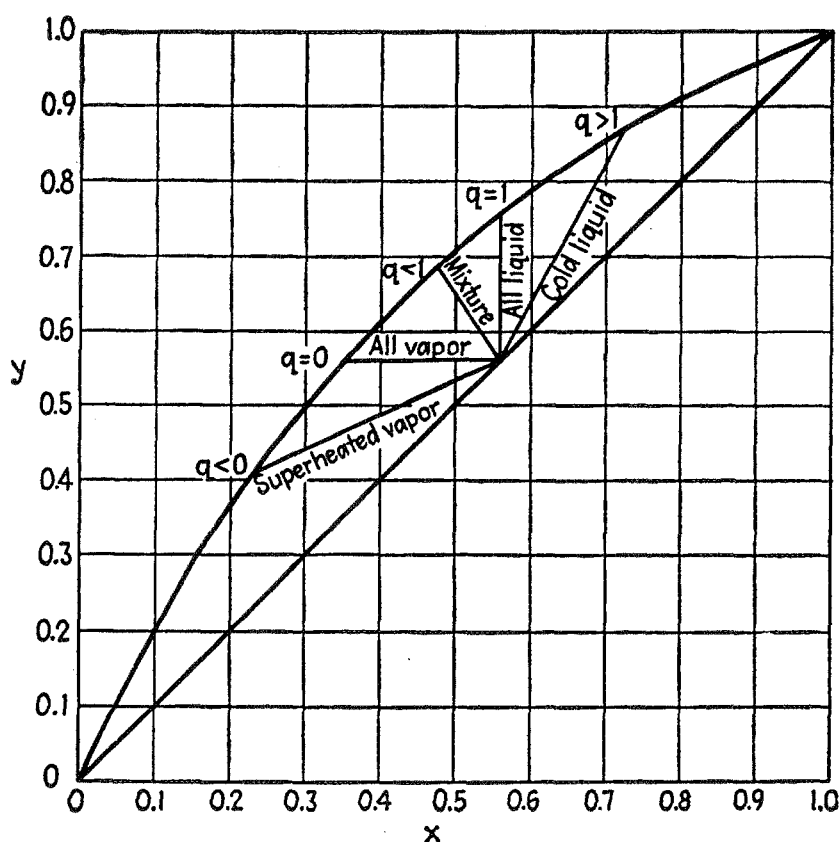


FIG. 15.22. Effect of thermal condition of feed.

indicates that for a given reflux ratio the colder the feed the fewer the number of plates required for the separation, since the reflux ratio in the lower section is increased by the condensation of the additional vapors at the feed plate. The increased reflux in the lower section must be offset by an increase in the size of the reboiler and the quantity of reboiler heat. This brings up the point of whether it is cheaper to supply the heat for cold feed in a feed preheater or in the reboiler. Very obviously it is cheaper to do so in nearly all cases in a separate preheater, since the temperature at which the feed absorbs the heat must be lower than the temperature at which the bottoms can absorb the additional heat in the

reboiler. This might suggest an advantage to the use of vapor feed, but this is offset by the increased cost of transporting the vapor in larger lines and by an increased number of theoretical plates. Furthermore, vapor feed cannot contain sufficient heat to offset the need for a reboiler, since almost all the heat entering as vapor feed must be removed in the condenser to give product and the latent heat required for reflux must still be provided by the reboiler. If feed is available as a vapor, it is unwise to condense it first, since the reduced reboiler duty and operating cost of vapor feed more than justify the addition of extra theoretical plates.

Example 15.7. Distillation of a Binary Mixture. 20,000 lb/hr of a 50 weight per cent mixture of benzene and toluene is to be distilled at an operating pressure of 5.0 psig to produce a distillate containing 99.0 weight per cent benzene and bottoms containing 95.0 weight per cent toluene. How many plates are required?

Solution. First determine the pounds of distillate and bottoms obtained as in Example 15.6. Basis: One hour.

Material balance, $20,000 = W_D + W_B$

Benzene balance, $20,000 \times 0.50 = 0.99W_D + 0.05W_B$

Solving simultaneously:

$$W_D = 9570 \text{ lb/hr}$$

$$W_B = 10,430 \text{ lb/hr}$$

COMPOSITIONS AND BOILING POINTS

	Lb/hr	Mol. wt.	Mol/hr	x_1	$p_{p1}, 214^\circ\text{F}$	$x_1 p_{p1}$	y_1
Feed:							
C_6H_6	10,000	78.1	128.0	0.543	1380	750	0.741
C_7H_8	10,000	93.1	107.5	0.457	575	262	0.259
	20,000		235.5	1.000	$\Sigma x p_p = 1012$		1.000
Distillate:					$p_{p1} 195^\circ\text{F}$		
C_6H_6	9,474	78.1	121.5	0.992	1025	1015	0.996
C_7H_8	96	93.1	1.0	0.008	410	4	0.004
	9,570		122.5	1.000	$\Sigma x p_p = 1019$		1.000
Bottoms:					$p_{p1} 246^\circ\text{F}$		
C_6H_6	520	78.1	6.7	0.059	2180	129	0.128
C_7H_8	9,910	93.1	106.3	0.941	940	882	0.872
	10,430		113.0	1.000	$\Sigma x p_p = 1011$		1.000

Minimum reflux: The equilibrium data plotted in Fig. 15.20 can be used.

$$\frac{W'_R}{V} = \frac{x_D - y_F}{x_D - x_F} = \frac{0.992 - 0.741}{0.992 - 0.543} = 0.558 \text{ mol/mol} \quad (15.42)$$

$$V = W'_R + W_D \quad \text{where} \quad W_D = 1$$

$$W'_R = 0.558W'_R + 0.558W_D$$

$$W'_R = 1.27 \text{ mol reflux/mol distillate}$$

Assume 200 per cent of the theoretical minimum reflux as an economic reflux.

$$W_R = 1.27 \times 2 = 2.54 \text{ mol/mol distillate}$$

The intercept for the upper operating line is

$$\frac{W_D}{W_R + 1} x_D = \frac{1}{2.54 + 1} \times 0.992 = 0.280$$

Connecting the corresponding lines in Fig. 15.23:

Thirteen plates are required.

The feed plate is seventh from the top.

Total reflux = $122.5 \times 2.54 = 310.5 \text{ mol}$

HEAT BALANCES

Enthalpies are computed above 0°F using specific heats from Fig. 3 and latent heats from Fig. 12 of the Appendix.

	Mol/hr	Mol. wt.	Lb/hr	Temp, °F	Btu/lb	Btu/hr
Heat balance around con- denser:						
Heat in:						
Top plate vapor.....	433	87.3	33,900	195	253.8	8,600,000
Heat out:						
Distillate.....	122.5	78.3	9,570	195	85.8	822,000
Reflux.....	310.5	78.3	24,330	195	85.8	2,090,000
Condenser duty, by differ- ence.....				...		5,688,000
						8,600,000
Overall heat balance:						
Heat in:						
Feed.....	235.5	84.8	20,000	214	92.0	1,840,000
Reboiler duty, by differ- ence.....				...		5,900,000
						7,640,000
Heat out:						
Distillate.....	122.5	78.3	9,570	195	85.8	822,000
Bottoms.....	113.0	92.8	10,430	246	108.0	1,130,000
Condenser duty.....				...		5,688,000
						7,640,000

For simplicity, in the case of a binary distillation it may be assumed that the trapout and the bottoms liquid are at nearly the same temperature and the latent heat of vaporization per pound is that of the bottoms.

At 246°F, $\lambda = 153 \text{ Btu/hr}$

Reboiler vapor = $5,800,000/153 = 37,900 \text{ lb/hr}$

Trapout = $37,900 + 10,430 = 48,330 \text{ lb/hr}$

Actually the trapout liquid on the lowest plate is at a lower temperature than the bottoms, and there is a boiling range in the reboiler. Since in this problem the aver-

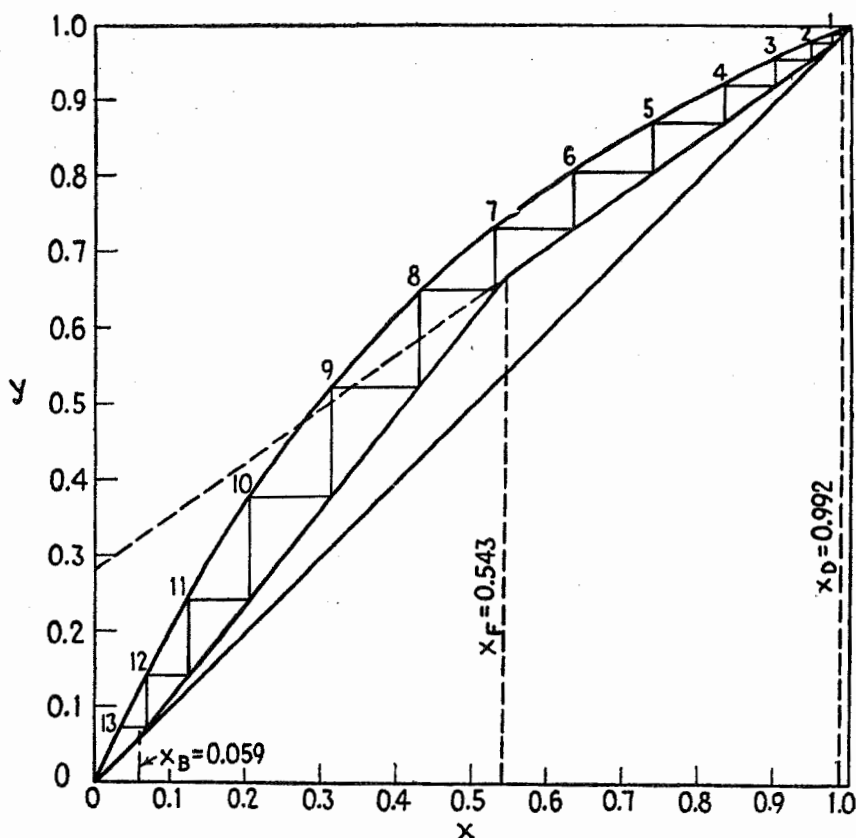


FIG. 15.23. Solution of Example 15.8.

age temperature change per plate is only $(246 - 195)/13 = 3.9^\circ\text{F}$, the boiling range can be neglected, although in distillations with greater average temperature change per plate it must be taken into account. This point is treated in Example 15.8.

HEAT BALANCE AROUND REBOILER

	Mol/hr	Mol. wt.	Lb/hr	Temp, °F	Btu/lb	Btu/hr
Heat in:						
Trapout.....	522	92.8	48,330	246	108.0	5,230,000
Reboiler duty, by difference	...			...		5,800,000
						11,030,000
Heat out:						
Reboiler vapor.....	409	92.8	37,900	246	261.0	9,900,000
Bottoms.....	113	92.8	10,430	246	108.0	1,130,000
						11,030,000

The reboiler requirements are

Total liquid to reboiler = 48,330 lb/hr

Vaporization = 37,900

Temperature (nearly isothermal) = 246°F

Pressure = 5 psig

Heat load = 5,800,000 Btu/hr

Reboiler Duty for a Multicomponent Mixture. The method of Example 15.7 may be extended to the calculation of the reboiler requirements for a multicomponent mixture (see Chap. 13). The feed quantity and composition as well as the distillate and bottoms compositions and the reflux ratio must be known. The steps in the calculation of the heat balance may be summarized:

1. Heat balance on condenser alone to determine Q_c

Into condenser: Top tray vapor

Out of condenser: Distillate and reflux

2. Overall heat balance on column to determine Q_R

Into column: Feed and reboiler heat

Out of column: Distillate, bottoms and condenser duty

3. Heat balance on reboiler alone: The quantity, composition, and temperature of the bottoms are known. The quantity, composition, and temperature of the trapout to the reboiler are *not* known. The heat required of the reboiler is known, but the quantity of the reboiler vapor is not known, although it has a composition which must be in equilibrium with the bottoms. The bottom-plate liquid will be at a lower temperature than the reboiler, and the liquid to the reboiler will be at this lower temperature. It is customary to assume this temperature difference and check the bottom-plate temperature afterward. For columns distilling a small range of components the difference will be as little as 5 to 10°F, and for wide ranges the liquid on the bottom plate may be 50°F or more below the reboiler temperature.

If X is the pounds of vapor formed and the subscript p refers to the bottom plate, the heat balance around the reboiler is

$$XH_{B(v)} + W_{B(l)}H_{B(l)} = XH_{P(l)} + W_{B(l)}H_{P(l)} + Q_R \quad (15.47)$$

from which X can be obtained (see Fig. 15.18).

4. Since there is a material balance across the bottom plate, the bottom-plate liquid is the sum of the mols of vapor and mols of bottoms. Having determined the mols of vapor and the bottoms being known, the trapout is the sum.

Having the total mols in the bottom-plate liquid, the temperature on the plate can be checked by a bubble-point calculation. If the temperature assumed in (3) does not check the bubble point of the trapout, a new bottom-plate temperature must be assumed and (3) reworked.

Example 15.8. The Reboiler Duty for a Multicomponent Mixture. The feed, distillate, and bottoms quantities and compositions are given below. A reflux ratio of 2:1 will be used. What is the reboiler duty?

	Feed, mol/hr	Distillate, mol/hr	Bottoms, mol/hr
C ₄	6.4	6.4	
C ₅	223.8	219.7	4.1
C ₆	51.6	2.3	49.3
C ₇	71.9		71.9
C ₈	52.5		52.5
C ₉	54.7		54.7
C ₁₀	82.5		82.5
C ₁₁	76.6		76.6
C ₁₂	22.4		22.4
	642.4	228.4	414.0

Solution:

DEW POINT OF OVERHEAD

	Mol/hr	$K_{40\text{ psia}}^{148^\circ\text{F}}$	$\frac{V}{K}$
C ₄	6.4	2.8	2.3
C ₅	219.7	1.01	217.5
C ₆	2.3	0.34	6.8
	228.4		228.4

BUBBLE POINT OF BOTTOMS

	Mol/hr	$K_{40\text{ psia}}^{330^\circ\text{F}}$	KL	Lb/hr
C ₅	4.1	5.8	23.8	1,700
C ₆	49.3	3.0	148.0	13,900
C ₇	71.9	1.68	120.8	13,030
C ₈	52.5	.98	51.4	6,260
C ₉	54.7	0.57	31.2	4,240
C ₁₀	82.5	0.35	28.9	4,330
C ₁₁	76.6	0.21	16.1	2,640
C ₁₂	22.4	0.13	2.9	520
	414.0		423.1	46,620

$$\text{Average mol. wt.} = 46,620/423.1 = 110.3$$

The pressure at the top and bottom of the column actually differs because of the pressure drop of the vapor in the column itself. Usually the distillate is taken at 5 to 10 psi lower than the bottom.

HEAT BALANCES

	Mol/hr	Mol. wt.	Lb/hr	Temp, °F	Btu/lb	Btu/hr
Heat Balance on Condenser						
Heat in:						
Top plate vapor	685.2	71.3	48,894	148	286	13,980,000
Heat out:						
Distillate	228.4	71.3	16,298	124	129	2,100,000
Reflux (2-1)	456.8	71.3	32,596	124	129	4,200,000
Condenser duty, by difference				...	...	7,680,000
						13,980,000
Overall heat balance:						
Heat in:						
Feed	642.4	111.8	71,775	323	275	19,700,000
Reboiler duty, by difference				...	...	4,280,000
						23,980,000
Heat out:						
Distillate	228.4	71.3	16,248	124	129	2,100,000
Bottoms	414.0	134.0	55,477	330	290	14,200,000
Condenser duty				...	...	7,680,000
						23,980,000

Heat Balance on Reboiler

Assume 30° difference between reboiler and bottom plate giving a bottom-plate temperature of 300°F

Heat in:						
Trapout	619.7*	126.6	78,177	300	234	18,300,000
Reboiler duty				...	...	4,280,000
						22,580,000
Heat out:						
Reboiler vapor	205.7*	110.3	22,700	330	369	8,380,000
Bottoms	414.0	134.0	55,477	330	256	14,200,000
						22,580,000

$$* \quad X_{369} + 14,200,000 = X_{234} + 55,477 \times 234 + 4,280,000$$

Eq. (15.47)

$$135X = 3,080,000$$

$$X = 22,700 \text{ lb/hr of vapor}$$

$$= \frac{22,700}{110.3} = 205.7 \text{ mol/hr}$$

$$\text{Trapout} = W_B + X = 414.0 + 205.7 = 619.7 \text{ mol/hr}$$

$$\text{Average molecular weight} = \frac{78,177}{619.7} = 126.6$$

CALCULATION OF BOTTOM PLATE TEMPERATURE

	y^*	Reboiler vapor $V = y205.7 + \text{Bottoms} = \text{Trapout}$			$K_{40 \text{ psi}}^{300^\circ\text{F}}$	Mol $\times K$
C_5	0.056	11.6	4.1	15.7	4.5	70.6
C_6	0.350	71.9	49.3	121.2	2.25	273.0
C_7	0.285	58.7	71.9	130.6	1.20	156.8
C_8	0.122	25.0	52.5	77.5	0.66	51.1
C_9	0.074	15.2	54.7	69.9	0.38	26.6
C_{10}	0.068	14.1	82.5	96.6	0.22	21.2
C_{11}	0.038	7.8	76.6	84.4	0.13	10.9
C_{12}	0.007	1.4	22.4	23.8	0.07	1.8
	1.000	205.7	414.0	619.7		612.0

which checks 619.7

* Mol fraction of vapor in equilibrium with bottoms.

This checks the 30°F assumption between reboiler and bottom plate.

The reboiler requirements are

Vaporization.....	22,700 lb/hr
Total liquor to reboiler.....	78,177 lb/hr
Heat load.....	4,280,000 Btu/hr
Temperature range.....	300–330°F
Operating pressure.....	40 psia

PROBLEMS

15.1. 20,000 lb/hr of hexane vapor is required from a liquid feed at 100°F and 85 psig. Heat is to be supplied by steam at 350°F.

Available for the service is a 17¼ in. ID 1-2 exchanger containing 160 ¾ in. OD, 16 BWG, 16'0" long tubes arranged for two tube passes on 1-in. square pitch. Shell baffles are spaced 16 in. apart.

What are the dirt factor and pressure drops?

15.2. 30,000 lb/hr of a mixture whose average properties correspond to hexane is vaporized from 40,000 lb/hr of feed at its bubble point at 105 psig. The boiling range is from 300 to 350°F, and heat is supplied by steam at 398°F.

Available for the service is a 19¼ in. ID 1-2 exchanger containing 132 1 in. OD, 14 BWG, 16'0" long tubes arranged for two tube passes on 1¼-in. square pitch. Shell baffles are spaced 12 in. apart.

What are the dirt factor and the pressure drops?

15.3. 45,000 lb/hr of 42°API kerosene enters a once-through horizontal thermosyphon reboiler. 32,000 lb/hr is vaporized between 285 and 330°F, using steam at 380°F. The required dirt factor is 0.0030, and the allowable pressure drop through the reboiler is 0.25 psi. Vaporization occurs at 5 psig, and the molecular weight of the vapor may be taken as 120. The latent heat of the vapor is 110 Btu/lb.

Available for the service is a 25 in. ID horizontal thermosyphon containing 324 ¾ in. OD, 16 BWG tubes, 12'0" long laid out on 1-in. square pitch for two passes.

Is the reboiler satisfactory?

15.4. 76,300 lb/hr of pentane bottoms (mol. wt. liquid = 77.2) is to be recirculated through a horizontal thermosyphon reboiler to provide 31,400 lb/hr of vapor (mol.

wt. vapor = 74.7, latent heat = 153 Btu/lb). The tower and exchanger operate at 85 psig, and vaporization occurs from 225 to 245°F. Heat is supplied by 28°API gas oil from 470 to 370°F with a maximum pressure drop of 15 psi.

A 23¼ in. ID horizontal thermosyphon reboiler contains 240 ¾ in. OD, 13 BWG, 8'0" tubes with six passes on 1-in. square pitch. A pressure drop of 0.25 is permissible.

What will the dirt factor and approximate recirculation ratio be?

15.5. A reboiler is to be designed for the vaporization service in Prob. 15.4 with a dirt factor of 0.004 except that the heat requirement will be supplied by a 50 psig steam line. Using ¾ in. OD, 13 BWG tubes, establish the layout of a reboiler which most reasonably fulfills the conditions.

15.6. A vertical thermosyphon reboiler must provide 120,000 lb/hr of butane vapor to a tower at 135 psig (boiling point = 178°F, latent heat = 122 Btu/lb) using steam at 12 psig in the shell.

Available for the service is a 35 in. ID 1-1 exchanger containing 900 ¾ in., 16 BWG, 16'0" long tubes on 1-in. triangular pitch. Shell baffles are half-circle supports on 24 in. centers.

What is the recirculation ratio?

15.7. For the process conditions in Prob. 15.6 determine the nearest vertical thermosyphon to yield approximately a 4:1 recirculation ratio and a dirt factor of 0.003 when the heat is supplied by steam at 250°F.

NOMENCLATURE FOR CHAPTER 15

A	Heat-transfer surface, ft ²
A_p, A_s, A_v	Heat-transfer surface for preheating, sensible heating, and vaporization, ft ²
a''	External surface per linear foot of pipe or tube, ft
a'_t	Flow area per tube, in. ²
B	Baffle spacing, in.
C	Specific heat of hot fluid, Btu/(lb)(°F)
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of tube, ft
D_s, D'_s	Equivalent diameter for heat transfer and pressure drop, ft
D_i	Inside diameter of shell, ft
d_e	Equivalent diameter, in.
e	Napierian base, dimensionless
F_c	Caloric fraction, dimensionless
F_T	Temperature-difference factor, $\Delta t = F_T \times \text{LMTD}$, °F
f	Friction factor, ft ² /in. ²
G	Mass velocity, lb/(hr)(ft ²)
g'	Acceleration of gravity, ft/sec ²
H	Enthalpy, Btu/lb
h	Heat-transfer coefficient in general, Btu/(hr)(ft ²)(°F)
h_i, h_o	Heat-transfer coefficient for inside fluid and for outside fluid, Btu/(hr)(ft ²)(°F)
h_{io}	Value of h_i when referred to the tube outside diameter, Btu/(hr)(ft ²)(°F)
h_p, h_s, h_v	Film coefficient for preheating, sensible-heat transfer and vaporization, Btu/(hr)(ft ²)(°F)
ID	Inside diameter of shell or tube, in.
j_H	Factor for heat transfer, dimensionless

K	Vapor-liquid equilibrium constant, dimensionless
K_c	Caloric constant, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F)/(ft)
L	Tube length, ft; liquid, mol/hr
LMTD	Logarithmic mean temperature difference, °F
m	Weight of a column of liquid and vapor, lb
N	Number of baffles
N_t	Number of tubes
n	Number of tube passes
OD	Outside diameter of tube, in.
P_T	Tube pitch, in.
ΔP	Pressure drop, psi
$\Delta P_T, \Delta P_t, \Delta P_r$	Total, tube, and return pressure drop, psi
p	Pressure, psi
p_p, p_t	Pressure of a pure component and total pressure on the system, atm or psi
q	Factor in distillation describing the condition of the feed, dimensionless
q_p, q_s, q_v	Heat flow for preheating, sensible heating, and vaporization, Btu/hr
Q	Heat flow, Btu/hr
Q_C, Q_R	Condenser and reboiler duties, Btu/hr
R	Reflux ratio; temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_d	Dirt factor, (hr)(ft ²)(°F)/Btu
Re, Re'	Reynolds number for heat transfer and pressure drop, dimensionless
s, s_{av}	Specific gravity, average specific gravity, dimensionless
T_c	Caloric temperature of hot fluid °F
T_s	Steam temperature, °F
t_s	Isothermal boiling temperature, °F
T_1, T_2	Inlet and outlet temperatures of hot fluid, °F
t_c	Caloric temperature of cold fluid, °F
$\Delta t_c, \Delta t_h$	Cold and hot terminal differences, °F
Δt	True temperature difference, °F
$\Delta t_p, \Delta t_s, \Delta t_v$	True temperature difference for preheating, sensible-heat transfer, and vaporization, °F
$(\Delta t)_w$	Temperature difference between tube wall and boiling liquid, °F
U, U_C, U_D	Overall coefficient of heat transfer, clean coefficient, and design coefficient, Btu/(hr)(ft ²)(°F)
V	Velocity, fps; vapor, mol/hr
v, v_v, v_l	Specific volume, specific volume of vapor, specific volume of liquid, ft ³ /lb
v_i, v_o, v_{av}	Inlet, outlet, and average specific volumes, ft ³ /lb
W	Weight flow of hot fluid, lb/hr
W'_R	Minimum reflux, lb/hr
w	Weight flow of cold fluid, lb/hr
X	Vapor formed in reboiler, lb/hr
x	Distance, ft
x, y	Mol fraction of liquid and vapor
x', y'	Mol fraction at intersection of operating lines
z	Static height or frictional head, ft
λ	Latent heat of vaporization, Btu/lb
μ	Viscosity at the caloric temperature, lb/(ft)(hr)

μ_w	Viscosity at the tube wall temperature, lb/(ft)(hr)
ρ	Density, lb/ft ³
ϕ'	Viscosity ratio, $(\mu/\mu_w)^{0.14}$

Subscripts (except as noted above)

B	Bottoms
D	Distillate
F	Feed
l	Liquid
m	Plate below the feed in a distilling column
$m + 1$	Plate below the m plate
n	Plate above the feed
$n + 1$	Plate below the n plate
p	Bottom plate
R	Reflux
s	Shell
t	Tube
v	Vapor

CHAPTER 16

EXTENDED SURFACES¹

Introduction. When additional metal pieces are attached to ordinary heat-transfer surfaces such as pipes or tubes, they extend the surface available for heat transfer. While the extended surface increases the total transmission of heat, its influence as surface is treated differently from simple conduction and convection.

Consider a conventional double pipe exchanger whose cross section is shown in Fig. 16.1*a*. Assume that hot fluid flows in the annulus and cold

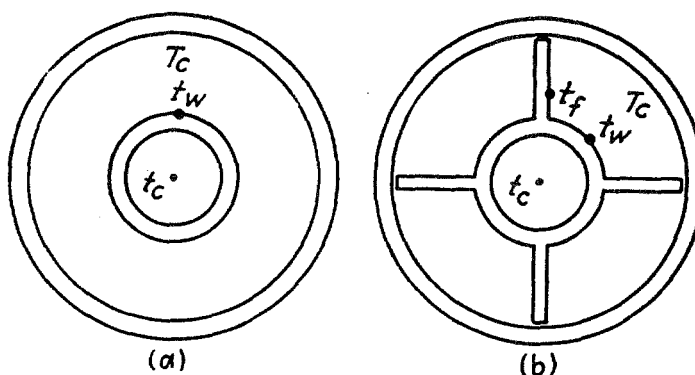


FIG. 16.1. Ordinary and finned tube.

fluid in the inner pipe, both in turbulent flow, and that the effective temperatures over the cross section are T_c and t_c , respectively. The heat transferred can be computed from the inner pipe surface, the annulus coefficient, and the temperature difference $T_c - t_w$, where t_w is the temperature of the outer surface of the inner pipe. Next assume that strips of metal are welded to the inner pipe as shown in Fig. 16.1*b*. Since the metal strips are attached to the cold tube wall, they serve to transfer additional heat from the hot fluid to the inner pipe. The total surface available for heat transfer no longer corresponds to the outer circumference of the inner pipe but is increased by the additional surface on the sides of the strips. If the metal strips do not reduce the conventional

¹ Comprehensive mathematical treatments of extended surfaces are given by Carslaw, H. S., and J. C. Jaeger, "Conduction of Heat in Solids," Clarendon Press, Oxford, 1947. Jakob, M., "Heat Transfer," John Wiley & Sons, Inc., New York, 1949.

annulus heat-transfer coefficient by appreciably changing the fluid-flow pattern, more heat will be transferred from the annulus fluid to the pipe fluid.

Pieces which are employed to extend the heat-transfer surfaces are known as *fins*. It will be shown in the case of pipes and tubes, however, that each square foot of extended surface is less effective than a square foot of unextended surface. Referring again to Fig. 16.1b, there is a temperature difference $T_c - t_f$ between the annulus fluid and the fin, and the heat which flows into the fin will be conducted by it to the inner

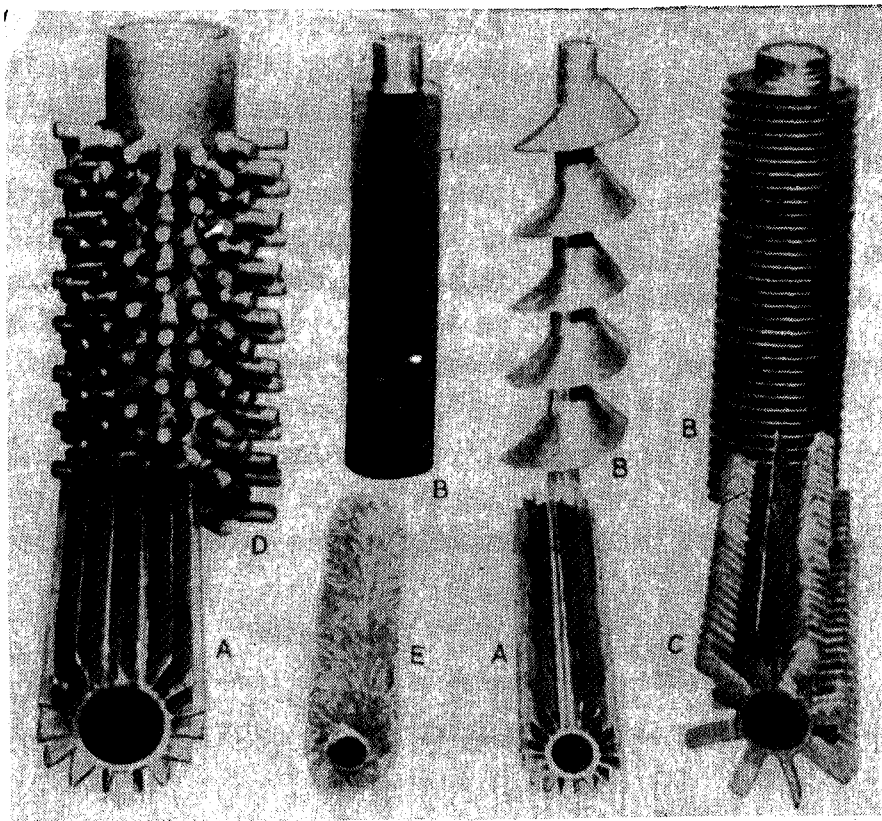


FIG. 16.2. Some commercial forms of extended surface. a. Longitudinal fins. (Griscom-Russell Co.) b. Transverse fins. (Griscom-Russell Co.) c. Discontinuous fins. (Babcock and Wilcox Co.) d. Pegs or studs. (Babcock and Wilcox Co.) e. Spines. (Thermek Corporation.) (Gardner, *Transactions of the ASME*.)

pipe. For heat to be conducted to the pipe, t_f must be greater than the pipe-wall temperature t_w . Then $T_c - t_f$ is less than $T_c - t_w$. Since the effective temperature difference between the fluid and fin is less than that between the fluid and pipe, it results in less heat transfer per square foot of fin than for the pipe. Furthermore, the temperature difference between the fluid and the fin changes continuously from the outer extremity to the base of the fin owing to the rate at which heat enters the fin by convection and the rate at which it is transferred to its base by conduction.

It will be found that two fundamental heat-transfer principles enter into the various relationships in fins: (1) to determine from the geometry and conductivity of the fin the nature of the temperature variation and (2) to determine the heat-transfer coefficient for the combination of fin and unextended surface. In the case of the double pipe exchanger, for example, the fins suppress the spiral eddies about the annulus, which in turn reduces the convection coefficient for the annulus below its conventional value as determined from Eq. (6.2).

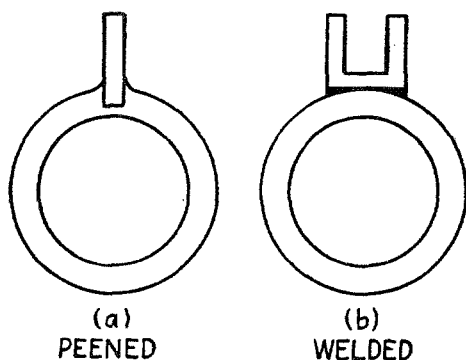


FIG. 16.3. Fin attachment.

Classification of Extended Surfaces.

Fins of a number of industrial types are shown in Fig. 16.2. Pipes and tubes with longitudinal fins are marketed by several manufacturers and consist of long metal strips or channels attached to the outside of the pipe. The strips are attached either by grooving and peening the tube as in Fig. 16.3a or by welding continuously along the base. When channels are attached, they are integrally welded to the tube as in Fig. 16.3b. Longitudinal fins of this type are commonly used in double pipe exchangers or in un baffled shell-and-tube exchangers when the flow proceeds along the axis of the tube. Longitudinal fins are most commonly employed in problems involving

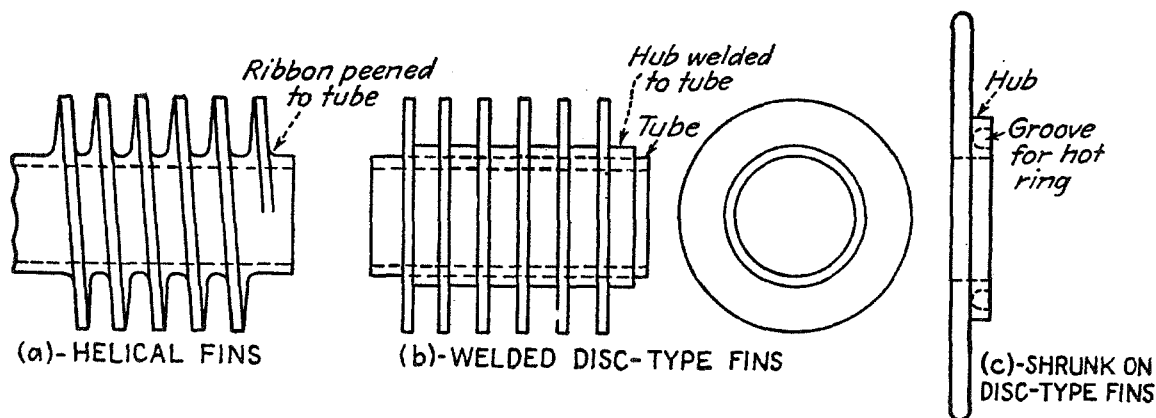


FIG. 16.4. Transverse fins.

gases and viscous liquids or when the smallness of one of a pair of heat-transfer streams causes streamline flow.

Transverse fins are made in a variety of types and are employed primarily for the cooling and heating of gases in *crossflow*. The helical fins in Fig. 16.4a are classified as transverse fins and are attached in a variety of ways such as by grooving and peening, expanding the tube metal itself to form the fin, or welding ribbon to the tube continuously. Disc-type

fins are also transverse fins and are usually welded to the tube or shrunk on to it as shown in Fig. 16.4*b* and *c*. In order to shrink a fin onto a tube a disc, with inside diameter slightly less than the outside diameter of the tube, is heated until its inside diameter exceeds the outside diameter of the tube. It is slipped onto the tube, and upon cooling, the disc shrinks to the tube and forms a bond with it. Another variation of the shrunk-on fin in Fig. 16.4*c* employs a hollow ring in its hub into which a hot metal ring is driven. Other types of transverse fins are known as *discontinuous fins*, and several shapes such as the *star fin* are shown in Fig. 16.5.

Spine- or peg-type fins employ cones, pyramids, or cylinders which extend from the pipe surface so that they are usable for either longitudinal flow or crossflow. Each type of finned tube has its own characteristics and effectiveness for the transfer of heat between the fin and the fluid inside the tube, and the remainder of this chapter deals with the derivation of the relationships and the applications of the commonest types. Perhaps the principal future uses will be in the field of atomic energy for the recovery of controlled heat of fission, in the reversing exchanger for commercial oxygen plants, in jet propulsion, and in gas-turbine cycles.

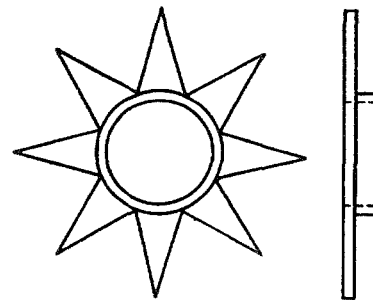
LONGITUDINAL FINS

Derivation of the Fin Efficiency. The simplest fin from the standpoint of manufacture as well as mathematics is the longitudinal fin of uniform thickness. For the derivation of its characteristics it is necessary to impose the limitations and assumptions given by Murray¹ and later by Gardner.²

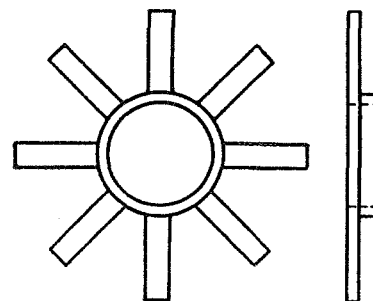
1. The heat flow and temperature distribution throughout the fin is independent of time; *i.e.*, the heat flow is steady.
2. The fin material is homogeneous and isotropic.
3. There is no heat source in the fin itself.
4. The heat flow to or from the fin surface at any point is directly proportional to the temperature difference between the surface at that point and the surrounding fluid.
5. The thermal conductivity of the fin is constant.

¹ Murray, W. M., *J. Applied Mechanics*, **5**, A78-80 (1938).

² Gardner, K. A., *Trans. ASME*, **67**, 621-632 (1945).



(a)- STAR-TYPE FIN



(b)-MODIFIED STAR FIN

FIG. 16.5. Discontinuous fins.

6. The heat-transfer coefficient is the same over the entire fin surface.
7. The temperature of the fluid surrounding the fin is uniform.
8. The temperature of the base of the fin is uniform.
9. The fin thickness is so small compared with its height that temperature gradients across the width of the fin may be neglected.
10. The heat transferred through the outermost edge of the fin is negligible compared with that passing into the fin through its sides.
11. The joint between the fin and the tube is assumed to offer no bond resistance.

At any cross section as in Fig. 16.6 let T_c be the constant temperature of the hot fluid everywhere surrounding the fin and let t be the tempera-

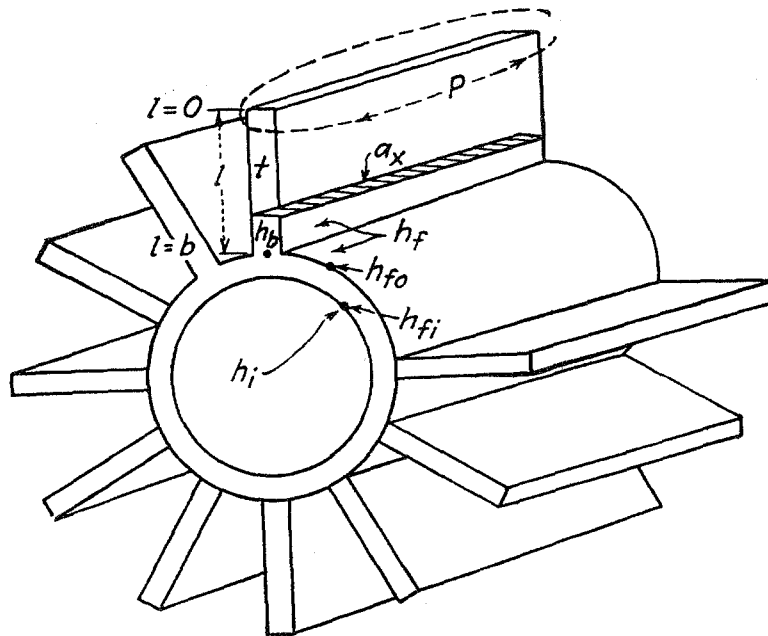


FIG. 16.6. Derivation of the longitudinal fin efficiency.

ture at any point in the fin and variable. Let Θ be the temperature difference driving heat from the fluid to the fin at any point in its cross section. Then

$$\Theta = T_c - t \quad (16.1)$$

If l is the height of the fin varying from 0 to b ,

$$\frac{d\Theta}{dl} = - \frac{dt}{dl} \quad (16.2)$$

The heat within the fin which passes through its cross section by conduction is

$$Q = ka_x \frac{d\Theta}{dl} \quad (16.3)$$

where a_x is the cross-sectional area of the fin. This is equal to the heat which passed into the fin through its sides from $l = 0$ down to the shaded cross section. If P is the perimeter of the fin, the area of the sides is $P dl$ and the film coefficient from liquid to fin side, whether on fin surface or tube surface, is h_f .

$$dQ = h_f \Theta P dl \quad \text{or} \quad \frac{dQ}{dl} = h_f P \Theta \quad (16.4)$$

Differentiating Eq. (16.3) with respect to l

$$\frac{dQ}{dl} = k a_x \frac{d^2 \Theta}{dl^2} \quad (16.5)$$

Equations (16.4) and (16.5) are equalities.

$$k a_x \frac{d^2 \Theta}{dl^2} - h_f P \Theta = 0 \quad (16.6)$$

Rearranging,

$$\frac{d^2 \Theta}{dl^2} - \frac{h_f P \Theta}{k a_x} = 0 \quad (16.7)$$

The direct solution of this equation is

$$\Theta = C_1 e^{\left(\frac{h_f P}{k a_x}\right)^{1/2} l} + C_2 e^{-\left(\frac{h_f P}{k a_x}\right)^{1/2} l} \quad (16.8)$$

Let

$$m = \left(\frac{h_f P}{k a_x}\right)^{1/2}$$

The general solution is

$$\Theta = C_1 e^{ml} + C_2 e^{-ml} \quad (16.9)$$

At $l = 0$

$$\Theta_e = C_1 + C_2 \quad (16.10)$$

where the subscript refers to the outer edge of the fin.

If no heat enters the fin at the extreme end, as qualified by assumption 10, $d\Theta/dl = 0$ when $\Theta = 0$ and $C_1 - C_2 = 0$.

$$C_1 = C_2 = \frac{\Theta_e}{2} \quad (16.11)$$

Equation (16.8) becomes

$$\frac{\Theta}{\Theta_e} = \frac{e^{ml} + e^{-ml}}{2} \quad (16.12)$$

or in general terms

$$\Theta = \Theta_e \cosh ml \quad (16.13)$$

At the base of the fin where $l = b$

$$\Theta_b = \Theta_e \cosh mb \quad (16.14)$$

where the subscript refers to the base of the fin.

Thus an expression has been obtained for the temperature difference between the constant fluid temperature and variable fin temperature in terms of the length of the fin. It is now necessary to obtain an expression for Q in terms of l . From Eq. (16.4) by differentiation with respect to the fin height l ,

$$\frac{d^2Q}{dl^2} = h_f P \frac{d\Theta}{dl} \quad (16.15)$$

Substituting in Eq. (16.3),

$$Q = \frac{ka_x}{h_f P} \frac{d^2Q}{dl^2} \quad (16.16)$$

$$\frac{d^2Q}{dl^2} - \frac{h_f P}{ka_x} Q = 0 \quad (16.17)$$

As before, the solution is

$$Q = C'_1 e^{ml} + C'_2 e^{-ml} \quad (16.18)$$

At $l = 0$

$$C'_1 + C'_2 = 0 \quad C'_1 = -C'_2$$

and

$$\frac{dQ}{dl} = 0$$

$$\frac{dQ}{dl} = h_f P \Theta_e = m C'_1 - m C'_2 = 0 \quad (16.19)$$

$$C'_1 = \frac{h_f P \Theta_e}{2m} \quad C'_2 = -\frac{h_f P \Theta_e}{2m}$$

$$Q = \frac{h_f P \Theta_e}{2m} e^{ml} - \frac{h_f P \Theta_e}{2m} e^{-ml} \quad (16.20)$$

In terms of hyperbolic functions

$$Q = \frac{h_f P \Theta_e}{m} \sinh ml \quad (16.21)$$

$$Q_b = \frac{h_f P \Theta_e}{m} \sinh mb \quad (16.22)$$

The ratio of heat load Q_b to the temperature difference Θ_b at the base is

$$\frac{Q_b}{\Theta_b} = \frac{h_f P \Theta_e \sinh mb}{m \Theta_e \cosh mb} \quad (16.23)$$

or

$$\frac{Q_b}{\Theta_b} = \frac{h_f P}{m} \tanh mb \quad (16.24)$$

Define h_b as the value of h_f to the fin surface alone when referred to the base area of the fin at $l = b$. Calling the ratio $h_b/h_f = \Omega$ the efficiency¹ of the fin, the average value of the heat-transfer coefficient at the base of the fin is given by Fourier's equation:

$$\frac{Q_b}{\Theta_b b P} = h_b \quad (16.25)$$

The fin efficiency h_b/h_f may be defined by Eqs. (16.24) and (16.25).

$$\Omega = \frac{h_b}{h_f} = \frac{Q_b/\Theta_b b P}{\frac{m Q_b}{\Theta_b P} \cdot \frac{1}{\tanh mb}} = \frac{\tanh mb}{mb} \quad (16.26)$$

Equation (16.26) applies only to the fin and not to the bare portion of the tube between the fins. To give the total heat removed by a finned tube, the heat flowing into the fin with a coefficient h_f must ultimately be combined with that flowing into the bare tube at the tube outside diameter. For this it is necessary to establish some reference surface at which the coefficient on the different parts can be reduced to the same heat flux. In an ordinary exchanger h_o is referred to the tube outside diameter. Because there is no simple reference surface on the outside of finned tubes it is convenient to use the inside diameter of the tube as the reference surface at which local coefficients are corrected to the same heat flux. By definition h_f is the coefficient to all the outside surface, whether it be the fin or bare tube. Beneath the bases of the fins there is naturally a greater heat flux than on the bare tube surface between fins, since the heat flowing through the bases of the fins is greater per unit of tube area. It may also be expected that part of the heat passing from the bases of the fins is conducted into the tube metal, so that the temperature difference between the annulus fluid and bare tube is not strictly constant. It is not usually necessary to correct for this effect, since the bare tube area so affected by the increased flux near the bases of the fins is ordinarily small compared with the total bare tube area. At the inside diameter of the tube, however, the heat from both the fins and the bare tube surface is assumed to have attained a uniform flux.

The total heat removed from the annulus liquid and arriving at the tube inside diameter is a composite of the heat transferred by the fins to the tube outside diameter and that transferred directly to the bare tube surface. These may be combined by means of the *weighted effi-*

¹ Another term which is used in the literature is the "fin effectiveness," which is the efficiency multiplied by the ratio of fin surface to fin base area.

ciency Ω' . If the heat transferred through the bare tube surface at the tube outside diameter is designated by Q_o , then

$$Q_o = h_f A_o \Theta_b \quad (16.27)$$

where A_o is the bare tube surface at the outside diameter exclusive of the area beneath the bases of the fins. If there are N_f fins on the tube, bPN_f is all of the fin surface. The total heat transfer at the outside diameter is given by

$$\begin{aligned} Q &= Q_b + Q_o = h_b bPN_f \Theta_b + h_f A_o \Theta_b \\ &= \left(\frac{h_b bPN_f A_o}{A_o} + \frac{h_f bPN_f A_o}{bPN_f} \right) \Theta_b \\ &= \left(\frac{h_b}{A_o} + \frac{h_f}{bPN_f} \right) bPN_f A_o \Theta_b \end{aligned} \quad (16.28)$$

Substituting Eq. (16.26) to eliminate h_b

$$Q = \left(bPN_f \frac{\tanh mb}{mb} + A_o \right) h_f \Theta_b \quad (16.29)$$

Calling h_{fo} the composite value of h_f to both the fin and bare tube surfaces when referred to the outside diameter of the tube, the weighted efficiency is by definition $\Omega' = h_{fo}/h_f$. Combining Eqs. (16.24) and (16.29)

$$\Omega' = \frac{h_{fo}}{h_f} = \left(\frac{bPN_f \frac{\tanh mb}{mb} + A_o}{bPN_f + A_o} \right) \quad (16.30)$$

But as in Eq. (6.2) the values of the coefficients vary inversely with the heat-flow area. If h_{fi} is the value of the composite coefficient h_{fo} referred to the tube inside diameter,

$$\frac{h_{fi}}{h_{fo}} = \frac{bPN_f + A_o}{A_i} \quad (16.31)$$

Substituting in Eq. (16.31)

$$h_{fi} = \left(bPN_f \frac{\tanh mb}{mb} + A_o \right) \frac{h_f}{A_i} \quad (16.32)$$

From Eq. (16.32)

$$h_{fi} = \left(\frac{\Omega A_f + A_o}{A_f + A_o} \right) \frac{(A_f + A_o)}{A_i} h_f \quad (16.33)$$

or simply

$$h_{fi} = (\Omega A_f + A_o) \frac{h_f}{A_i} \quad (16.34)$$

Thus an equation has been obtained which gives directly the heat-transfer coefficient on the inside of an extended tube which is the equivalent of a value of h_f on the outside surface of the tube. By substituting the physical and geometrical factors for a given tube and fin arrangement a *weighted efficiency curve* can be developed relating h_f to h_{fi} based on the inside tube surface.

Bonilla¹ has presented a graphical solution of Eq. (16.34) for the rectangular fin of uniform thickness. The method of derivation employed here is applicable to other types of longitudinal fins as well, although it is rather awkward for the longitudinal fin of triangular cross section. A table of the hyperbolic tangents is given in Table 16.1.

TABLE 16.1. HYPERBOLIC TANGENTS

mb	$\text{Tanh } mb$	$\frac{\text{Tanh } mb}{mb}$
0.0	0.0000	1.000
0.1	0.0997	0.997
0.2	0.1974	0.987
0.3	0.2913	0.971
0.4	0.380	0.950
0.5	0.462	0.924
0.6	0.537	0.895
0.7	0.604	0.863
0.8	0.664	0.830
0.9	0.716	0.795
1.0	0.762	0.762
1.1	0.801	0.728
1.2	0.834	0.695
1.3	0.862	0.663
1.4	0.885	0.632
1.5	0.905	0.603
2.0	0.964	0.482
3.0	0.995	0.333
4.0	0.999	0.250

Example 16.1. Calculation of the Fin Efficiency and a Weighted Efficiency Curve. To illustrate the use of Eq. (16.34) a weighted fin efficiency curve will be developed for one of the common types of finned tubes used in double pipe exchangers.

A double pipe exchanger employs a 1½ in. OD, 13 BWG steel inner tube to which are attached 20 fins 20 BWG, ¾ in. high. Through suitable bushings the inner pipe is inserted into a 3-in. IPS outer pipe. Determine (a) the efficiency of the fins for various values of h_f , (b) the weighted efficiency curve.

¹ Bonilla, C. F., *Ind. Eng. Chem.*, **40**, 1098-1101 (1948).

Solution:

Total outside surface, A :

Fin surface, $A_f = (20 \times 0.75 \times 12 \times 2)/144 = 2.50 \text{ ft}^2/\text{lin ft}$

Bare tube surface, $A_o = (3.14 \times 1.25 - 20 \times 0.035)^{1/2}/144 = 0.268 \text{ ft}^2/\text{lin ft}$

$A_f + A_o = 2.50 + 0.268 = 2.77 \text{ ft}^2/\text{lin ft}$

Total inside surface, $A_i = (3.14 \times 1.06 \times 12)/144 = 0.277 \text{ ft}^2/\text{lin ft}$

(a) Fin efficiencies:

$$\text{Efficiency, } \Omega = \frac{\tanh mb}{mb} \quad (16.26)$$

$$m = \left(\frac{h_f P}{k a_x} \right)^{1/2} = h_f^{1/2} \left(\frac{P}{k a_x} \right)^{1/2} = h_f^{1/2} \left(\frac{2 \times 20 \times 144}{25 \times 0.035 \times 12 \times 20} \right)^{1/2} = 5.24 h_f^{1/2}$$

h_f	$5.24 h_f^{1/2}$	mb	$\tanh mb$	$\Omega = \frac{\tanh mb}{mb}$
4	10.48	0.655	0.5717	0.872
16	20.96	1.31	0.864	0.660
36	31.44	1.97	0.962	0.498
100	52.4	3.28	0.995	0.303
400	104.8	6.55	1.00	0.152
625	131	8.19	1.00	0.122
900	157	9.82	1.00	0.102

The fin efficiencies are given in the last column. Note that, when the outside coefficient h_f is 4.0, both sides of the fin contribute heat-transfer surface which is 87.2 per cent as effective as bare tube surface. When $h_f = 100$, the surface is only 30.3 per cent as effective, and when $h_f = 900$ (water or steam), the extended surface is only 10.2 per cent as effective. This is because at the higher film coefficient the fin metal more nearly attains the temperature of the hot fluid and the temperature difference between the fluid and fin is correspondingly less for given values of T_c and t_c .

(b) The weighted efficiency curve directly gives the value of h_{fi} obtained at the tube inside diameter when the coefficient to the fin and the bare tube is h_f .

$$h_{fi} = (\Omega A_f + A_o) \frac{h_f}{A_i} \quad (16.34)$$

$$\begin{aligned} h_f \times (\Omega A_f + A_o) / A_i &= h_{fi} \\ 4 \times (0.872 \times 2.50 + 0.268) / 0.277 &= 35.4 \\ 16 \times (0.660 \times 2.50 + 0.268) / 0.277 &= 110.8 \\ 36 \times (0.498 \times 2.50 + 0.268) / 0.277 &= 193.5 \\ 100 \times (0.303 \times 2.50 + 0.268) / 0.277 &= 370 \\ 400 \times (0.152 \times 2.50 + 0.268) / 0.277 &= 935 \\ 625 \times (0.122 \times 2.50 + 0.268) / 0.277 &= 1295 \\ 900 \times (0.102 \times 2.50 + 0.268) / 0.277 &= 1700 \end{aligned}$$

These values of h_f and h_{fi} are plotted in Fig. 16.7. If the film coefficient to the fin is large, as with condensation or the cooling or heating of water in the annulus, there is little advantage to be gained from the use of fins particularly when a dirt factor must be included which may be the controlling resistance. Actually, a fin of a desired metal

can be designed to give a high efficiency for any reasonable value of h_f , but as h_f increases, the size and cost of the fin also increases. Greater efficiencies can also be obtained by using fins which are not uniformly thick such as are treated later on.

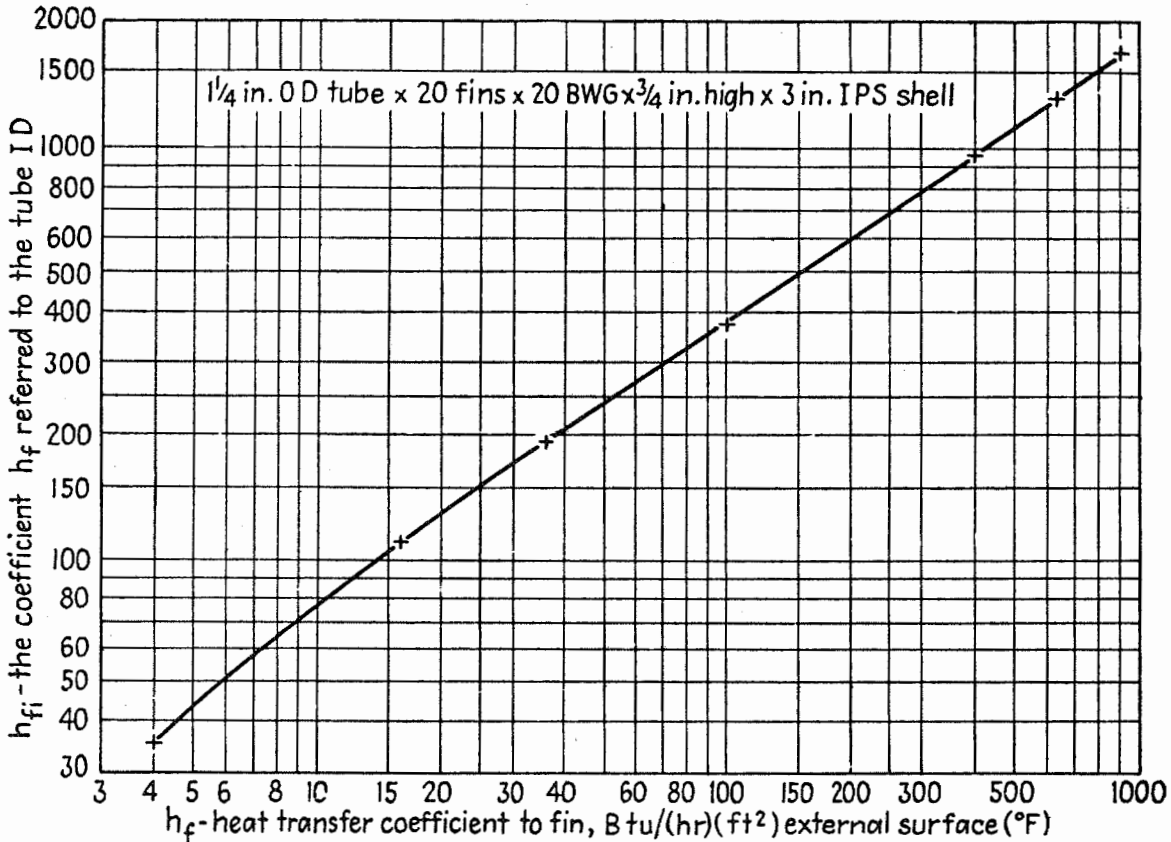


FIG. 16.7. Weighted fin-efficiency curve.

There are several manufacturers who supply completely assembled double pipe exchangers with fins on the inner pipe. Because the finned pipe or tube cannot be inserted through a packed gland as in Fig. 6.1,

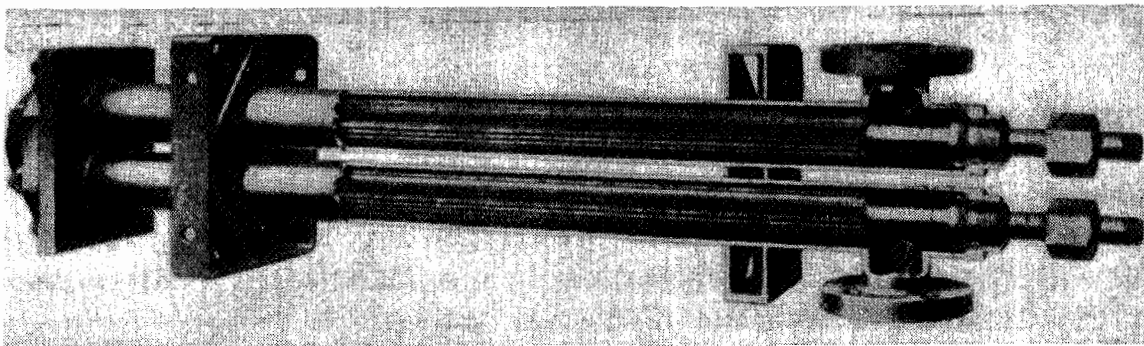


FIG. 16.8. Longitudinal fin double pipe exchanger. (Griscom-Russell Co.)

the method of confining the inner pipe is somewhat more complicated. A typical example of a standard preassembled double pipe finned exchanger is shown in Fig. 16.8, and as in Chap. 6, a single unit is called a *hairpin*. The cost of these units is extremely moderate, and they are preferable

when one of the fluids is a gas, a viscous liquid, or a small stream. Operated in series or in parallel they are often superior to shell-and-tube exchangers, notwithstanding the increased number of leakable joints they introduce. When connected in any other manner but in series the true temperature difference is computed for series-parallel flow by Eq. (6.35). When operated with cooling water in the inner pipe a nonferrous pipe may be selected with either steel or nonferrous fins attached. Three

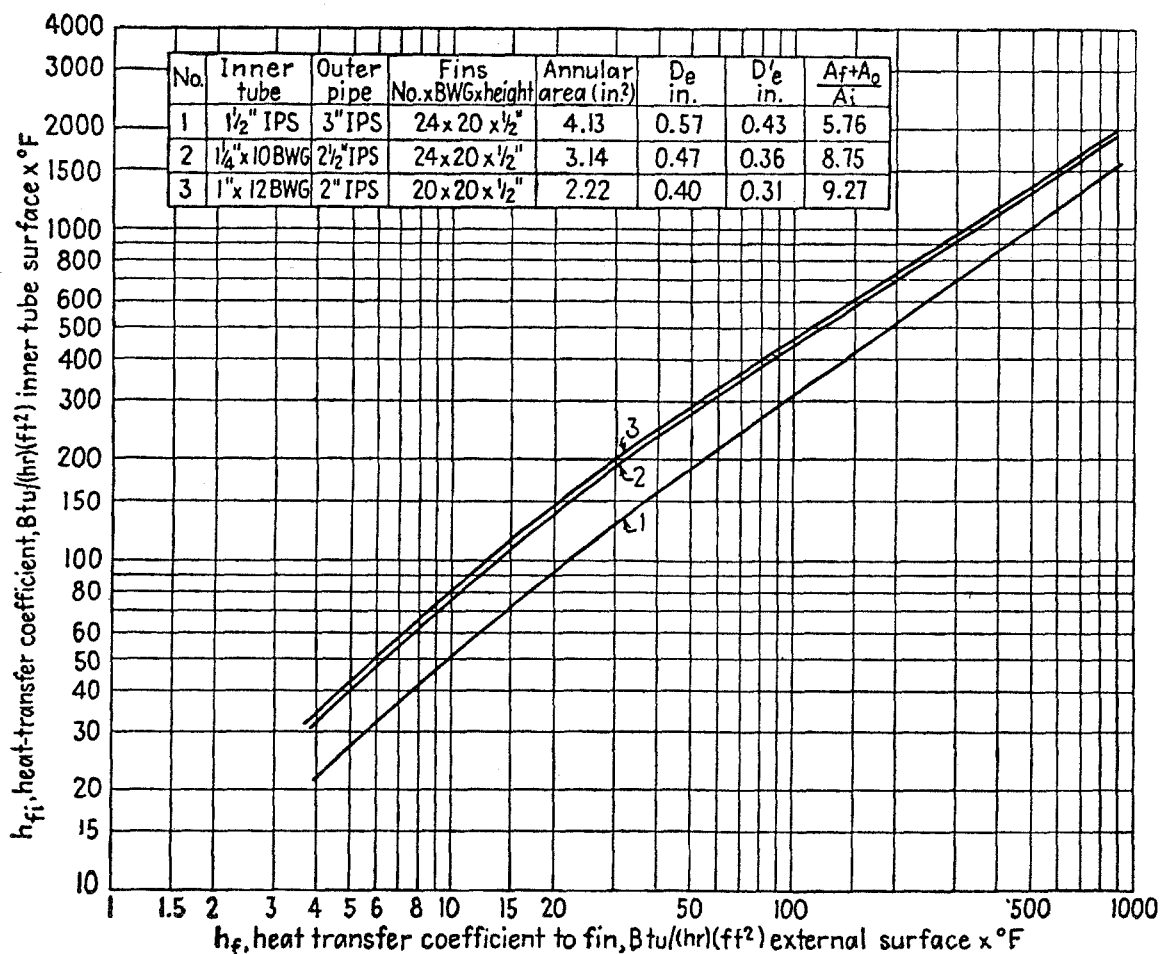


FIG. 16.9. Weighted fin-efficiency curves for steel double pipe exchangers.

weighted efficiency curves for three of the commonest sizes of double pipe finned exchangers are given in Fig. 16.9. These do not correspond to the offerings of any single manufacturer but are a composite of the three sizes ordinarily available from manufacturers' stocks.

Double-pipe-exchanger Coefficients and Pressure Drops. The addition of fins to the outsides of the inner pipes of double pipe exchangers generally reduces the value of the annulus film coefficient by smoothing the flow of fluid on the outside of the tube or pipe. Data obtained for bare tube surfaces cannot be used for the calculation of double pipe extended-surface exchangers. While a design curve or equation relating

h_f to the flow variables and heat-transfer properties can be determined by experiment in the usual manner (see Chap. 3), the correlation differs for extended surfaces. Starting with a double pipe finned exchanger, a fluid is passed through the annulus and heated by the dropwise condensation of steam, whose condensing coefficient is very high compared with the annulus coefficient. Any other heating medium can be substituted

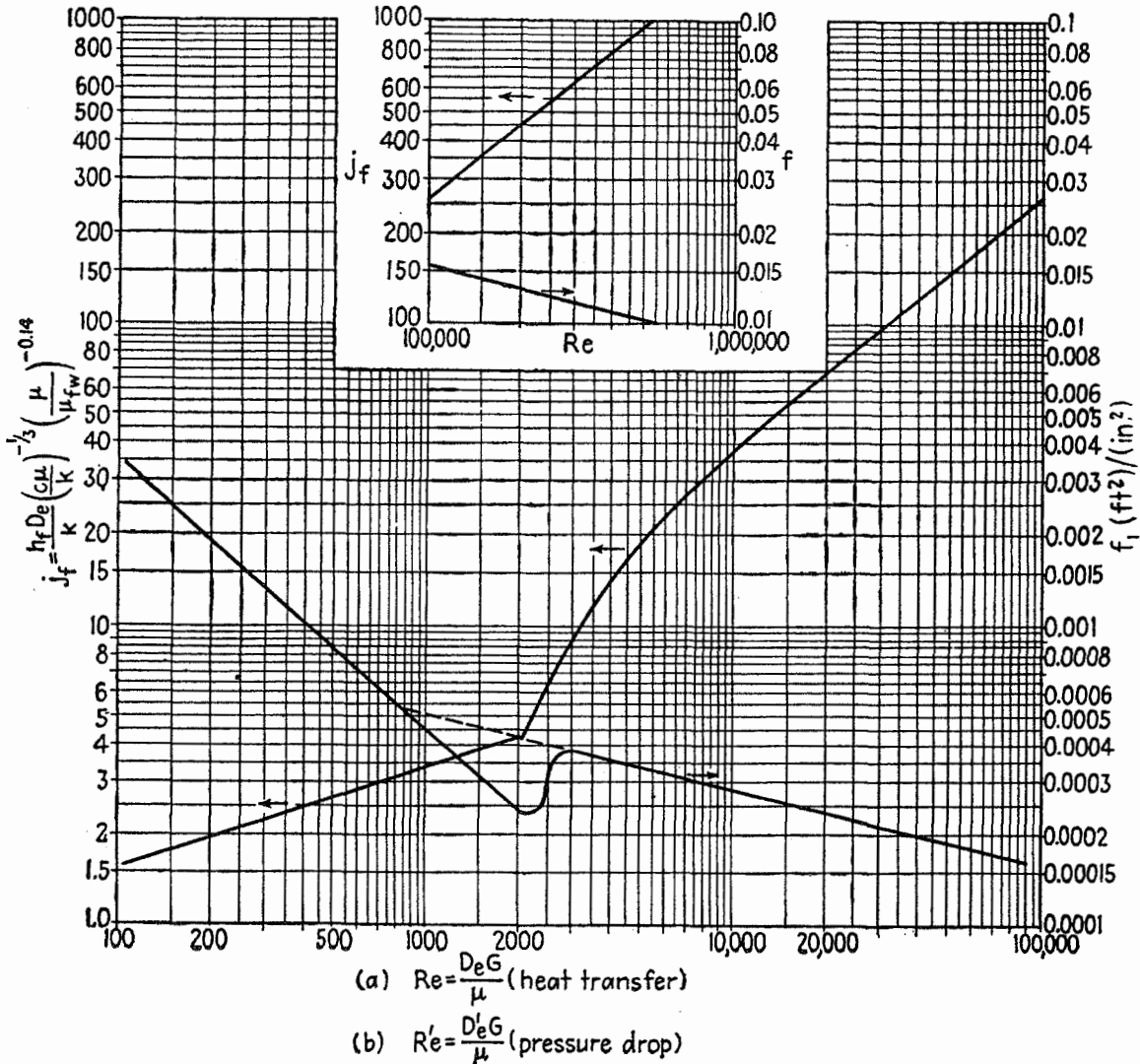


FIG. 16.10. Longitudinal fin heat transfer and pressure drop.

if its individual coefficient can be or has been determined accurately. For cooling experiments water is usually employed. Readings are taken of the oil-temperature change, the steam or water temperatures, and the fluid flow. The surface A_i is known from the inside diameter and length of the inner pipe, and the heat load, the LMTD, and U_i , the clean overall coefficient, can also be computed. The film coefficient for the annulus h_{fi} can then be determined from the following equation:

$$\frac{1}{U_i} = \frac{1}{h_i} + \frac{1}{h_{fi}} \quad (16.35)$$

The value of h_{fi} so obtained represents h_f to the fin and bare tube already combined and weighted. If four or more fins are attached to the inner pipe, the addition of a greater number does not appear to affect the heat-transfer coefficient h_f significantly. Naturally the weighted efficiency and the total heat transfer are directly influenced by the number of fins. To make the experimental data general for other fin and tube arrangements than those tested, the value of h_{fi} must be transferred from the inside diameter of the inner tube to the annulus. In the annulus it may then be resolved between the fin and bare tube surface by converting from h_{fi} to h_f and plotting a heat-transfer factor j_f , which includes the average value of h_f to both types of surface.

Figure 16.10a is an isothermal curve which was obtained in this manner from numerous heating and cooling experiments on different fin and tube double pipe arrangements and agrees with the published data of DeLorenzo and Anderson.¹ Friction factors are given in Fig. 16.10b. The equivalent diameter D'_e for pressure drop differs from D_e for heat transfer because of the inclusion of the perimeter of the outer pipe in the wetted perimeter. Both equivalent diameters are computed as in Chap. 6 for double pipe exchangers using four times the hydraulic radius.

To illustrate the method of treating experimental data on pipes or tubes with extended surface, a typical test point will be computed below. This is in the reverse order of the calculation of the surface requirements for the fulfillment of process conditions which will be demonstrated later.

Example 16.2. Calculation of a Heat-transfer Curve from Experimental Data. Experimental double pipe exchanger, steel hairpin.

Shell, 3 in. IPS

Finned inner tube: $1\frac{1}{4}$ in. OD, 13 BWG to which are attached 20 fins, 20 BWG, $\frac{3}{4}$ in. high

Length of finned inner tube, 10'0" per leg between return bends.

Data and observations:

Hot fluid, steam, $T_s = 302^\circ\text{F}$ (in inner tube)

Cold fluid, 40.8°API kerosene, $t_1 = 151^\circ\text{F}$

$t_2 = 185^\circ\text{F}$

Weight flow, $w = 15,200$ lb/hr

The dropwise condensation of steam was promoted with oil.

Calculations:

ID of 3 in. IPS pipe = 3.068 in.

Annulus flow area, $a_a = \frac{\pi(3.068^2 - 1.25)^2}{4 \times 144} - \frac{20 \times 0.035 \times 0.75}{144} = 0.0395 \text{ ft}^2$

¹ DeLorenzo, B., and E. D. Anderson, *Trans. ASME*, **67**, 697-702 (1945).

$$\text{Wetted perimeter} = \pi \times \frac{1.25}{12} - 20 \times \frac{0.035^*}{12} + 20 \times 0.75 \times \frac{2}{12} = 2.77 \text{ ft}$$

$$\text{Equivalent diameter, } D_e = \frac{4 \times \text{flow area}}{\text{wetted perimeter}} = 4 \times \frac{0.0395}{2.77} = 0.057 \text{ ft}$$

$$\text{Heat load, } Q = 15,200 \times 0.523(185 - 151) = 271,000 \text{ Btu/hr}$$

$$\text{LMTD} = 133^\circ\text{F}$$

$$\text{Internal surface/lin ft, } A_i = 0.277 \text{ ft}^2/\text{ft}$$

$$U_i = \frac{271,000}{0.277 \times 20 \times 133} = 368 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

$$h_i = 3000 \text{ (assumed value for dropwise condensation of steam)}$$

$$\frac{1}{U_i} = \frac{1}{h_i} + \frac{1}{h_{fi}} \quad \frac{1}{368} = \frac{1}{3000} + \frac{1}{h_{fi}}$$

$$h_{fi} = 418 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

From the efficiency curve (Fig. 16.7) for $h_{fi} = 418$, $h_f = 120$.

At the average temperature $\mu = 0.80 \text{ cp} \times 2.42 = 1.94 \text{ lb/(ft)(hr)}$

$$\left(\frac{c\mu}{k}\right)^{1/3} = 2.34$$

$$j_f = h_f \frac{D_e}{k} \left(\frac{c\mu}{k}\right)^{-1/3} = \frac{120 \times 0.057}{0.079 \times 2.34} = 37.0 \quad (16.36)$$

$$G_a = \frac{15,200}{0.0395} = 385,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$Re_a = 0.057 \times \frac{385,000}{1.94} = 11,300$$

This single point, $Re_a = 11,300$ vs. $j_f = 37$, does not fall exactly on the isothermal curve of Fig. 16.10a. It is off by more than is first apparent (37 vs. 41 or 9.8 per cent), since it represents a heating run and j_f should be divided by a viscosity correction to permit its representation on an isothermal curve which is applicable to both heating and cooling. The general implications of such a procedure have been discussed in Chaps. 5 and 6. In the correlations covered in Fig. 16.10a the viscosity correction was found to be $\phi_a = (\mu/\mu_{fw})^{0.14}$, where μ_{fw} is the viscosity taken at the temperature of the "wall" t_{fw} . Thus μ_{fw} replaces μ_w when the pipe wall has fins.

The Wall Temperature t_{fw} of a Finned Tube. The wall temperature of an extended-surface pipe or tube influences the values of the heat-transfer coefficients which are obtained when a fluid is heated and cooled over the same operating range. Since the fin and tube metals will not be at the same temperature, the use of any single temperature such as t_{fw} to replace the fin temperature t_f and the tube-wall temperature t_w is naturally fictitious. Kayan¹ has developed an ingenious experimental method of analyzing for the wall temperature which simulates heat transfer by means of an electrical analogy. Theoretical methods of obtaining the wall temperature are quite complex, and even the most typical applications of extended surfaces have not been covered by adequate derivations. The method employed below is semiempirical. Its principal

* This is the area of the bare tube beneath the bases of the fins. The extremities of the fins were not considered to receive heat in the derivation, and their lengths are not included in the perimeter.

¹ Kayan, C. F., *Trans. ASME*, **67**, 713-718 (1945); *Ind. Eng. Chem.*, **40**, 1044-1049 (1948).

advantage lies in the brevity with which a solution can be obtained. It is presented here in the sequence which is most adaptable to design when an isothermal performance curve such as Fig. 16.10a is available. If an isothermal curve is to be developed from heating and cooling experiments, the sequence need merely be rearranged:

The true temperature difference Δt corresponds to the total resistance $1/U_{Di}$ in the design equation $(Q/A_i) = U_{Di} \Delta t$. The sum of the smaller temperature differences through each of the component resistances between the annulus and inner tube fluids must equal the true temperature difference. The dirt factors for the annulus and tube may not be combined in extended surface equipment, since they are applicable to such widely different actual surfaces that flux corrections must be employed. The component resistances are then the (1) annulus film, (2)

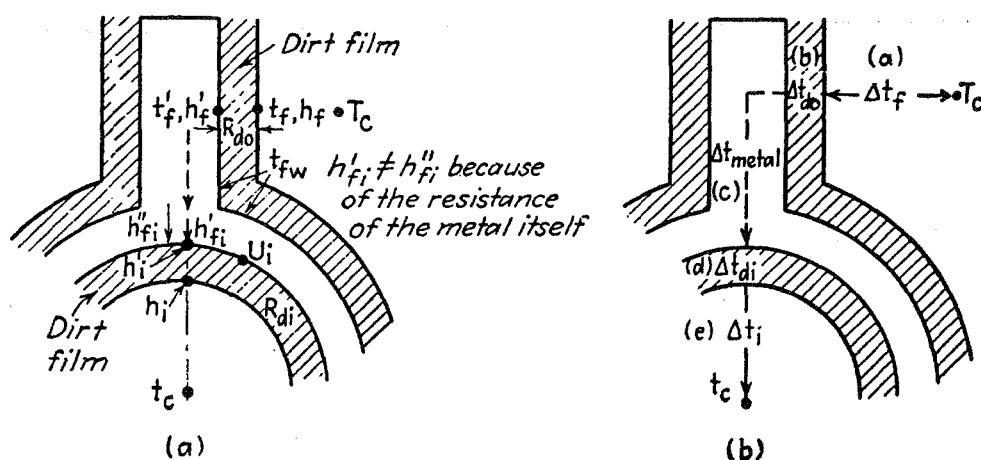


FIG. 16.11. The location of coefficients and temperature drops in longitudinal fins.

annulus dirt factor, (3) fin and tube metal, compositely, (4) tube-side dirt factor and (5) tube-side film. The component temperature differences are obtained by multiplying the component resistances by the flux Q/A_i when the latter is corrected to the appropriate surface at each point. The wall temperature t_{fw} occurs between temperature drops (2) and (3), and by obtaining this temperature μ_{fw} can be evaluated. While the clean coefficient U_i is obtained from individual coefficients calculated for properties at T_c and t_c , the summation of the component temperature differences in the design equation corresponds to Δt and not $T_c - t_c$.

The value of the wall t_{fw} is obtained by trial and error. First assume a value of t_{fw} which permits the calculation of ϕ_a , the annulus viscosity correction, and the annulus film coefficient corrected for heating or cooling. The reciprocal of the annulus film coefficient is the resistance in (1). The fouling factor is the resistance in (2). If the temperature after passing through these two resistances corresponds to the assumed value of t_{fw} and if the total temperature drop over the five resistances corre-

sponds to Δt , the assumed value was correct. If no check is obtained, a new value of t_{fw} must be assumed and the procedure repeated. The location of the various temperatures and resistances involved in this method are shown in Fig. 16.11. The nomenclature differs from Eq. (16.35) in which there were no dirt factors. The steps in summing up the resistances and correcting for the different surfaces per linear foot of extended surface pipe or tube should be evident in the outline below.

1. Assume a temperature t_{fw} to establish a temperature difference between the annulus fluid and the wall.
2. Obtain $\phi_a = (\mu/\mu_{fw})^{0.14}$ where μ_{fw} is obtained at t_{fw} .
3. Obtain h_f/ϕ_a from a design curve and correct for the viscosity ratio ϕ_a and obtain h_f by $h_f = (h_f/\phi_a)\phi_a$. Then $R_f = 1/h_f$.
4. To obtain the value of h_f effective at the fin surface add to it the resistance through the annulus dirt factor R_{do} . For a dirt factor of R_{do} ,

$$\frac{1}{h'_f} = R_f + R_{do} \quad (16.37)$$

where h'_f is the effective value of h_f at the fin.

5. Obtain h'_{fi} for the value of h'_f in 4 from an appropriate weighted efficiency curve such as Fig. 16.9.
6. Multiply h'_f in 4 by the surface ratio $(A_f + A_o)/A_i$ giving h''_{fi} , which is h'_f corrected for flux but not for the resistance of the fin and wall metals.

$$h''_{fi} = \frac{h'_f(A_f + A_o)}{A_i} \quad (16.38)$$

The difference between h'_{fi} and h''_{fi} may be considered due to the resistance of the fin and tube wall metals.

$$R_{\text{metal}} = \frac{1}{h'_{fi}} - \frac{1}{h''_{fi}} \quad (16.39)$$

7. Obtain h_i for the tube fluid from any suitable tube-side data such as Fig. 24 or 25, where $\phi_i = (\mu/\mu_{fw})^{0.14}$ instead of $(\mu/\mu_w)^{0.14}$, and combine with the tube-side dirt factor, R_{di} . $R_i = 1/h_i$, and if h'_i is the value of h_i at the tube wall,

$$\frac{1}{h'_i} = R_i + R_{di} \quad (16.40)$$

8. From h'_i and h'_{fi} obtain U_{Di} , the corrected overall design coefficient based on the inside tube diameter.

$$\frac{1}{U_{Di}} = \frac{1}{h'_{fi}} + \frac{1}{h'_i} \quad (16.41)$$

9. Obtain the flux for the surface actually employed.

$$\frac{Q}{A_i} = U_{Di} \Delta t \quad (16.42)$$

10. Impose the flux from 9 on the individual resistances and determine whether the summation of the first two individual differences is the same as the assumed value

of $T_c - t_{fw}$ and if the sum of all differences equals the true temperature difference. In the same order as before, the temperature drops are

a. Annulus film:

$$\Delta t_f = \frac{Q/A_i}{h_{fi}} \quad (16.43a)$$

b. Annulus dirt, corrected to the inside diameter:

$$\Delta t_{do} = \left(\frac{Q}{A_i} \right) R_{do} \frac{A_i}{A_f + A_o} \quad (16.43b)$$

c. Fin and tube metal:

$$\Delta t_{\text{metal}} = \left(\frac{Q}{A_i} \right) R_{\text{metal}} \quad (16.43c)$$

d. Tube-side dirt:

$$\Delta t_{di} = \left(\frac{Q}{A_i} \right) R_{di} \quad (16.43d)$$

e. Tube-side film:

$$\Delta t_i = \frac{Q/A_i}{h_i} \quad (16.43e)$$

When the tube fluid is a gas, water, or a similar nonviscous fluid, the tube-side viscosity correction can be omitted and ϕ_a can be taken as 1.0. The metal of which the fin is fabricated greatly affects the efficiency, particularly since k for steel is 26 and for copper 220 Btu/(hr)(ft²)(°F/ft).

The Calculation of a Double Pipe Finned Exchanger. The method employed here closely follows that of the outline in Chap. 6. The true temperature difference for a double pipe exchanger is the LMTD calculated for true counterflow or a value computed from Eq. (6.35) for series-parallel flow. The effective tube length does not include the return bend between legs of a hairpin or any of the unfinned portion of the inner tube. The inside of the tube is used as the reference surface. Most manufacturers prefer to use the outside as the reference surface, since it has a larger numerical value. The solutions of problems can be converted to manufacturers' data by multiplying the inside surface by the ratio $(A_f + A_o)/A_i$ and dividing the overall coefficient U_{Di} by this ratio. Tube lengths of 12, 15, 20, and 24 ft are considered reasonable for extended-surface hairpins. Large tube lengths are permissible, since the fins on the inner pipe rest snugly on the outer pipe and there is no sagging. As explained previously, fouling factors cannot be combined as in ordinary double pipe exchangers because they are effective on widely different surfaces and each must be treated separately.

Example 16.3. Calculation of a Double Pipe Extended-surface Gas Oil Cooler. It is desired to cool 18,000 lb/hr of 28°API gas oil from 250 to 200°F in double pipe exchangers consisting of 3-in. IPS shells with 1½-in. IPS inner pipes on which are mounted 24 fins ½ in. high by 0.035 in. (20 BWG) wide. Water from 80 to 120°F will serve as the cooling medium. Pressure drops of 10.0 psi are allowable on both streams, and fouling factors of 0.002 for the gas oil and 0.003 for the water are required.

How many 20-ft hairpins will be required?

Solution:

- (1) Heat balance: Gas oil,
- $Q = 18,000 \times 0.53(250 - 200) = 477,000 \text{ Btu/hr}$

(Fig. 4)

$$\text{Water, } Q = 11,950 \times 1.0(120 - 80) = 477,000 \text{ Btu/hr}$$

- (2)
- Δt
- : Assume true counterflow for the first trial. Unless the allowable pressure drop on either stream is exceeded, it will not be necessary to consider series-parallel arrangements calculated by Eq. (6.35).

$$\Delta t = \text{LMTD} = 124^\circ\text{F} \quad (5.14)$$

- (3) Caloric temperatures
- T_c
- and
- t_c
- :

$$\frac{\Delta t_c}{\Delta t_h} = \frac{120}{130} = 0.92 \quad K_c = 0.27 \quad F_c = 0.47 \quad (\text{Fig. 17})$$

$$T_c = 200 + 0.47(250 - 200) = 224^\circ\text{F} \quad (5.28)$$

$$t_c = 80 + 0.47(120 - 80) = 99^\circ\text{F} \quad (5.29)$$

Hot fluid: annulus, gas oil

- (4') 3 in. IPS, ID = 3.068 in. [Table 11]

- 1½ in. IPS, OD = 1.90 in. [Table 11]

Fin cross section, 20 BWG, ½ in. high

$$= 0.035 \times 0.5 = 0.0175 \text{ in.}^2$$

[Table 10]

$$a_a = \left(\frac{\pi}{4} \times 3.068^2 - \frac{\pi}{4} \times 1.90^2 - 24 \right. \\ \left. \times 0.0175 \right)$$

$$= 4.13 \text{ in.}^2$$

$$= 4.13/144 = 0.0287 \text{ ft}^2$$

$$\text{Wetted perimeter}^* = (\pi \times 1.90 - 24 \\ \times 0.035 + 24 \times 2 \times 0.5)$$

$$= 29.13 \text{ in.}$$

$$d_e = 4 \times 4.13/29.13 = 0.57 \text{ in.}$$

$$D_e = 0.57/12 = 0.0475 \text{ ft}$$

$$(5') G_a = W/a_a \\ = 18,000/0.0287 \\ = 628,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6') \text{ At } T_c = 224^\circ\text{F,} \\ \mu = 2.50 \times 2.42 = 6.05 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$Re_a = D_e G_a / \mu \\ = 0.0475 \times 628,000 / 6.05 = 4930$$

$$(7') j_f = 18.4 \quad [\text{Fig. 16.10}]$$

$$(8') \text{ At } T_c = 224^\circ\text{F} \\ k(c\mu/k)^{1/2} = 0.25 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Fig. 16}]$$

$$(9') h_f = j_f \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/2} \phi_a \quad [\text{Eq. (6.15)}] \\ \frac{h_f}{\phi_a} = 18.4 \times \frac{0.25}{0.0475} = 96.7$$

Cold fluid: inner pipe, water

- (4)
- $D = 1.61/12 = 0.134 \text{ ft}$
- [Table 11]

$$a_i = \pi D^2/4 = \pi \times 0.134^2/4 = 0.0142 \text{ ft}^2$$

$$(5) G_i = w/a_i \\ = 11,950/0.0142 \\ = 842,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$V = G_i/3600\rho = 842,000/3600 \times 62.5 = \\ 3.75 \text{ fps}$$

$$(6) \text{ At } t_c = 99^\circ\text{F,} \\ \mu = 0.72 \times 2.42 = 1.74 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$Re_i = DG_i/\mu (\text{Re}_i \text{ is for pressure drop only}) \\ = 0.134 \times 842,000/1.74 = 65,000$$

$$(9) h_i = 970 \times 0.82 \\ = 795 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad [\text{Fig. 25}]$$

* In the derivation, the outermost edge of the fins was assumed to have zero heat transfer.

Calculation of t_{fw} (numbers refer to the outline procedure):

- (1) Assume $T_c - t_{fw} = 40^\circ\text{F}$

$$t_{fw} = 224 - 40 = 184^\circ\text{F}$$

(Fig. 14)

- (2) At 184°F , $\mu_{fw} = 3.5$ cp

$$\phi_a = \left(\frac{\mu}{\mu_{fw}}\right)^{0.14} = \left(\frac{2.5}{3.5}\right)^{0.14} = 0.95$$

- (3) $h_f = 96.7 \times 0.95 = 91.8$ Btu/(hr)(ft²)(°F)

- (4) $R_{do} = 0.002$, $R_f = \frac{1}{91.8} = 0.0109$ (hr)(ft²)(°F)/Btu

$$\frac{1}{h'_f} = 0.002 + 0.0109 \quad (16.37)$$

$$h'_f = 77.5$$

- (5) $h'_{fi} = 255$ (Fig. 16.9)

- (6) $h''_{fi} = h'_f \frac{A_f + A_o}{A_i} = 77.5 \times 5.76 = 447$ (16.38) and (Fig. 16.9)

$$R_{\text{metal}} = \frac{1}{h'_{fi}} - \frac{1}{h''_{fi}} = \frac{1}{255} - \frac{1}{447} = 0.00169 \quad (16.39)$$

- (7) Assume $\phi_i = 1.0$ for cooling water:

$$R_{di} = 0.003 \quad R_i = \frac{1}{795} = 0.00126$$

$$\frac{1}{h'_i} = 0.00126 + 0.003 \quad (16.40)$$

$$h'_i = 235$$

- (8) $\frac{1}{U_{Di}} = \frac{1}{255} + \frac{1}{235}$ (16.41)

$$U_{Di} = 122$$

- (9) To obtain the true flux the heat load must be divided by the *actual* heat-transfer surface.

For a 1½-in. IPS pipe there are 0.422 ft²/lin foot (Table 11)

Trial:

$$A_i = \frac{Q}{U_{Di} \Delta t} = \frac{47,000}{122 \times 124} = 31.5 \text{ ft}^2$$

$$\text{Length of pipe required} = \frac{31.5}{0.422} = 74.8 \text{ lin ft}$$

Use two 20-ft hairpins = 80 lin ft

$$A_i = 80 \times 0.422 = 33.8 \text{ ft}^2$$

$$\frac{Q}{A_i} = \frac{477,000}{33.8} = 14,100 \text{ Btu/(hr)(ft}^2\text{)}$$

$$(a) \text{ Annulus film } \Delta t_f = \left(\frac{Q}{A_i}\right) / h''_{fi} = \frac{14,100}{447} = 31.6^\circ$$

$$(b) \text{ Annulus dirt, } \Delta t_{do} = \left(\frac{Q}{A_i}\right) R_{do} \frac{A_i}{A_f + A_o} = 14,100 \times \frac{0.002}{5.76} = 4.9$$

$$T_c - t_{fw} = 36.5^\circ$$

$$(c) \text{ Fin and tube metal, } \Delta t_{\text{metal}} = \left(\frac{Q}{A_i}\right) R_{\text{metal}} = 14,100 \times 0.00169 = 23.8$$

$$(d) \text{ Tube-side dirt, } \Delta t_{di} = \left(\frac{Q}{A_i}\right) R_{di} = 14,100 \times 0.003 = 42.3$$

$$(e) \text{ Tube-side film, } \Delta t_i = \left(\frac{Q}{A_i}\right) / h_i = \frac{14,100}{795} = 17.7$$

$$120.3^\circ$$

Assumed $T_c - t_{fw} = 40.0^\circ$ Calculated $T_c - t_{fw} = 36.5^\circ$ LMTD = 124°

The difference of $40 - 36.5 = 3.5^\circ\text{F}$ will not materially change the value of $(\mu/\mu_{fw})^{0.14}$.

Note: The temperature drop of 120.3°F corresponds to a fouling factor based on 74.8 lin ft while the flux corresponds to 80 lin ft. The correction appears at the end of the solution.

Pressure Drop

(1') $d'_e = 4 \times 4.13 / (29.13 + \pi \times 3.07) = 0.43 \text{ in.}$ $D'_e = 0.43/12 = 0.0359 \text{ ft}$ $Re'_a = D'_e G_a / \mu = 0.0359 \times 628,000 / 6.05 = 3730$ $f = 0.00036 \text{ ft}^2/\text{in.}^2$ [Fig. 16.10] $s = 0.82$ [Fig. 6] (2') $\Delta P_a = \frac{f G_a^2 L n}{5.22 \times 10^{10} D_e'^3 s \phi_a}$ [Eq. (7.45)] $= \frac{0.00036 \times 628,000^2 \times 80}{5.22 \times 10^{10} \times 0.0359 \times 0.82 \times 0.95} = 7.9 \text{ psi}$	(1) $Re_t = 65,000$ (pipe) $f = 0.000192 \text{ ft}^2/\text{in.}^2$ [Fig. 26] (2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)] $= \frac{0.000192 \times 842,000^2 \times 80}{5.22 \times 10^{10} \times 0.134 \times 1.0 \times 1.0} = 1.4 \text{ psi}$
---	---

Summary

0.002	Dirt Factor	0.003
255	h inside	235
$U_{Di} \quad 122$		
7.9	Calculated ΔP	1.4
10.0	Allowable ΔP	10.0

Adjustment of the fouling factor. $U_{Di} = 122$ was based on 74.8 lin ft. Based on 80 lin ft $U_{Di} = 114$ and the difference is excess dirt factor equal to 0.00057. This can readily be added to the tube side to give a dirt factor of 0.00357, or it can be added to the annulus by taking $0.00057 \times 5.76 = 0.00328$. The total annulus dirt factor for the latter will then be $0.002 + 0.00328 = 0.00528$. For the new value of h'_f

$$\frac{1}{h'_f} = 0.00528 + \frac{1}{91.8} \quad (16.37)$$

$$h'_f = 62$$

$$h'_{fs} = 220 \quad (\text{Fig. 16.9})$$

$$U_{Di} = \frac{h'_{fs} h'_i}{h'_{fs} + h'_i} = \frac{220 \times 235}{220 + 235} = 114 \quad (16.41)$$

The corrected summary is

0.0053	Dirt Factor	0.003
220	h inside	235
U_{Di}	114	
7.9	Calculated ΔP	1.4
10.0	Allowable ΔP	10.0

Remarks: There is no need to readjust the calculation of t_{fw} , since only item (b) will be affected and this is usually insignificant in its contribution to the value of Δt . Thus the corrected value is

$$\Delta t_{do} = 14,100 \times \frac{0.0053}{5.76} = 13.0^\circ$$

This gives a computed Δt of 128.4° vs. 124.0° , but this difference does not justify a new trial value of t_{fw} . The need for readjusting the fouling factor is of particular value when checking the performance of an existing hairpin or battery of hairpins for a new service.

As reported by a manufacturer the total surface would be $33.8 \times 5.76 = 194 \text{ ft}^2$. The reported overall coefficient would be $114/5.76 = 19.8 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$.

Extended-surface Shell-and-tube Exchangers. The use of extended surfaces in double pipe exchangers permits the transfer of a great deal of heat in a compact unit. The same advantages may be obtained from the use of longitudinally finned tubes in shell-and-tube arrangements equivalent to the 1-1, 1-2 or 2-4 exchanger. Because they are relatively uncleanable, extended-surface tubes are usually laid out on triangular pitch and are never spaced so closely that the fins of adjacent tubes intermesh. To prevent sagging and the possibility of tube vibration which might result from intermeshing, each tube in a bundle is supported individually. This cannot be done with conventional support plates because they introduce a certain amount of flow across the bundle which cannot be accomplished very well with longitudinal finned tubes. Support is accomplished, however, by welding or shrinking small circumferential rings about each tube which enclose the fins but at different points along the length of each tube. The rings prevent any tubes from intermeshing and at the same time afford a positive elimination of vibration damage. At several points along its length the entire bundle is then bound with circumferential bands which keep all of the finned tubes firmly pressed against the rings of adjacent tubes.

Longitudinal fin exchangers are relatively expensive and, since they are

not cleanable, can be used only for fluids which ordinarily have very low film coefficients and which are clean or form dirt that can be boiled out. This makes them ideal for gases at low pressure where the density is low and the allowable pressure drop is accordingly small.

The prototype of this exchanger is the 1-2 exchanger without baffles as calculated in Example 7.8. Longitudinal fin shell-and-tube exchangers are computed in the same manner as double pipe extended-surface exchangers using the same efficiency curves for identical tubes. Only the equivalent diameters for heat transfer and pressure drop differ. These are computed in the conventional manner for the entire shell using four times the hydraulic radius as discussed in Chap. 6 and demonstrated in Example 7.8.

Example 16.4. Calculation of a Longitudinal Fin Shell-and-tube Exchanger. 30,000 lb/hr of oxygen at 3 psig pressure and 250°F is to be cooled to 100°F using water from 80 to 120°F. The maximum allowable pressure drop for the gas is 2.0 psi and for the water 10.0 psi. Fouling factors of not less than 0.0030 should be provided for each stream.

Available for the service is a 19¼ in. ID 1-2 exchanger equipped with 70 16'0" tubes each with 20 fins ½ in. high of 20 BWG (0.035 in.) steel. The tubes are 1 in. OD, 12 BWG and are laid out on 2-in. triangular pitch for four passes.

Will the exchanger fulfill the service? What is the final gas side dirt factor?

Solution:

Exchanger:

<i>Shell side</i>		<i>Tube side</i>	
ID = 19¼ in.		Number and length = 70, 16'0", 20 fins, 20 BWG, in.	
Baffle Space = ring supports		OD, BWG, pitch = 1 in., 12 BWG, 2-in. tri.	
Passes = 1		Passes = 4	

(1) Heat balance:

Oxygen at 17.7 psia, $Q = 30,000 \times 0.225(250 - 100) = 1,010,000$ Btu/hr

Water, $Q = 50,500 \times 1(100 - 80) = 1,010,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	
250	Higher Temp	100	150
100	Lower Temp	80	20
150	Difference	20	130

$$\text{LMTD} = 64.6^\circ\text{F} \quad (5.14)$$

$$R = \frac{150}{20} = 7.5 \quad S = \frac{20}{250 - 80} = 0.1175 \quad (7.18) \quad F_T = 0.87 \quad (\text{Fig. 18})$$

$$\Delta t = 0.87 \times 64.6 = 56.2^\circ\text{F}$$

(3) T_o and t_o : The average temperatures of 175 and 90°F will be adequate.

Hot fluid: shell side, oxygen

$$(4') a_s = \frac{\pi}{4} \times 19.25^2 \\ - 70 \left(\frac{\pi}{4} \times 1^2 + 20 \times 0.035 \times 0.5 \right) \\ = 211.5 \text{ in.}^2 \\ = 211.5/144 = 1.47 \text{ ft}^2$$

$$\text{Wetted perimeter} = 70(\pi \times 1 - 20 \\ \times 0.035 + 20 \times 2 \times 0.5) \\ = 1570 \text{ in.}$$

$$d_e = 4 \times 211.5/1570 = 0.54 \text{ in.}$$

[Eq. (6.4)]

$$D_e = 0.54/12 = 0.045 \text{ ft}$$

$$(5'') G_s = W/a_s \\ = 30,000/1.47 \\ = 20,400 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6') \text{ At } T_a = 175^\circ\text{F}, \mu = 0.0225 \times 2.42 \\ = 0.0545 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Fig. 15}]$$

$$Re_s = D_e G_s / \mu \\ = 0.045 \times 20,400 / 0.0545 = 16,850$$

$$(7') j_H = 59.5 \quad [\text{Fig. 16.10a}]$$

$$(8') \text{ At } 175^\circ\text{F}, c = 0.225; k = 0.0175 \\ (c\mu/k)^{1/4} = (0.225 \times 0.0545 / 0.0175)^{1/4} \\ = 0.89$$

$$\phi_s = 1.0 \text{ (for gases)}$$

$$(9') h_f = j_f \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/4} \phi_s \quad [\text{Eq. (6.15)}] \\ = 59.5 \times 0.0175 \times 0.89 / 0.045 \\ = 20.5$$

$$R_{do} = 0.003, h_{do} = 1/0.003 = 333$$

$$h'_f = \frac{h_{do} h_f}{h_{do} + h_f} = \frac{333 \times 20.5}{333 + 20.5} = 19.3$$

[Eq. (16.37)]

$$h'_{fi} = 142 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad [\text{Fig. 16.9}]$$

Cold fluid: tube side, water

$$(4) a'_t = 0.479 \text{ in.}^2 \quad [\text{Table 10}]$$

$$a_t = N a'_t / 144n \quad [\text{Eq. (7.48)}]$$

$$= 70 \times 0.479 / 144 \times 4 = 0.0582 \text{ ft}^2$$

$$D = 0.782/12 = 0.0652 \text{ ft}$$

$$(5) G_t = w/a_t \\ = 50,500/0.0582 \\ = 868,000 \text{ lb}/(\text{hr})(\text{ft}^2) \\ V = G_t/3600\rho = 868,000/3600 \times 62.5 \\ = 3.86 \text{ fps}$$

$$(6) \text{ At } t_a = 90^\circ\text{F}, \\ \mu = 0.80 \times 2.42 = 1.94 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Fig. 14}]$$

$$Re_t = D G_t / \mu \text{ (for pressure drop only)} \\ = 0.0652 \times 868,000 / 1.94 = 29,100$$

$$(9) h_i = 940 \times 0.96 = 903 \quad [\text{Fig. 25}] \\ R_{di} = 0.003, h_{di} = 1/0.003 = 333$$

$$h'_i = \frac{h_{di} h_i}{h_{di} + h_i} = \frac{333 \times 903}{333 + 903} \\ = 243 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad [\text{Eq. (16.40)}]$$

Overall design coefficient based on inside of tube U_{Di} :

$$U_{Di} = \frac{h'_{fi} h'_i}{h'_{fi} + h'_i} = \frac{142 \times 243}{142 + 243} = 89.6 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (16.41)$$

Actual overall coefficient based on inside of tube:

$$\text{Internal surface per lin ft} = 0.2048 \text{ ft} \quad (\text{Table 10})$$

$$A_i = 70 \times 0.2048 \times 16'0'' = 230 \text{ ft}^2$$

$$U_{Di} = \frac{Q}{A_i \Delta t} = \frac{1,010,000}{230 \times 56.2} = 78.2$$

Adjustment of the fouling factor:

$$\text{Excess fouling factor} = \left(\frac{1}{U_{D\text{actual}}} \right) - \frac{1}{U_{Di}} = \frac{1}{78.2} - \frac{1}{89.6} = 0.00165$$

$$\text{Adding to the outside fouling factor: } \frac{A_f + A_o}{A_i} = 9.27$$

$$R_{do} = 0.003 + 9.27 \times 0.00165 = 0.0183$$

$$h'_{f,\text{actual}} = 0.0183 + \frac{1}{20.5} = 14.9$$

$$h'_{fi} = 113$$

Check of actual overall coefficient:

$$U_{Di} = \frac{h'_{fi} h'_i}{h'_{fi} + h'_i} = \frac{113 \times 243}{113 + 243} = 77.3 \quad \text{Check vs. 78.2}$$

Pressure Drop

$$(1') \quad d'_e = 4 \times 211.5 / (1570 + \pi \times 19.25) = 0.52 \text{ in.} \quad [\text{Eq. (6.5)}] \quad (1) \quad \text{For } Re_t = 29,100$$

[Fig. 26]

$$D'_e = 0.52 / 12 = 0.0433 \text{ ft}$$

$$Re'_e = D'_e G_s / \mu$$

$$= 0.0433 \times 20,400 / 0.0545 = 16,200$$

$$f = 0.00025 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 16.10b)}]$$

$$\text{Mol. wt. oxygen} = 32$$

$$\rho = \frac{32}{359 \times \frac{635}{492} \times 14.7 / 17.7} = 0.083 \text{ lb/ft}^3$$

$$s = 0.083 / 62.5 = 0.00133$$

$$(2') \quad \Delta P_s = \frac{f G_s^2 L n}{5.22 \times 10^{10} D'_e s \phi_s} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00025 \times 20,400^2 \times 16}{5.22 \times 10^{10} \times 0.0433 \times 0.00133 \times 1.0}$$

$$= 0.6 \text{ psi}$$

$$(2) \quad \Delta P_t = \frac{f G_t L n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00021 \times 868,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0652 \times 1.0 \times 1.0} = 3.0 \text{ psi}$$

Summary

0.018	Dirt Factor	0.003
113	h inside	243
U_{Di}	78.2	
0.6	Calculated ΔP	3.0
2.0	Allowable ΔP	10.0

The large shell-side dirt factor might suggest that the exchanger is considerably oversized. However, the number of finned tubes which can be fitted on 2-in. triangular

pitch into a 17¼-in. shell is only 54. This gives a required value of U_{Di} of 102 which is slightly greater than can be performed in a 17¼-in. exchanger.

TRANSVERSE FINS

Derivation of the Fin Efficiency. The different types of transverse finned pipes and tubes have been discussed at the beginning of the chapter. Expressions for their efficiencies are somewhat more difficult to derive than for longitudinal fins, since even the transverse fin of uniform

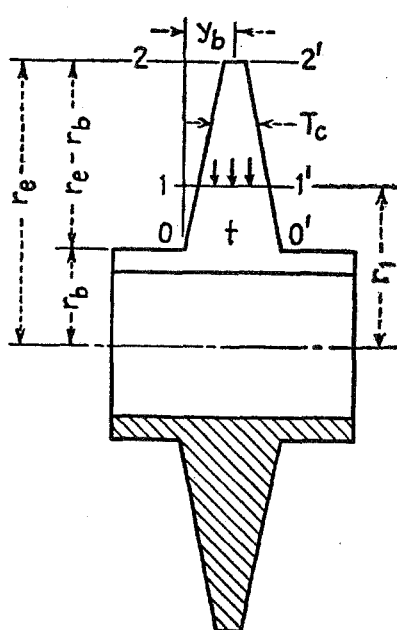


FIG. 16.12. Derivation of the transverse-fin efficiency.

cross section does not reduce to the simple differential equations of the longitudinal fin of uniform cross section. The derivations given here are those of Gardner,¹ which are ingenious because they develop a general expression that is applicable on modification to all the types of manufactured fins including the longitudinal fins. For the derivation of a general case, a transverse fin of varying cross section will be considered. The fluid surrounding the fin is again assumed to be hotter than the fin itself, and the flow of heat is from the outer fluid to the fin. The same assumptions apply as before. Referring to Fig. 16.12, let $\Theta = T_c - t$ where T_c is the constant hot-fluid temperature and t is the metal temperature at any point in the fin. The heat entering the two sides of the fin between 2-2' and 1-1' is dependent upon the surface between the two radii r_e and r_1 . Thus the total surface between 2-2' and 0-0' is a function of r

$$dQ = h_f \Theta dA \quad (16.44)$$

The heat which enters the fin between 2-2' and 1-1' flows toward its base through the cross section of the fin at 1-1'. Applying the Fourier equation

$$Q = -ka_x \frac{d\Theta}{dr} \quad (16.45)$$

where k is the thermal conductivity and a_x is the cross-sectional area of the fin which in this case also varies with r

Differentiating Eq. (16.45),

$$-\frac{dQ}{dr} = \frac{d}{dr} \left(ka_x \frac{d\Theta}{dr} \right) = ka_x \frac{d^2\Theta}{dr^2} + \frac{k da_x}{dr} \frac{d\Theta}{dr} \quad (16.46)$$

¹ Gardner, *op. cit.*

Equating Eqs. (16.44) and (16.46)

$$\frac{d^2\Theta}{dr^2} + \left(\frac{1}{a_x} \frac{da_x}{dr}\right) \frac{d\Theta}{dr} + \left(\frac{h_f}{ka_x} \frac{dA}{dr}\right) \Theta = 0 \quad (16.47)$$

This second-order differential equation is somewhat more difficult to evaluate than Eq. (16.6), which had simple roots, and can be solved by *Bessel functions*.

A number of second-order differential equations arise in the solution of engineering problems which can be solved by a power series. The solution can be written as the sum of two arbitrary functions and two arbitrary constants as suggested by Eq. (16.9). In a general form the Bessel equation is

$$r^2 \frac{d^2\Theta}{dr^2} + r \frac{d\Theta}{dr} + (r^2 - n^2)\Theta = 0 \quad (16.48)$$

where n is a constant. Many types of functions which are independent solutions of the Bessel equation have been developed, and their properties are tabulated.¹ Douglass has provided the following solution for Eq. (16.48) into which Eq. (16.47) may be transformed by multiplying by r^2 .

$$r^2 \frac{d^2\Theta}{dr^2} + [(1 - 2m)r - 2\alpha_1 r] \frac{d\Theta}{dr} + [p^2 C_3^2 r^{2p} + \alpha_1^2 r^2 + \alpha_1(2m - 1)r + m^2 - p^2 n^2]\Theta = 0 \quad (16.49)$$

where α_1 , C_3 , p , m , and n are constants, the last being the *order* of the Bessel function.

Equations (16.48) and (16.49) will have the same form when

$$a_x = C_4 r^{1-2pn} \quad (16.50)$$

and

$$\frac{dA}{dr} = C_5 r^{2p(1-n)-1} \quad (16.51)$$

where C_4 and C_5 are positive constants. If the cross section of the fin can be described by Eq. (16.50) and the surface from the outer edge to the cross section by Eq. (16.51), the general solution is found from the boundary conditions at the outer edge of the fin and at the base of the fin.

At the outer edge, $r = r_e$ and $\Theta = \Theta_e$

At the base, $r = r_b$ and $\Theta = \Theta_b$

For n equal to zero or an integer

$$\Theta = \Theta_b \left(\frac{u}{u_b}\right)^n \left[\frac{I_n(u) + \beta_1 K_n(u)}{I_n(u_b) + \beta_1 K_n(u_b)} \right] \quad (16.52)$$

¹ Sherwood, T. K., and C. E. Reed, "Applied Mathematics in Chemical Engineering," p. 211, McGraw-Hill Book Company, Inc., New York, 1939.

where

$$\beta_1 = -\frac{I_{n-1}(u_e)}{K_{n-1}(u_e)} \quad (16.53)$$

and

$$u = -iC_3 r^p = r \sqrt{\frac{h_f}{ka_x} \frac{dA}{dr}} \quad (16.54)$$

where u_b and u_e are found by substituting r , a_x , and dA/dr for the edge and base, respectively, and $i = \sqrt{-1}$. Since Θ is a function of A as well as r , the heat transferred to the entire fin surface from the fluid is

$$Q = h_f \int_0^{A_f} \Theta dA \quad (16.55)$$

where A_f is the total fin surface. The same heat when transferred through the base of the fin to the tube is

$$Q = h_b \Theta_b A_f \quad (16.56)$$

Defining the fin efficiency as before,

$$\Omega = \frac{h_b}{h_f} = \frac{\int_0^{A_f} \Theta dA}{\Theta_b A_f} = \frac{2(1-n)}{u_b[1 - (u_e/u_b)^{2(1-n)}]} \left[\frac{I_{n-1}(u_b) - \beta_1 K_{n-1}(u_b)}{I_n(u_b) + \beta_1 K_n(u_b)} \right] \quad (16.57)$$

For a fin of given contour the exponent of r is known in Eq. (16.50). Equation (16.51) may be eliminated accordingly to permit the solution for n , the order of the Bessel function.

When $n = 0$, the introduction of Eqs. (16.52) and (16.53) and

$$dA = C_6 u^{1-2n} du$$

as obtained from Eqs. (16.51) and (16.54) into Eq. (16.57) gives the value of Ω for the annular fin of constant width.

Equation (16.57) reduces to

$$\Omega = \frac{2}{u_b[1 - (u_e/u_b)]^2} \left[\frac{I_1(u_b) - \beta_1 K_1(u_b)}{I_0(u_b) + \beta_1 K_0(u_b)} \right] \quad (16.58)$$

where

$$\beta_1 = \frac{I_1(u_e)}{K_1(u_e)} \quad (16.59)$$

and

$$u_b = \frac{(r_e - r_b) \sqrt{h_f/ky_b}}{\left(\frac{r_e}{r_b} - 1\right)} \quad (16.60)$$

$$u_e = u_b \left(\frac{r_e}{r_b}\right) \quad (16.61)$$

Plots of these equations are given in Fig. 16.13a.

For the case where n equals a fraction,

$$\Theta = \Theta_b \left(\frac{u}{u_b} \right)^n \left[\frac{I_n(u) + \beta_1 I_{1-n}(u)}{I_n(u_b) + \beta_1 I_{1-n}(u_b)} \right] \quad (16.62)$$

where

$$\beta_1 = \frac{I_{n-1}(u_e)}{I_{1-n}(u_e)} \quad (16.63)$$

If half the thickness of the fin, y , is given by

$$y = y_b \left(\frac{r}{r_b} \right)^{-\frac{2n}{1-n}} \quad (16.64)$$

$$u = (1 - n) \left(\frac{r}{r_b} \right)^{\frac{1}{1-n}} \sqrt{\frac{h_f}{ky_b}} r_b \quad (16.65)$$

When n equals a fraction, Eq. (16.57) becomes

$$\Omega = \frac{2(1 - n)}{u_b[1 - (u_e/u_b)^{2(1-n)}]} \left[\frac{I_{n-1}(u_b) + \beta_1 I_{1-n}(u_b)}{I_n(u_b) + \beta_1 I_{1-n}(u_b)} \right] \quad (16.66)$$

When $n = 1/3$, corresponding to annular fins with constant heat flux at every cross section from r_b to r_e

$$\Omega = \frac{4}{3u_b[1 - (u_e/u_b)^{4/3}]} \left[\frac{I_{-2/3}(u_b) + \beta_1 I_{2/3}(u_b)}{I_{1/3}(u_b) + \beta_1 I_{-1/3}(u_b)} \right] \quad (16.67)$$

$$\beta_1 = \frac{-I_{-2/3}(u_e)}{I_{2/3}(u_e)} \quad (16.68)$$

$$u_e = u_b \left(\frac{r_e}{r_b} \right)^{3/2} \quad (16.69)$$

$$y = y_b \left(\frac{r_b}{r} \right) \quad (16.70)$$

These values are plotted in Fig. 16.13*b*. For longitudinal fins and spines the efficiency curves and the fundamental equations are included in Fig. 16.13*c* and *d*. When a fouling factor is present replace h_f by h'_f .

In these graphs the efficiency Ω has been plotted as the ordinate and the principal shape element influencing the value of n in the Bessel equation has been plotted as the abscissa. The variations in the types of construction of transverse finned tubes are numerous and less standardized than for longitudinal fins, and no attempt will be made here to compute weighted efficiency curves of h'_f vs. $h_{f,i}$. A weighted efficiency curve can be computed, however, using Eq. (16.34) with the values of Ω being taken from Fig. 16.13*a* and *b*, in the same manner as before, when the frequent use of given fin and tube arrangements justifies their preparation.

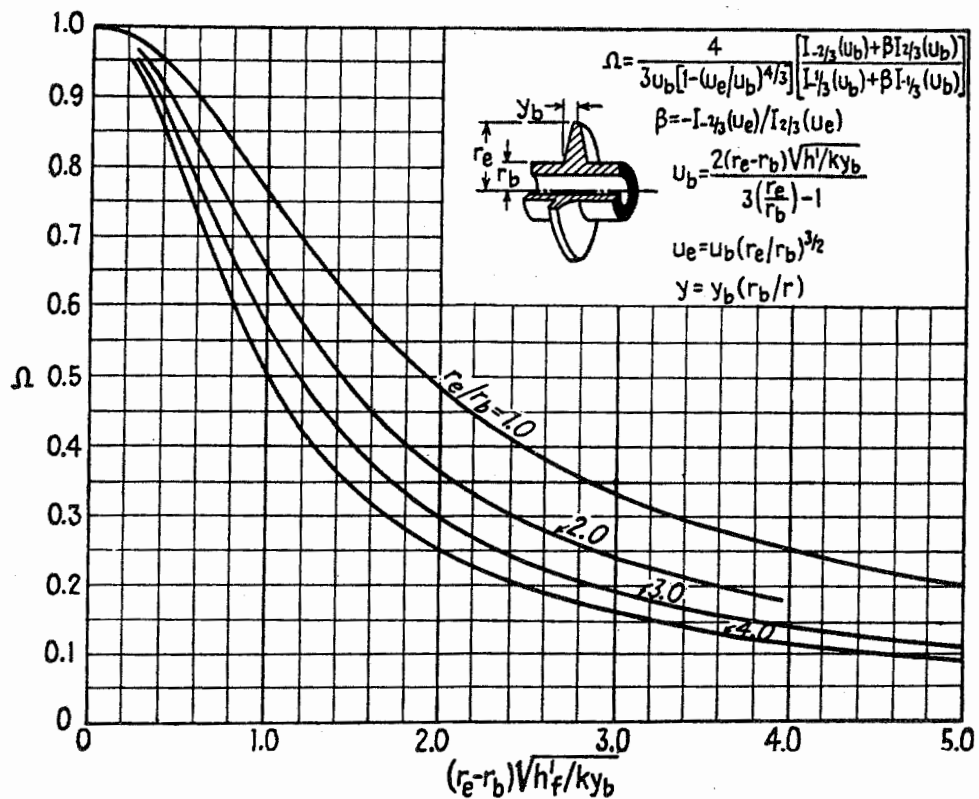
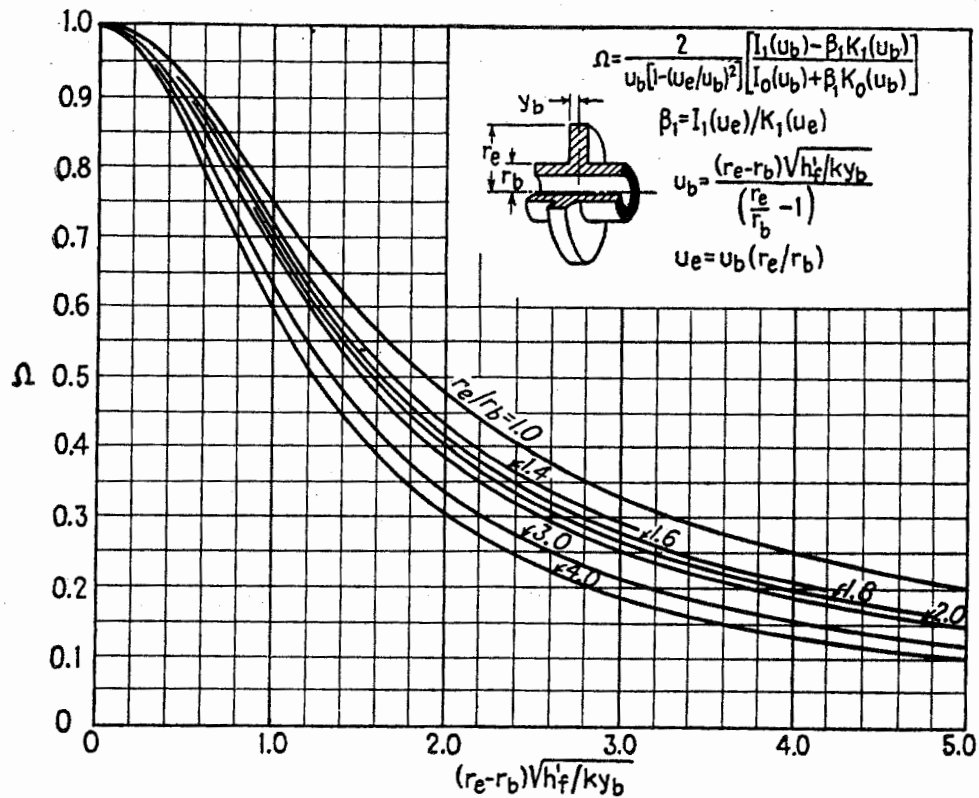
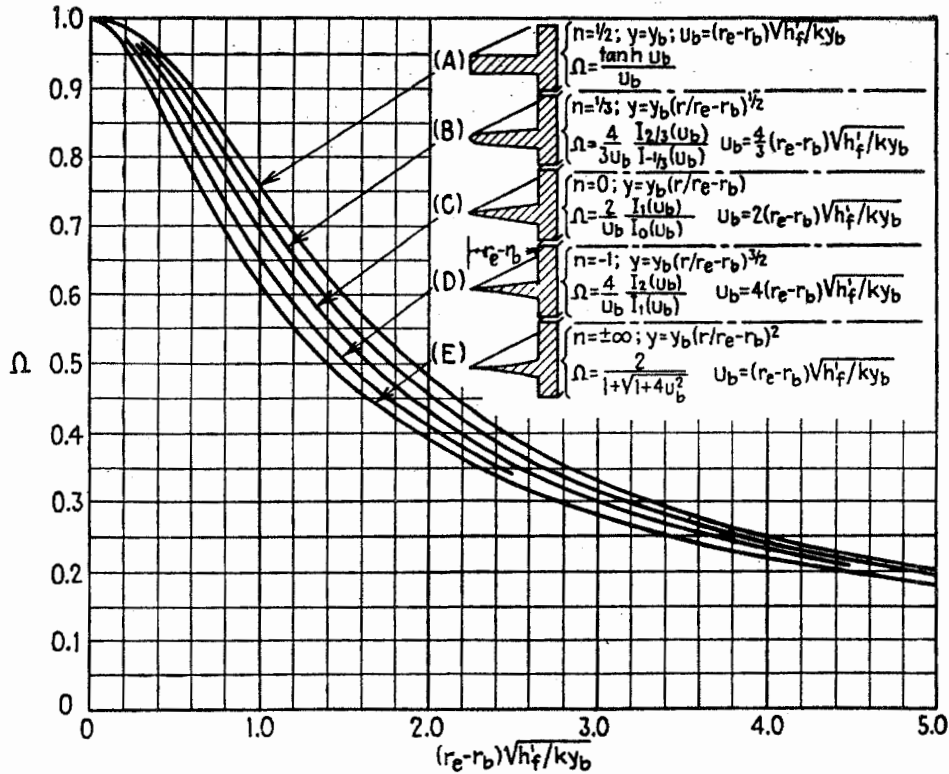
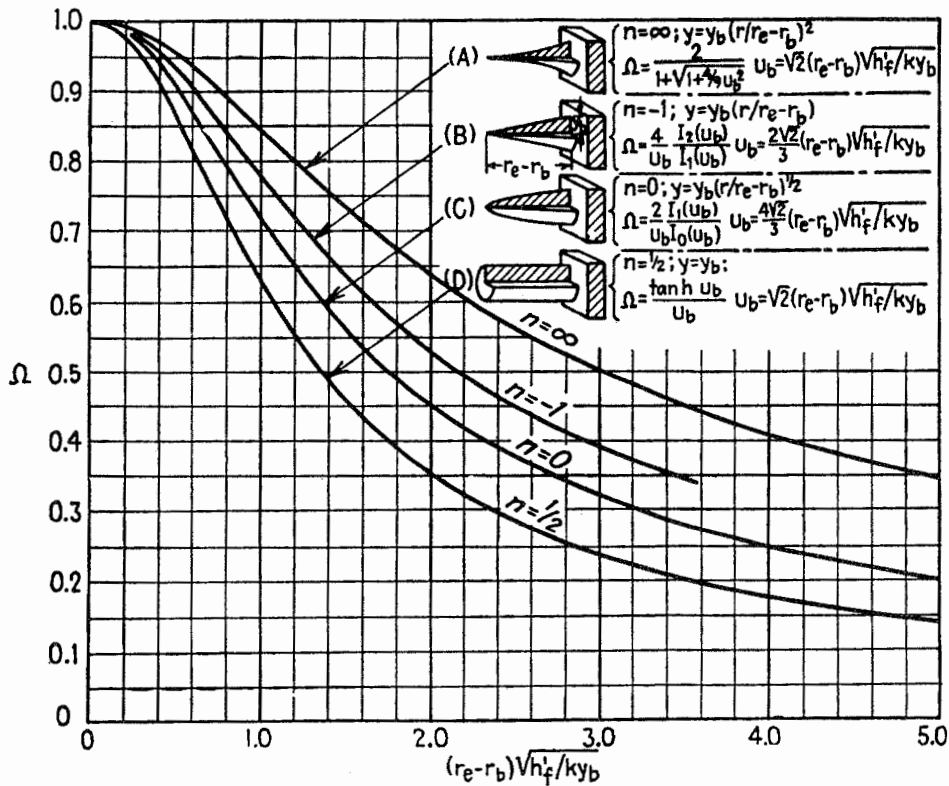


FIG. 16.13. Fin efficiencies. Where there is no foul-



(c)-FIN EFFICIENCY OF SEVERAL TYPES OF STRAIGHT FIN



(d)-EFFICIENCY CURVES FOR FOUR TYPES OF SPINE

ing factor replace h_f' by h_f . (Gardner, Transactions of the ASME.)

The Optimum Thermal Fin. Schmidt¹ has undertaken the derivations for the shapes of various types of fins which give the highest heat transfer for the minimum amount of metal. As a result, the optimum fin may be considered the one which has a constant heat flux at any cross section between the outer edge and the base. For the common longitudinal and transverse fins these would correspond to tall narrow shapes with small base areas. Furthermore, the sides of the fins should have parabolic curvatures. Although the thermal efficiency for ideal fins may be high, the cost of their manufacture is usually excessive and they are not often structurally adaptable to industrial applications. The calculations of the optimum shapes have led, however, to the present type of manufacture using 20 BWG and lighter metal ribbons except where the conditions of heat transfer requires a more rugged construction.

The metal used in their manufacture greatly affects the economics of the fins. Schmidt has prepared the following table comparing the ratio of the quantity of metal required for fins with identical shapes where the weight and volume of a copper fin is taken as unity.

TABLE 16.2. OPTIMUM VOLUMES AND WEIGHTS OF METAL FINs

Metal	k	Gravity	Volume	Weight
Copper.....	222	8.9	1.0	1.0
Aluminum.....	121	2.7	1.83	0.556
Steel.....	30	7.8	7.33	0.43

Jakob² presents an excellent survey of optimum fins defined by other criteria.

Transverse Fin Exchangers. Transverse fin exchangers in crossflow are used only when the film coefficients of the fluids passing over them are low. This applies particularly to gases and air at low and moderate pressures. Tubes are also available which have many very small fins integrally shaped from the tube metal itself and which are usable in conventional 1-2 exchangers with baffled side-to-side flow. These can be calculated using a suitable shell-side heat-transfer curve and an appropriate efficiency curve.

Perhaps the most interesting applications of transverse fins are found in the larger gas cooling and heating services such as on furnaces (economizers), tempering coils for air conditioning, air-cooled steam condensers for turbine and engine work, and miscellaneous special services. An application which is growing in popularity is the air-cooled steam con-

¹ Schmidt, E., *Z. Ver. deut. Ing.* 885, 70 (1926).

² Jakob, *op. cit.*

denser as shown in Fig. 16.14 for localities with inadequate cooling-water supply. The steam enters the tubes, and an induced-draft fan circulates air over the transverse finned tubes. In this way it is possible to attain a closer approach to the atmospheric temperature than could be done with a reasonable surface composed entirely of bare tubes. In

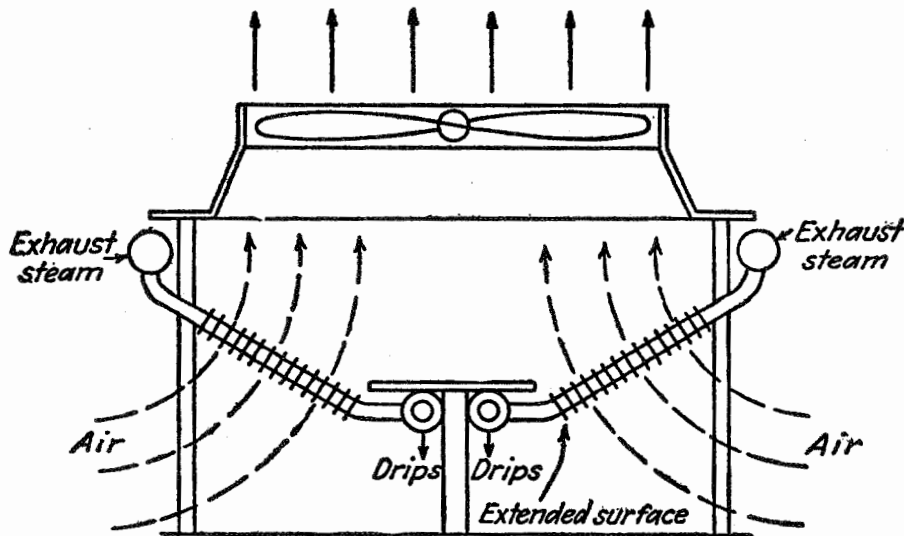


FIG. 16.14. Transverse-fin atmospheric condenser.

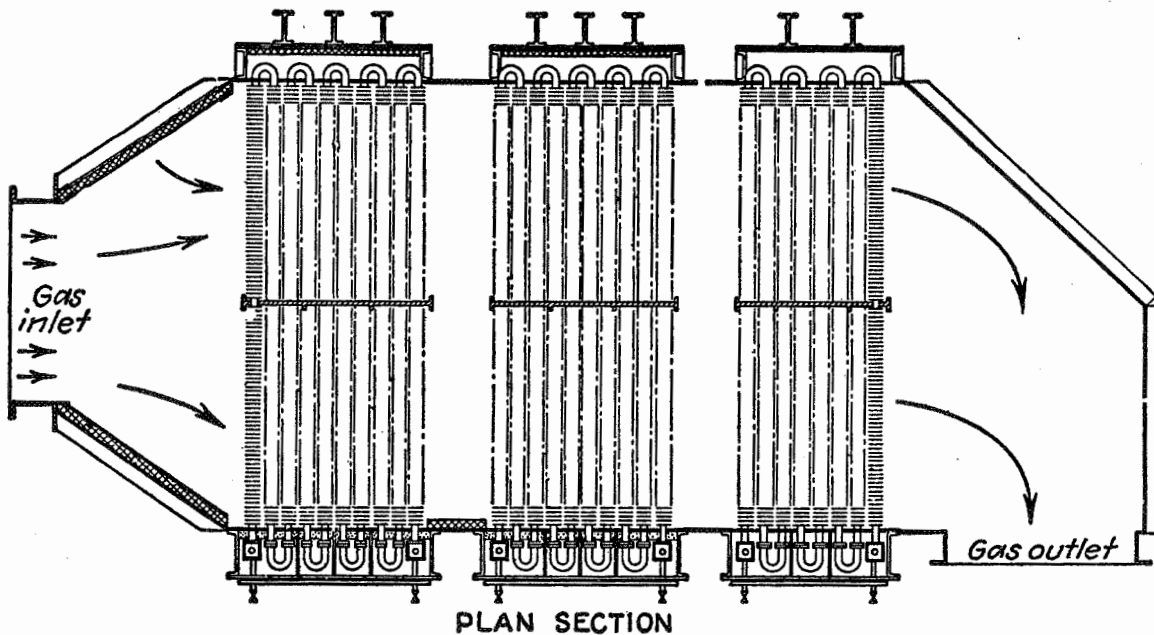


FIG. 16.15. Transverse-fin gas cooler. (Foster Wheeler Corp.)

Fig. 16.15 a plan is shown of a high-temperature gas cooler used in an aviation testing laboratory. With the exception noted before, all other transverse fin exchangers operate in crossflow. The true temperature differences for systems operating in crossflow naturally differ from other flow patterns except when one fluid is isothermal; then all systems are in true counterflow. Before proceeding to the calculation of a typical

extended-surface gas exchanger, an investigation will be made of the influence of flow pattern on the true temperature difference in crossflow.

True Temperature Difference for Crossflow Arrangements. Consider a duct in which a hot gas (or other fluid) passes at right angles to a bundle carrying a cold tube liquid in a single pass as in Fig. 16.16. Assume that baffles indicated by the vertical lines are placed across the bundle in the direction of gas flow to prevent mixing over the length of the tubes. At point *A* in the first horizontal row of tubes and in the plane 1-1' there is an unique temperature difference between the gas-inlet temperature and

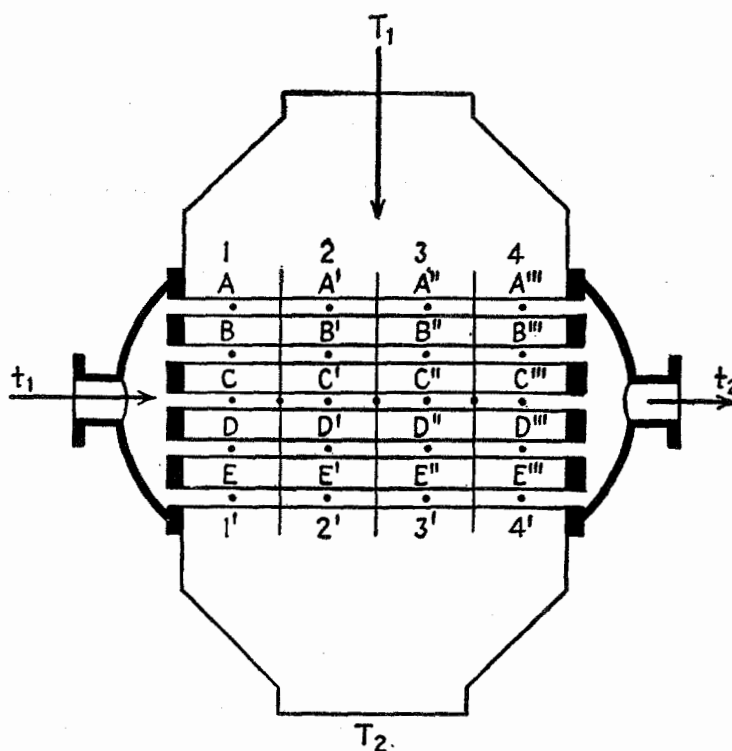


FIG. 16.16. Crossflow.

the liquid-inlet temperature. Moving to plane 2-2' there is a smaller temperature difference at *A'* between the gas-inlet temperature and the liquid at some higher temperature. Similarly at 3-3' and the point *A''* the temperature difference is smaller than *A'*, and further reductions of the temperature difference may be expected over the entire length of the tubes in the first tube row. Carrying the analysis to a lower row of tubes, the gas temperature off the first horizontal-tube row varies over the tube length, since the quantity of heat transferred in passing over the first row decreased from point to point with the decreasing temperature difference. Over the second horizontal row the temperature differences also vary but the temperatures of the gas leaving the row are different from those on leaving the preceding row of tubes. This picture is unlike that used in the derivation of the true temperature difference for a 1-2

exchanger as given in Chap. 7, in which the shell-side fluid was considered mixed and of homogeneous temperature at every cross section and where it was assumed that all the tube rows comprising each tube pass were at the same temperature.

Suppose, on the other hand, that the baffles in the exchanger in Fig. 16.16 are removed and that the tubes are not very long so that the gas passing over a row of tubes may be considered thoroughly mixed. This is much different from the preceding analysis, but it also differs from that of the 1-2 exchanger. In the 1-2 exchanger the uniform cross-section temperature varied along the length of the tube. In the present case the gas temperature at a cross section varies from row to row, and it can be seen that the true temperature difference is influenced considerably by whether one, both, or neither stream is mixed.

When a gas passes at right angles to a bundle composed of a single row of single-pass tubes, both fluids may be considered unmixed. However, it is not always possible to determine clearly if both streams are mixed, the definition being somewhat arbitrary in the case of tube bundles with long and short tube lengths. In the study below, four common theoretical mixing possibilities are treated to permit an estimation of the error resulting from the assumption of any particular one of them. Not all the possible crossflow temperature differences have been studied, but the principal derivations are available.

The derivations below are essentially those of Nusselt¹ and Smith,² and the graphs of the final equations are in the form developed by Nagle and Bowman, Mueller and Nagle.³ These derivations involve the usual assumptions except as qualified for mixing. If T refers to the hot fluid, t to the cold fluid, and subscripts 1 and 2 to the inlet and outlet, respectively, it is convenient to define three parameters as follows:

$$\left. \begin{aligned} K &= \frac{T_1 - T_2}{T_1 - t_1} \\ S &= \frac{t_2 - t_1}{T_1 - t_1} \end{aligned} \right\} \quad R = \frac{K}{S} = \frac{T_1 - T_2}{t_2 - t_1}$$

$$(r) = \frac{\Delta t}{T_1 - t_1}$$

From Eq. (7.39) for counterflow define

$$(r)_{\text{counterflow}} = \frac{R - 1}{\ln(1 - S)/(1 - RS)} = \frac{(K - S)/S}{\ln(1 - S)/(1 - K)}$$

¹ Nusselt, W., *Tech. Mech. u. Thermodynam.*, I, 417-422 (1930).

² Smith, D. M., *Engineering*, **138**, 474-481, 606-607 (1934).

³ See Chap. 7 for references.

Employing the factor F_T to correct the LMTD to the true temperature difference Δt in crossflow by the method introduced in Chap. 7,

$$F_T = \frac{(r)}{(r)_{\text{counterflow}}}$$

where $\Delta t = F_T \times \text{LMTD}$. Without a subscript (r) refers to the true value in crossflow.

Referring to Fig. 16.16 and applying the heat balances,

$$UA\Theta = \iint U(T - t) dx dy \quad (16.71)$$

$$WC(T_1 - T_2) = wc(t_2 - t_1) \quad (16.72)$$

$$\text{LMTD} = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln (T_1 - t_2)/(T_2 - t_1)} = \frac{\Delta t_h - \Delta t_c}{\ln \Delta t_h / \Delta t_c} = C_7 \Delta t_h \quad (16.73)$$

and where $C_7 = \left(1 - \frac{\Delta t_c}{\Delta t_h}\right) / \ln \frac{\Delta t_h}{\Delta t_c}$ is a function only of $\frac{\Delta t_c}{\Delta t_h}$. Let XY be the total tube area swept out as the shell fluid moves downward in the x direction and the tube fluid moves from left to right in the y direction.

Case A. Both Fluids Unmixed. Referring to Fig. 16.17a,

$$U(T - t) dx dy = -\frac{WC}{Y} dy \frac{\partial T}{\partial x} dx = \frac{wc}{X} dx \frac{\partial t}{\partial y} dy \quad (16.74)$$

If

$$x' = \frac{x}{X} \cdot \frac{UA}{WC} \quad \text{and} \quad y' = \frac{y}{Y} \frac{UA}{wc}$$

then

$$T - t = -\frac{\partial T}{\partial x'} = +\frac{\partial t}{\partial y'} \quad (16.75)$$

Solve when $T = T_1$ at $x' = 0$ and $t = t_1$ at $y' = 0$.

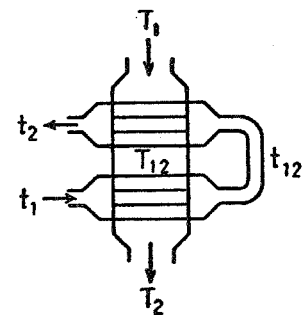
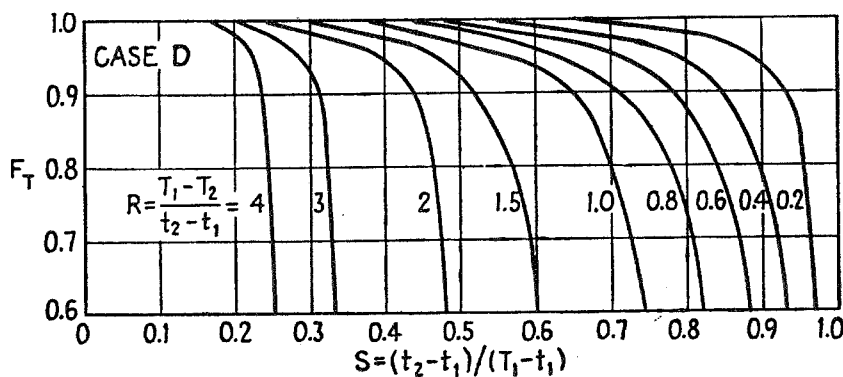
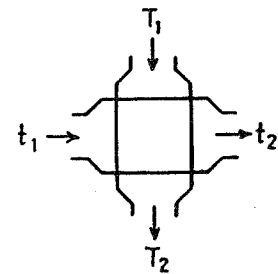
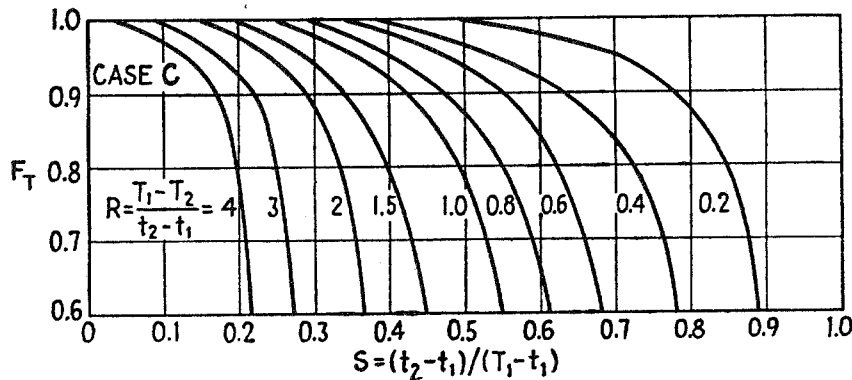
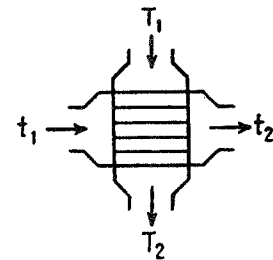
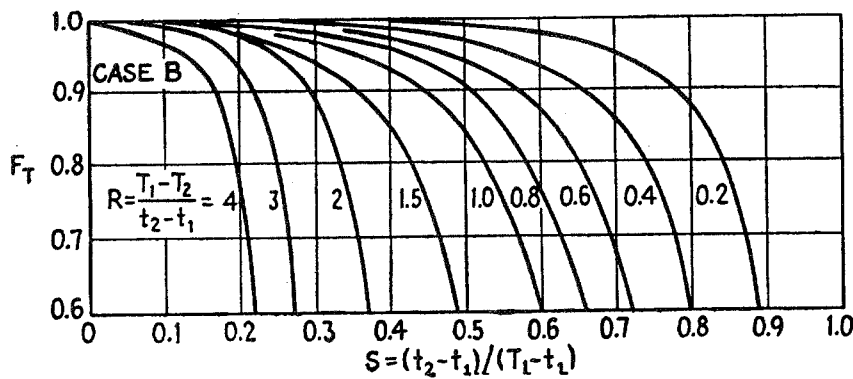
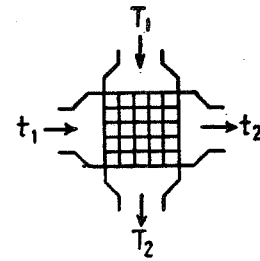
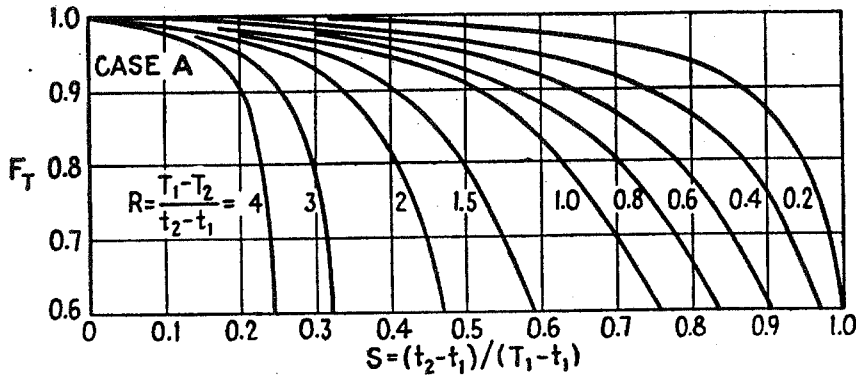
The solution is expressed in a doubly infinite series.

$$\frac{T - t_1}{T_1 - t_1} = 1 + \sum_{u=1}^{\infty} \sum_{v=1}^{\infty} \left[(-1)^{u+v} \frac{(u+v-1)!}{(u!v!)^2} (x')^u (y')^v \right] \quad (16.76)$$

T_2 is obtained by integration as the mean of T when $x = X$ and y goes from 0 to Y . Then $WC(T_1 - T_2) = Q$, $x = X$, $x' = K/(r)$, $y = Y$, and $y' = S/(r)$.

$$(r) = \sum_{u=0}^{\infty} \sum_{v=0}^{\infty} \left\{ (-1)^{u+v} \frac{(u+v)!}{u!(u+1)!v!(v+1)!} \left[\frac{K}{(r)} \right]^u \left[\frac{S}{(r)} \right]^v \right\} \quad (16.77)$$

Values of F_T are plotted for $F_T = (r)/(r)_{\text{counterflow}}$ in terms of R and S in Fig. 16.17a.



(d) Two-pass counterflow, shell fluid mixed, tube fluid unmixed.

FIG. 16.17. Crossflow-temperature-difference correction factors. (Bowman, Mueller, and Nagle, Transactions of the ASME.)

Case B. One Fluid Mixed, Other Fluid Unmixed. Referring to Fig. 16.17b let the shell fluid be mixed. T is a function only of x , but t is a function of both x and y . Consider a strip at x_1 of width dx . The fluid flowing across the strip is at the constant shell temperature T_1 , but the fluid flowing along the strip is unmixed and varies. If t is the unmixed varying temperature, at the inlet where $t = t_1$ at $y = 0$,

$$\frac{dt}{dy} = \frac{UX}{wc} (T - t) \quad (16.78)$$

At the outlet of the strip the temperature is given by

$$\frac{T - t_{x_1}Y}{T - t_1} = e^{-UXY/wc} \quad (16.79)$$

where t_{x_1} refers to the strip. The total heat transfer in the strip to the mixed fluid is

$$wc \frac{dx}{X} (t_1 - t_{x_2}Y) = WC dT \quad (16.80)$$

Substituting for $t_{x_2}Y$ in Eq. (16.80), integrating, and allowing for the boundary condition $t = t_1$ at $x = 0$, an expression is obtained for T which gives T_2 when $x = X$. Thus

$$\frac{T_1 - T_2}{T_1 - t_1} = 1 - e^{wc/WC} (1 - e^{-UXY/wc}) \quad (16.81)$$

The parameters have been chosen so that

$$\frac{wc}{WC} = \frac{K}{S} \quad \text{and} \quad \frac{UXY}{wc} = \frac{S}{(r)} \\ K = 1 - e^{-\frac{K}{S}(1 - e^{-\frac{S}{(r)}})} \quad (16.82)$$

Expressing (r) as an implicit function

$$(r) = \frac{S}{\ln 1 / \left(1 - \frac{S}{K} \ln \frac{1}{1 - K} \right)} \quad (16.83)$$

Factors for this equation are plotted in Fig. 16.17b.

Case C. Both Fluids Mixed. Referring to Figure 16.17c, T is a function of x only and t is a function of y only. The total heat flow across a section dy is

$$U dy \int_0^x (T - t) dx$$

where T is variable and t is constant. The heat transferred is $wc dt$ and

$$\frac{wc dt}{UX dy} + t = \frac{1}{X} \int_0^X T dx \quad (16.84)$$

Similarly,

$$\frac{wc dt}{UY dy} + t = \frac{1}{Y} \int_0^Y t dy \quad (16.85)$$

Let $\frac{1}{X} \int_0^X T dx = \alpha_2$, $\frac{1}{Y} \int_0^Y t dy = \beta_2$ where α_2 and β_2 are constants.

Substituting, integrating, and allowing for the boundary conditions $T = T_1$ at $x = 0$ and $t = t_1$ at $y = 0$, equations are obtained in α_2 and β_2 .

$$\alpha_2 = \beta_2 - \frac{(r)}{K} (T_1 - \beta_2)(e^{-K/(r)} - 1) \quad (16.86)$$

$$\beta_2 = \alpha_2 - \frac{(r)}{S} (t_1 - \alpha_2)(e^{-S/(r)} - 1) \quad (16.87)$$

The integration also gives the temperature change for one fluid.

$$t_2 - t_1 = (\alpha_2 - t_1)(1 - e^{-S/(r)}) \quad (16.88)$$

Solving simultaneously and using the temperature change times wc to obtain the heat transmitted,

$$(r) \left[\frac{K/(r)}{1 - e^{-K/(r)}} + \frac{S/(r)}{1 - e^{-S/(r)}} \right] = 1 \quad (16.89)$$

Factors for this equation are plotted in Fig. 16.17c. Many of the cases for one crossflow and several series passes have not been solved so far. The case of greatest value is that of one crossflow and two series passes in parallel flow-counterflow when the shell fluid is unmixed and the series fluid is mixed between passes.

Case D. Shell Fluid Mixed, Two Pass, Counterflow, Fluid Unmixed Except between Passes. Let t_{12} be the mixed temperature between passes and T_{12} the corresponding shell temperature as shown in Fig. 16.17d. Call the upper pass I and the lower pass II, each with surface $A/2$.

From Case B

$$1 - K_I = e^{-K_I/(S)I}(1 - e^{-S_I/(r)I}) \quad (16.90)$$

and

$$-K_{II} = e^{-K_{II}/S_{II}}(1 - e^{-S_{II}/(r)II}) \quad (16.91)$$

From the identity of flow in each path

$$\frac{K_I}{S_I} = \frac{K}{S} = \frac{K_{II}}{S_{II}} \quad (16.92)$$

From the constancy of the heat-transfer coefficient and area in each pass

$$\frac{(r)_I}{K_I} = \frac{(r)_{II}}{K_{II}} \quad (16.93)$$

Let $M = (T_1 - t_{12})/(T_1 - t_1)$ and $N = (T_{12} - t_1)/(T_1 - t_1)$. Since the total heat transfer is the sum of both passes,

$$UA \Delta t = \frac{UA}{2} \Delta t_I + \frac{UA}{2} \Delta t_{II} \quad (16.94)$$

and

$$(r) = \frac{1}{2}M(r)_I + \frac{1}{2}N(r)_{II} \quad (16.95)$$

Also

$$T_1 - T_2 = (T_1 - T_{12}) + (T_{12} - T_2) \quad (16.96)$$

$$K = MK_I + NK_{II}$$

$$T_1 - t_1 = (T_1 - t_{12}) + (t_{12} - t_1) \quad (16.97)$$

$$1 = M + NS_{II}$$

$$T_1 - t_1 = (T_1 - T_{12}) + (T_{12} - t_1) \quad (16.98)$$

$$1 = MK_I + N$$

Solving all nine equations simultaneously and eliminating the eight unknowns K_I , S_I , $(r)_I$, K_{II} , S_{II} , $(r)_{II}$, M , and N produces a solution in terms of K , S , and (r) . From Eqs. (16.92) and (16.93)

$$\frac{K_I}{K_{II}} = \frac{S_I}{S_{II}} = \frac{(r)_I}{(r)_{II}} \quad (16.99)$$

From Eqs. (16.90) and (16.91)

$$K_I = K_{II} \quad S_I = S_{II} \quad (r)_I = (r)_{II}$$

From Eqs. (16.94) and (16.96)

$$\frac{K_I}{(r)_I} = \frac{1}{2} \frac{K}{(r)} \quad \frac{S_I}{(r)_I} = \frac{1}{2} \frac{S}{(r)}$$

Eliminating M and N between Eqs. (16.96) to (16.98) and applying Eq. (16.92),

$$K_I^2 \left(1 + \frac{S}{K} - S \right) - 2K_I + K = 0 \quad (16.100)$$

$$K_I = \frac{K}{1 \pm \sqrt{(1-K)(1-S)}} \quad (16.101)$$

Substituting in Eq. (16.90) using only the positive root,

$$\frac{\sqrt{(1-S)/(1-K)} - S/R}{1 - S/K} = e^{K/S}(1 - e^{1/2 S/(r)}) \quad (16.102)$$

or

$$(r) = \frac{S}{2 \ln \frac{1}{1 - (S/K) \ln \sqrt{(1-S)/(1-K)} - (S/K)}} \quad (16.103)$$

$$1 - S/K$$

Factors for this equation are plotted in Fig. 16.17*d*.

Dunn and Bonilla¹ have treated the case of fin transfer with no transverse mixing parallel to the extended surface and present graphs to permit a rapid solution for the system.

Comparison of Thermal Efficiency in Different Arrangements. Examination of the standard types of fluid-flow arrangements will indicate the relative thermal efficiency of each. There are three true fluid-flow patterns: counterflow, crossflow, and parallel flow. Consider a fluid being cooled from 400 to 200°F by a fluid being heated from 100 to 200°F so that the approach in terms of a 1-2 exchanger is zero or 200°F for a counterflow exchanger.

<i>Hot fluid</i>	<i>Cold fluid</i>
400	200 = 200 = Δt_c
200	100 = 100 = Δt_h
$T_1 - T_2 = 200$	$t_2 - t_1 = 100$
$R = 200/100 = 2.0$	$S = 100/300 = 0.333$

<i>Type of Flow</i>	<i>F_T</i>
Counterflow, 1-1 exchanger.....	1.00
Parallel flow-counterflow, 2-4 exchanger.....	0.95
Crossflow, shell mixed and two unmixed tube passes in series.....	0.98
Crossflow, both fluids unmixed.....	0.90
Crossflow, shell fluid mixed and tube fluid unmixed.....	0.87
Parallel flow-counterflow, 1-2 exchanger.....	0.81
Crossflow, both fluids mixed.....	0.77
Crossflow, shell mixed and two unmixed parallel-flow tube passes.....	0
Parallel flow.....	0

The deductions which may be made from above are (1) that the presence of parallel flow greatly decreases the utilization of the thermal potential for heat transfer and (2) that mixing the fluid streams also decreases the utilization. The curves of F_T vs. R and S for crossflow with two parallel tube passes have not been included in Fig. 16.17, since it is impractical for even moderate temperature approaches and has little value as a means of heat recovery.

Film Coefficients to Transverse Fins. Almost all the data available on commercial scale runs have been made with air or flue gas. The heat-

¹ Dunn, W. E., and C. F. Bonilla, *Ind. Eng. Chem.*, **40**, 1101-1104 (1948).

transfer curve employed here has been transposed from the data of Jameson,¹ which agree within reasonable limits with the published results of Tate and Cartinhour² on economizers. It has been found that the heat-transfer coefficient is not influenced by the spacing of succeeding rows although nearly all the data were obtained for triangular pitch.

The actual heat-flow pattern in a transverse fin probably differs somewhat from the idealized picture used in deriving the fin efficiency. In triangular pitch the air or gas strikes the front and sides of the annular fins but not the back of the fin. As determined experimentally h_f is actually only an average value. The concentration of heat on the lead side of an annular fin probably introduces a potential for heat flow around the fin metal which was not considered in the derivation. Discontinuous fins such as stars or other types give generally higher coefficients than helical or disc fins, and this may be attributed in part to the greater ease with which the gas penetrates the space between adjacent discontinuous fins.

Because so many of the applications involve gases, it has become a common convention in extended-surface heat transfer to use Colburn's heat-transfer factor $j_h = (h/cG)(c\mu/k)^{3/4}$. The advantages of Colburn's form of the factor particularly with gases have been discussed in Chap. 9. For the sake of a consistent presentation the data of Jameson have been converted to the Sieder-Tate heat-transfer factor as shown in Fig. 16.18a with the value of the viscosity correction ϕ taken as 1.0 for gases. The equivalent diameter in Jameson's correlation has been defined by

$$D_e = \frac{2(A_f + A_o)}{\pi (\text{projected perimeter})} \quad (16.104)$$

The projected perimeter is the sum of all the external distances in the plan view of a transverse finned tube. The mass velocity is computed from the free flow area in a single bank of tubes at right angles to the gas flow.

Pressure Drop for Transverse Fins. Unlike the heat transfer coefficient, the pressure drop is greatly influenced by the spacing of the succeeding rows of tubes, their layout and closeness. It is often possible in transverse-fin equipment that the vertical and transverse pitches will be unequal. Of a number of excellent correlations of pressure drop in cross-flow that of Gunter and Shaw³ is used here. It is equally satisfactory for crossflow calculations over bare tubes and the correlation is based on oils, water, and air. While there has been some objection to the broad-

¹ Jameson, S. L., *Trans. ASME*, **67**, 633-642 (1945).

² Tate, G. E., and J. Cartinhour, *Trans. ASME*, **67**, 687-692 (1945).

³ Gunter, A. Y., and W. A. Shaw, *Trans. ASME*, **67**, 643-660 (1945).

ness of the correlation it gives relatively safe values for the pressure drop. The Reynolds numbers are computed on the basis of a *volumetric* equivalent diameter which reflects the proximity and arrangement of the succeeding rows, and the pressure-drop equation contains two dimensionless configuration factors. The volumetric equivalent diameter is defined by

$$D'_{ev} = \frac{4 \times \text{net free volume}}{A_f + A_o} \quad (16.105)$$

The net free volume is the volume between the center lines of two vertical banks of tubes less the volumes of the half-tubes and fins within the

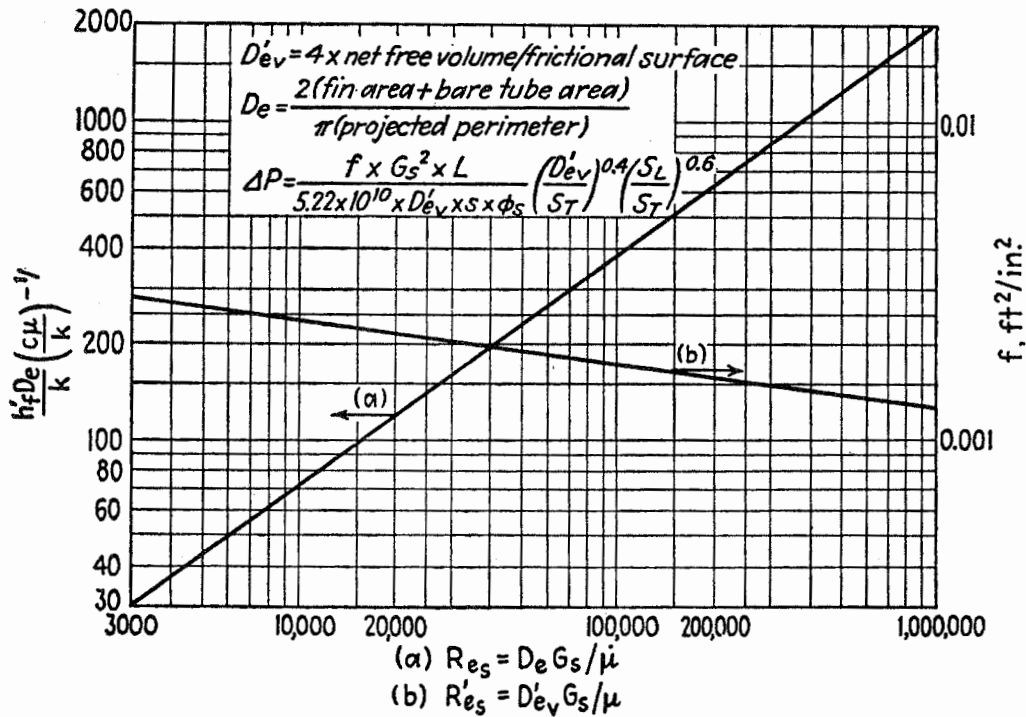


FIG. 16.18. Transverse-fin heat transfer and pressure drop. ((a) Jameson (b) Gunter and Shaw, *Transactions of the ASME*.)

center lines. A configuration factor is a dimensionless group which is ordinarily lost in the derivation but which may be included at the discretion of the experimenter. The factors used here are S_T and S_L where S_T is the pitch in a transverse bank and S_L is the center-to-center distance to the nearest tube in the next bank. The pressure-drop equation is thus

$$\Delta P = \frac{f G_s^2 L_p}{5.22 \times 10^{10} \times D'_{ev} s \phi_s} \left(\frac{D'_{ev}}{S_T} \right)^{0.4} \left(\frac{S_L}{S_T} \right)^{0.6} \quad (16.106)$$

where L_p is the length of the path. Friction factors are given in Fig. 16.18b.

Example 16.5. Calculation of a Transverse-fin Air Cooler. A duct 4'0" by 4'0" carries 100,000 lb/hr of air from a kiln drier at 250°F. The gas is to be used to preheat treated water from 150 to 190°F by revamping a section of the duct and installing 1 in. OD, 14 BWG tubes with $\frac{3}{8}$ in. high annular brass fins of 20 BWG metal spaced $\frac{1}{8}$ in. apart. Assume the spacing for the tubes is on $2\frac{1}{4}$ -in. triangular pitch.

Use dirt factors of 0.0030 on both streams. The pressure head on the gas side is only a fraction of a pound per square inch, so that the overall pressure drop must be low.

How many tubes should be installed? How shall they be arranged?

Solution:

(1) Heat balance:

Air at 14.7 psia, $Q = 100,000 \times 0.25(250 - 200) = 1,250,000$ Btu/hr

Water, $Q = 31,200 \times 1.0(190 - 150) = 1,250,000$ Btu/hr

(2) Δt : Assume that in the final arrangement the water will flow in series from one vertical tube bank to the next, so that there will be as many passes as banks. If there are more than several banks in a 4- by 4-ft duct, the air may be considered mixed. Figure 16.17d will apply.

Hot Fluid		Cold Fluid	
250	Higher Temp	190	60
200	Lower Temp	150	50
50	Difference	40	10

$$\text{LMTD} = 54.6^\circ\text{F} \quad (5.14)$$

$$R = \frac{50}{40} = 1.25 \quad S = \frac{40}{250 - 150} = 0.40 \quad F_T = 0.985 \quad (\text{Fig. 16.17d})$$

$$\Delta t = 54.6 \times 0.985 = 53.7^\circ\text{F}$$

(3) T_c and t_c : The average temperatures of 225 and 170°F will be adequate.

Calculation of the duct equivalent diameter and flow area:

Equivalent diameter (refer to Fig. 16.19a):

$$d_e = \frac{2 (\text{fin area} + \text{bare tube area})}{\pi (\text{projected perimeter})} \quad (16.104)$$

$$\text{Fin area, } A_f = \frac{\pi}{4} (1.75^2 - 1^2) \times 2 \times 8 \times 12 = 310 \text{ in.}^2/\text{ft}$$

$$\text{Bare tube area, } A_o = \pi \times 1 \times 12 - \pi \times 1 \times 8 \times 0.035 \times 12 = 27.2$$

$$\text{Total} = 337.2 \text{ in.}^2/\text{ft}, (2.34 \text{ ft}^2/\text{ft})$$

$$\text{Projected perimeter} = 2 \times \frac{3}{8} \times 2 \times 8 \times 12 + 2(12 - 8 \times 0.035 \times 12) = 161.3 \text{ in.}/\text{ft}$$

$$d_e = \frac{2 \times 337.2}{\pi \times 161.3} = 1.33 \text{ in.}$$

$$D_e = 1.33/12 = 0.111 \text{ ft}$$

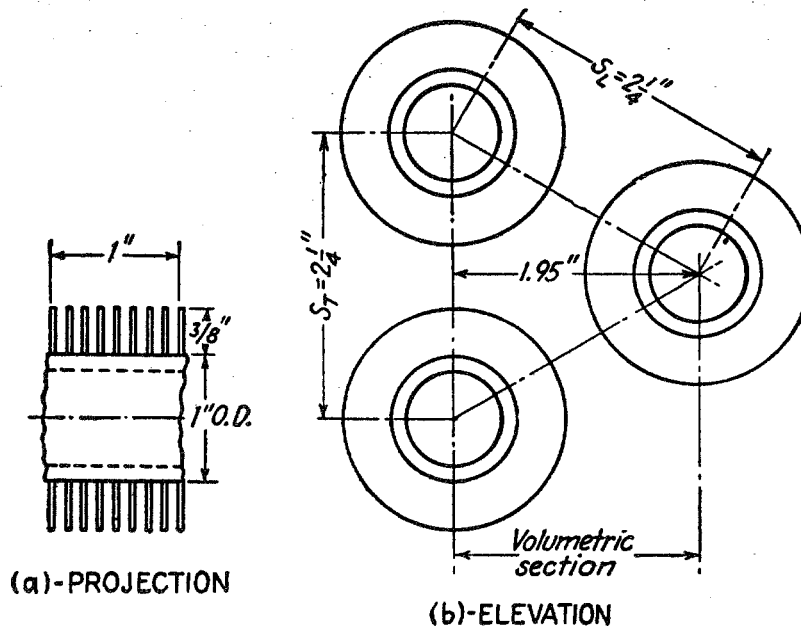


FIG. 16.19.

Flow area:

21 tubes may be fit in one vertical bank

(Fig. 16.19b)

20 tubes in alternating banks for triangular pitch

$$a_s = (4 \times 12)(4 \times 12) - 21 \times 1 \times 48 - 21(2 \times 0.035 \times \frac{3}{8} \times 8 \times 48) = 1079 \text{ in.}^2$$

$$= 1079/144 = 7.50 \text{ ft}^2$$

Hot fluid: duct, air

$$(4') a_s = 7.50 \text{ ft}^2$$

$$D_e = 0.111 \text{ ft}$$

$$(5') G_s = W/a_s$$

$$= 100,000/7.50$$

$$= 13,300 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6') \text{ At } T_a = 225^\circ\text{F,}$$

$$\mu = 0.0215 \times 2.42 = 0.052 \text{ lb/(ft)(hr)}$$

[Fig. 15]

$$Re_s = D_e G_s / \mu$$

$$= 0.111 \times 13,300 / 0.052 = 28,400$$

$$(7') j_f = 157$$

[Fig. 16.18a]

$$(8') \text{ At } T_a = 225^\circ\text{F,}$$

$$k = 0.0183 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$$

$$(c\mu/k)^{1/3} = (0.25 \times 0.052 / 0.0183)^{1/3} = 0.89$$

$$\phi_s = 1.0 \text{ (for air)}$$

$$(9') h_f = j_f (k/D_e) (c\mu/k)^{1/3} \quad [\text{Eq. (6.15)}]$$

$$= 157 \times \frac{0.0183}{0.111} \times 0.89 = 23.1$$

$$R_{do} = 0.003; h_{do} = 1/0.003 = 333$$

$$h'_f = \frac{h_{do} h_f}{h_{do} + h_f} = \frac{333 \times 23.1}{333 + 23.1}$$

$$= 21.5 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

[Eq. (16.37)]

Cold fluid: tubes, treated water

$$(4) a'_t = 0.546 \text{ in.}^2 \quad [\text{Table 10}]$$

Assume each bank carries all of the water.

$$a_t = N_t (a'_t / 144n) \quad [\text{Eq. (7.48)}]$$

$$= 21 \times 0.546 / 144 = 0.0795 \text{ ft}^2$$

$$D = 0.834/12 = 0.0695 \text{ ft}$$

$$(5) G_t = w/a_t$$

$$= 31,200 / 0.0795$$

$$= 392,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$V = G_t / 3600\rho = 392,000 / 3600 \times 62.5$$

$$= 1.74 \text{ fps}$$

$$(6) \text{ At } t_a = 170^\circ\text{F,}$$

$$\mu = 0.37 \times 2.42 = 0.895 \text{ lb/(ft)(hr)}$$

[Fig. 14]

$$Re_t = DG_t / \mu \text{ (for pressure drop only)}$$

$$= 0.0695 \times 392,000 / 0.895 = 30,400$$

$$(9) h_i = 710 \times 0.94 = 667 \quad [\text{Fig. 25}]$$

$$R_{di} = 0.003, h_{di} = 1/0.003 = 333$$

$$\phi_t = 1.0$$

$$h'_i = \frac{h_{di} h_i}{h_{di} + h_i} = \frac{333 \times 667}{333 + 667} = 222$$

[Eq. (16.40)]

h'_{fi} (see Fig. 16.13a) k for brass = 60 Btu/(hr)(ft²)(°F/ft) (Table 3)

$$y_b = \frac{0.035}{2 \times 12} = 0.00146 \text{ ft}$$

$$(r_e - r_b) \sqrt{h'_f / ky_b} = \left(\frac{0.875 - 0.5}{12} \right) \sqrt{\frac{21.5}{60 \times 0.00146}} = 0.49$$

$$\frac{r_e}{r_b} = \frac{0.875}{0.50} = 1.75$$

$$\Omega = 0.91$$

(Fig. 16.13a)

$$h'_{fi} = (\Omega \times A_f + A_o) \frac{h'_f}{A_i} \quad (16.34)$$

$$A_i = 0.218 \text{ ft}^2/\text{ft}$$

$$h'_{fi} = \left(0.91 \times \frac{310}{144} + \frac{27.2}{144} \right) \frac{21.5}{0.218} = 212$$

$$U_{Di} = \frac{h'_{fi} h'_i}{h'_{fi} + h'_i} = \frac{212 \times 222}{212 + 222} = 108 \quad (16.41)$$

$$\text{Inside surface/bank} = 21 \times 4 \times 0.218 = 18.3 \text{ ft}^2$$

$$A_i = \frac{Q}{U_{Di} \Delta t} = \frac{1,250,000}{108 \times 53.7} = 215 \text{ ft}^2$$

Use $215/18.3 = 11.8$ banks, say 12 banks.¹

Pressure Drop

$$(1) D'_{e_v} = \frac{4 \times \text{net free volume}}{\text{frictional surface}}$$

Refer to Fig. 16.19b

$$\begin{aligned} \text{Net free volume} &= 4 \times 4 \times \frac{1.95}{12} - \frac{1}{2} (21 + 20) \frac{\pi}{4} \times \frac{1^2 \times 4}{144} - \frac{1}{2} (21 + 20) \frac{\pi}{4} \\ &\quad (1.75^2 - 1^2) \frac{0.035}{144} \times 8 \times 4 \\ &= 1.91 \text{ ft}^3 \end{aligned}$$

$$\text{Frictional surface}^* = \frac{1}{2} (21 + 20) 2.34 \times 4 = 192 \text{ ft}^2$$

$$D'_{e_v} = 4 \times \frac{1.91}{192} = 0.040 \text{ ft}$$

$$G_s = 13,300 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$Re_s = 0.040 \times 13,300 / 0.052 = 10,200$$

$$f = 0.0024 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 16.18b}]$$

$$\rho = \frac{29}{359 \times 68.5/492} = 0.058 \text{ lb}/\text{ft}^3$$

$$s = 0.058 / 62.5 = 0.000928$$

$$L_p = 12 \times 1.95 / 12 = 1.95 \text{ ft}$$

$$\left(\frac{D'_{e_v}}{S_T} \right)^{0.4} = \left(\frac{0.040}{2.25/12} \right)^{0.4} = 0.538$$

$$\left(\frac{S_L}{S_T} \right)^{0.6} = 1.0$$

$$(1) Re_t = 30,400$$

$$f = 0.00020 \text{ ft}^2/\text{in.}^2$$

[Fig. 26]

¹ The effective path for pressure drop will be approximately $12 \times 1.95 \text{ in.} = 23.4 \text{ in.}$

* The walls of the duct may be neglected.

$$\begin{aligned}
 (2') \Delta P_s &= \frac{f G_s^2 L_p}{5.22 \times 10^{10} D_{e_v} s \phi_s} \left(\frac{D'_{e_v}}{S_T} \right)^{0.4} \left(\frac{S_L}{S_T} \right)^{0.6} \\
 &= \frac{0.0024 \times 13,300^2 \times 1.95 \times 0.538}{5.22 \times 10^{10} \times 0.040 \times 0.000928 \times 1} \\
 &= 0.23 \text{ psi}
 \end{aligned}
 \quad
 \begin{aligned}
 (2) \Delta P_t &= \frac{f G_t^2 L_n}{5.22 \times 10^{10} D_s \phi_t} \\
 &= \frac{0.00020 \times 392,000^2 \times 4 \times 12}{5.22 \times 10^{10} \times 0.0695 \times 1 \times 1} \\
 &= 0.41 \text{ psi}
 \end{aligned}$$

Return losses will be negligible.

Double Extended Surface. Another type of heating element, which at first inspection appears to offer unlimited possibilities, is the double extended surface. Suppose two flowing fluids are separated by a metal wall and the surface is extended into both fluids by means of spines or pegs whose bases are superimposed. Per square foot of wall area it should be possible to add as much surface as desired as long as there are no restrictions along the axes of the spines. Such an arrangement will be called for only when the coefficients from both fluids to their respective fins is small. Under these conditions it is usually found that an unpractical length of fin is required for even a moderate total heat transfer. The fin efficiencies and total transfer can be determined by the methods already treated and by the use of Fig. 16.13*c* or *d*. The heat-transfer coefficients generally must be approximated from more conventional arrangements.

PROBLEMS

16.1. A double pipe exchanger consists of a 3-in. IPS outer pipe and a 1½-in. IPS inner pipe on which several arrangements of fins ½ in. high are under consideration. Compare the weighted efficiency curves with that in Fig. 16.9 when the following are used: (a) 18 steel fins, 20 BWG; (b) 18 copper fins, 20 BWG; (c) 18 steel fins, 16 BWG.

16.2. 4620 lb/hr of a 28°API gas oil will be used to preheat 5700 lb/hr of a 110°API butane reactor feed at elevated pressure from 260 to 400°F. The gas oil will enter at 575°F and leave at 350°F. A pressure drop of 10 psi is permissible on the gas oil, but on the reactor feed it should not exceed 2 to 3 psi. Dirt factors of 0.002 should be provided on each side. The gas oil is the controlling fluid and should flow in the annulus.

Available for the purpose are a number of 20-ft hairpin double pipe exchangers consisting of 2½-in. IPS outer pipes and 1¼ in. OD, 10 BWG inner tubes with 24 fins 20 BWG, ½ in. high.

How many sections should be used, and how shall they be arranged?

16.3. In a regenerative gas-absorption process 10,300 lb/hr of 15°Bé caustic soda ($s = 1.115$) leaves the regenerator at 240°F and is cooled to 170°F. The heat is absorbed by 10,300 lb/hr of 15°Bé caustic soda at 100°F being sent to the regenerator.

Available for the service are a number of 20-ft hairpins consisting of 3-in. IPS shells and 1½-in. IPS inner tubes with 24 fins 20 BWG, ½ in. high. Pressure drops of 10 psi are allowable. Dirt factors of 0.002 should be provided on each side.

How many sections are required, and how shall they be arranged?

16.4. 20,000 lb/hr of nitrogen at 0 gage pressure is to be heated from 100 to 175°F using exhaust steam at 212°F. Fouling factors of 0.002 should be provided for both. Available for the service is a 19¼ in. ID 1-2 exchanger containing 56 1¼ in. OD tubes, 12'0" long and having 24 fins 20 BWG, ½ in. high arranged for two passes.

Is the exchanger satisfactory? What are the pressure drops?

16.5. A textile impregnating room measures 50 by 100 by 12 ft. Because of the possibility of developing an explosive and toxic concentration it is necessary to change the air eight times an hour. To provide comfortable conditions the room should be kept at 75°F, although no provision will be made for humidity control. The lowest winter temperature to be anticipated is 30°F (Middle Atlantic). The air intake will be through an existing 4 by 4 ft duct in which a tempering coil is to be provided using exhaust steam at 212°F.

Available are 4'0" long helical fin tubes with male and female threads. These are ¾ in. OD, 16 BWG with ¾ in. high brass fins 0.018 in. thick spaced six to the inch. Assume that they will be installed on 1¾-in. triangular pitch.

How many tubes are required? What static pressure must be available in the duct?

NOMENCLATURE FOR CHAPTER 16

A	Total outside heat-transfer surface, ft ²
A_f	Surface of thin fins (both sides), ft ²
A_i, A_o	Surface on inside of tubes and bare surface on the outside of a finned tube, respectively, ft ²
a	Flow area, ft ²
a'_i	Flow area/tube, in ²
a_x	Cross-sectional area of fin at right angles to the heat flow, ft ²
b	Length of fin from outer edge to the base, ft
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
$C_1, C_2, C_3, C_4, C_5, C_6, C_7$	Constants, dimensionless
C'_1, C'_2	Constants, dimensionless
c	Specific heat of cold fluid, Btu/(lb)(°F); a constant in the solution of the Bessel equation, dimensionless
D	Inside diameter of tubes or pipes, ft
d	Inside diameter of tubes, in.
D_e, D'_e	Equivalent diameter for heat transfer and pressure drop, ft
D'_{eo}	Volumetric equivalent diameter for crossflow, ft
e	Napierian base
F_T	Temperature-difference factor, $\Delta t = F_T \times \text{LMTD}$, dimensionless
f	Friction factor, ft ² /in ²
G	Mass velocity, lb/(hr)(ft ²)
g	Acceleration of gravity, 4.18×10^8 ft/hr ²
h	Heat-transfer coefficient in general, Btu/(hr)(ft ²)(°F)
h_b	Heat-transfer coefficient h'_f or h_f to fin delivered at the base of the fin, Btu/(hr)(ft ²)(°F)
h_{di}, h_{do}	Dirt coefficient equivalent to the reciprocal of the dirt factor inside the tube R_{di} and outside the tube, R_{do} , respectively, Btu/(hr)(ft ²)(°F)
h_f, h'_f	Heat-transfer coefficient on fin side of a pipe or tube, heat-transfer coefficient on fin side of a pipe or tube corrected for the dirt factor, Btu/(hr)(ft ²)(°F)

h_{fi}, h'_{fi}	h_f and h'_f corrected to the inside surface of a pipe or tube, respectively, Btu/(hr)(ft ²)(°F)
h''_{fi}	Value of h'_{fi} if there were no resistance in the fin itself, Btu/(hr)(ft ²)(°F)
h_{fo}	The average film coefficient of the outside diameter of the tube weighting h_f to the fin surface and to the tube outside surface, Btu/(hr)(ft ²)(°F)
h_i, h'_i	Heat-transfer coefficient for the fluid inside the tube, h_i corrected for the dirt factor, Btu/(hr)(ft ²)(°F)
h_o	Heat-transfer coefficient for outside fluid, Btu/(hr)(ft ²)(°F)
$I_n(u)$	Modified Bessel function of the first kind and order n , dimensionless
i	$\sqrt{-1}$, dimensionless
j_f	Factor for heat transfer to finned pipes and tubes, dimensionless
j_H, j_h	The Sieder-Tate and Colburn heat-transfer factors, dimensionless
K	Temperature group for crossflow $(T_1 - T_2)/(T_2 - t_1)$, dimensionless
$K_n(u)$	Modified Bessel function of the second kind and order n , dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Length of tube, ft
L_p	Length of path, ft
LMTD	Logarithmic mean temperature difference, °F
l	Height of fin at any point, ft
m	Either of two constants used in the derivation of longitudinal or annular fins, dimensionless
N_f	Number of fins (per tube)
N_t	Number of tubes in bundle
n	Number of tube passes; a constant denoting the order of the Bessel function, dimensionless
P	Perimeter, ft
ΔP	Pressure drop, psi
p	A constant in the Bessel equation, dimensionless; pressure, atm
Q	Total heat flow or total heat flow through bases of fins and bare tube surface, Btu/hr
Q_o	Heat flow, heat flow through bare tube surface, Btu/hr
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_{do}, R_{di}	Dirt or fouling factor, outside tube and inside tube, (hr)(ft ²)(°F)/Btu
Re, Re	Reynolds number for heat transfer and pressure drop, dimensionless
r	Radius, ft
(r)	Temperature group in crossflow, $\Delta t/(T_1 - t_1)$, dimensionless
$(r)_{\text{counterflow}}$	Temperature group in counterflow, LMTD/($T_1 - t_1$), dimensionless
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless

s	Specific gravity, dimensionless
T	Temperature of hot fluid, °F
T_s	Temperature of steam, °F
t	Temperature of fin or cold fluid in general, °F
T_e, t_e	Caloric temperature of hot fluid and cold fluid, respectively, from Eqs. (5.28) and (5.29), °F
t_w, t'_w, t_{fw}	Temperature of tube wall, temperature in fin metal, temperature of wall in finned tubes, °F
t_f, t'_f	Temperature of clean fin, temperature of fin beneath dirt film, °F
Δt	True temperature difference, °F
$\Delta t_c, \Delta t_h$	Hot and cold terminal differences, °F
$\Delta t_f, \Delta t_{do}, \Delta t_{di}, \Delta t_i$	Temperature drops across fin coefficient, outside dirt, inside dirt, and the inside fluid film corrected for viscosity and dirt, °F
U_{Di}	Overall design coefficient of heat transfer based on the inside tube surface, Btu/(hr)(ft ²)(°F)
U_i	Overall clean coefficient of heat transfer based on inside tube surface, Btu/(hr)(ft ²)(°F)
u	A function
V	Velocity of flow, fps
v	A function
W	Weight flow in general, weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
X, Y, x, y	Distances, ft
y	Half the width of a fin, ft
α	Constant in transverse fin and crossflow derivations
β	Constant in transverse fin and crossflow derivations
Θ	Temperature difference between fluid and fin $T - t$, °F
μ	Viscosity at the caloric temperature, lb/(ft)(hr)
μ_w, μ_{fw}	Viscosity at the tube wall or fin temperatures, t_w or t_{fw} , lb/(ft)(hr)
ρ	Density, lb/ft ³
ϕ	Viscosity ratio $(\mu/\mu_{fw})^{0.14}$, dimensionless
Ω	Fin efficiency h_b/h_f , dimensionless
Ω'	Weighted efficiency for fin and bare tube, dimensionless

Subscripts (except as noted above)

a	Annulus
b	Base (of fin)
e	Edge (of fin)
i	Interface of film or inside of pipe or tube
s	Shell
t	Tube
I	First of two passes
II	Second of two passes

CHAPTER 17

DIRECT-CONTACT TRANSFER: COOLING TOWERS

Introduction. In all preceding chapters the hot and cold fluids were separated by impervious surfaces. In tubular equipment, the tube limits the intimacy of contact between the hot and cold fluids and also serves as a surface upon which resistances accumulate as fouling and scale films. In order that a turbulent fluid in a tube may receive heat, particles in the eddying fluid body must contact a warm film at the tube wall, take in heat by conduction, and then mix with the eddying fluid body. A similar process occurs on the shell side, and the net heat exchange may occur through as many as seven individual resistances.

One of the principal reasons for employing tubes is to prevent the contamination of the hot fluid by the cold fluid. When one of the fluids is a gas and the other a liquid, an impervious surface is often unnecessary, since there may be no problem of mutual contamination, the gas and liquid being readily separable after mixing and exchanging heat. Fouling resistances are automatically eliminated by the absence of a surface on which they can collect and permit a direct-contact apparatus to operate indefinitely with uniform thermal performance. The greater intimacy of the direct contact generally permits the attainment of greater heat-transfer coefficients than are usual in tubular equipment.

Perhaps the outstanding application of an apparatus operating with a direct contact between a gas and a liquid is the cooling tower. It is usually a boxlike redwood structure with redwood internals. Cooling towers are employed to contact hot water coming from process cooling systems with air for the purpose of cooling the water and allowing its reuse in a process. The function of the wooden internals, or *fill*, is to increase the contact surface between the air and the water. A cooling tower ordinarily reduces the fresh cooling-water requirement by about 98 per cent, although there is some mutual contamination caused by the saturation of the air with water vapor.

The prospective use for direct-contact equipment in other services requiring rapid rates of heat transfer is perhaps greater than for any other type of heat-transfer apparatus. Although applied now almost exclusively to the humidification of air or the cooling of water, the principles of direct-contact heat transfer may be applied to the cooling or heating of

other insoluble gases or liquids. This is especially true in the cooling of gases over long temperature ranges. The jet condenser referred to in Chap. 14 is also an example of direct contact as applied to condensation in which a large heat load may be condensed in an apparatus of small volume. A modification of the same principle might readily be applied to the condensation of organic vapors by a spray of water and particularly to the problem of condensing an oil vapor in the presence of a non-condensable gas. Future developments may well be anticipated for the recovery of atomic energy in commercial fission processes, in jet propulsion, and in gas-turbine cycles. This chapter treats the theory and develops the principal calculations applicable to direct-contact heat transfer.

Diffusion. If dry air at constant temperature is saturated by water at the same temperature in a direct-contact apparatus, the water vapor entering the air carries with it its latent heat of vaporization. The humidity of the air-water-vapor mixture increases during saturation because the vapor pressure of water out of the liquid is greater than it is in the unsaturated air and vaporization is the result. When the vapor pressure of water in the air equals that of the liquid, the air is saturated and vaporization ceases. The temperature of the water can be kept constant during air saturation if heat is supplied to it to replace that lost from it to the gas as latent heat of vaporization. Clearly, then, the heat transfer during the saturation of a gas with a liquid can be made to proceed without a temperature difference, although such a limitation is rarely encountered. It is seen, however, that there is a fundamental difference between this type of heat transfer and conduction, convection, or radiation.

When a movement of material is promoted between two phases by a vapor pressure (or concentration) difference, it is *diffusion* and is characterized by the fact that material is transferred from one phase to another or between both phases. This behavior is called *mass* or *material transfer* to set it apart from the ordinary concepts of heat transfer. While the phase-rule definitions apply to systems at equilibrium, if a phase is not homogeneous, *self-diffusion* may occur as the phase approaches homogeneity.

For the condensation of a vapor in the presence of a noncondensable gas it was expedient in Chap. 13 to introduce Colburn's analogy between heat transfer and mass transfer. This in turn was compared with Reynold's analogy between heat transfer and fluid friction as discussed in Chap. 3. The Reynold's analogy holds more closely for heat transfer and fluid friction than do the Colburn and other analogies for heat transfer and mass transfer. In the condensation of a vapor from a noncon-

densable gas an inert film near the tube wall retards the condensable vapor from reaching the condensate film at the tube wall. The rate at which the vapor passes through the inert film is a function of the pressure of the vapor in the gas body and in the condensate film adjoining the tube wall and follows the mechanism of diffusion.

Several of the illustrative examples solved in this chapter can be understood without a thorough background in the theory of diffusion, but the advantages of a good foundation are evident. The elements of simple diffusion are presented here in abbreviated form, drawing, wherever possible, upon the similarity of diffusion to convection heat transfer. Excellent treatments of the subject may be found in Sherwood¹ and Perry².

Diffusion Theory. Diffusion involves the passage of one fluid through another. Consider a gas, such as air, containing a small amount of acetone vapor which is soluble in water while the air may be considered insoluble in water. Suppose the air-acetone mixture is fed to a tower which has fresh water flowing continuously down its walls so that any acetone molecules which might bound into the water are removed by it from the gas body. How fast will acetone molecules be removed from the gas body?

An idealized picture of the problem is shown in Fig. 17.1. It may be assumed that a relatively stagnant air film forms at the liquid surface owing to the loss of momentum of the air molecules striking the liquid film and being dragged along by it. This is represented between 1-1 and 2-2'. The liquid film may also be considered relatively stagnant compared with the air body. This is the basis of the "two-film" theory. It may be conceded, because of the mutual solubility of acetone in water, that the rate at which acetone molecules can pass through the liquid film is exceedingly great. Thus the acetone molecules in the air film which arrive at the liquid film are depleted so rapidly by solution in the liquid film that the concentration of acetone molecules in the air film is less than it is in the gas body. This establishes a pressure or concentration

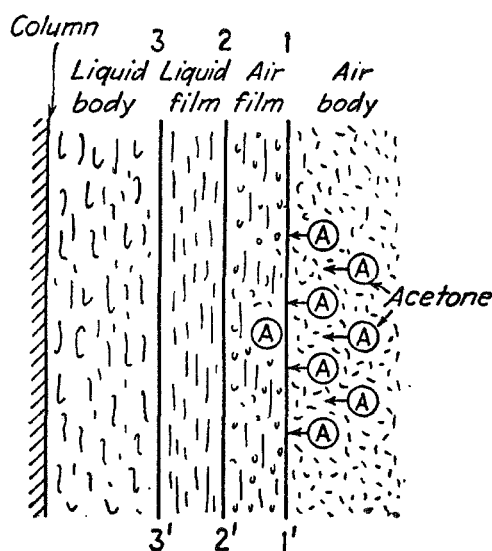


FIG. 17.1. Film theory showing principal resistances.

¹ Sherwood, T. K., "Absorption and Extraction," McGraw-Hill Book Company, Inc., New York, 1937.

² Perry, J. H., "Chemical Engineers' Handbook," 3d ed., Sec. 8, pp. 523-559, A. P. Colburn and R. L. Pigford, McGraw-Hill Book Company, Inc., New York, 1950.

gradient between the air body and the air film which continuously promotes the movement of acetone molecules in the direction of the liquid. This picture is analogous to that of the steady-state film theory of Chap. 3 in which a temperature gradient promoted heat transfer.

The air and liquid films are similar to thermal resistances in series. In the case of diffusion from 1-1' to 2-2' the concentration difference of the diffusing air-acetone mixture is the potential for the mass transfer of the acetone across the several resistances in series. Designating the diffusing gas by the subscript A and the inert or insoluble gas (air) by the subscript B the concentration of each gas may be expressed individually by its molar density, δ mols/ft³. The rate at which the concentration of A in the air body decreases, $-d\delta_A$, depends upon four factors: (1) the number of mols of A , δ_A ; (2) the number of mols of B , δ_B ; (3) the relative difference of movement between the velocity of A , u_A and the velocity of B , u_B in the direction of diffusion; and (4) the length of the air film dl .

$$-d\delta_A = \alpha_{AB}\delta_A\delta_B(u_A - u_B) dl \quad (17.1)$$

where α_{AB} is the proportionality constant. If the net diffusion of the insoluble gas is zero, $u_B = 0$ and Eq. (17.1) reduces to

$$-d\delta_A = \alpha_{AB}\delta_A\delta_B u_A dl \quad (17.2)$$

If N_A is the number of pound-mols per hour transferred through A ft² of film surface,

$$\frac{N_A}{A} = u_A\delta_A \quad \text{lb-mol/(hr)(ft}^2\text{)} \quad (17.3)$$

$$-d\delta_A = \alpha_{AB}\delta_B \frac{N_A}{A} dl \quad (17.4)$$

and since

$$-d\delta_A = d\delta_B \quad (17.5)$$

$$N_A = \frac{A}{\alpha_{AB}\delta_B} \frac{d\delta_B}{dl} = \frac{A}{\alpha_{AB}} \frac{d \ln \delta_B}{dl} \quad (17.6)$$

and for equimolar diffusion of A into B .

$$\frac{N_A}{A} = -\frac{N_B}{A} = u_A\delta_A = -u_B\delta_B \quad (17.7)$$

From Eq. (17.1)

$$-d\delta_A = (\alpha_{AB}\delta_B u_A \delta_A - \alpha_{AB}\delta_A u_B \delta_B) dl = \alpha_{AB} u_A \delta_A (\delta_A + \delta_B) dl \quad (17.8)$$

$$u_A \delta_A = -\frac{1}{\alpha_{AB}(\delta_A + \delta_B)} \frac{d\delta_A}{dl} \quad (17.9)$$

From the continuity equation, input - output = accumulation. At steady state there will be no accumulation of A in the gas film. In pounds per cubic foot per unit time

$$-M_A \frac{\partial \delta_A}{\partial \theta} - M_A \frac{\partial (u_A \delta_A)}{\partial l} = 0 \quad (17.10)$$

where M_A is the molecular weight of gas A and θ is time.

$$\frac{\partial \delta_A}{\partial \theta} = \frac{\partial}{\partial l} \left[\frac{1}{\alpha_{AB}(\delta_A + \delta_B)} \frac{d\delta_A}{dl} \right] \quad (17.11)$$

Call $\delta_A + \delta_B = \delta$ the total number of mols per cubic feet.

$$\frac{\partial \delta_A}{\partial \theta} = \frac{\partial}{\partial l} \left(k_d \frac{\partial \delta_A}{\partial l} \right) \quad (17.12)$$

Defining the constant $k_d = 1/\alpha_{AB}\delta$ or

$$\alpha_{AB} = \frac{1}{k_d \delta} \quad (17.13)$$

k_d is the *diffusivity* or diffusion coefficient introduced in Chap. 13. It is basically defined, however, by Eqs. (17.12) and (17.13). Returning to Eq. (17.6),

$$N_A = k_d \delta A \frac{d \ln \delta_B}{dl} \quad (17.14)$$

And for a perfect gas, which the insoluble gas usually approximates,

$$\delta = \frac{n}{v} = \frac{p_t}{RT} \quad (17.15)$$

where n = total number of mols

p_t = total pressure

R = gas constant

$$N_A dl = \frac{k_d p_t A}{RT} d \ln \delta_B \quad (17.16)$$

Integrating over the length of the gas film from the gas body to the gas-film-liquid-film interface,

$$N_A = \frac{k_d p_t A}{RTl} \ln \frac{\delta_B}{\delta_{Bi}} = \frac{k_d p_t A}{RTl} \ln \frac{p_B}{p_{Bi}} \quad (17.17)$$

where the concentrations have also been expressed by the partial pressures and where the subscript B_i refers to the value at the interface 2-2'. For a

driving force across the gas film consisting of the two pressures p_B and p_{Bi} , let p_{Bm} be the log mean driving pressure of the inert B . Then

$$p_{Bm} = \frac{p_B - p_{Bi}}{\ln p_B / p_{Bi}} \quad (17.18)$$

and

$$\ln \frac{p_B}{p_{Bi}} = \frac{p_B - p_{Bi}}{p_{Bm}} \quad (17.19)$$

$$N_A = \frac{k_d p_t A}{RT l p_{Bm}} (p_B - p_{Bi}) \quad (17.20)$$

This is similar to the ordinary transfer equation $Q = h_o A (T_c - t_w)$, using the nomenclature of Chap. 6, where $T_c - t_w$ is the temperature difference at a single cross section and the tube-wall temperature t_w corresponds to the interface concentration. Since the partial pressure of a component is proportional to its mol fraction in a mixture of perfect gases,

$$N_A = k_G A (p - p_i) = k_G A p_t (y - y_i) \quad (17.21)$$

where y and y_i are the mol fractions in the gas body and at the interface and

$$k_G = \frac{k_d p_t}{RT l p_{Bm}} \quad (17.22)$$

k_G may be compared with h for one of two fluids engaging in heat transfer.

The diffusivity can be determined experimentally by measuring the rate of evaporation of the diffusing gas A from a volumetric container into an inert which is passed over the vessel. Diffusivities can be computed by the Gilliland equation [Eq. (13.31)]. It will be noted that in Eq. (13.31) k_d is inversely proportional to the total pressure on the system p_t so that in Eq. (17.20) no correction need be made for pressure, since the product $k_d p_t$ in the numerator will be constant.

In an ordinary diffusion tower the potential difference $p - p_i$ or $y - y_i$ differs at every cross section or height of the tower as material is transferred out of the gas body. If the total transfer is the sum of a number of transfers through incremental surfaces with a differential potential on each increment, the differential equation for the entire height in which all the surface is contained becomes

$$dN_A = k_G (p - p_i) dA = k_G p_t (y - y_i) dA \quad (17.23)$$

Just as the total heat transfer Q can be calculated from a single individual film coefficient, the total surface, and the temperature difference between the fluid and the tube wall, so the material transfer can be determined from the change in the gas phase alone, using Eq. (17.21). The total material

leaving the gas phase is obviously the same as that which enters the liquid phase. By a similar analysis to that above it can be shown that, for the liquid-film-liquid-body interface,

$$dN_A = k_L([c]_{Ai} - [c]_A) dA = k_L[c]_{av}(x_i - x) dA \quad (17.24)$$

where $[c]$ is the concentration in the liquid with subscripts as above, x the mol fraction of the diffusing material in the liquid, and k_L is the liquid-side diffusion coefficient. Expressed as an equality,

$$dN_A = k_G p_i(y - y_i) dA = k_L[c]_{av}(x_i - x) dA \quad (17.25)$$

k_G and k_L are the reciprocals of the two resistances in series comprising the gas and liquid films. In terms of overall coefficients and the overall potential for diffusion

$$\begin{aligned} dN_A &= K_G(p - p') dA = K_L([c]'_A - [c]_A) dA \\ &= K_G p_i(y - y') dA = K_L[c]_{av}(x' - x) dA \end{aligned} \quad (17.26)$$

where p' = partial pressure of the diffusing vapor which would be in equilibrium with liquid of the concentration of the liquid body, atm

$[c]'_A$ = concentration of the diffusing vapor which would be in equilibrium with the partial pressure of the diffusing vapor in the gas body, lb-mols/ft³

x' and y' = mol fractions corresponding to $[c]'_A$ and p' , respectively, dimensionless.

The overall coefficients are expressed in either of the two ways

K_G = overall coefficient of mass transfer, lb-mols/(hr)(ft²) (atmosphere of partial pressure difference)

K_L = overall coefficient of mass transfer, lb-mols/(hr)(ft²) (concentration difference/ft³)

K_G and K_L may be related to the individual films and to each other by means of Henry's law.

$$p' = C_H[c]'_A$$

where p' is the equilibrium partial pressure of A in the gas phase corresponding to a liquid concentration of $[c]'_A$ lb-mols/ft³. C_H is the Henry's law proportionality constant. The law holds only for relatively dilute solutions.

Then

$$K_G = \frac{1}{(C_H/k_L) + (1/k_G)} \quad (17.27)$$

$$K_L = \frac{1}{(1/k_L) + (1/C_H k_G)} \quad (17.28)$$

K_G and K_L are really the same except for the dimensional differences in the respective equations in which they are employed. Equations (17.27) and (17.28) will be seen to resemble Eq. (6.7) for obtaining the overall coefficient of heat transfer from two individual film coefficients.

Colburn¹ has introduced the idea of a unit of mass transfer which is a measure of the number of interphase equilibrium changes required to effect a given amount of diffusion. It is identical with the concept of a theoretical plate in distillation under a particular condition. If G_m is the gas rate in mols per hour per square foot of tower cross section and dy is the change of concentration of the diffusing component, $dN_A = G_m dy$ and Eq. (17.26) can be written

$$G_m dy = K_G p_i (y - y') dA \quad (17.29)$$

If the surface per cubic foot of tower is $a \text{ ft}^2/\text{ft}^3$, the total surface is $dA = a dV$, where V is the volume of the tower per square foot of tower cross section.

$$n_t = \int \frac{dy}{y - y'} = K_G a \frac{V}{G_m} \quad (17.30)$$

The integral of $dy/(y - y')$ over the entire height of tower gives the number of times the average potential may be divided into the total desired concentrate change. This is an index of the size of the absorption or desorption task which must be accomplished, and n_t is called the *number of transfer units*. When n_t is multiplied by G_m , it gives $K_G a V$, which is the total number of mols of material transferred. Different heights of various diffusion towers are required for the accomplishment of one transfer unit of diffusion depending upon how a particular tower is constructed and how much surface it contains per cubic foot of volume. From experiments on a particular type of tower of overall height Z it is possible to determine experimentally the number of transfer units of diffusion accomplished by it, and the height of a single transfer unit HTU will be

$$HTU = \frac{Z}{n_t} \quad (17.31)$$

The Wet-bulb Temperature. Humidification is a form of heat transfer as well as a form of diffusion. In air conditioning systems it is the increase in the moisture content of air and is usually accomplished by a spray washer. In a cooling tower air is also humidified but cold water and not moist air is its principal product. Particularly relative to the cooling tower, it is customary in the United States to define the moisture content of air by its wet-bulb temperature. This is a valuable concept,

¹ Colburn, A. P., *Trans. AIChE*, **35**, 211 (1934).

since it will be shown later that the wet-bulb temperature is also the lowest water temperature which can possibly be obtained by adiabatic humidification. The wet-bulb temperature is described here because it is a simple concept from which an excellent picture of simultaneous mass and heat transfer can be drawn. Refer to Fig. 17.2 which consists of a thermometer surrounded by a wick dipping into water at the same temperature as the surrounding air. The wick is always wet. A second thermometer is suspended in the surrounding air to indicate the *dry-bulb* temperature.

Assume that unsaturated air at any dry bulb is circulated over the wick. Because the wick is wet and the air is unsaturated, the partial pressure of the water vapor out of the wick is greater than that of the water vapor in the circulated air and water evaporates from the wick to the air. But the evaporation of water from the wick requires many Btu of latent heat, which must come from somewhere. When the air is initially circulated over the wick, the Btu could come from the wick itself by lowering the temperature of the wick below its initial temperature. If the initial wick temperature was the same as the dry bulb, any temperature drop in the wick would establish a temperature difference between the dry-bulb temperature and the lower temperature of the wick.

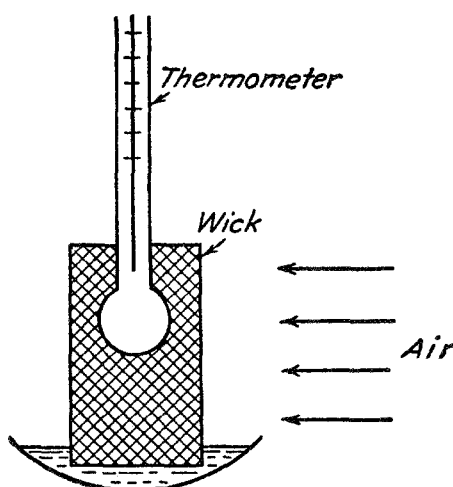


FIG. 17.2. The wet-bulb thermometer.

This causes sensible heat to flow from the air into the wick, thereby reducing the air temperature. An additional depression of the wick temperature continues as long as air is circulated, until a point is reached at which the temperature difference between the dry bulb and the wick causes just enough heat to flow into the wick to counterbalance the loss of heat from the wick by evaporation. Under this condition an equilibrium is established whereby the vaporization into a cubic foot or a pound of air is offset by the sensible heat removed from a cubic foot or a pound of air. Thus it should make no difference whether the air passes slowly or rapidly over the wick when the equilibrium is attained; the temperature of the wick should be depressed to the same extent, since the amount of water evaporating into each pound of dry air is offset by an equal sensible-heat removal from the same pound of air. The faster air is circulated over the wick, however, the greater will be the weight of water evaporated from the wick. This convenient temperature is called the *wet-bulb temperature*. The wet bulb naturally varies for air of a given dry bulb according

to the extent to which it is saturated before contacting the wick, since the degree of saturation affects the amount of water vapor which can be evaporated into the air and consequently the amount of heat which can be removed from it.

The depression of the temperature of the wet-bulb thermometer, in the experiment described above, started when the water, presumably at the dry-bulb temperature, evaporated into the air at the same temperature. If the air had been saturated at the dry-bulb temperature, no water would have evaporated at all. If instead of humidifying air our primary interest was in a process for obtaining cold water by evaporating part of the water from the wick, the lowest water temperature attainable would be the wet-bulb temperature. Thus a cooling tower or any other humidifying apparatus which transferred no heat to or from its surroundings while the air and liquid were in contact could cool the water only to the wet-bulb temperature of the air, which is a function of the degree of saturation of the air or inert gas with the liquid being evaporated. In the northern part of the temperate zone, where a wet-bulb temperature of 75°F occurs in the summertime, it is impossible to cool water lower than 75°F except by refrigeration.

The process between the air and the wick can be described rather simply for the equilibrium condition when the wick has presumably reached the wet-bulb temperature. Let X be the absolute humidity of the original air, pounds of water per pound of dry air, which is related to the partial pressure of the water vapor in the original air by

$$X = \frac{p_{\text{water vapor}}}{p_t - p_{\text{water vapor}}} \times \frac{M_w}{M_a}$$

where M_w and M_a are the molecular weights of water and air. Let X' be the humidity (or vapor pressure) of the water out of the wick at the wet-bulb temperature. For simplicity, a diffusion overall mass-transfer coefficient K_x will be used in which the evaporative potential from the wick to the air is expressed in terms of the two humidities X' and X , whereas K_G employs the pressure in atmospheres to express the diffusion potential. Then if λ is the latent heat of vaporization, the heat leaving the wick per square foot of surface is given by

$$\frac{Q}{A} = K_x(X' - X)\lambda \quad (17.32)$$

The heat of vaporization per square foot of wick is offset by the heat transferred from the air to the wick, which is given by

$$\frac{Q}{A} = h(t_{DB} - t_{WB}) \quad (17.33)$$

where t_{DB} and t_{WB} are the dry-bulb and wet-bulb temperatures of the air, respectively, and h is the sensible-heat-transfer coefficient between the air and the wick.

Since Eqs. (17.32) and (17.33) are identities, and can be equated.

$$h(t_{DB} - t_{WB}) = K_X(X' - X)\lambda \quad (17.34)$$

or

$$\frac{h}{K_X} = \frac{(X' - X)\lambda}{t_{DB} - t_{WB}} \quad (17.35)$$

and the potential for K_X is pounds of water/pound of dry air. Using K_G for K_X the potential is atmospheres of water/atmosphere of air, which employs the *molar* ratio of water to air for a perfect gas. Weight and mol ratios are obviously not equal, although both are dimensionless. Thus a 1:1 mol ratio of water to air is equal to an $18\frac{1}{2}$:1 or 0.62:1 weight ratio. The net effects of weight transfer must be the same as the product of molar transfer and the molecular weight of the diffusing vapor.

$$K_X(X' - X) = K_G M_A (p' - p) \quad (17.36)$$

$$X' = \frac{p'}{p - p'} \frac{M_A}{M_B} = \frac{p'}{p_{Bm}} \frac{M_A}{M_B} = \frac{p'}{p_t} \frac{M_A}{M_B} \quad (17.37)$$

The last is very nearly true for the temperature ranges in humidification

$$X = \frac{p}{p - p'} \frac{M_A}{M_B} = \frac{p}{p_{Bm}} \frac{M_A}{M_B} = \frac{p}{p_t} \frac{M_A}{M_B} \quad (17.38)$$

$$K_X \left(\frac{p' - p}{p_{Bm}} \right) \frac{M_A}{M_B} = K_G M_A (p' - p) \quad (17.39)$$

$$K_X = K_G p_{Bm} M_B \quad (17.40)$$

Relation between the Wet-bulb and the Dew-point Temperatures.

There is an interesting distinction between the wet-bulb and dew-point temperatures. Both are used in diffusion calculations. To obtain the wet bulb temperature the following is considered: As bone-dry air plus its accompanying vapor is circulated continuously over a wick at a given initial temperature, the air and vapor mixture are cooled by the passage of sensible heat from it into the wick. The air and vapor mixture get the heat back, however, in the form of vaporized water whose latent heat is equal to the heat which passed into the wick although the final temperature of the mixture is lower after circulation. The volume or weight of vapor per pound of *dry* air increases. Since no heat enters or leaves the system during the direct contact, it is an adiabatic process, and the wet bulb is obviously a temperature in such an adiabatic process.

The dew point is the temperature at which a gas of given vapor content deposits the first drop of condensate when cooled by a *constant*-

pressure process. In the temperature-entropy plot of Fig. 17.3 the saturation curve for water is described by *BEF*. The constant pressure cooling of water vapor occurs along the constant pressure path *AB*. Consider next the adiabatic contact between water and air. Being adiabatic, it is described by a fictitious path *A''C''*, which is an adiabat although not isentropic, since the mixture of air and water vapor is irreversible. The cooling of the air and evaporation of the water separately are not true adiabatic processes. Only the gross effects between the two are truly adiabatic. For the process of adiabatic saturation the path of the water vapor *alone* can be determined by experiment and is found to be along the line *AC*. The point *C* is the wet bulb, and since water vapor is added during the adiabatic saturation the partial pressure of the water vapor

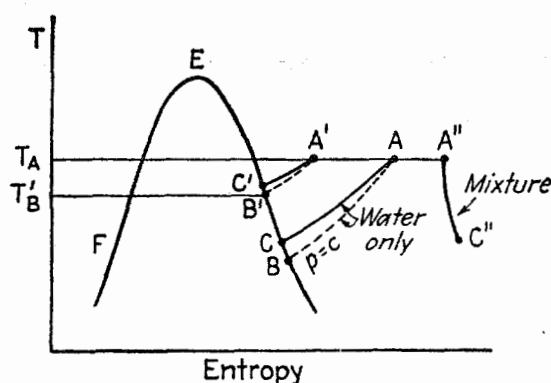


FIG. 17.3. Relation between wet-bulb and dew-point temperatures.

at the wet bulb *C*, is somewhat higher than *B*. When air is initially 70 per cent saturated or more, the wet bulb exceeds the dew point by less than 2 per cent of the difference between the dry bulb and the wet bulb. If gas at the dry-bulb temperature T_A possesses a higher degree of saturation, the point *A* moves to *A'* and the dew point is at the obviously higher temperature T'_B .

The Lewis Number. The relationship between the heat-transfer coefficient and the mass-transfer coefficient, h/K_x , enters into nearly every direct-contact problem. The analogy between heat transfer and fluid friction in Chap. 3 was arrived at by mathematical similarity. Heat transfer and mass transfer may also be related by comparing both with fluid friction. References to the subject will be found in Chap. 13. Only the most general application is developed here, although it appears that greater accuracies may be obtained from other analogies. For heat transfer Chilton and Colburn give

$$j_h = \frac{f}{2} = \frac{hD/k}{(DG/\mu)(c\mu/k)} \left(\frac{c\mu}{k} \right)^{2/3} = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{2/3}$$

where f is the dimensionless friction factor.

For mass transfer (diffusion)

$$j_d = \frac{f}{2} = \frac{K_x D / \rho k_d}{(DG/\mu)(\mu/\rho k_d)} \left(\frac{\mu}{\rho k_d} \right)^{2/3} = \frac{K_x}{G} \left(\frac{\mu}{\rho k_d} \right)^{2/3} \quad (17.41)$$

Dividing,

$$\frac{j_d}{j_h} = \frac{K_{xc}(\mu/\rho k_d)^{2/3}}{h(c\mu/k)^{2/3}} = 1 \quad (17.42)$$

or

$$\frac{h}{K_{xc}} = \frac{(\mu/\rho k_d)^{2/3}}{(c\mu/k)^{2/3}} = \left(\frac{k}{c\rho k_d}\right)^{2/3} = 1 \quad (17.43)$$

and

$$\frac{h}{K_{xc}} = 1 \quad (17.44)$$

Lewis¹ called attention to this relation, and the dimensionless group h/K_{xc} is known as the Lewis number Le . It conveys the extremely important information that the heat-transfer coefficient is to the mass-transfer coefficient as the value of the specific heat of the medium which serves for both heat transfer and mass transfer. For the case of the wick

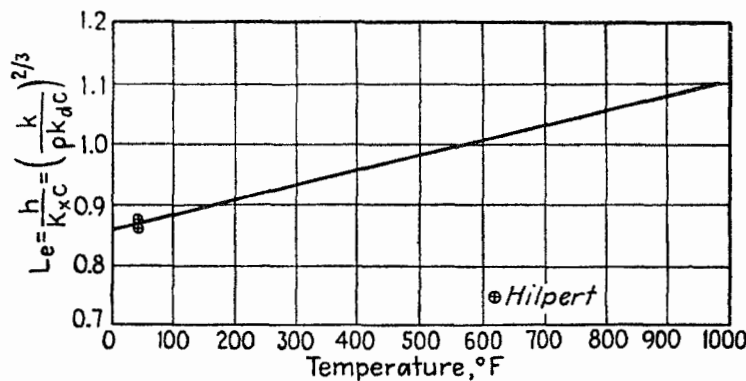


FIG. 17.4. Lewis numbers for the air-water system.

in a wet-bulb experiment if the mean specific heat of the moist air is 0.25 Btu/(lb)(°F), then the heat-transfer rate will be but one-quarter of the mass-transfer rate. It thus requires four times as great a potential for a given number of Btu to be transferred by sensible heat as for an equal number of Btu by mass transfer. The plot of Lewis number vs. temperature for the system air-water calculated from $Le = (k/\rho k_d c)^{2/3}$ is shown in Fig. 17.4. Two points obtained experimentally by Hilpert² are also included in the plot. Values of h/K_x for other systems are tabulated in Table 17.1.

The Effective Film. In the wet-bulb experiment it was seen that the same wet-bulb temperature would be attained regardless of the rate of gas flow over the wick. It has also been shown in Chap. 11 that the rate of heat transfer varies with the 0.8 power of the gas velocity. The same should be true for mass transfer. In the Eq. (17.44), if the ratio of the

¹ Lewis, W. K., *Trans. ASME*, **44**, 329 (1922).

² Hilpert, R., *Forschungsheft*, **3**, 355 (1932).

coefficients is constant, the film for both heat transfer and mass transfer must be affected to the same degree by a change in the gas rate. Sherwood¹ has presented a correlation for the vaporization of numerous liquids into air, including water, and obtained

$$\frac{D}{l} = 0.023 \left(\frac{DG}{\mu} \right)^{0.83} \left(\frac{\mu}{\rho k_d} \right)^{0.44} \quad (17.45)$$

where D is the inside diameter of the tube and l is the thickness of the film. This may be compared with Eq. (6.2) of Sieder and Tate for heat transfer:

$$\frac{h_i D}{k} = 0.027 \left(\frac{DG}{\mu} \right)^{0.8} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu_w}{\mu} \right)^{0.14}$$

TABLE 17.1. AVERAGE VALUES OF h/K_X CALCULATED FROM WET-BULB DETERMINATIONS*
Diffusion into Air

Vapor	$\frac{h}{K_X}$	Calculated from Chilton-Colburn analogy, Eq. (17.43)
Benzene	0.41	0.44
Carbon tetrachloride	0.44	0.49
Chlorobenzene	0.44	0.48
Ethyl acetate	0.42	0.46
Ethylene tetrachloride	0.50	0.51
Toluene	0.44	0.47
Water	0.26	0.21

* Sherwood, T. K., *Trans. AIChE*, **28**, 107 (1932).

when $(\mu/\mu_w)^{0.14}$ is substantially 1.0 for gases. Other interesting extensions of the theory have been made by Arnold² and Chilton and Colburn.³

HUMIDIFICATION AND DEHUMIDIFICATION

Humidifiers: The Cooling Tower. The largest present-day uses of diffusional heat transfer are found in the cooling tower, the air-conditioning spray chamber, the spray drier, the spray tower, and the spray pond. The use of cooling towers has grown tremendously in the last twenty years owing to an increasing necessity. In many industrial localities cold fresh water is too scarce to permit its unlimited use as a cooling medium. The problem of supplying sufficient surface and subsurface cooling water has grown to such an extent that new plants are often

¹ Sherwood, *op. cit.*, p. 39.

² Arnold, J. H., *Physics*, **4**, 255 (1933).

³ Chilton, T. H., and A. P. Colburn, *Ind. Eng. Chem.*, **26**, 1183 (1934).

required to develop the continual reuse of the limited water they may obtain from public or private sources. In some communities even river water, which may be present in abundance, requires precooling as discussed in Chap. 7. This is especially true of some of the rivers in the southern part of the United States having northern sources and which are heated to the dry-bulb temperature by the time they reach the South.

The available temperature of cooling water has been shown to be an important economic factor in the design of modern chemical and power plants. In the chemical plant it fixes the operating pressure on the condensers of distillation and evaporation processes and consequently on the equipment preceding them. In the power plant it fixes the turbine- or engine-discharge pressure and the ultimate recovery of heat. For these vital reasons the study of the cooling tower and the temperature of the water made available by it is of great importance in the planning of a process. The cooling tower is also the simplest member of a class of apparatus whose potentialities have scarcely been realized.

Classification of Cooling Towers. Modern cooling towers are classified according to the means by which air is supplied the tower. All employ stacked horizontal rows of fill to provide increased contact surface between the air and water. In *mechanical-draft* towers the air is supplied in either of the two ways shown in Fig. 17.5a and b. If the air is sucked into the tower by a fan at the top through louvers at the bottom, it is *induced draft*. If the air is forced in by a fan at the bottom and discharges through the top, it is *forced draft*. *Natural-circulation* towers are of two types, *atmospheric* and *natural draft* as in Fig. 17.5c and d.

Mechanical-draft Towers. Towers of this class are the commonest erected in the United States at present, and of these the vast majority are now induced-draft towers. The trend toward the induced-draft tower has been pronounced only in the last ten years, but it represents a logical transition, since there are advantages to its use which exceed all others except under very special conditions. In the forced-draft type as in Fig. 17.5b the air enters through a circular fan opening and a relatively large ineffective height and volume of tower must be provided as air-inlet space. The air distribution is relatively poor, since the air must make a 90° turn while at high velocity. In the induced-draft tower, on the other hand, the air can enter along one or more entire lengths of wall, and as a result the height of tower required for air entry is very small.

In the forced-draft tower the air is discharged at low velocity from a large opening at the top of the tower. Under these conditions the air possesses a small velocity head and tends to settle into the path of the fan intake stream. This means that the fresh intake air is contaminated by partially saturated air which has already passed through the tower

before. When this occurs, it is known as *recirculation* and reduces the performance capacity of the cooling tower. In the induced-draft tower the air discharges through the fan at a high velocity so that it is driven up into the natural air currents which prevent it from settling at the air intake. In induced-draft towers, however, the pressure drop is on the intake side of the fan, which increases the total fan-power requirements. The higher velocity of discharge of the induced-draft tower also causes a

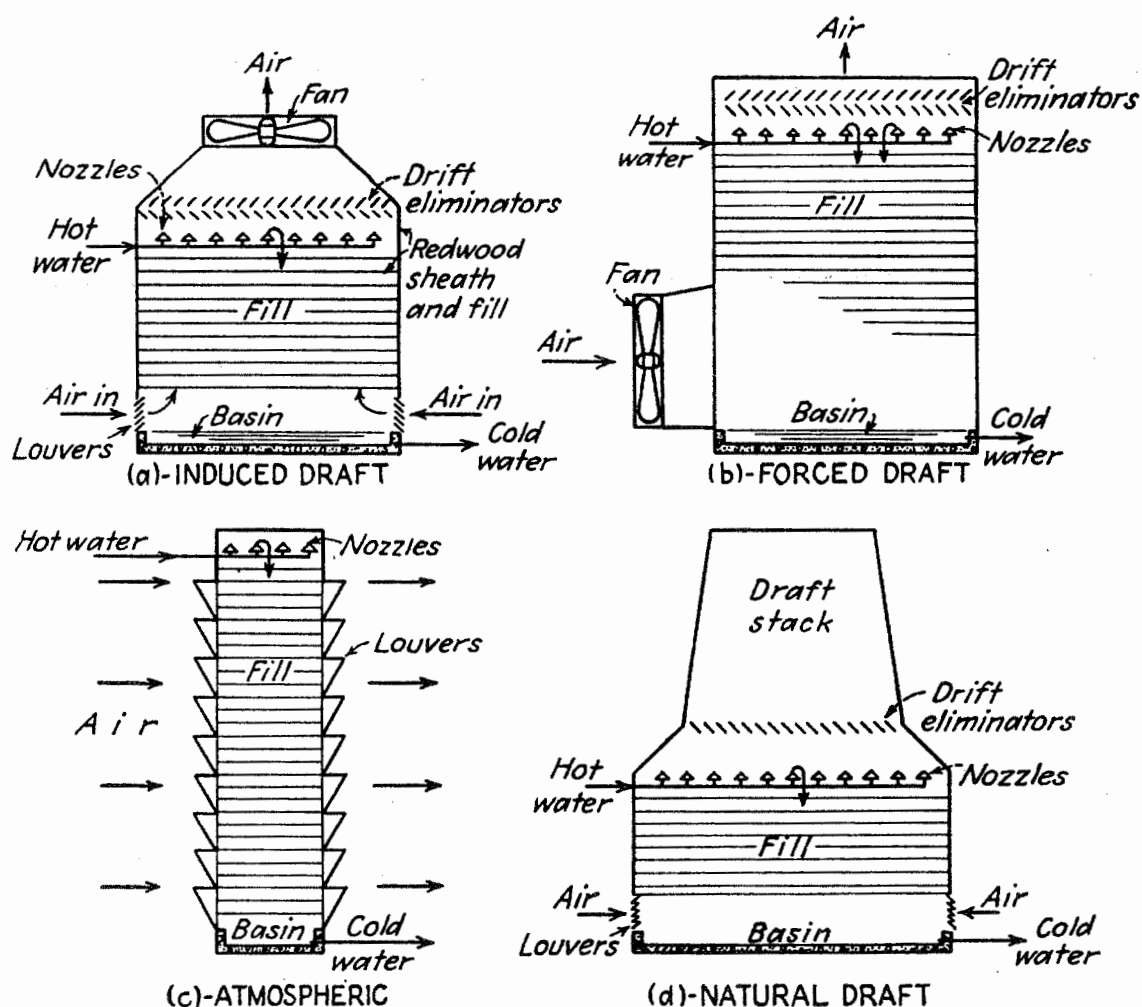


FIG. 17.5. The common types of cooling towers.

somewhat greater entrainment or *drift* loss of water droplets carried from the system by the air stream.

Natural-circulation Towers. In use in Europe and the Orient, the popularity of the types in Fig. 17.5c and d is declining in the United States. This is particularly true of the natural-draft tower.

The atmospheric tower avails itself of atmospheric wind currents. Air blows through the louvered sides in one direction at a time, shifting with the time of year and other atmospheric conditions. In exposed places having average wind velocities of 5 or 6 mph the atmospheric tower may

prove to be the most economical type, and where power costs are high, it may even be preferable with wind velocities as low as $2\frac{1}{2}$ to 3 mph. Since the atmospheric currents must penetrate the entire width of the tower, the towers are made very narrow by comparison with other types and must be very long to afford equal capacity. Towers of this type have been built which are over 2000 ft long. Drift losses occur over the entire side and are greater than for other types. These towers make less efficient use of the available potential, since they operate in crossflow whereas it was shown in Chap. 16 that the most effective use of potentials occurs in counterflow. When cooling water is desired at a temperature close to the wet bulb, this type is incapable of producing it. Atmospheric towers are consequently extremely large and have high initial costs, and when the air is becalmed, they may cease to operate. They have one great advantage, however, in that they eliminate the principal operating cost of mechanical draft towers, *i.e.*, the cost of fan power. In areas with low average wind velocities the fixed charges and pumping costs offset the advantage. Representative average July wind velocities in the United States are shown in Fig. 17.6. An average velocity exceeding 5 or 6 mph is not a sufficient indication that an atmospheric tower is the best. With an average wind velocity of 5 mph the tower will operate at less than design capacity for part of the time. The placement of the tower in a 5-mph wind locality must be such that it is free of obstruction and can avail itself very fully of existing currents.

Natural-draft towers operate in the same way as a furnace chimney. Air is heated in the tower by the hot water it contacts, so that its density is lowered. The difference between the density of air in the tower and outside it causes a natural flow of cold air in at the bottom and the rejection of less dense, warm air at the top. Natural-draft towers must be tall for sufficient buoyancy and must have large cross sections because of the low rate at which the air circulates compared with mechanical draft. Natural-draft towers consume more pumping power. However, they eliminate the cost of fan power and may be more reliable in some localities than are atmospheric towers. In atmospheric towers emphasis must be placed on wind characteristics. In natural-draft towers primary consideration must be given to the temperature characteristics of the air. If it is customary for the air to rise to a high temperature during the day, at least relative to the hot-water temperature, the natural-draft tower will cease to operate during the hot portion of the day. The initial cost and fixed charges on these towers are rather great, and they seem to be passing out of use.

Closely allied to natural-circulation cooling towers is the spray pond consisting of a number of up-spray nozzles, which spray water into the

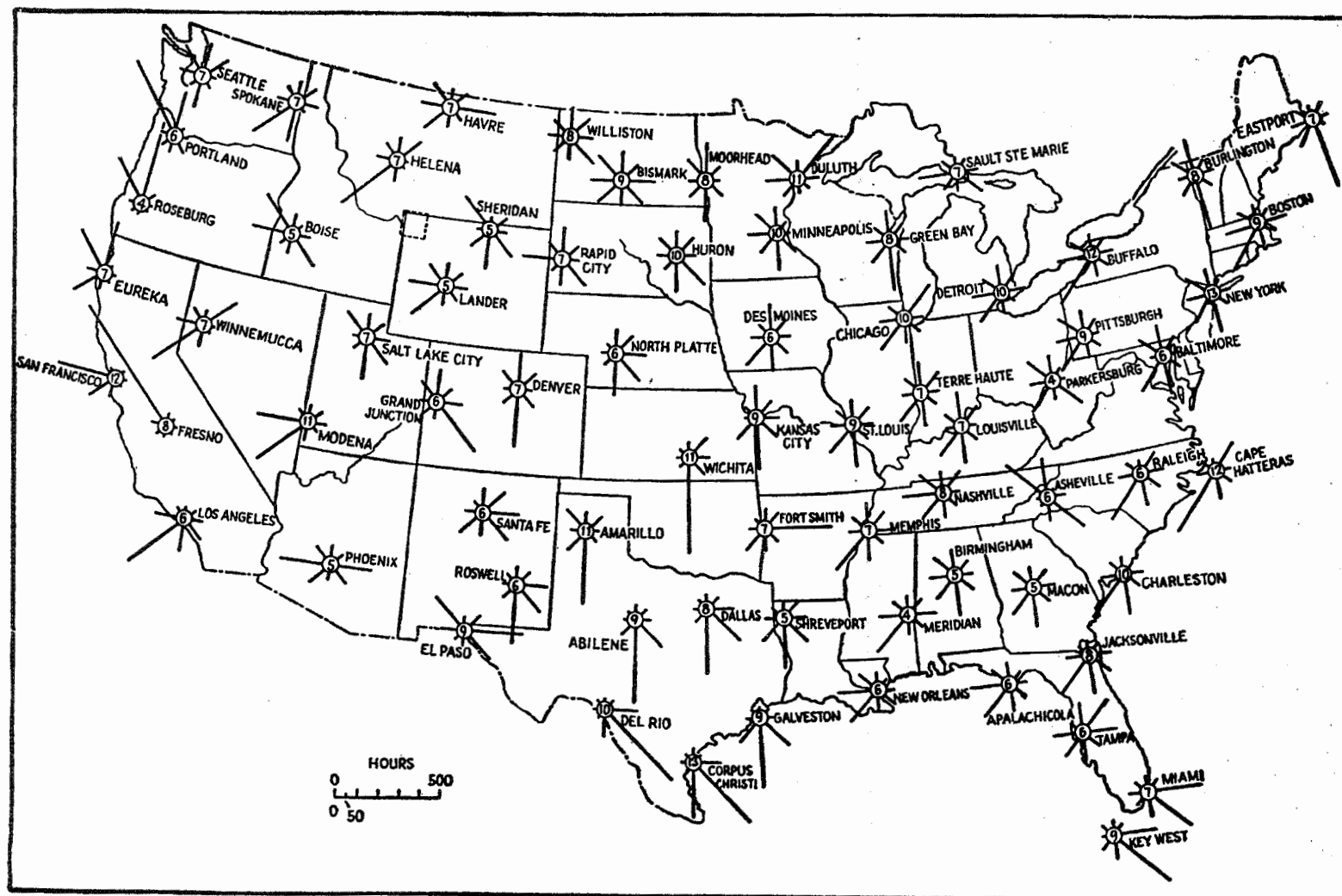


FIG. 17.6. Summer wind data. The length of each line indicates the number of hours in a normal July that the wind blows toward the center circles at each of 71 stations. Figures in circles are average July velocities in miles per hour. (The Marley Company.)

air without inducing air currents. These do not operate with an orderly attempt at air flow and consequently are not capable of producing water approaching the wet bulb as closely as cooling towers. Where the water must be cooled over a short range and without a close approach to the wet-bulb temperature, spray ponds may provide the most economical solution to a water-cooling problem. Drift losses in spray ponds are relatively high. Spray towers are also used widely. They are similar to atmospheric towers except that they employ little or no fill.

Cooling towers are occasionally equipped with bare-tube bundles, which are inserted just above the water basin at the bottom of the tower. These are referred to as *bare-tube* or *atmospheric coolers*. The primary cooling water flows inside the cooler while tower water is continuously circulated over it. The primary cooling water is thus contained in a totally closed system. The calculation of this type of apparatus will be treated in Chap. 20.

Cooling-tower Internals and the Role of Fill. If water passes through a nozzle capable of producing small droplets, a large surface becomes available for air-water contact. Since the water-air interface is also the heat-transfer surface, the use of a nozzle permits the attainment of considerable performance per cubic foot of contact apparatus. This is the principle of the spray pond and the spray tower. Consider a hypothetical spray tower as shown in Fig. 17.7a. Liquid fed to it falls through by gravity. If the tower is 16 ft high and no

initial velocity is imparted to the droplet, it falls in approximate accord with the free-fall law, $z = \frac{1}{2}g\theta^2$, where z is the height, g the acceleration of gravity, and θ the time. A droplet of water will fall through the height of the tower in 1 sec. If liquid is fed at the rate of one droplet per second and there is no obstruction, one droplet will always be present in the tower and one droplet may be continuously removed from the tower per second. The effective surface in the tower of Fig. 17.7a is that of one droplet.

Now suppose that by introducing some geometrical forms on which a droplet may impinge or be deflected it is possible to make a droplet take 4 sec to fall through the height of the tower. Then as shown in Fig. 17.7b, one droplet is fed per second at the top and one droplet is

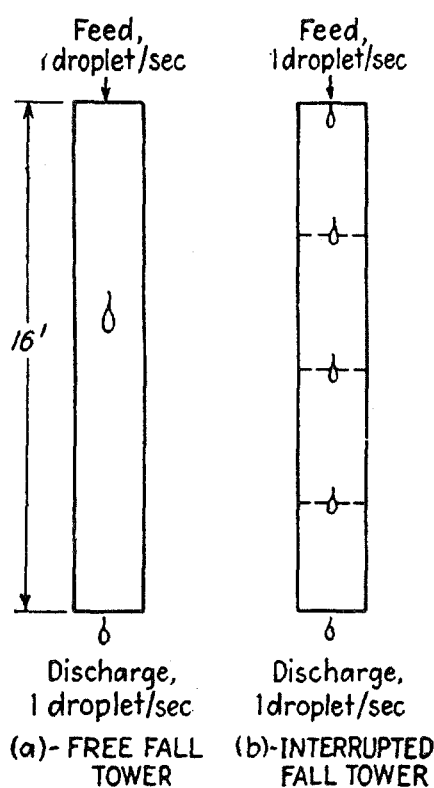


FIG. 17.7. Free and interrupted fall.

continuously removed at the bottom but four droplets remain in the tower. The effective surface in the latter is that of four droplets or four times the free-fall surface. The function of fill is to increase the available surface in the tower either by spreading the liquid over a greater surface or by retarding the rate of fall of the droplet surface through the apparatus. In ordinary diffusion towers such as chemical absorbers the packing is introduced in the form of Raschig rings, Berl saddles, or other objects which are very compact and provide a surface on which the liquid spreads and exposes a large film. This is *film surface*. In the cooling tower, because of the requirement of a large air volume and small allowable pressure drop, it is customary to use spaced wooden slats of triangular or rectangular cross section, leaving the tower substantially unob-

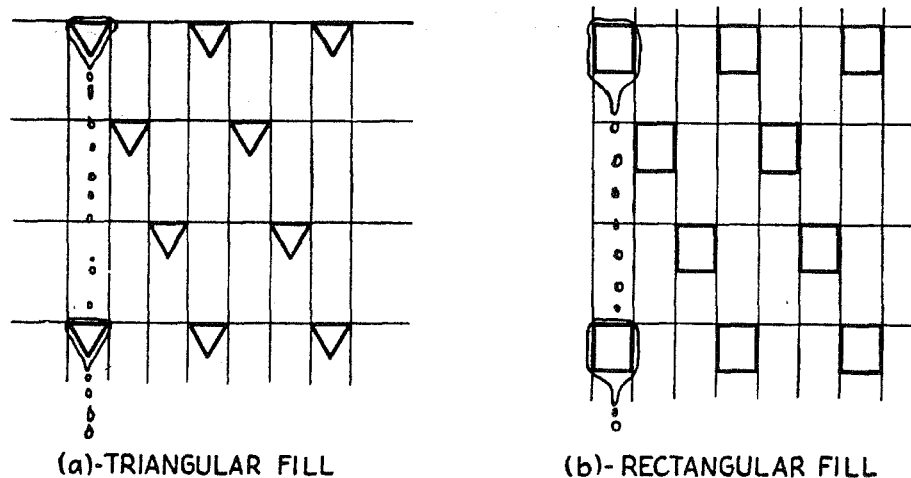


FIG. 17.8 Common types of cooling-tower fill.

structed. The packing, or *fill*, in a cooling tower is almost entirely fabricated in either of the forms of Fig. 17.8, and its purpose is to interrupt the descent of liquid. Although the free space between adjoining fill slats is relatively large, the horizontal projection of the fill does not permit the droplets of liquid to fall through the tower without impinging repeatedly on successively lower slats. Some of the liquid striking the top of a slat spatters, but a large part flows about it and breaks into turbulent flow at the bottom so as to form new droplets automatically and recreate new *droplet surface*. The surface on the sides of the fill is comparable to film surface in packed absorbers. There is, in addition, a large amount of *droplet surface*. More recently the trend has been toward the use of small rectangular slats. They are considerably cheaper to fabricate and install than those of triangular cross section and cause lower pressure drops. The mechanism of producing droplets below each horizontal row relies upon the draining liquid breaking into turbulent flow. Consequently, the method by which the droplets are formed at the top of the tower is of

little consequence to the overall formation of surface as long as there is a uniform distribution of liquid over the entire cross section of the tower.

To demonstrate the effectiveness of these types of fill the free-fall analysis can be carried still further. According to the free-fall equation a droplet will fall 16 ft in 1 sec. In $\frac{1}{2}$ sec a droplet with zero velocity at the top falls approximately 4 ft, and its average velocity is 8 fps. In the second $\frac{1}{2}$ sec it falls the remaining 12 ft with an average velocity of 24 fps. The droplets traverse the lower three-quarters of the tower so quickly that the time of contact in the last three-quarters of the tower equals only that of the first quarter. The advantage of interrupted fall thus becomes apparent: Each time the fall is interrupted (say at each quarter of the tower in Fig. 17.7b) it is as if a droplet with zero velocity were starting to fall anew, and the interrupted tower is equivalent to the effective first quarters of four spray towers in series.

In most cooling towers the liquid is introduced by spraying the water upward at the top so that it travels up and then down before striking the first row of fill. This provides effective contact inexpensively, since the velocity of a droplet traveling upward must decrease to zero when its direction reverses. Another means of increasing the surface or time of contact in spray towers and cooling towers

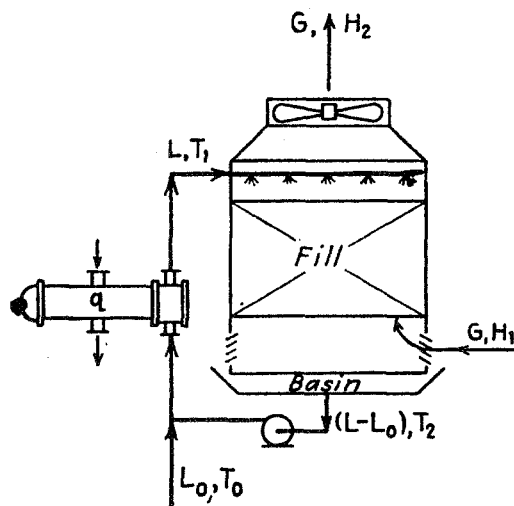


FIG. 17.9. Cooling-tower water and air flow.

is by atomizing the water instead of forming droplets. The same can be achieved with a nozzle instead of an atomizer by using a water-soluble wetting agent. This is not very practical, however, since very fine droplets cannot be caught in the drift eliminators without the expenditure of a large pressure drop. In the construction of cooling towers it is customary to employ droplets of such a size that drift losses can be guaranteed not to exceed 0.25 per cent of the total water circulated over the tower. With the expenditure of a greater fan power this loss can be reduced to 0.10 per cent.

The Heat Balance. Consider the flow diagram in Fig. 17.9. It consists of a cooling tower operating with a heat source in a closed circuit. Water from the basin of the tower is pumped through a battery of surface condensers, where its temperature is raised. The heated water passes back onto the tower along with make-up water, which compensates for the evaporation from the system owing to the saturation of the air passing through the tower.

For convenience, cooling towers are analyzed on the basis of 1 ft² of tower internal ground area. The air rate or *air loading* is taken as the *bone-dry* gas rate per unit of ground area G lb/(hr)(ft²). The *water loading* at the top of the tower is the water rate per unit of ground area L lb/(hr)(ft²), and the make-up rate is L_0 lb/(hr)(ft²). The heat load per hour per square foot q is accordingly the total hourly rate Q divided by the internal ground dimensions of the tower. Using the subscript 1 for the inlet and 2 for the outlet, the balance around the tower in terms of the gas for a 0°F datum is

$$q + L_0CT_0 = G(H_2 - H_1) \quad (17.46)$$

where C is the specific heat of water and H is the enthalpy per pound of bone-dry air and including the heat of the vapor associated with the pound of bone-dry air. In terms of the water loading the heat balance is

$$q = LC(T_1 - T_2) + L_0C(T_2 - T_0) \quad (17.47)$$

Combining both equations,

$$G(H_2 - H_1) = LC(T_1 - T_2) + L_0CT_2 \quad (17.48)$$

The quantity of make-up water required to compensate for evaporation is

$$L_0 = G(X_2 - X_1) \quad (17.49)$$

Dividing Eq. (17.48) by Eq. (17.49)

$$\frac{G(H_2 - H_1)}{G(X_2 - X_1)} = \frac{LC(T_1 - T_2) + L_0CT_2}{L_0} \quad (17.50)$$

$$L_0 \frac{H_2 - H_1}{(X_2 - X_1)} = LC(T_1 - T_2) + L_0CT_2 \quad (17.51)$$

Combining Eq. (17.51) with Eq. (17.47)

$$LC(T_1 - T_2) + L_0CT_2 = q + L_0CT_0 \quad (17.52)$$

$$L_0 = \frac{q}{(H_2 - H_1)/(X_2 - X_1) - CT_0} \quad (17.53)$$

Equation (17.53) gives the quantity of make-up corresponding to any fixed terminal conditions. The enthalpy for saturated air appears on ordinary psychrometric charts. Caution is suggested in the use of these charts, since they invariably have differing datums. Some employ a 0°F air datum and a 32°F water vapor datum to permit the use of the Keenan and Keyes steam tables (Table 7). When a single chart is used, the enthalpy differences are all of sufficient accuracy for heat-transfer calculations. When part of the data is taken from one chart and part from another, serious error may result. For the solution of all air-water

mixtures in this chapter the enthalpy of saturated air above 0°F can be computed by

$$H' = X't + X'\lambda + 0.24t \quad (17.54)$$

where 0.24 is the specific heat of air.

For unsaturated air

$$H = Xt_{DP} + X\lambda_{DP} + 0.45X(t - t_{DP}) + 0.24t \quad (17.55)$$

where 0.45 is the specific heat of water vapor from 0 to 120°F and the subscript *DP* refers to the dew point.

Example 17.1. Calculation of the Enthalpy of Saturated Air. What is the saturation enthalpy of air at 75°F?

At 75°F the saturation partial pressure of water is 0.4298 psia (Table 7).

$$\text{Humidity, } X' = \frac{p_w}{p_t - p_w} \frac{M_w}{M_a} = \frac{0.4298}{14.696 - 0.4298} \times \frac{18}{29} = 0.0187 \text{ lb water/lb air}$$

$$\begin{aligned} \text{Enthalpy above 0°F, } H' &= 0.0187 \times 75 + 0.0187 \times 1051.5 + 0.24 \times 75 \\ &= 39.1 \text{ Btu/lb dry air} \end{aligned} \quad (17.54)$$

The enthalpies of Tables 17.2 and 17.3 have been computed in this manner.

TABLE 17.2. ENTHALPIES AND HUMIDITIES OF AIR-WATER MIXTURES AT 14.7 PSIA

Temp, °F	Vapor pressure, psia	Humidity, lb H ₂ O/lb air	Enthalpy, Btu/lb air	<i>v</i> air, ft ³ /lb	<i>v</i> air + H ₂ O, ft ³ /lb
40	0.1217	0.005	15.15	12.59	12.70
45	0.1475	0.0063	17.8	12.72	12.85
50	0.1781	0.0076	20.5	12.84	13.00
55	0.2141	0.0098	23.8	12.97	13.16
60	0.2563	0.0110	26.7	13.10	13.33
65	0.3056	0.0130	30.4	13.23	13.51
70	0.3631	0.0160	34.5	13.35	13.69
75	0.4298	0.0189	39.1	13.48	13.88
80	0.5069	0.0222	44.1	13.60	14.09
85	0.5959	0.0262	50.0	13.73	14.31
90	0.6982	0.0310	56.7	13.86	14.55
95	0.8153	0.0365	64.2	13.99	14.81
100	0.9492	0.0430	72.7	14.11	15.08
105	1.1016	0.0503	82.5	14.24	15.39
110	1.2748	0.0590	93.8	14.36	15.73
115	1.4709	0.0691	106.7	14.49	16.10
120	1.6924	0.0810	121.5	14.62	16.52
125	1.9420	0.0948	138.8	14.75	16.99
130	2.2225	0.1108	158.5	14.88	17.53
135	2.5370	0.1300	181.9	15.00	18.13
140	2.8886	0.1520	208.6	15.13	18.84
145	3.2810	0.1810	243.8	15.26	19.64
150	3.7180	0.2160	286.0	15.39	20.60

TABLE 17.3. ENTHALPIES AND HUMIDITIES AT VARIOUS ELEVATIONS

Elevations are feet above sea level

Humidity, X' , lb water/lb dry air; enthalpies, H' , Btu/lb dry air

Temp., °F	2000		4000		6000		8000	
	X'	H'	X'	H'	X'	H'	X'	H'
40	0.0056	15.8	0.0061	16.4	0.0065	16.8	0.0070	17.3
45	0.00680	18.4	0.0073	18.9	0.0079	19.6	0.0085	20.3
50	0.0082	21.1	0.0088	21.8	0.0095	22.6	0.0103	23.4
55	0.0099	24.3	0.0106	25.0	0.0115	26.0	0.0124	27.0
60	0.0119	27.7	0.0128	28.7	0.0138	29.8	0.0149	31.0
65	0.01420	31.5	0.0153	32.8	0.0165	34.1	0.0178	35.6
70	0.0170	35.9	0.0183	37.3	0.0197	38.9	0.0212	40.6
75	0.0202	40.7	0.0217	42.2	0.0234	44.4	0.0253	46.5
80	0.0239	46.1	0.0253	47.7	0.0278	50.5	0.0300	53.0
85	0.0284	52.5	0.0306	55.0	0.0330	57.7	0.0358	60.9
90	0.0334	59.4	0.0361	62.4	0.0390	65.7	0.0425	69.6
95	0.0394	67.5	0.0425	71.0	0.0460	75.0	0.0499	80.4
100	0.0465	76.8	0.0501	81.0	0.0544	85.7	0.0590	91.0
105	0.0545	87.2	0.0590	92.4	0.0640	98.1	0.0695	104.3
110	0.0645	99.9	0.0692	105.4	0.0751	111.9	0.0819	119.8
115	0.0750	113.4	0.0812	120.6	0.0885	128.8	0.0964	137.8
120	0.0880	129.8	0.0955	130.3	0.1040	147.0	0.1137	158.8
125	0.1029	148.0	0.1120	158.5	0.1220	170.0	0.1340	183.5
130	0.1208	169.8	0.1317	182.4	0.1440	196.7	0.1580	212.7
135	0.1412	195.1	0.1548	210.6	0.1698	227.9	0.1870	247.6
140	0.1665	225.7	0.1830	244.8	0.2010	265.5	0.2220	289.7
145	0.1965	261.8	0.2190	287.8	0.2420	314.8	0.2690	345.8
150	0.2320	304.5	0.2560	332.3	0.2842	365.0	0.3180	404.0

Heat Transfer by Simultaneous Diffusion and Convection. In the wet-bulb experiment the air and water were initially at the same temperature, although this was not a necessary requirement. By simply referring to Fig. 17.3 it can be seen that the equilibrium represented by the wet-bulb temperature is influenced by the absolute vapor content of the air.

In a cooling tower hot water is cooled by cold air. When water passes through the tower, the water temperature may fall below the dry bulb of the *inlet* air but not lower than the wet bulb of the inlet air. Consider the tower to be divided into two portions. In the upper portion hot water contacts outlet air which is colder than the water. Unlike the wet-bulb experiment, in this case the partial pressure of the water out of the liquid is greater than that in the exit air while the temperature of the water is *also* greater than that of the exit air. *Both* potentials work to decrease the water temperature by evaporation and sensible transfer to the air, thereby increasing the air enthalpy. In this manner,

depending upon the amount of air and the amount of evaporation it is possible for the temperature of the water to fall to or *below* the dry bulb of the inlet air before reaching the bottom of the tower where the air enters. It is the fact that both potentials may operate adiabatically in the same direction while saturating air which makes the cooling tower so effective for cooling water. In the lower portion of the tower the water may possess a temperature equal to or lower than the dry bulb of the air it contacts, and sensible heat and mass transfer are in opposite directions, identical with the wet-bulb experiment. The limit to which the outlet water temperature may then fall in a cooling tower is that which is adiabatically in equilibrium with the inlet air, namely, the wet bulb.

The derivation of the performance of a cooling tower given below is essentially that of Merkel.¹ Since the total heat transfer in a cooling tower is the passage of heat by diffusion and convection from the water to the air,

$$q = q_d + q_c \quad (17.56)$$

where q_d Btu/(hr)(ft²) is the portion transferred by diffusion and q_c Btu/(hr)(ft²) is the portion transferred by convection. In the definition of q it should be remembered that the area implied by its dimensions is the ground area of the tower and not the heat-transfer surface.

If λ is the average latent heat of vaporization of all the water vaporized in the tower,

$$q_d = L_0\lambda \quad (\text{nearly}) \quad (17.57)$$

Combining with Eq. (17.47)

$$q_c = LC(T_1 - T_2) + LC(T_2 - T_0) - L_0\lambda \quad (17.58)$$

and

$$\frac{q_c}{q_d} = \frac{LC(T_1 - T_2) + L_0C(T_2 - T_0) - L_0\lambda}{L_0\lambda} = \frac{LC(T_1 - T_2) + L_0C(T_2 - T_0)}{L_0\lambda} - 1 \quad (17.59)$$

But

$$L_0 = G(X_2 - X_1) \quad (17.60)$$

and

$$G(H_2 - H_1) = LC(T_1 - T_2) + L_0C(T_2 - T_0) \quad (17.48)$$

from which

$$\frac{q_c}{q_d} = \frac{1}{\lambda} \left(\frac{H_2 - H_1}{X_2 - X_1} \right) - 1 \quad (17.61)$$

This is an interesting relationship, since it states that the ratio of the heat transferred by convection to that transferred by diffusion, both in the

¹ Merkel, F., *Forschungsarb.*, **275**, 1-48 (1925).

same direction, is fixed by the inlet and outlet air conditions, which are either known or can be obtained by calculation. While Eq. (17.61) establishes the quantities of heat transfer by convection and diffusion, the ratio of the rates of heat and mass transfer has been set by the Lewis number.

Based on the overall rather than the individual films, the sensible-heat transfer from the water at temperature T to the air at temperature t is given by

$$dq_c = h(T - t)a dV \quad (17.62)$$

where a is the surface of the water per cubic foot of tower as both droplet and film surface and dV is the differential tower volume in which the surface exists. From this $a dV = dA$, where A is the heat-transfer surface. If c is the humid heat of the air defined by $c = 0.24 + 0.45X$,

$$dq_c = Gc dt \quad (17.63)$$

$$dq_a = \lambda dL \quad (17.64)$$

Since dL is the rate at which material diffuses, the differential form of Eq. (17.26) for weight flow is

$$\frac{1}{18} dL = K_g(p' - p)a dV \quad (17.65)$$

where p' is the partial pressure of water corresponding to the water temperature T and p is the vapor pressure in the air. For all practical purposes the humidity can be considered proportional to the partial pressure, at least in the range encountered in cooling-tower applications. Equation (17.65) becomes

$$dL = K_x(X' - X)a dV \quad (17.66)$$

where X' is the humidity at the water temperature T and X the humidity of the air.

Substituting in Eq. (17.64),

$$dq_a = K_x \lambda (X' - X)a dV \quad (17.67)$$

The water evaporated dL increases the humidity of the air stream above its value at the inlet by

$$dL = G dX \quad (17.68)$$

The combined heat transfer dq is then the sum of both modes of transfer

$$dq = dq_c + dq_a = h(T - t)a dV + K_x \lambda (X' - X)a dV \quad (17.69)$$

and

$$dq = G dH \quad (17.70)$$

Equation (17.70) is useful if it can be combined with Eq. (17.69), since it expresses the total heat transfer in the system in heat units alone. In order to avoid the appearance of both X and H in the same equation, the

X values can be factored out. Using an average value for the humid heat c and the latent heat λ and neglecting superheat, all of which are permissible in the relatively close range in which the cooling tower operates, for an air-water mixture consisting of 1 lb of air and X lb of water vapor,

$$H = 1ct + \lambda X \quad (17.71)^*$$

$$G dH = G(c dt + \lambda dX) \quad (17.72)$$

Regrouping Eq. (17.69),

$$dq = K_x a dV \left[\left(\frac{hT}{K_x} + \lambda X' \right) - \left(\frac{ht}{K_x} + \lambda X \right) \right] \quad (17.73)$$

Add and subtract $c(T - t)$.

$$dq = K_x a dV \left[(cT + \lambda X') - (ct + \lambda X) + c(T - t) \left(\frac{h}{K_x c} - 1 \right) \right] \quad (17.74)$$

Substitute Eq. (17.71) into Eq. (17.74).

$$dq = K_x a dV \left[(H' - H) + c(T - t) \left(\frac{h}{K_x c} - 1 \right) \right] \quad (17.75)$$

dq can be expressed in terms of the enthalpy decrease of the total water quantity or the enthalpy increase of the total air mixture, both of which are equal.

$$dq = d(LCT) = G dH \quad (17.76)$$

The gas loading G remains constant throughout the tower because it is based on the bone-dry gas only. The liquid loading is not quite constant, however, owing to the evaporation of water into the bone-dry air. The saturation loss from water to air amounts to less than 2 per cent of the water circulated over the tower and may be considered constant without introducing a serious error.

Hence

$$d(LCT) = LC dT \quad (17.77)$$

and

$$LC dT = G dH \quad (17.78)$$

From Table 17.1, for water diffusing into air the Lewis number is approximately $h/K_{xc} = 1$, and the last term of Eq. (17.75) drops out, thus

$$LC dT = G dH = K_x (H' - H) a dV \quad (17.79)$$

By introducing the mass-transfer-rate equivalent $K_x = h/c$ both modes of heat transfer can be combined by the use of either coefficient, whichever is the easier to obtain. Actually h/K_{xc} for the system air-water is not 1.0 as predicted by the Lewis number. The data of several investi-

* This is an approximation for Eqs. (17.54) and (17.55).

gators indicate that the value of the Lewis number is nearer 0.9. In practice Eq. (17.79) is always evaluated from diffusion potentials, which means that only the convection heat-transfer coefficient is in error if the theoretical value of the Lewis number is used. In most cooling-tower services the convection-heat transfer alone represents less than 20 per cent of the total heat load. The development of equations without the convenient simplification that the Lewis number equals 1.0 will be treated later.

The Analysis of Cooling-tower Requirements. Equation (17.79) is the key equation for the calculation of the design and the analysis of performance in cooling towers. K_x is the overall transfer-rate term analogous to U_c in exchangers, and it should be remembered that there is no dirt factor in direct-contact heat transfer. However, in tubular exchangers the heat-transfer surface is usually known or can be readily computed. In the cooling tower the value of a cannot be determined directly, since it is composed of a random disposition of droplet and film surface. The film surface is nearly independent of the thickness of the film, while the droplet surface is dependent upon both the portion of the liquid loading forming droplets and the average size of the droplets formed. In a tower having interrupted fall other factors will obviously enter. The inability to calculate a is overcome by experimentally determining the product $K_x a$ as a whole for a particular *type* of tower fill and at specific rates of flow for the diffusing fluids comprising the system.

In the development of diffusion theory it was shown that the number of transfer units n_t obtained from Eq. (17.30) provided a useful means of sizing the task to be accomplished in fulfilling a required amount of mass transfer by diffusion. For a particular type of fill if the height of one transfer unit (HTU) is known, the total height of tower required for the task per square foot of ground area is obtained as the product of $n_t(HTU)$. The units of mols and atmospheres are convenient for absorption calculations but the pound is more convenient in diffusion-heat transfer. Equation (17.79) is seen to be the analogue of Eq. (17.29) except that the portion of the total Btu transferred by sensible-heat transfer has first been converted by the Lewis number into an equivalent quantity of Btu mass transfer and then combined with the actual mass transfer since it is a fixed proportion of it. Consequently, $G dH$ is the total heat transfer in the diffusion tower. Rearranging Eq. (17.79),

$$\int \frac{dH}{H' - H} = K_{xa} \frac{V}{G} \quad (17.80)$$

$$n_d = \int \frac{dT}{H' - H} = K_{xa} \frac{V}{L} \quad (17.81)$$

While Eq. (17.80) resembles Eq. (17.30) except for its dimensions, it is not very convenient for use in cooling-tower calculations where the principal interest lies in the temperature of the water produced. Equation (17.80) can be transformed to Eq. (17.81) when multiplied by the ratio G/L and recalling that $C = 1.0$ for water. It is more convenient to use Eq. (17.81), whose value will be called the *number of diffusion units* n_d to avoid confusion with the number of transfer units n_t . If the height of one diffusion unit HDU is known for a given type of fill, the total height of tower required for a given service can be computed.

Determination of the Number of Diffusion Units. The number of diffusion units calculated by Eq. (17.81) is equal to $\int dT/(H' - H)$ and is determined only by the process conditions imposed upon a tower and not the

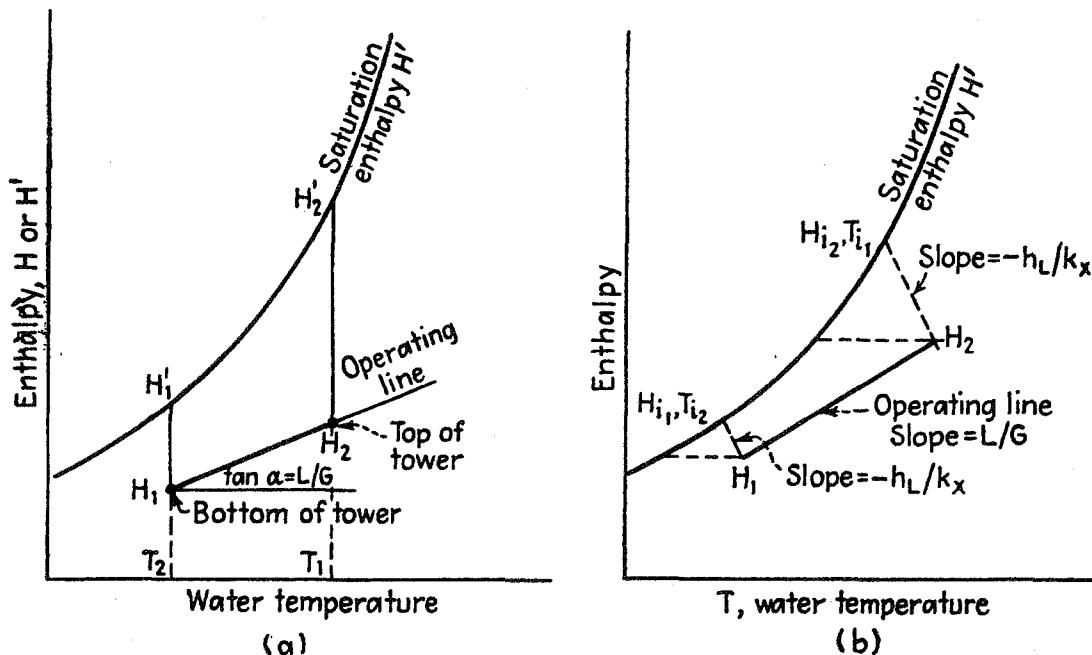


FIG. 17.10. (a) Graphical integration of $dT/(H' - H)$. (b) Correction for the liquid film resistance.

performance of the tower itself. Only the HDU is obtained experimentally. If a quantity of water of a given inlet temperature and a quantity of air of a given wet bulb are to be contacted, it will require a certain number of diffusion units as determined by the integration of Eq. (17.81) to reduce the water to any desired temperature. The number of diffusion units so obtained will be required in any type of tower, whether it be filled, packed, or empty. The height of the tower which is capable of providing the predetermined number of diffusion units varies for each type of fill and with the liquid and gas loadings.

Since the temperature of the water T is not a simple function of H' and H , it is more convenient to carry out the integration of Eq. (17.81) graphically or numerically. Referring to Fig. 17.10a, the tabulated

saturation enthalpies of air H' are plotted against the water temperature T throughout the water range in the tower. The saturation values of H' form a curve which are the values of air saturated at the water temperature and which may be regarded to exist at the air film on the surface of the water. Since the saturation enthalpies also include the saturation humidities, this line is equivalent to the vapor pressure of water out of the water.

The next requirement is to determine the actual enthalpy at any point in the tower. Equation (17.78) states that $LC dT = G dH$, where $C = 1.0$ for water. It relates the change in enthalpy in the gas phase dH to the accompanying change in the water temperature dT in contact with the gas. This change can be represented by rearranging Eq. (17.78) to give

$$\frac{L}{G} = \frac{dH}{dT} \quad (17.82)$$

which is the equation of a straight line in Fig. 17.10a whose slope is the ratio of the liquid to air loading L/G . The value of H at any point on the operating line is given by

$$H_2 = H_1 + \frac{L}{G} (T_2 - T_1) \quad (17.83)$$

since the enthalpy of the inlet air H_1 is known or can be readily determined. As a visual aid it should be understood that the area on the graph between the saturation curve and the operating line is an indication of the size of the potential promoting the total heat transfer. A change in process conditions such that the operating line is moved downward to include a greater area between itself and the saturation line means that fewer diffusion units and a lesser height of any type of tower are ultimately required. At any temperature T in the column between T_1 and T_2 the potential driving heat out of the saturated film at the water surface and into the unsaturated air is the difference between the value of H' and H at that point. By taking small increments of T and dividing by the average difference of $H' - H$ for the increment, the number of diffusion units required to change the temperature of the water is obtained. When the increments are summed up, the total change in the water temperature gives the total value of n_d . As mentioned above, the starting point for the operating line H_1 is obtained from the condition of the atmospheric air entering the tower at the bottom.

When the number of diffusion units is obtained from Eq. (17.80) instead of Eq. (17.81), $(H' - H)$ can be obtained as the log mean of the enthalpy difference at the bottom and top of the tower. In Fig. 17.10a this is equivalent to taking the area between the operating line and a

straight line drawn between H'_1 and H'_2 . The potential obtained in this manner is greater than the actual potential and causes errors which may be on the unsafe side. To obtain the number of diffusion units from the $\int dH/(H' - H)$, the expression must be multiplied by G/L .

Correction for the Liquid-film Resistance. From Eqs. (17.46) to (17.82) the derivation was based upon the assumption that the overall coefficient and the gas-side coefficient were identical. This, of course, implied that the liquid film does not offer any significant resistance to diffusion and that the gas side is controlling. This may be in error, particularly when the humidifying liquid is an aqueous solution. McAdams¹ has devised an excellent graphic method of allowing for the resistance of the liquid film. In writing Eq. (17.62) it was assumed that h was identical with h_g , which is the heat-transfer coefficient from the liquid-film-air-film interface to the air. When there is an appreciable liquid-film resistance, h in Eq. (17.62) should more correctly be written as an overall coefficient U , since it is the resultant of h_g and h_L , the latter being the convection coefficient from the liquid film to the interface. h without a subscript will be retained for cooling-tower applications in place of U because it is consistent with much of the literature. For the actual relationship when there is a significant liquid-film resistance, rewrite Eq. (17.62).

$$LC dT = h_L(T - T_i)a dV \quad (17.62a)$$

which is the rate of transfer from the liquid body through the liquid film to the liquid-air interface. The rate of sensible transfer from the interface through the gas film to the gas body is given by

$$Gc dt = h_g(T_i - t)a dV$$

The analogue of Eq. (17.79) may be obtained in terms of the interface.

$$G dX = k_x(X_i - X)$$

Applying the Lewis number $h_g/k_xc = 1$, where k_x is the gas-side diffusion coefficient,

$$G dH = k_x(H_i - H)a dV \quad (17.79a)$$

Equating (17.62a) and (17.79a) gives

$$\frac{h_L}{k_x} = \frac{H_i - H}{T_i - T}$$

Referring to Fig. 17.10b, a line of negative slope $-h_L/h_x$ is drawn from each of the terminal points of the operating line and the potential is the area included between the saturation line, the operating line, and the

¹ McAdams, W. H., "Heat Transmission," 2d ed., p. 290, McGraw-Hill Book Company, Inc., New York, 1942.

two lines of negative slope. The interface values of the enthalpies and temperatures are marked accordingly.

Cooling-tower Process Conditions. From the standpoint of tube corrosion 120°F is the maximum temperature at which cooling water ordinarily emerges from tubular equipment. If some of the liquid products in a plant are cooled below 120°F, the outlet-water temperature is usually lower than 120°F in order to prevent an appreciable temperature cross inside the tubular equipment. The temperature of the water to a cooling tower rarely exceeds 120°F and is usually less. When the temperature of the water from a process is over 120°F, the increased evaporation may justify the use of an atmospheric cooler to prevent the direct contact between hot water and air.

The lowest possible temperature to which water can be cooled in a cooling tower corresponds to the wet-bulb temperature of the air. This is *not* a practical limit, however, since the vapor pressure out of the water and in the air will be the same when the water attains the wet-bulb temperature, resulting in a zero diffusion potential for which an infinite tower is required. The difference between the water-outlet temperature from the tower T_2 and the wet-bulb temperature is called the *approach* in a cooling tower. Since most cooling towers operate over nearly the same water range, the approach is the principal index of how difficult the task will be as well as an indication of the number of diffusion units which will be required.

In Fig. 17.11 is shown a map of the United States on which are recorded the wet-bulb temperatures which are not exceeded for more than 5 per cent of the total hours for the four summer months from June through September. In the northeastern portion of the country the 5 per cent wet-bulb temperature is about 75°F, and it is customary in this region to cool the water in a tower to a 10° approach, or 85°F. On the Gulf Coast the 5 per cent wet-bulb temperature is about 80°F, and in these regions it is customary to invest more in cooling towers by cooling to a 5° approach, or again 85°F. Countless installations have justified the economics of these approaches. It must be realized, however, that, when a tower is designed for a definite approach to the 5 per cent wet bulb, there will be some hours during which the water issuing from the tower will be warmer than originally specified in the process conditions.

When a process involves volatile materials or vacuum operation, the 5 per cent wet bulb is discarded for another basis of selection. The approach may be selected as a compromise between the values in Fig. 17.11 and a compilation of reasonable maximum temperatures. Such a compilation is contained in Table 17.4. These are not the all-time maximums but temperatures which are rarely exceeded. As an example, dur-

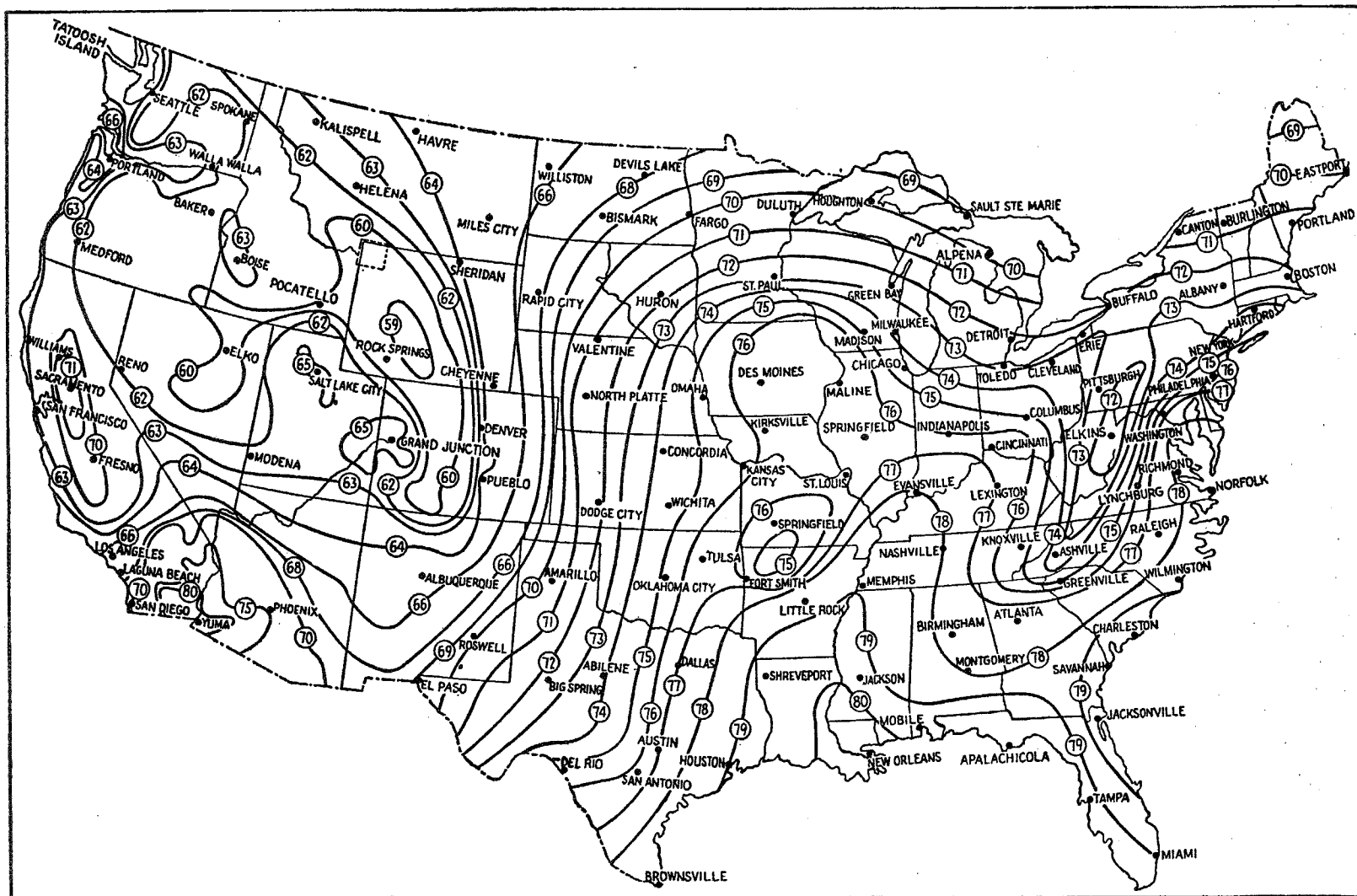


FIG. 17.11. Summer wet-bulb temperature data. The wet-bulb temperatures shown will be exceeded not more than 5 per cent of the total hours during June to September, inclusive, of a normal summer. (*The Marley Company.*)

TABLE 17.4. REASONABLE MAXIMUM TEMPERATURES AND ABSOLUTE MAXIMUM WIND VELOCITIES IN THE UNITED STATES*

State	City	Reasonable maximum, Temp, °F			Max recorded wind velocity, mph
		Dry bulb	Wet bulb	Dew point	
Alabama.....	Mobile	95	82	80	87
Arizona.....	Phoenix	113	78	74	40
Arkansas.....	Little Rock	103	83	78	49
California.....	Fresno	110	75	66	41
	Laguna Beach	82	70	68	
	Oakland	94	68	67	50
	San Diego	88	74	72	43
	Williams	110	80	71	
Colorado.....	Denver	99	68	64	53
	Grand Junction	102	68	63	
Connecticut.....	Hartford	...	82	..	58
District of Columbia.....	Washington	99	84	80	55
Florida.....	Jacksonville	99	82	79	58
	Miami	92	81	79	87
Georgia.....	Atlanta	101	82	79	51
Idaho.....	Boise	109	71	65	43
Illinois.....	Chicago	104	80	77	65
	Moline	103	83	80	
Indiana.....	Evansville	102	82	79	60
Kansas.....	Wichita	110	79	74	68
Louisiana.....	New Orleans	95	83	81	66
	Shreveport	102	83	79	50
Massachusetts.....	Boston	96	78	76	60
Michigan.....	Detroit	101	79	76	67
Minnesota.....	St. Paul	103	79	75	78
Mississippi.....	Jackson	103	83	80	49
Missouri.....	Kansas City	109	79	76	57
	Kirksville	108	82	79	
	St. Louis	108	81	76	91
	Springfield	98	79	76	52
Montana.....	Helena	97	70	60	54
Nebraska.....	North Platte	104	76	71	73
	Omaha	108	80	75	53
Nevada.....	Elko	101	64	58	
	Reno	102	66	56	46
New Jersey.....	Camden	102	82	78	68
	Newark	99	81	77	
New Mexico.....	Albuquerque	98	68	66	63
New York.....	Albany	97	78	75	59
	Buffalo	93	77	74	73
	New York	100	81	78	73

TABLE 17.4. REASONABLE MAXIMUM TEMPERATURES AND ABSOLUTE MAXIMUM WIND VELOCITIES IN THE UNITED STATES.*—(Continued)

State	City	Reasonable maximum, Temp, °F			Max recorded wind velocity, mph
		Dry bulb	Wet bulb	Dew point	
North Carolina.....	Raleigh	98	82	80	45
	Wilmington	94	81	79	53
North Dakota.....	Fargo	105	..	..	58
Ohio.....	Cincinnati	106	81	78	54
	Cleveland	101	79	76	60
Oklahoma.....	Tulsa	106	79	77	
Oregon.....	Portland	99	70	68	43
Pennsylvania.....	Bellefonte	96	78	75	
	Pittsburgh	98	79	74	56
South Carolina.....	Charleston	98	82	80	81
South Dakota.....	Huron	106	76	74	56
	Rapid City	103	71	66	
Tennessee.....	Knoxville	100	79	76	59
	Memphis	103	83	80	58
Texas.....	Amarillo	101	75	70	65
	Brownsville	96	80	79	80
	Dallas	105	80	76	63
	El Paso	101	72	69	60
	Houston	100	81	79	63
	San Antonio	102	83	82	56
	Modena	97	66	61	
Utah.....	Salt Lake City	102	68	64	53
Washington.....	Seattle	86	70	67	59
	Spokane	106	68	58	41
Wisconsin.....	Green Bay	99	79	75	53
	La Crosse	100	83	81	69
Wyoming.....	Rock Springs	91	62	58	

* Courtesy the Marley Company.

ing a 14-year period the wet-bulb temperature in New York City only twice exceeded 80°F for 7 hr duration or more. On one occasion a momentary wet bulb of 88°F was recorded. The two occasions on which the wet bulb exceeded 80°F were so rare that it is simpler to justify reducing the plant capacity or shutting down altogether during these infrequent occurrences rather than to use 80+°F as the basis for design. Since the 5 per cent wet bulb for New York is 75°F, only critical requirements can justify the use of a temperature between 75°F and the reasonable maximum of 81°F. Cooling towers are usually designed to withstand a wind velocity of 100 mph, which is equivalent to 30 lb/ft².

In the study of cooling towers the impression sometimes arises that the cooling tower cannot operate when the inlet air is at its wet-bulb temperature. This, of course, is not so. When air at the wet-bulb temperature enters the tower, it receives sensible heat from the hot water, and its temperature is raised thereby so that it is no longer saturated. Water then evaporates continuously into the air as it travels upward in the tower.

One of the objectionable characteristics of cooling towers is known as *fogging*. When the hot and saturated exit air discharges into a cold atmosphere, condensation will occur. It may cause a dense fog to fall over a portion of the plant with attendant safety hazards. If provision is made during the initial design, condensation can be reduced by any means which reduces the outlet temperature of the air. If it is desired to maintain a fixed range for the cooling water in coolers and condensers, fogging can be reduced by the recirculation of part of the basin water back to the top of the tower where it combines with hot water from the coolers and condensers. This reduces the temperature of the water to the tower, while the heat load is unchanged. The principal expense of the operation, aside from initial investment, will be that of the pumping power for the recirculation water, which does not enter the coolers and condensers.

Humidification Coefficients. An apparatus in which air and water can be brought into intimate contact may serve as a cooling tower or air humidifier. Considerable information can be found in the literature on the performance of a variety of packing and fills. Simpson and Sherwood¹ have given an excellent review of some of the pertinent data in the literature as well as some original data on fills suitable for cooling towers. Because of the outgrowth of modern diffusion calculations from absorption practices in the chemical industries, much of the data in the literature is represented by a plot of $K_G a$ vs. G for systems in which the gas film is controlling. This method now appears to be giving way to plots of the HTU vs. G . Colburn² has recalculated much of the early data on this basis. The relation between the HTU and $K_G a$ is

$$HTU = \frac{Z}{K_G a} \frac{V}{G}$$

where Z is the total height containing n_t transfer units. Both calculations are based on the use of the pound-mols of water transferred and a driving potential expressed in atmospheres. There seems little reason to employ these units in humidification calculations, since the pounds of

¹ Simpson, W. M., and T. K. Sherwood, *Refrig. Eng.*, 535 (1946).

² Colburn, A. P. *Trans. AIChE*, 29, 174 (1939).

water transferred and a driving potential in humidity units are so convenient. No data have appeared in the literature so far in which HDU , the height of a diffusion unit, is plotted against G for the humidification of air, but data have appeared with K_xa plotted against G . The relation between the HDU and K_xa is

$$HDU = \frac{Z}{K_xa} \frac{V}{L} \quad (17.84)$$

and the relation between the K_xa and K_ga can be obtained from Eq. (17.40). The HDU or K_xa are the performance characteristics of the given fill, or packing, and n_d is the size of the job required for the fulfillment of process conditions. In the case of packed towers containing small objects it is possible to report the data over wide ranges by an equation of the form

$$K_xa = C_1 G^\gamma \quad (17.85)$$

If the value of K_xa is multiplied by the ratio V/L , the number of diffusion units in a given height can be obtained, since $V = 1Z$.

If a tower is in operation and it is desired to determine its performance characteristics such as the HDU or K_xa , the number of diffusion units being performed must be calculated first from the observed inlet and outlet temperatures, humidities, and flow rates. The total packed, or filled, height divided by the value of n_d calculated from observed data will give the HDU . Of the data available in the literature only those for Raschig rings and Berl saddles are given here, since other types of packings and fill are less standardized and in some instances difficult to reproduce. The data in Table 17.5 have been published in part by McAdams¹ from tests by Parekh² and have been made available through the courtesy of Dean T. K. Sherwood. Data on fills producing droplet surface cannot be represented quite so readily as film-type packings, since the total droplet surface changes greatly with the number of droplets formed. This in turn is influenced by the liquid loading.

Caution is emphasized in the application of the information in the literature to use in actual services. Many of the data have been obtained on small packed, or filled, heights or on laboratory-scale apparatus of small cross section. A problem in the design of large towers is the attainment of a uniform distribution of both air and water over the entire cross section and height of the tower. A tower of large cross section should be easier to control than a small one because of the lower ratio of wall perimeter to cross section, which, in a sense, is a rough index of the fraction of the liquid flowing down the tower walls. Since the water is

¹ McAdams, W. H., *op. cit.*, p. 289.

² Parekh, M., Sc. D. Report in *Chem. Eng.*, MIT, (1941).